



Lecture
Notes

2026

Introduction to Elementary Particles

Prof. Yonit Hochberg

77609 · HUJI Physics · Eran Rehani

Contents

I	Field Theory Foundations	4
1	Lecture 01 — Introduction: Particle Physics, Classical Field Theory, and the Limits of Non-Relativistic Quantum Mechanics	4
2	Lecture 02 — Euler–Lagrange Equations for Fields, Basic Examples, and Relativistic Notation	9
3	Lecture 03 — Canonical Quantisation of the Real Scalar Field	17
II	Feynman Diagrams & Scattering	23
4	Lecture 04 — Perturbation Theory and Feynman Diagrams for the Harmonic Oscillator	23
5	Lecture 05 — Feynman Rules for Scalar Fields and Mandelstam Variables	30
6	Lecture 06 — Connected Diagrams, Loop Divergences, 1PI, and Decay Rates	39
7	Lecture 07 — Two-Body Decays, Cross-Sections, Symmetry Factors, and Fermions	48
III	Symmetries & the Standard Model	59
8	Lecture 08 — Fermion Bilinears, Renormalizability, and the Road to the Standard Model	59
9	Lecture 09 – Model Building, Global Symmetries, and Charges	68
10	Lecture 10 – Product Groups, Accidental Symmetries, and Local Gauge Invariance	77
11	Lecture 11 – QED: Interactions, Processes, and Two Generations	85
IV	Non-Abelian Gauge Theory & QCD	95
12	Lecture 12 – SM Charged Fermions, Accidental Symmetries, and Introduction to Non-Abelian Symmetries	95
13	Lecture 13 – Non-Abelian Symmetries: Representations, Invariants, and Gauge Theory	102
13.1	From $U(1)$ to $SU(N)$: Multiplets and Generators	102
13.2	The Two Representations You Must Know	103
13.3	Product Groups: One Index per Factor	104
13.4	Building Invariants: the Singlet Rule	105
13.5	$SU(3)$ Colour and the Bound States of QCD	106
13.6	Local Non-Abelian Invariance: Gauging $SU(N)$	107

14 Lecture 14 – QCD Interactions, Running Couplings, Confinement, and Accidental Symmetries	112
14.1 Running Coupling Constants	112
14.2 Defining QCD	113
14.3 The QCD Lagrangian	114
14.4 Interactions in QCD	115
14.5 The Parameters of QCD	116
14.6 Confinement and the Running of g_s	116
14.7 Accidental Symmetries of QCD	118
15 Lecture 15 – QCD + QED and Spontaneous Symmetry Breaking	121
15.1 Overview	121
15.2 Combining QCD with Electromagnetism	121
15.3 What Spontaneous Symmetry Breaking Means	124
15.4 Discrete Example: A Real Scalar with $\mathbb{Z}_2$	124
15.5 Continuous Example: Global $U(1)$	127
15.6 Tachyons and the Meaning of Negative Mass Squared	129
15.7 Goldstone Bosons	129
16 Lecture 16 – Non-Abelian SSB, Fermion Masses, and the Higgs Mechanism	131
16.1 Overview	131
16.2 Non-Abelian Global SSB: $SO(3) \rightarrow SO(2)$	131
16.3 Fermion Masses from Spontaneous Symmetry Breaking	134
16.4 The Higgs Mechanism: SSB of a Local $U(1)$	136
V Homework Solutions	138
17 Homework Set 1	138
17.1 Brief review of relativity	138
17.2 Natural Units	139
17.3 Relativistic kinematics: Colliders vs. fixed targets	140
17.4 Lagrangian formalism of E&M	141
18 Homework Set 2	145
19 Homework Set 3	149
20 Homework Set 4	156
21 Homework Set 5	161

Part I

Field Theory Foundations

1 Lecture 01 — Introduction: Particle Physics, Classical Field Theory, and the Limits of Non-Relativistic Quantum Mechanics

Date: 22/03/2026

Overview

This lecture opens the course by framing the central question of particle physics — *what is the Lagrangian of nature?* — and presenting the Standard Model as the current, symmetry-based answer. Hochberg argues that non-relativistic quantum mechanics is fundamentally insufficient because it cannot describe the creation and annihilation of particles; the correct framework is relativistic quantum field theory (QFT), valid up to the energies probed so far (a few TeV). The lecture then introduces the building block that makes this transition possible: the *classical field*, a quantity depending on both space and time. By reviewing the Lagrangian formalism of classical mechanics and generalizing the role of the generalized coordinate from a particle position $x(t)$ to a field $\phi(\mathbf{x}, t)$, Hochberg sketches the logical path from classical mechanics $\rightarrow$ classical field theory $\rightarrow$ quantum field theory that the rest of the course will follow.

Definitions

Physical Concept 1.1: Particle physics / high-energy physics

The branch of physics that asks: *what are the fundamental laws of nature?* The working assumption is that nature obeys the *principle of minimal action*, so the question reduces to: what is the Lagrangian (or Hamiltonian) of nature?

Physical Concept 1.2: Standard Model (SM)

The current best answer to the above question: a relativistic quantum field theory built on axioms and symmetry principles that describes all known elementary particles and three of the four fundamental forces.

Physical Concept 1.3: Field

A *field* is a quantity ϕ that depends on *both* space and time:

$$\phi = \phi(\mathbf{x}, t). \quad (1)$$

A field may be:

- a *scalar field* — a single number at each spacetime point (e.g. temperature $T(\mathbf{x}, t)$, density $\rho(\mathbf{x}, t)$);
- a *vector field* — a vector at each spacetime point (e.g. wind velocity $\mathbf{W}(\mathbf{x}, t)$, magnetic field $\mathbf{B}(\mathbf{x}, t)$);
- a *tensor field* — a tensor at each spacetime point (examples deferred to relativistic notation).

Physical Concept 1.4: Vector field (precise)

A collection of component fields that *transforms as a vector under spatial rotations*. While one may formally decompose $\mathbf{B} = (B_x, B_y, B_z)$ into three independent scalar functions, it is the rotational transformation property

$$B_i \xrightarrow{\text{rotation}} R_{ij} B_j \quad (2)$$

that promotes them to a single vector field.

Physical Concept 1.5: UV cutoff

The energy scale (equivalently, the shortest length scale) at which a given field description ceases to be valid. For the electromagnetic field and QED, the UV cutoff is believed to be at or below the *Planck scale*. For macroscopic fields (temperature, density), the UV cutoff is set by the microscopic structure (e.g. interatomic spacing).

Physical Concept 1.6: IR cutoff

The complementary scale: the *largest* distance (lowest energy) at which a given field description remains valid.

Physical Concept 1.7: Classical field theory

A dynamical theory whose *generalized coordinates are fields* $\phi(\mathbf{x}, t)$ rather than particle positions $x_i(t)$. Solving a classical field theory means finding $\phi(\mathbf{x}, t)$ for all $\mathbf{x}$ and t , given initial data — a function rather than a finite set of numbers.

Key Results

Physical Law 1.1 Lagrangian of classical mechanics: standard form

In fundamental physics, the Lagrangian for a system with generalized coordinates $\{x_i\}$ takes the form

$$L = L(x_i, \dot{x}_i), \quad (3)$$

with *no* explicit time dependence (to ensure energy conservation) and dependence on *at most* first time-derivatives (standard assumption). The equation of motion follows from the *Euler–Lagrange equation*:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} - \frac{\partial L}{\partial x_i} = 0. \quad (4)$$

Each degree of freedom x_i requires *two* initial conditions (e.g. $x_i(t_0)$ and $\dot{x}_i(t_0)$).

The central boxed statement of the lecture — the key conceptual shift:

$$\underbrace{x_i(t)}_{\text{mechanics}} \longrightarrow \underbrace{\phi(\mathbf{x}, t)}_{\text{field theory}} \quad (\text{generalized coordinate becomes the field}) \quad (5)$$

Physical Law 1.2 Why non-relativistic QM is insufficient

Non-relativistic quantum mechanics conserves particle number: the normalization $\int |\psi|^2 d^3x = 1$ encodes the assumption that the particle is *always present*. In reality, particles are created and annihilated (e.g. $e^+e^- \rightarrow \gamma\gamma$, nuclear decays, collider events). Describing such processes requires a framework where particle number is not conserved — quantum field theory.

Physical Law 1.3 Scope of QFT

Relativistic quantum field theory is a good description of nature at all energy scales probed so far, up to $\mathcal{O}(\text{few TeV})$. Beyond this scale (towards the Planck scale), the framework may break down, but no experimental violation has been found.

Derivations

Review: principle of minimal action and the Euler–Lagrange equation

The *action* is defined as

$$S[x_i] = \int_{t_1}^{t_2} L(x_i, \dot{x}_i) dt. \quad (6)$$

The *principle of minimal action* states that the physical trajectory $x_i(t)$ is the one that makes S stationary, $\delta S = 0$. Varying (6):

$$\begin{aligned} \delta S &= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x_i} \delta x_i + \frac{\partial L}{\partial \dot{x}_i} \delta \dot{x}_i \right) dt \\ &= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} \right) \delta x_i dt + \left[\frac{\partial L}{\partial \dot{x}_i} \delta x_i \right]_{t_1}^{t_2}. \end{aligned} \quad (7)$$

With fixed endpoints $\delta x_i(t_1) = \delta x_i(t_2) = 0$, the boundary term vanishes. Since δx_i is arbitrary, $\delta S = 0$ requires (4).

Physical meaning of no explicit time dependence. If L has no explicit t , then by Noether’s theorem the associated conserved charge is the *Hamiltonian* (energy). Hence, the assumption $\partial L / \partial t = 0$ is the mathematical statement that energy is conserved.

Symmetry constraints on the Lagrangian: examples

Hochberg illustrated how symmetry requirements constrain the allowed form of the Lagrangian. Take $L = \frac{1}{2}m\dot{x}^2 - V(x)$ in one spatial dimension.

Example 1: $\mathbb{Z}_2$ symmetry $x \rightarrow -x$. Requiring invariance of L under $x \rightarrow -x$ means $V(-x) = V(x)$, i.e. V must be an *even* function of x (equivalently, a function of $|x|$).

Example 2: $SO(3)$ rotational invariance in 3D. Promoting $x \rightarrow \mathbf{x} \in \mathbb{R}^3$ and requiring invariance under $\mathbf{x} \rightarrow R\mathbf{x}$ for all $R \in SO(3)$ forces V to depend only on $|\mathbf{x}|$:

$$V = V(|\mathbf{x}|) \quad (\text{central potential}). \quad (8)$$

The general lesson: *symmetry principles constrain and ultimately determine the allowed form of the Lagrangian*. This will be the central strategy for model building later in the course.

Examples

Worked Example 1.1 (Familiar fields from everyday life)

- **Temperature $T(\mathbf{x}, t)$:** scalar field. Valid only at scales larger than the interatomic spacing (UV cutoff $\sim \text{\AA}$).
- **Wind velocity $\mathbf{W}(\mathbf{x}, t)$:** vector field.
- **Population density $\rho(\mathbf{x}, t)$:** scalar field. Valid only at scales larger than the typical size of a person.
- **Magnetic field $\mathbf{B}(\mathbf{x}, t)$:** vector field. Its three components (B_x, B_y, B_z) transform as a vector under rotations, which is why we bundle them together rather than treating them as three independent scalars.

Worked Example 1.2 (Solving mechanics vs. solving field theory)

In *classical mechanics* with one degree of freedom, solving the problem means finding $x(t)$ — a function of one argument — given two initial conditions $x(t_0)$ and $\dot{x}(t_0)$.

In *classical field theory* in D spatial dimensions, solving the problem means finding $\phi(\mathbf{x}, t)$ — a function of $D+1$ arguments. The “initial condition” is now an *entire function* $\phi(\mathbf{x}, t_0)$ (and its time derivative), i.e. infinitely many numbers.

Connections

- **Analytical Mechanics.** Hochberg explicitly stated that the course assumes knowledge of analytical mechanics at the level of the Lagrangian formalism, Euler–Lagrange equations, and the principle of minimal action. Classical field theory is presented as a direct generalization of that framework.
- **Quantum Mechanics (two semesters).** The course also assumes two semesters of QM. The limitations of non-relativistic QM (fixed particle number, non-covariant Schrödinger equation) motivate the move to QFT.
- **Electromagnetism.** Basic knowledge of E&M is assumed. The electromagnetic field $(\mathbf{E}, \mathbf{B})$ is used as the canonical example of a physical vector field.
- **Quantum Field Theory.** This course builds a “baby version” of QFT: classical field theory $\rightarrow$ second quantization $\rightarrow$ perturbation theory $\rightarrow$ Feynman diagrams. Students are not assumed to have taken QFT previously.

- **Standard Model.** The SM is the long-range target. The course works towards it by building QED, QCD, and spontaneous symmetry breaking from first principles.
- **Grossman & Nir (draft/book).** From roughly weeks 5–6 onward, the course follows a draft by Yossi Nir and Yuval Grossman (now also a published book), available on Moodle as a PDF for internal course use only.

To Remember

- The central question of particle physics is: *what is the Lagrangian of nature?* The answer, as of now, is the Standard Model.
- Nature obeys the *principle of minimal action*; all dynamics follows from $\delta S = 0$ with $S = \int L dt$.
- Non-relativistic QM cannot describe particle creation and annihilation. The correct framework is relativistic QFT.
- In QFT, particles are *excitations of fields*, not fundamental objects in their own right.
- A field $\phi(\mathbf{x}, t)$ is the generalized coordinate of field theory; it replaces the particle position $x(t)$ of mechanics.
- Every field description has a **UV cutoff** (shortest scale of validity) and an **IR cutoff** (longest scale). For QED, the UV cutoff is at or below the Planck scale.
- Symmetry requirements constrain the form of the Lagrangian: this will be the main tool for model building.

Exam / HW Hints

- **HW is mandatory for exam eligibility:** all sets must be submitted; at most one set may be skipped. If an extension is needed, email Hochberg in advance — she is flexible if asked.
- The **first HW set** will include: (a) deriving additional constraints on Lagrangians from symmetry requirements (parity, rotational invariance, etc.), extending the examples worked in class; (b) elements of the Lagrangian construction for classical mechanics. Hochberg explicitly pointed to the in-lecture examples ($x \rightarrow -x$ symmetry, rotational invariance) as representative of what the homework will ask.
- Hochberg explicitly warned against submitting AI-generated solutions without understanding them; the exam requires genuine mastery of the material.
- **Next topic:** relativistic notation and natural units. A COVID-era Panopto recording (2 hours) may be posted if the Thursday session cannot be held live.

Further Reading

No specific textbook was cited in this lecture. The following are recommended:

- **Griffiths, *Introduction to Elementary Particles* (2nd ed.), Chapter 1** — “Historical Introduction to the Elementary Particles”: the cast of characters (quarks, leptons, bosons), the four forces, and the logic behind the Standard Model. Ideal companion to this introductory lecture.

- **Griffiths, Chapter 6.1** — Brief derivation of Fermi’s Golden Rule in the particle-physics context, motivating the $(2\pi)^4\delta^{(4)}$ phase-space factor that appears throughout the course.
- **Halzen & Martin, *Quarks and Leptons*, Chapter 1** — “Quarks and Leptons”: a concise survey of the known particles and their quantum numbers (I , J^{PC} , colour). Good reading alongside the SM overview from this lecture.
- **Tong, *Lectures on Particle Physics*, Chapter 1** (freely available at damp.cam.ac.uk/user/tong) — a streamlined account of what a relativistic quantum field theory is and why particles arise from fields.
- **PDG Summary Tables** — pdg.lbl.gov: the definitive up-to-date listing of all known particles, masses, widths, and quantum numbers. Useful to skim even at this early stage to get a concrete sense of what the Standard Model contains.

2 Lecture 02 — Euler–Lagrange Equations for Fields, Basic Examples, and Relativistic Notation

Date: 26/03/2026

Overview

This lecture continues the construction of *classical field theory* as the direct generalization of analytical mechanics in which the **generalized coordinates are fields** $\phi(x, t)$ rather than particle positions $x(t)$. Starting from an *action* $S = \int \mathcal{L} dx dt$ built from a *local* (single space-time point) and *real* Lagrangian density $\mathcal{L}$, we impose the principle of minimal action ($\delta S = 0$) to obtain the Euler–Lagrange equations for fields. Several instructive examples are worked out: the wave equation, the Klein–Gordon equation (adding a “mass term”), and a nonlinear interacting equation from a cubic interaction $\sim \phi^3$, which motivates perturbation theory.

The second half of the lecture is a compact but thorough recap of **relativistic notation** needed for Lorentz-invariant field theories: four-vectors, upper/lower indices, Einstein summation, the mostly-negative metric $g_{\mu\nu}$, raising and lowering indices, the Lorentz-invariant scalar product, four-derivatives ∂_μ , and the d’Alembertian (box) operator $\square$. These tools are then used to rewrite the earlier Lagrangians and equations of motion in manifestly Lorentz-covariant form. The lecture closes with a brief discussion of the Lorentz group condition $\Lambda^T g \Lambda = g$, the classification timelike/lightlike/spacelike via A^2 , the transformation of tensors, and the convention of **natural units** $\hbar = c = 1$.

Foundational viewpoint: fields as generalized coordinates

Remark 1 (The conceptual shift). In classical mechanics we solve for a finite number of generalized coordinates $q(t)$ (e.g. $x(t)$). In classical field theory the generalized coordinate is a *field* $\phi(x, t)$; solving the theory means determining ϕ for *all* spacetime points (x, t) . In d spatial dimensions, ϕ depends on $d + 1$ arguments.

Axioms/assumptions for the classical field theories we study

We consider a theory with (possibly) several fields $\phi_i(x, t)$, but for most of the lecture we restrict to a **single real scalar field** $\phi(x, t)$ in $1 + 1$ **dimensions** (one space coordinate x and time t).

Physical Concept 2.1: Action and Lagrangian density

The action functional is

$$S[\phi] = \int dt dx \mathcal{L}(\phi, \partial_t \phi, \partial_x \phi; x, t), \quad (9)$$

where $\mathcal{L}$ is the *Lagrangian density* (denoted $\mathcal{L}$ in lecture).

The lecture adopts simplifying assumptions that mirror standard choices in mechanics:

- **No explicit spacetime dependence:** we focus on $\mathcal{L} = \mathcal{L}(\phi, \dot{\phi}, \phi')$ with no explicit x or t dependence. This is the field-theory analogue of “no explicit time dependence” in mechanics and encodes **translational invariance**, ensuring conservation of energy and momentum (via Noether’s theorem).
- **First derivatives only:** we take $\mathcal{L}$ to depend on ϕ and at most $\dot{\phi} \equiv \partial_t \phi$ and $\phi' \equiv \partial_x \phi$. Higher-derivative theories exist but are beyond the baseline framework introduced here.
- **Locality:** $\mathcal{L}$ depends on fields evaluated at the *same* spacetime point (x, t) . This defines a **local field theory**.
- **Reality:** $\mathcal{L} \in \mathbb{R}$. Physically, this ensures a real Hamiltonian and (in the quantum theory) unitary time evolution / probability conservation.

Euler–Lagrange equation for fields

Physical Law 2.1 Euler–Lagrange equation in 1 + 1 dimensions

For an action

$$S[\phi] = \int dt dx \mathcal{L}(\phi, \dot{\phi}, \phi') \quad (10)$$

the principle of minimal action ($\delta S = 0$) gives the field equation

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}}{\partial \phi'} \right). \quad (11)$$

Remark 2 (Analogy with mechanics). Compare (11) with $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0$. Field theory introduces an additional “evolution-like” variable x , so stationarity of the action produces *both* a t -derivative term and an x -derivative term. This is a concrete sense in which **space and time appear on similar footing** in the variational principle.

Worked examples: free waves, mass term, and nonlinearity

Example 1: the wave equation

Take

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - v^2 (\phi')^2. \quad (12)$$

Here v is a constant with units of velocity (the wave speed).

- Since $\mathcal{L}$ has no explicit ϕ dependence, $\partial \mathcal{L} / \partial \phi = 0$.
- $\partial \mathcal{L} / \partial \dot{\phi} = \dot{\phi}$, so $\partial_t (\partial \mathcal{L} / \partial \dot{\phi}) = \ddot{\phi}$.

- $\partial\mathcal{L}/\partial\phi' \propto -v^2\phi'$, so the spatial contribution yields a term proportional to $v^2\phi''$.

The resulting equation of motion is the **wave equation**

$$\ddot{\phi} = v^2\phi''. \quad (13)$$

A plane-wave solution is

$$\phi(x, t) = e^{i(kx - \omega t)}, \quad \omega = kv. \quad (14)$$

Remark 3 (Why this is “free”). The equation is linear, so superpositions of solutions are solutions. There is no term nonlinear in ϕ ; physically, different waves do not interact with each other.

Example 2: adding a “mass term” $-m^2\phi^2$

Now take

$$\mathcal{L} = \frac{1}{2}\dot{\phi}^2 - v^2(\phi')^2 - m^2\phi^2, \quad (15)$$

where the additional term $-m^2\phi^2$ is called the **mass term**.

Now $\partial\mathcal{L}/\partial\phi \neq 0$, and the Euler–Lagrange equation becomes

$$\ddot{\phi} = v^2\phi'' - m^2\phi. \quad (16)$$

This is the **Klein–Gordon equation** (in the $1 + 1$ notation used in class; later it will be written in a Lorentz-covariant form).

Try the same plane-wave ansatz $\phi = e^{i(kx - \omega t)}$. Plugging into (16) yields the modified dispersion relation

$$\omega^2 = (kv)^2 + m^2. \quad (17)$$

Remark 4 (Physical intuition for the mass term). Compared to the pure wave equation, the additional $-m^2\phi$ term acts like a restoring force even at $k = 0$: the field can oscillate locally with frequency $\omega = m$. In the quantum theory, this same parameter will set the rest energy of the corresponding particle.

Example 3: an interaction $\sim \phi^3$ and nonlinearity

Consider

$$\mathcal{L} = \frac{1}{2}\dot{\phi}^2 - v^2(\phi')^2 + \frac{\lambda}{3}\phi^3. \quad (18)$$

The Euler–Lagrange equation becomes

$$\ddot{\phi} = v^2\phi'' + \lambda\phi^2. \quad (19)$$

This is **nonlinear** because of the ϕ^2 term. Unlike the previous two examples, there is generally **no closed-form exact solution** for arbitrary initial data.

Remark 5 (Why this matters for the course: perturbation theory). Interactions corresponding to terms ϕ^n with $n \geq 3$ typically produce nonlinear equations. The course strategy will be **perturbation theory**: expand around a minimum of an effective potential and treat higher powers of the field as small corrections. This will be the seed of the perturbative expansion that later becomes Feynman diagrams in the quantum theory.

Relativistic notation: four-vectors and indices

Up to this point we wrote x and t explicitly. For most of the systems of interest in particle physics we want **Lorentz invariance** to be manifest, so it is efficient to adopt relativistic notation.

Spacetime interval

Special relativity implies invariance of

$$s^2 = c^2 t^2 - x_i x_i, \quad i = 1, 2, 3, \quad (20)$$

between inertial frames.

Four-vectors x^μ and general A^μ

Physical Concept 2.2: Position four-vector

The spacetime coordinate of an event is

$$x^\mu = (ct, x, y, z), \quad \mu = 0, 1, 2, 3. \quad (21)$$

All four components have dimensions of length.

For a boost in the x -direction,

$$x^{0'} = \gamma(x^0 - \beta x^1), \quad x^{1'} = \gamma(x^1 - \beta x^0), \quad x^{2'} = x^2, \quad x^{3'} = x^3, \quad (22)$$

where $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$.

Physical Concept 2.3: Four-vector

A four-vector is any object $A^\mu = (A^0, A^1, A^2, A^3)$ that transforms under boosts/rotations in the same way as x^μ .

Worked Example 2.1 (Four-momentum)

$p^\mu = (E/c, \mathbf{p})$ is a four-vector.

Scalar (dot) product

Physical Concept 2.4: Lorentz-invariant scalar product

The scalar product of four-vectors A^μ and B^μ is defined as

$$A \cdot B \equiv A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3. \quad (23)$$

With this definition, $A \cdot B$ is Lorentz invariant.

Lower indices and Einstein summation

Physical Concept 2.5: Lower-index components

Define the lower-index version of the position four-vector by

$$x_\mu = (ct, -x, -y, -z), \quad (24)$$

and similarly for any four-vector A^μ ,

$$A_\mu = (A^0, -A^1, -A^2, -A^3). \quad (25)$$

Physical Concept 2.6: Einstein summation convention

Any index appearing twice is summed from 0 to 3.

Then

$$A^\mu B_\mu = A^0 B_0 + A^1 B_1 + A^2 B_2 + A^3 B_3 = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3 = A \cdot B. \quad (26)$$

Remark 6 (Upper vs. lower indices). Quantities like $A^\mu B^\mu$ (both indices upstairs) are not Lorentz invariant and should not appear in physical expressions. In sensible Lorentz-covariant formulas, repeated indices appear once upstairs and once downstairs.

Remark 7 (Index sanity checks from lecture). • Never repeat an index more than twice in the same product ($A_\mu B^\mu C^\mu$ is ambiguous).

- The name of a dummy summed index does not matter: $A_\mu B^\mu = A_\nu B^\nu$.
- All indices contracted $\Rightarrow$ scalar; one free index $\Rightarrow$ vector; two free indices $\Rightarrow$ rank-2 tensor; etc.

Metric tensor and raising/lowering

Physical Concept 2.7: Metric tensor

The Minkowski metric (“mostly negative”) is

$$g_{\mu\nu} = \text{diag}(1, -1, -1, -1), \quad (27)$$

often denoted $\eta_{\mu\nu}$.

It lowers indices:

$$A_\mu = g_{\mu\nu} A^\nu, \quad (28)$$

and raises them with $g^{\mu\nu}$, which has the same numerical entries:

$$A^\mu = g^{\mu\nu} A_\nu, \quad g^{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (29)$$

The scalar product can be written as

$$A \cdot B = A^\mu B_\mu = A^\mu g_{\mu\nu} B^\nu. \quad (30)$$

Four-derivatives and the box operator

Physical Concept 2.8: Four-derivative

Define

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right), \quad (31)$$

and

$$\partial^\mu = g^{\mu\nu} \partial_\nu = \left(\frac{1}{c} \frac{\partial}{\partial t}, -\frac{\partial}{\partial x}, -\frac{\partial}{\partial y}, -\frac{\partial}{\partial z} \right). \quad (32)$$

Worked Example 2.2 (Derivative of a scalar is a four-vector)

If $f(x)$ is a scalar function of spacetime, then $\partial_\mu f$ is a four-vector.

Worked Example 2.3 (Divergence of a four-vector is a scalar)

If $h^\mu(x)$ is a four-vector field, then $\partial_\mu h^\mu$ is a Lorentz scalar.

Physical Concept 2.9: D'Alembertian / box operator

$$\square \equiv \partial^\mu \partial_\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}. \quad (33)$$

Rewriting the scalar field theory in covariant form

The free scalar Lagrangian with a mass term can be written as

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - m^2 \phi^2. \quad (34)$$

The $\partial_\mu \phi \partial^\mu \phi$ term is called the **kinetic term**, and $-m^2 \phi^2$ is the **mass term**.

The Klein–Gordon equation becomes

$$(\square + m^2)\phi = 0. \quad (35)$$

The Euler–Lagrange equation in relativistic notation was given as

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = \frac{\partial \mathcal{L}}{\partial \phi}. \quad (36)$$

Lorentz group, invariants, and tensors (recap)

Write a Lorentz transformation as

$$x^{\mu'} = \Lambda^\mu{}_{\nu'} x^\nu. \quad (37)$$

Invariance of $x^2 = x^\mu x_\mu$ implies

$$\Lambda^T g \Lambda = g, \quad (38)$$

equivalently $g \Lambda^T g = \Lambda^{-1}$.

Physical Concept 2.10: Timelike / lightlike / spacelike

For any four-vector A^μ ,

$$A^2 \equiv A \cdot A = (A^0)^2 - |\mathbf{A}|^2. \quad (39)$$

If $A^2 > 0$, A^μ is *timelike*. If $A^2 = 0$, it is *lightlike*. If $A^2 < 0$, it is *spacelike*.

Remark 8 (One Λ per Lorentz index). Tensors transform with one Lorentz matrix Λ for each Lorentz index. For example, a rank-2 tensor $A^{\mu\nu}$ transforms as

$$A^{\mu'\nu'} = \Lambda^\mu{}_{\sigma} \Lambda^\nu{}_{\rho} A^{\sigma\rho}, \quad (40)$$

and a rank-3 tensor similarly with three factors of Λ .

Natural units

Physical Concept 2.11: Natural units

We often work in units where

$$\hbar = c = 1. \quad (41)$$

Then

$$[\text{length}] = [\text{time}] = [E^{-1}] = [m^{-1}]. \quad (42)$$

Remark 9 (Electron mass example). The lecture emphasized that in natural units, a mass can be expressed as an energy (rest energy) and also as an inverse length scale (inverse Compton wavelength). For the electron, $m_e \simeq 0.511 \text{ MeV}$, and this corresponds to a characteristic inverse length scale.

To Remember

- Classical field theory treats fields $\phi(x, t)$ as generalized coordinates; the action is $S = \int dt dx \mathcal{L}$.
- Under the assumptions (locality, reality, at most first derivatives), $\delta S = 0$ gives $\frac{\partial \mathcal{L}}{\partial \phi} = \partial_t(\partial \mathcal{L} / \partial \dot{\phi}) + \partial_x(\partial \mathcal{L} / \partial \phi')$.
- $\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - v^2(\phi')^2$ gives the wave equation $\ddot{\phi} = v^2 \phi''$ and plane waves with $\omega = kv$.
- Adding $-m^2 \phi^2$ yields the Klein–Gordon equation with $\omega^2 = (kv)^2 + m^2$.
- Interactions ϕ^n with $n \geq 3$ generically lead to nonlinear equations, motivating perturbation theory.
- Lorentz invariance is made manifest with four-vectors, the metric $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$, and contractions of one upper with one lower index.
- The box operator $\square = \partial^\mu \partial_\mu$ packages the wave operator, and $(\square + m^2)\phi = 0$ is the covariant KG equation.
- In natural units $\hbar = c = 1$, energies, masses, inverse lengths, and inverse times are interchangeable as units.

Further Reading

The lecturer explicitly recommended the following:

- **Griffiths, *Introduction to Elementary Particles*, Chapter 3** — Relativistic kinematics and four-vector notation. The lecturer suggested this as a reference for the special-relativity review and the $g_{\mu\nu}$ conventions used in this course.
- **Peskin & Schroeder, *An Introduction to QFT*, §2.2** — “The Klein–Gordon Field as Harmonic Oscillators”: covers the classical field Lagrangian, the Euler–Lagrange equation for fields, and the Klein–Gordon equation. The lecturer described P&S as “one of the Bibles of particle physics”.
- **Goldstein, Poole & Safko, *Classical Mechanics* (3rd ed.), Chapter 13** — Lagrangian field theory from a classical-mechanics perspective, bridging the gap between particle mechanics (generalised coordinates $q_i(t)$) and field theory (the field $\phi(x, t)$ as a continuous generalised coordinate).

Additional recommended reading:

- **Tong, *Quantum Field Theory* (lecture notes), Chapter 1** (damtp.cam.ac.uk/user/tong/qft free): “Classical Field Theory” — Lagrangian densities, Noether’s theorem, energy–momentum tensor. Closely parallels the material of this lecture.
- **Halzen & Martin, *Quarks and Leptons*, Chapter 1** — Four-vector notation and relativistic kinematics in the particle-physics style, with worked examples of Lorentz invariants.

3 Lecture 03 — Canonical Quantisation of the Real Scalar Field

Date: 12/04/2026

Overview

Having established classical field theory in Lecture 02, we now perform the pivotal step of **canonical quantisation**: promoting the classical real scalar field $\phi(x, t)$ and its conjugate momentum density $\pi(x, t)$ to *operators* satisfying the **equal-time canonical commutation relations** (ET-CCR).

$$\boxed{[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i \delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad [\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = [\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0.} \quad (43)$$

The Klein-Gordon equation, linear in ϕ , calls for a Fourier decomposition in momentum space. For each Fourier mode $\mathbf{p}$, the field satisfies a simple harmonic oscillator equation with frequency $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$ (the relativistic dispersion relation). The reality condition on the classical field, and the hermiticity condition on the quantum field, constrain the Fourier coefficients: they become the familiar creation and annihilation operators $a_{\mathbf{p}}^\dagger$ and $a_{\mathbf{p}}$. This lecture thereby establishes the bridge between the classical field and the Fock-space picture of free particles.

Step 1: Conjugate momentum density from the Lagrangian

We work in natural units ($\hbar = c = 1$) and in $3 + 1$ spacetime dimensions. The free real scalar field Lagrangian density is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \frac{1}{2} (\dot{\phi}^2 - |\nabla \phi|^2) - \frac{1}{2} m^2 \phi^2. \quad (44)$$

Physical Concept 3.1: Conjugate momentum density

By analogy with analytical mechanics—where the conjugate momentum to q is $p = \partial L / \partial \dot{q}$ —the conjugate momentum density associated with the field ϕ is

$$\pi(x, t) \equiv \frac{\partial \mathcal{L}}{\partial (\partial_t \phi)} = \dot{\phi}(x, t). \quad (45)$$

Remark 10 (Density vs. total momentum). The quantity π defined above is a *density* (it derives from the Lagrangian density $\mathcal{L}$, not the integrated Lagrangian L). This is the standard convention in QFT. The momentum p in the Fourier decomposition below is the *ordinary* (kinematic) 3-momentum label; it should not be confused with π .

Step 2: Hamiltonian density

The Hamiltonian density (via the Legendre transform of $\mathcal{L}$) is

$$\begin{aligned} \mathcal{H} &= \pi \dot{\phi} - \mathcal{L} \\ &= \dot{\phi}^2 - \left[\frac{1}{2} \dot{\phi}^2 - \frac{1}{2} |\nabla \phi|^2 - \frac{1}{2} m^2 \phi^2 \right] \\ &= \frac{1}{2} \dot{\phi}^2 + \frac{1}{2} |\nabla \phi|^2 + \frac{1}{2} m^2 \phi^2. \end{aligned} \quad (46)$$

Using $\pi = \dot{\phi}$ we write this in the canonical form

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{1}{2}m^2\phi^2, \quad (47)$$

and the total Hamiltonian is $H = \int d^3x \mathcal{H}(\mathbf{x}, t)$.

Step 3: The quantisation prescription

Physical Concept 3.2: Canonical quantisation of fields

We *promote* ϕ and π from classical functions to Hermitian operators on a Hilbert space and impose the **equal-time canonical commutation relations**:

$$[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}), \quad (48)$$

$$[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0, \quad (49)$$

$$[\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0. \quad (50)$$

These are the direct field-theory analogues of $[q_i, p_j] = i\delta_{ij}$.

Remark 11 (Why equal-time?). The commutators are imposed at the *same time* t . This singles out a time slice (a Cauchy surface) and is the natural generalization of the single-time quantum mechanics condition $[q(t), p(t)] = i\hbar$. Although equal-time quantisation breaks manifest Lorentz covariance, the resulting QFT is nonetheless fully Lorentz invariant once interactions are included properly. A Lorentz-covariant quantisation exists but is more involved.

Step 4: Fourier decomposition and the Klein-Gordon spectrum

The Klein-Gordon equation arising from (44) via the Euler-Lagrange prescription is

$$(\square + m^2)\phi = 0, \quad \square \equiv \partial^\mu \partial_\mu = \partial_t^2 - \nabla^2. \quad (51)$$

Fourier decomposition in momentum space

Write the field as a Fourier integral over 3-momentum:

$$\phi(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} \tilde{\phi}(\mathbf{p}, t) e^{i\mathbf{p}\cdot\mathbf{x}}. \quad (52)$$

Substituting into (51):

$$\begin{aligned} 0 &= (\partial_t^2 - \nabla^2 + m^2)\phi \\ &= \int \frac{d^3p}{(2\pi)^3} [\ddot{\tilde{\phi}}(\mathbf{p}, t) + |\mathbf{p}|^2 \tilde{\phi}(\mathbf{p}, t) + m^2 \tilde{\phi}(\mathbf{p}, t)] e^{i\mathbf{p}\cdot\mathbf{x}}. \end{aligned} \quad (53)$$

Since this must hold for every mode independently,

$$\ddot{\tilde{\phi}}(\mathbf{p}, t) + \underbrace{(|\mathbf{p}|^2 + m^2)}_{\omega_{\mathbf{p}}^2} \tilde{\phi}(\mathbf{p}, t) = 0. \quad (54)$$

Physical Concept 3.3: Relativistic dispersion relation

Define the *on-shell frequency* for each mode:

$$\omega_{\mathbf{p}} \equiv \sqrt{|\mathbf{p}|^2 + m^2}. \quad (55)$$

This is precisely the relativistic energy-momentum relation $E^2 = p^2 + m^2$ (in natural units) for a particle of mass m and 3-momentum $\mathbf{p}$.

Equation (54) is the equation of a **simple harmonic oscillator** with frequency $\omega_{\mathbf{p}}$ for each Fourier mode. The general solution is

$$\tilde{\phi}(\mathbf{p}, t) = A(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t} + B(\mathbf{p}) e^{+i\omega_{\mathbf{p}} t}. \quad (56)$$

Step 5: Reality (hermiticity) condition on the field

The *classical* field $\phi(\mathbf{x}, t)$ is real: $\phi^*(\mathbf{x}, t) = \phi(\mathbf{x}, t)$. Taking the complex conjugate of the Fourier representation and requiring equality gives

$$\begin{aligned} \phi^*(\mathbf{x}, t) &= \int \frac{d^3 p}{(2\pi)^3} \tilde{\phi}^*(\mathbf{p}, t) e^{-i\mathbf{p} \cdot \mathbf{x}} \\ &= \int \frac{d^3 p}{(2\pi)^3} \tilde{\phi}^*(-\mathbf{p}, t) e^{+i\mathbf{p} \cdot \mathbf{x}} \quad [\text{re-label } \mathbf{p} \rightarrow -\mathbf{p}], \end{aligned} \quad (57)$$

where in the last line we used the substitution $\mathbf{p} \rightarrow -\mathbf{p}$ to match the $e^{i\mathbf{p} \cdot \mathbf{x}}$ Fourier basis. Comparing with (52) we need

$$\tilde{\phi}(\mathbf{p}, t) = \tilde{\phi}^*(-\mathbf{p}, t). \quad (58)$$

Applying this to (56):

$$A(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t} + B(\mathbf{p}) e^{+i\omega_{\mathbf{p}} t} = A^*(-\mathbf{p}) e^{+i\omega_{\mathbf{p}} t} + B^*(-\mathbf{p}) e^{-i\omega_{\mathbf{p}} t}. \quad (59)$$

Matching the $e^{-i\omega t}$ and $e^{+i\omega t}$ coefficients separately:

$$B(\mathbf{p}) = A^*(-\mathbf{p}). \quad (60)$$

Hence the general real solution contains only one independent complex amplitude per mode, which we rename $a(\mathbf{p}) \equiv A(\mathbf{p})$:

$$\tilde{\phi}(\mathbf{p}, t) = a(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t} + a^*(-\mathbf{p}) e^{+i\omega_{\mathbf{p}} t}. \quad (61)$$

Step 6: Full mode expansion of the quantum field

Substituting (61) back into the Fourier integral (52) and relabelling $\mathbf{p} \rightarrow -\mathbf{p}$ in the second term:

$$\begin{aligned} \phi(\mathbf{x}, t) &= \int \frac{d^3 p}{(2\pi)^3} \left[a(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a^*(-\mathbf{p}) e^{+i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} \right] \\ &= \int \frac{d^3 p}{(2\pi)^3} \left[a(\mathbf{p}) e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a^*(\mathbf{p}) e^{+i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} \right]. \end{aligned} \quad (62)$$

It is conventional to include a Lorentz-invariant normalisation factor $1/(2\omega_{\mathbf{p}})^{1/2}$, giving the standard mode expansion:

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left[a_{\mathbf{p}} e^{-i(\omega_{\mathbf{p}} t - \mathbf{p} \cdot \mathbf{x})} + a_{\mathbf{p}}^\dagger e^{+i(\omega_{\mathbf{p}} t - \mathbf{p} \cdot \mathbf{x})} \right]. \quad (63)$$

In the quantum theory $a(\mathbf{p}) \rightarrow a_{\mathbf{p}}$ and $a^*(\mathbf{p}) \rightarrow a_{\mathbf{p}}^\dagger$ become **operators** on the Fock space; the reality condition $\phi = \phi^*$ becomes the hermiticity condition $\phi = \phi^\dagger$.

Similarly, the conjugate momentum $\pi = \dot{\phi}$ has the expansion

$$\pi(\mathbf{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{(-i)\sqrt{\omega_{\mathbf{p}}}}{\sqrt{2}} \left[a_{\mathbf{p}} e^{-i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} - a_{\mathbf{p}}^\dagger e^{+i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right], \quad (64)$$

obtained by differentiating (63) with respect to t .

Step 7: Commutation relations for $a_{\mathbf{p}}$ and $a_{\mathbf{p}}^\dagger$

Imposing the ET-CCR (43) on the mode expansions (63)–(64), one finds (by inverting the Fourier transform and using the completeness of plane waves) that the creation and annihilation operators satisfy

$$[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad (65)$$

$$[a_{\mathbf{p}}, a_{\mathbf{q}}] = [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] = 0. \quad (66)$$

These are precisely the commutation relations of an **infinite collection of decoupled quantum harmonic oscillators**, one for each momentum mode $\mathbf{p}$.

Remark 12 (Physical interpretation of $a_{\mathbf{p}}$). Before quantisation, $a(\mathbf{p})$ is merely a complex Fourier coefficient—a *c-number*. After quantisation it becomes an *operator*: $a_{\mathbf{p}}$ **annihilates** a particle of momentum $\mathbf{p}$, while $a_{\mathbf{p}}^\dagger$ **creates** one. The Fock vacuum $|0\rangle$ is defined by $a_{\mathbf{p}}|0\rangle = 0$ for all $\mathbf{p}$.

Examples

Worked Example 3.1 (Explicit derivation of π from ϕ)

Starting from the mode expansion of ϕ in (63), differentiate with respect to t :

$$\begin{aligned} \pi(\mathbf{x}, t) = \dot{\phi} &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \frac{\partial}{\partial t} \left[a_{\mathbf{p}} e^{-i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} + a_{\mathbf{p}}^\dagger e^{+i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left[(-i\omega_{\mathbf{p}}) a_{\mathbf{p}} e^{-i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} + (i\omega_{\mathbf{p}}) a_{\mathbf{p}}^\dagger e^{+i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right] \\ &= -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{p}}}{2}} \left[a_{\mathbf{p}} e^{-i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} - a_{\mathbf{p}}^\dagger e^{+i(\omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x})} \right]. \end{aligned} \quad (67)$$

This confirms the expression in (64).

Worked Example 3.2 (Each mode as a harmonic oscillator)

Fix a mode $\mathbf{p}$ and define the “position” and “momentum” of that mode:

$$Q_{\mathbf{p}} \propto a_{\mathbf{p}} + a_{\mathbf{p}}^\dagger, \quad P_{\mathbf{p}} \propto -i(a_{\mathbf{p}} - a_{\mathbf{p}}^\dagger). \quad (68)$$

From (65) one recovers $[Q_{\mathbf{p}}, P_{\mathbf{p}}] = i$, and the Hamiltonian contribution of this mode is

$$H_{\mathbf{p}} = \omega_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \frac{1}{2} \right). \quad (69)$$

This is identical to a quantum harmonic oscillator of frequency $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$. The total Hamiltonian is therefore

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \frac{1}{2} \right), \quad (70)$$

where the $\frac{1}{2}$ term gives the (formally infinite) zero-point energy of the vacuum, which is later handled by normal ordering.

Worked Example 3.3 (Dispersion relation as the relativistic energy-momentum relation)

The on-shell frequency defined in (55), $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$, is equivalent to $E^2 = p^2 + m^2$ (natural units). For a massless field ($m = 0$) this reduces to $\omega_{\mathbf{p}} = |\mathbf{p}|$ (linear dispersion, i.e. photons or massless scalar). For a non-zero mass it implies a **minimum energy** $E_{\min} = m$ at $\mathbf{p} = 0$, corresponding to a particle at rest.

To Remember

- The conjugate momentum density of the real scalar field is $\pi = \partial\mathcal{L}/\partial\dot{\phi} = \dot{\phi}$.
- Canonical quantisation promotes ϕ and π to Hermitian operators satisfying the ET-CCR: $[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i\delta^{(3)}(\mathbf{x} - \mathbf{y})$.
- The Klein-Gordon equation in each Fourier mode reduces to a simple harmonic oscillator equation with frequency $\omega_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$ (relativistic dispersion).
- The reality (hermiticity) condition on ϕ constrains the two Fourier coefficients to be complex conjugates: $B(\mathbf{p}) = A^*(-\mathbf{p})$, leaving one independent operator per mode.
- The mode expansion is $\phi = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-ipx} + a_{\mathbf{p}}^\dagger e^{ipx})$, where $px \equiv \omega_{\mathbf{p}}t - \mathbf{p} \cdot \mathbf{x}$.
- Imposing the ET-CCR on the mode expansion yields the harmonic-oscillator algebra: $[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})$.
- The full Hamiltonian is a sum (integral) of harmonic oscillators: $H = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} (a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \frac{1}{2})$; the infinite zero-point energy is removed by normal ordering.

Further Reading

No specific textbook was cited in this lecture. The following cover the same material at varying depths:

- **Peskin & Schroeder, §2.1–2.3** — The canonical quantization of the real Klein–Gordon field from first principles: mode expansion, creation/annihilation operators, Fock space, normal ordering, and the Feynman propagator. This is the primary QFT reference for the material in this lecture.
- **Tong, *QFT Notes*, Chapter 2** (damtp.cam.ac.uk/user/tong/qft.html): “Free Fields” — covers the same quantization procedure in a more pedagogical style, with explicit derivations of the commutation relations and the Hamiltonian as a sum of harmonic oscillators.

- **Griffiths, *Introduction to Elementary Particles*, Appendix** — Brief summary of creation and annihilation operators and their algebra, useful for a quick reference on the operator formalism.
- **Lancaster & Blundell, *Quantum Field Theory for the Gifted Amateur*, Chapters 11–12 (OUP)**: a very readable step-by-step quantization of the scalar field, explicitly connecting each step to the quantum harmonic oscillator the student already knows.

Part II

Feynman Diagrams & Scattering

4 Lecture 04 — Perturbation Theory and Feynman Diagrams for the Harmonic Oscillator

Date: 19/04/2026

Overview

This lecture introduces **perturbation theory** in the context of coupled harmonic oscillators — a tractable system where results can be verified explicitly. The central objects are:

1. **Fermi's Golden Rule:** the transition rate between states expressed in terms of a transition *amplitude* and a *phase-space* factor.
2. **Selection rules:** which transitions are kinematically and dynamically allowed for a given interaction Hamiltonian.
3. **Second-order perturbation theory:** explicit computation of a $y \rightarrow 3x$ decay amplitude via a sum over intermediate states.
4. **Feynman diagrams:** a pictorial — and ultimately computational — shorthand that encodes the same information with far less effort.

The motivating philosophy is to *prove* that Feynman diagrams work on a familiar system (the harmonic oscillator), so that one can later *trust* the same machinery when applied to quantum field theory.

Setup: Coupled Harmonic Oscillators

Consider a two-dimensional system with potential

$$V(x, y) = \frac{k x^2}{2} + \frac{4k^2 y^2}{2} + \alpha x^2 y. \quad (71)$$

- The first two terms describe two *decoupled* harmonic oscillators: an x -mode with angular frequency $\omega_x = \sqrt{k}$ and a y -mode with $\omega_y = 2\omega_x$.
- The third term $\alpha x^2 y$ is an *anharmonic* (interaction) term that couples the two modes and transfers energy between them.
- The coupling constant α is assumed small, so this term is treated as a perturbation.

Remark 13 (Physical picture of the interaction term). The anharmonic term $\alpha x^2 y$ can be thought of as an *interaction vertex*: it can annihilate one y -quantum and create two x -quanta (or vice versa). This is the harmonic-oscillator analogue of a particle decay in QFT. The parameter α controls the *rate* of energy transfer between the modes — just as a coupling constant controls an interaction rate in field theory.

Fermi's Golden Rule

Physical Concept 4.1: Fermi's Golden Rule

The transition *rate* W from an initial state $|i\rangle$ to a final state $|f\rangle$ is

$$W_{i \rightarrow f} \propto |\mathcal{A}|^2 \times (\text{phase space}), \quad (72)$$

where $\mathcal{A} = \langle f | H_1 | i \rangle$ is the **transition amplitude** (sometimes written $\mathcal{M}$) and the phase-space factor encodes the kinematic constraints.

- The amplitude $\mathcal{A}$ contains **all dynamical information** about the interaction. It is computed via perturbation theory or, equivalently, via Feynman diagrams.
- The **phase-space** factor accounts for kinematics: how much “room” is available in the space of final states. For example, if a heavy particle decays into products whose combined mass is close to its own mass, the phase space is small and the decay rate is suppressed, regardless of the coupling strength.

Ladder Operators and Matrix Elements

In the quantum harmonic oscillator with coordinate q (standing for x or y or z), the position operator is expressed in terms of creation and annihilation operators as

$$q = \frac{1}{\sqrt{2\omega_q}} (a_q + a_q^\dagger), \quad (73)$$

where ω_q is the frequency of that mode. The standard ladder-operator actions are

$$a_q^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad a_q |n\rangle = \sqrt{n} |n-1\rangle. \quad (74)$$

Selection Rules from the Interaction Hamiltonian

The interaction Hamiltonian for the two-oscillator system is

$$H_1 = \alpha x^2 y. \quad (75)$$

Using (73), $x^2 \propto (a_x + a_x^\dagger)^2$ and $y \propto (a_y + a_y^\dagger)$. Expanding these products reveals which changes in occupation numbers Δn_x and Δn_y are possible:

Physical Concept 4.2: Selection rules for $H_1 = \alpha x^2 y$

A transition $|i\rangle \rightarrow |f\rangle$ has a non-zero amplitude *only* if

$$\Delta n_y = \pm 1 \quad \text{and} \quad \Delta n_x \in \{0, \pm 2\}. \quad (76)$$

Remark 14 (Generalisation). For a general interaction $H_1 \propto x^p y^q$, the selection rules become $\Delta n_x \in \{-p, -p+2, \dots, p\}$ and $\Delta n_y \in \{-q, -q+2, \dots, q\}$. For example, $H_1 \propto x^2 y^3$ gives $\Delta n_x \in \{0, \pm 2\}$ and $\Delta n_y \in \{\pm 1, \pm 3\}$.

First Example: $y \rightarrow 2x$ Decay (Two-Oscillator System)

Kinematic setup

With $\omega_y = 2\omega_x$, energy conservation requires that the final state has $n_y = 0$ and $n_x = 2$. Starting from $|i\rangle = |n_x, n_y\rangle = |0, 1\rangle$, the allowed final state is $|f\rangle = |2, 0\rangle$.

First-order amplitude

$$\begin{aligned}\mathcal{A} &= \langle 2, 0 | \alpha x^2 y | 0, 1 \rangle \\ &= \alpha \langle 2, 0 | \frac{(a_x + a_x^\dagger)^2}{2\omega_x} \cdot \frac{(a_y + a_y^\dagger)}{\sqrt{2\omega_y}} | 0, 1 \rangle.\end{aligned}\quad (77)$$

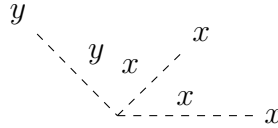
The y factor $a_y + a_y^\dagger$ acting on $|n_y = 1\rangle$ gives $a_y|1\rangle = |0\rangle$ (the relevant term, since $a_y^\dagger|1\rangle = \sqrt{2}|2\rangle$ does not lead to $|0\rangle$). The x factor $(a_x + a_x^\dagger)^2$ acting on $|0\rangle$ produces $|2\rangle$ via two applications of $a_x^\dagger$:

$$(a_x^\dagger)^2|0\rangle = \sqrt{2}|2\rangle. \quad (78)$$

This process is described pictorially as:

One y -particle enters the vertex and is annihilated; two x -particles are created and exit.

The Feynman diagram



The vertex carries a factor α (the coupling constant).

Decay width and lifetime

$$\Gamma \equiv \Gamma_{y \rightarrow 2x} \propto |\mathcal{A}|^2 \times \text{phase space} \propto \alpha^2. \quad (79)$$

The **lifetime** of the y particle is

$$\tau = \frac{1}{\Gamma} \propto \frac{1}{\alpha^2}. \quad (80)$$

Second Example: $y \rightarrow 3x$ Decay — Three-Oscillator System

System and Hamiltonian

Now consider a *three-mode* system with coordinates x, y, z and frequencies $\omega_x, \omega_y = 3\omega_x, \omega_z \gg \omega_x, \omega_y$. The interaction Hamiltonian has two terms:

$$H_1 = \lambda_1 xyz + \lambda_2 x^2 z, \quad (81)$$

where λ_1 and λ_2 are small coupling constants treated perturbatively.

States are labelled $|n_x, n_y, n_z\rangle$. We wish to compute the transition amplitude for

$$|i\rangle = |0, 1, 0\rangle \longrightarrow |f\rangle = |3, 0, 0\rangle. \quad (82)$$

Energy conservation: $E_i = \omega_y = 3\omega_x = E_f$. ✓

Why second-order perturbation theory is needed

Neither $\lambda_1 xyz$ nor $\lambda_2 x^2 z$ alone can connect $|0, 1, 0\rangle$ to $|3, 0, 0\rangle$ in a single step. Each interaction changes occupation numbers by limited amounts (e.g. λ_1 changes n_x, n_y, n_z by ± 1 each; λ_2 changes n_x by ± 2 and n_z by ± 1). Therefore, one must act with H_1 *twice* (second order), passing through an intermediate state $|n\rangle$.

Physical Concept 4.3: Second-order perturbation theory amplitude

$$\mathcal{A}_{i \rightarrow f}^{(2)} = \sum_n \frac{\langle f | H_1 | n \rangle \langle n | H_1 | i \rangle}{E_i - E_n}, \quad (83)$$

where the sum runs over all intermediate states $|n\rangle$ with $E_n \neq E_i$.

Remark 15 (Virtual intermediate states). The intermediate state $|n\rangle$ in (83) need *not* satisfy energy conservation ($E_n \neq E_i$ is generically the case). These are called **virtual** (or *off-shell*) intermediate states. Only the initial and final states must be on-shell (energy-conserving). This is the quantum mechanical precursor to virtual particles in QFT.

Identifying intermediate states

We must find all $|n\rangle$ such that both $\langle n | H_1 | i \rangle \neq 0$ and $\langle f | H_1 | n \rangle \neq 0$.

Intermediate state 1: $|n_1\rangle = |1, 0, 1\rangle$

- Step 1 (act on $|i\rangle = |0, 1, 0\rangle$ with $\lambda_1 xyz$): annihilate the y and create one x and one z , reaching $|1, 0, 1\rangle$.
- Step 2 (act on $|1, 0, 1\rangle$ with $\lambda_2 x^2 z$): annihilate the z and create two more x , reaching $|3, 0, 0\rangle = |f\rangle$.

Intermediate state 2: $|n_2\rangle = |2, 1, 1\rangle$

- Step 1 (act on $|0, 1, 0\rangle$ with $\lambda_2 x^2 z$): create two x and one z ; y is untouched, reaching $|2, 1, 1\rangle$.
- Step 2 (act on $|2, 1, 1\rangle$ with $\lambda_1 xyz$): annihilate the y and the z , create one more x , reaching $|3, 0, 0\rangle = |f\rangle$.

These are the *only* two contributing intermediate states.

Matrix elements

Using $q = \frac{1}{\sqrt{2\omega_q}}(a_q + a_q^\dagger)$ and defining the shorthand

$$C_1 \equiv \frac{1}{(2)^{3/2} \sqrt{\omega_x \omega_y \omega_z}}, \quad C_2 \equiv \frac{1}{(2)^{3/2} \omega_x \sqrt{\omega_z}}, \quad (84)$$

the relevant matrix elements evaluate to:

$$\langle 1, 0, 1 | \lambda_1 xyz | 0, 1, 0 \rangle = \lambda_1 C_1, \quad (85)$$

$$\langle 3, 0, 0 | \lambda_2 x^2 z | 1, 0, 1 \rangle = \lambda_2 C_2 \cdot \sqrt{6}, \quad (86)$$

$$\langle 2, 1, 1 | \lambda_2 x^2 z | 0, 1, 0 \rangle = \lambda_2 C_2 \cdot \sqrt{2}, \quad (87)$$

$$\langle 3, 0, 0 | \lambda_1 xyz | 2, 1, 1 \rangle = \lambda_1 C_1 \cdot \sqrt{6}. \quad (88)$$

The $\sqrt{6}$ and $\sqrt{2}$ factors arise from the ladder-operator actions $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$.

Full second-order amplitude

The energy denominators for the two channels are:

$$E_i - E_{n_1} = 3\omega_x - (\omega_x + \omega_z) = 2\omega_x - \omega_z, \quad (89)$$

$$E_i - E_{n_2} = 3\omega_x - (2\omega_x + \omega_y + \omega_z) = -(\omega_x + \omega_z) \quad [\text{using } \omega_y = 3\omega_x]. \quad (90)$$

Substituting into (83):

$$\mathcal{A}^{(2)} = \frac{(\lambda_2 C_2 \sqrt{6})(\lambda_1 C_1)}{2\omega_x - \omega_z} + \frac{(\lambda_1 C_1 \sqrt{6})(\lambda_2 C_2 \sqrt{2})}{-(\omega_x + \omega_z)}. \quad (91)$$

Factoring out $\lambda_1 \lambda_2 C_1 C_2 \sqrt{6}$:

$$\mathcal{A}^{(2)} = \lambda_1 \lambda_2 C_1 C_2 \sqrt{6} \left[\frac{1}{2\omega_x - \omega_z} - \frac{\sqrt{2}}{\omega_z + 2\omega_x} \right]. \quad (92)$$

Taking a common denominator ($\omega_z^2 - 4\omega_x^2$):

$$\mathcal{A}^{(2)} = -\frac{\lambda_1 \lambda_2 \sqrt{2}}{4\omega_x^2} \cdot \frac{1}{\omega_z^2/(4\omega_x^2) - 1} \quad (93)$$

(after substituting $\omega_y = 3\omega_x$ and simplifying). This is the complete first-nontrivial-order transition amplitude for $y \rightarrow 3x$.

Introduction to Feynman Diagrams

Motivation

The second-order calculation above requires:

1. Identifying all contributing intermediate states by inspection.
2. Computing each matrix element using ladder operators.
3. Combining all terms with the correct energy denominators.

For homework problems with many interactions or many modes, this becomes *very* cumbersome. Feynman diagrams provide a systematic pictorial method that automatically generates the correct answer.

Elements of a Feynman diagram

Physical Concept 4.4: External states

External states (also called *external lines*) represent the physical, on-shell initial and final particles. In the $y \rightarrow 3x$ example there is one external initial line (y) and three external final lines (x, x, x).

Physical Concept 4.5: Vertices

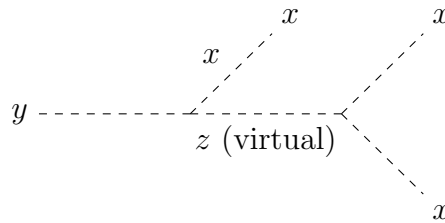
Vertices are points where lines meet; they correspond to interaction terms in H_1 . Each vertex contributes a factor of the coupling constant (λ_1 or λ_2).

Physical Concept 4.6: Internal lines / propagators

Internal lines (also called *propagators* or *virtual lines*) connect two vertices and represent the virtual intermediate state. They correspond to the energy denominator $1/(E_i - E_n)$ in second-order perturbation theory. Unlike external lines, internal lines need not be on-shell.

Feynman diagram for $y \rightarrow 3x$

The process requires one λ_1 vertex (the only one that can annihilate a y) and one λ_2 vertex (to build up the remaining x quanta via x^2z). The z mode plays the role of the internal (virtual) line:



- Vertex v_1 : corresponds to $\lambda_1 xyz$. The y is annihilated; one x and one z emerge.
- Internal line: the z particle propagates from v_1 to v_2 . It is *virtual* — its energy need not equal ω_z .
- Vertex v_2 : corresponds to $\lambda_2 x^2z$. The z is annihilated; two x particles emerge.

There is a second diagram corresponding to the other ordering (acting with λ_2 first, then λ_1), which gives the second term in (92).

From diagram to amplitude — preview of Feynman rules

Each element of the Feynman diagram maps to a mathematical factor. The full set of **Feynman rules** (to be derived in Lecture 05) will specify:

Diagram element	Mathematical factor
External line (particle q)	$\frac{1}{\sqrt{2\omega_q}}$
Internal line (particle q)	$\frac{1}{E_i - E_q}$ (energy denominator)
Vertex with coupling λ	λ
Symmetry factor	S (combinatorial, diagram-dependent)

Multiplying these factors together and summing all topologically distinct diagrams reproduces the perturbation-theory result — in this case (93) — without ever writing down an explicit sum over intermediate states.

To Remember

- **Fermi's Golden Rule:** $W \propto |\mathcal{A}|^2 \times \text{phase space}$. The amplitude $|\mathcal{A}|$ carries all dynamical information; phase space carries kinematics.
- For a coupling α at a single vertex, the decay rate goes as $\Gamma \propto \alpha^2$ and the lifetime as $\tau \propto 1/\alpha^2$.
- **Selection rules** from $H_1 \propto x^p y^q$: only transitions with $|\Delta n_x| \leq p$ (same parity as p) and $|\Delta n_y| \leq q$ (same parity as q) are allowed.
- **Second-order perturbation theory:** $\mathcal{A}^{(2)} = \sum_n \langle f | H_1 | n \rangle \langle n | H_1 | i \rangle / (E_i - E_n)$. Sum over *all* intermediate states reachable from $|i\rangle$ and from which $|f\rangle$ can be reached.
- **Virtual (intermediate) states** need not conserve energy; only external states must.
- A **Feynman diagram** has external lines (initial/final), vertices (couplings), and internal lines (propagators/virtual particles).
- The **Feynman rules** (next lecture) translate each diagram element into a factor; the product of all factors gives the amplitude without explicit sums over intermediate states.

Further Reading

No specific textbook was cited in this lecture. The following are recommended:

- **Griffiths, Chapter 6.1** — “The Theory of Ionization and Excitation by Impact”: derives Fermi's Golden Rule in the particle-physics context and explains the role of the matrix element and the phase-space factor. This is the closest Griffiths chapter to the perturbation-theory content of this lecture.
- **Peskin & Schroeder, §4.1–4.3** — Interaction picture, time-ordered products, and Wick's theorem. These sections provide the formal QFT justification for the diagrammatic rules that were here derived for the harmonic oscillator and extended heuristically to field theory in Lecture 05.
- **Tong, *QFT Notes*, Chapter 3, §3.1–3.2** (damtp.cam.ac.uk/user/tong/qft.html): “Interacting Fields” — introduces the interaction picture and perturbation expansion in a QFT context, culminating in the same diagrammatic logic.
- **Sakurai & Napolitano, *Modern Quantum Mechanics*, Chapter 5** — Time-dependent perturbation theory and Fermi's Golden Rule in the non-relativistic QM setting. A useful review of the underlying formalism before the field-theory generalisation.
- **Feynman, *QED: The Strange Theory of Light and Matter* (Princeton UP)**: a non-technical but surprisingly deep account of how Feynman diagrams encode quantum amplitudes. Recommended for conceptual grounding.

5 Lecture 05 — Feynman Rules for Scalar Fields and Mandelstam Variables

Date: 23/04/2026

Overview

In Lecture 04 we proved that Feynman diagrams give the correct perturbation-theory amplitudes for coupled harmonic oscillators. Today we make the leap to quantum *field* theory. The motivating observation is that a quantum field is nothing but an infinite collection of harmonic oscillators (one per Fourier mode), so the same diagrammatic logic carries over — with one key upgrade: particles now carry *four-momenta* rather than occupation numbers, and the propagator (internal-line factor) becomes $i/(q^2 - m^2)$ rather than $1/(E_i - E_n)$.

The lecture develops the complete **Feynman rules for real scalar fields**, illustrates them through a series of worked examples, introduces the tree/loop-level distinction, identifies the three kinematic **channels** (s, t, u) for $2 \rightarrow 2$ scattering, and derives the celebrated **Mandelstam sum rule** $s + t + u = \sum_i m_i^2$.

Bridge: From Harmonic Oscillators to Fields

Recall from Lecture 03 that a free real scalar field ϕ is expanded in Fourier modes as

$$\phi(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(a_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} + a_{\mathbf{k}}^\dagger e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega_{\mathbf{k}} t)} \right), \quad (94)$$

where $\omega_{\mathbf{k}} = \sqrt{|\mathbf{k}|^2 + m^2}$ and $a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger$ are ladder operators satisfying the canonical commutation relation $[a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}')$.

Remark 16 (Why the harmonic-oscillator rules extend to fields). Compare (94) with the harmonic-oscillator expression $q = (a + a^\dagger)/\sqrt{2\omega}$. They are structurally identical: each Fourier mode of ϕ is an independent harmonic oscillator. When we write an interaction term ϕ^n in the Lagrangian, it creates/annihilates particles through the same ladder-operator algebra we studied for the harmonic oscillator. The Feynman rules are therefore the same machinery — just with four-momenta replacing occupation numbers, and a Lorentz-invariant propagator replacing the energy-denominator.

Momentum eigenstates. We label single-particle states by their four-momentum $p = (E_p, \mathbf{p})$:

$$|p\rangle \propto a_{\mathbf{p}}^\dagger |0\rangle, \quad (95)$$

normalised so that these states are Lorentz invariant. For the present course, all that matters is that a state with momentum p is created by $a_{\mathbf{p}}^\dagger$ acting on the vacuum.

What generates transitions? Terms in the Lagrangian with *three or more fields* are interaction terms — they produce matrix elements that connect different particle-number states, giving rise to decays and scattering processes. The decay rate and scattering cross-section are both computed via Fermi's Golden Rule:

$$\Gamma, \sigma \propto |\mathcal{A}|^2 \times (\text{phase space}), \quad (96)$$

where the amplitude $\mathcal{A}$ is read off from the Feynman diagram.

The Feynman Rules for Scalar Fields

For a theory of real scalar fields with a given interaction Lagrangian $\mathcal{L}_{\text{int}}$, the transition amplitude for any process is computed by the following systematic procedure.

Step 1 — Draw all Feynman diagrams

For the process of interest, draw every topologically distinct diagram built from the available *vertices*. Each vertex corresponds to an interaction term in $\mathcal{L}_{\text{int}}$: an interaction $\lambda \phi_1^a \phi_2^b \cdots$ gives a vertex with a legs of type ϕ_1 , b legs of type ϕ_2 , etc. Different line styles (solid, dashed, ...) are used to distinguish different fields.

Step 2 — Label all momenta

Label the incoming and outgoing four-momenta of every external state $(p_1, p_2, p_3, \dots)$, including a direction arrow. Assign a label (say q) and direction to each internal line; the direction is arbitrary but must be chosen and held fixed.

Step 3 — Apply the Feynman rules

Physical Concept 5.1: Feynman rules for real scalar fields

Multiply together a factor for each element of the diagram:

Element	Factor	Notes
Vertex with coupling λ	$i\lambda$	one factor per vertex
Internal line (propagator) with momentum q , mass m	$\frac{i}{q^2 - m^2}$	$q^2 \equiv q^\mu q_\mu$
External line (initial or final state)	—	no factor; just label p
Momentum conservation at vertex	$\delta^{(4)}(\sum p_{\text{in}} - \sum p_{\text{out}})$	imposed at every vertex
Undetermined (loop) momentum ℓ	$\int \frac{d^4\ell}{(2\pi)^4}$	one integral per loop
Symmetry factor	S	combinatorial; see example

Sum the contributions of **all** distinct diagrams at the given order to obtain the total amplitude $i\mathcal{A}$.

Remark 17 (Physical interpretation of the propagator). The harmonic-oscillator energy denominator $1/(E_i - E_n)$ from second-order perturbation theory becomes $i/(q^2 - m^2)$ in the field theory. In relativistic notation, $q^2 = E_q^2 - |\mathbf{q}|^2$. For a real, on-shell particle, $q^2 = m^2$, so the denominator would vanish — but internal (virtual) particles are *off-shell* ($q^2 \neq m^2$), so the propagator is finite and well-defined.

On-shell versus off-shell particles

Physical Concept 5.2: On-shell and off-shell

A particle with four-momentum p is **on-shell** (physical) if

$$p^2 = E^2 - |\mathbf{p}|^2 = m^2. \quad (97)$$

External (initial and final) particles are always on-shell. Internal lines in a Feynman diagram represent **virtual** (or **off-shell**) particles: they carry four-momentum forced upon them by energy-momentum conservation at each vertex, and this four-momentum need not satisfy (97).

The physical picture is this: a real particle in the lab has a definite mass. A virtual particle is a quantum-mechanical intermediate state that exists only for a brief time, limited by the uncertainty principle, and therefore need not satisfy the mass-shell condition. The heavier the virtual particle relative to the energy available, the more suppressed the process — this suppression is captured by the propagator $i/(q^2 - m^2)$.

Vertices from Interaction Lagrangians

Before doing full calculations, it is useful to be able to read off the vertex structure from $\mathcal{L}_{\text{int}}$. For a coupling $\lambda \phi_1^a \phi_2^b$, the vertex has a legs of type ϕ_1 and b legs of type ϕ_2 . Some examples:

Interaction	Vertex legs	Diagram
$\lambda \phi^3$	3 identical legs	$\phi\phi\phi$
$\lambda \phi_1^2 \phi_2$	2 of type ϕ_1 + 1 of type ϕ_2	$\phi_1\phi_1\phi_2$
$\lambda \phi^4$	4 identical legs	$\phi\phi\phi\phi$
$\lambda \phi ^4 = \lambda \phi^\dagger \phi \phi^\dagger \phi$	2 incoming + 2 outgoing	(complex scalar)
$\lambda \phi_1^2 \phi_2^2$	2 of type ϕ_1 + 2 of type ϕ_2	$\phi_1\phi_1\phi_2\phi_2$
$\lambda \phi_1^3 \phi_2$	3 of type ϕ_1 + 1 of type ϕ_2	$\phi_1\phi_1\phi_1\phi_2$

Remark 18 (Directionality). Right now we are thinking of vertices as abstract building blocks. When we analyse a specific process, we connect these building blocks to external legs (initial/final states) and to each other via internal lines. The orientation (which leg is incoming, which outgoing) is determined by the process: there is no directionality in the vertex itself — only in the process diagram.

Worked Example 1: $\phi_1 \rightarrow \phi_2 + \phi_3$ with $\mathcal{L}_{\text{int}} = \lambda \phi_1 \phi_2 \phi_3$

Setup

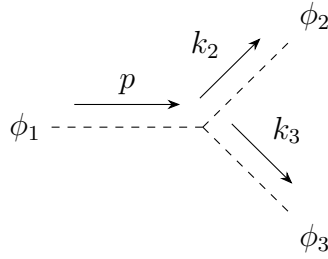
The interaction Lagrangian $\mathcal{L}_{\text{int}} = \lambda \phi_1 \phi_2 \phi_3$ gives a single vertex with one leg of each type. We wish to compute the amplitude for $\phi_1(p) \rightarrow \phi_2(k_2) + \phi_3(k_3)$.

Kinematic constraint

For this decay to be kinematically allowed, the mass of ϕ_1 must be at least $m_2 + m_3$ (the threshold condition from four-momentum conservation):

$$m_1 \geq m_2 + m_3. \quad (98)$$

Feynman diagram



Amplitude

This diagram has:

- One vertex: factor $i\lambda$.
- No internal lines (all three legs are external).
- No undetermined momenta.

Momentum conservation at the vertex gives $p = k_2 + k_3$. The amplitude is simply

$$i\mathcal{A}(\phi_1 \rightarrow \phi_2\phi_3) = i\lambda. \quad (99)$$

The decay rate therefore scales as $\Gamma \propto \lambda^2$.

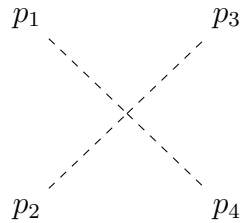
Worked Example 2: $\phi\phi \rightarrow \phi\phi$ with $\mathcal{L}_{\text{int}} = \lambda\phi^4$

Setup

With a single ϕ^4 vertex, label the incoming momenta p_1, p_2 and outgoing momenta p_3, p_4 . Momentum conservation for the whole process: $p_1 + p_2 = p_3 + p_4$.

Leading order: $\mathcal{O}(\lambda)$

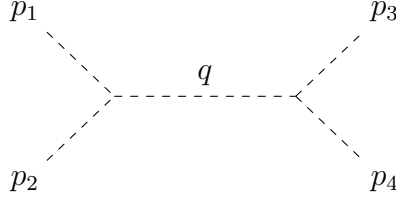
A single vertex connects all four external legs. There are no internal lines and no undetermined momenta.



$$i\mathcal{A}^{(1)} = i\lambda. \quad (100)$$

Next order: $\mathcal{O}(\lambda^2)$ — Tree-level

At order λ^2 we connect two ϕ^4 vertices. There are three inequivalent ways to pair up the four external momenta through the internal line (we will name them the s , t , and u channel diagrams below). The first is:

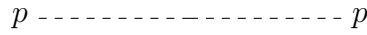


Here $p_1 + p_2$ flows into the internal line, so $q = p_1 + p_2$. The amplitude contribution is:

$$i\mathcal{A}_s^{(2)} = (i\lambda) \frac{i}{q^2 - m^2} (i\lambda) = \frac{-\lambda^2}{(p_1 + p_2)^2 - m^2}. \quad (101)$$

Next order: $\mathcal{O}(\lambda^2)$ — Loop level (tadpole)

The ϕ^4 vertex also allows a *loop* diagram for $\phi \rightarrow \phi$ where one particle emits and reabsorbs a virtual $\phi\phi$ pair:



This **loop diagram** has one undetermined loop momentum ℓ . The amplitude contains:

$$i\mathcal{A}_{\text{loop}} = (i\lambda) \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2}. \quad (102)$$

Notice that this integral over all ℓ is not obviously finite — it requires regularisation and renormalisation, topics beyond this course. The key distinction here is:

Physical Concept 5.3: Tree-level vs. loop-level diagrams

A **tree-level diagram** contains no closed loops; all internal momenta are fully determined by four-momentum conservation at each vertex. A **loop diagram** contains at least one closed loop, leading to an undetermined loop momentum that must be integrated over. At each loop order, the diagram is suppressed by an additional factor of the coupling squared relative to tree level.

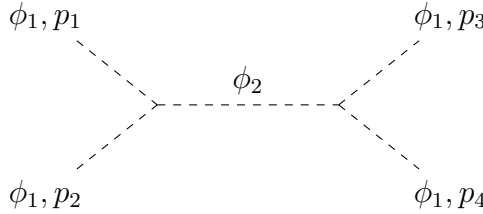
Worked Example 3: $\phi_1\phi_1 \rightarrow \phi_1\phi_1$ with $\mathcal{L}_{\text{int}} = \lambda\phi_1^2\phi_2$

Why there is no order- λ contribution

The vertex $\lambda\phi_1^2\phi_2$ has two ϕ_1 legs and one ϕ_2 leg. A single vertex would require one ϕ_2 leg to connect to an external state, but the process $\phi_1\phi_1 \rightarrow \phi_1\phi_1$ has *only* external ϕ_1 lines. Therefore, there is no diagram at order λ : the leading contribution is at order λ^2 .

Leading order diagram

Two vertices are connected by an internal ϕ_2 line:



The internal ϕ_2 carries momentum $q = p_1 + p_2$ and is off-shell. The amplitude is

$$i\mathcal{A} = (i\lambda) \frac{i}{(p_1 + p_2)^2 - m_2^2} (i\lambda) = \frac{-\lambda^2}{(p_1 + p_2)^2 - m_2^2}. \quad (103)$$

Remark 19 (Physical analogy). This is structurally identical to the harmonic-oscillator $y \rightarrow 3x$ decay computed in Lecture 04, with $\phi_1 \leftrightarrow x$, $\phi_2 \leftrightarrow z$. The same virtual-particle mechanism underlies, for example, the *muon decay* $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ through a virtual W^- boson.

Worked Example 4: $\phi_2 \rightarrow \phi_1 \phi_1 \phi_1$ with Two Interactions

Setup

Consider the theory

$$\mathcal{L}_{\text{int}} = \lambda_1 \phi_1 \phi_2 \phi_3 + \lambda_2 \phi_1^2 \phi_3, \quad (104)$$

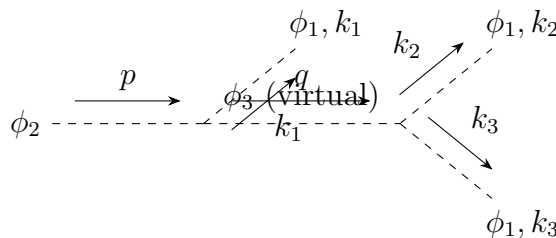
and the decay $\phi_2(p) \rightarrow \phi_1(k_1) + \phi_1(k_2) + \phi_1(k_3)$.

Why second order is needed

The λ_1 vertex has legs $\phi_1 \phi_2 \phi_3$ and the λ_2 vertex has legs $\phi_1 \phi_1 \phi_3$. Neither vertex alone can connect a single ϕ_2 to three ϕ_1 lines in one step. We need *two* vertices, so the leading contribution is $\mathcal{O}(\lambda_1 \lambda_2)$.

Diagram and amplitude

The process proceeds through a virtual ϕ_3 :



Momentum conservation at v_1 : $p = k_1 + q$, so $q = p - k_1$. The amplitude is:

$$i\mathcal{A} = (i\lambda_1) \frac{i}{q^2 - m_3^2} (i\lambda_2) = \frac{-\lambda_1 \lambda_2}{(p - k_1)^2 - m_3^2}. \quad (105)$$

(There is also a second diagram in which v_1 emits k_2 and v_2 emits k_1 and k_3 , plus a third; all must be summed.)

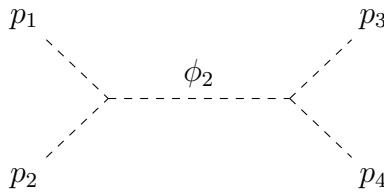
The s , t , u Channels

Setup

Consider $\phi_1\phi_1 \rightarrow \phi_1\phi_1$ in the theory $\mathcal{L}_{\text{int}} = \lambda \phi_1^2 \phi_2^2$ (two ϕ_1 legs and two ϕ_2 legs at each vertex). At order λ^2 , the two λ vertices are connected by a virtual ϕ_2 . There are *three topologically inequivalent ways* to connect the external legs, corresponding to three different momentum channels.

Remark 20 (Why diagrams with relabelled momenta are not equivalent). Even if two diagrams look graphically identical, they are *not equivalent* if different combinations of external momenta are flowing through the internal line. The propagator $i/(q^2 - m^2)$ is a function of q^2 , so different q means a different amplitude contribution. Each distinct pairing must be summed separately.

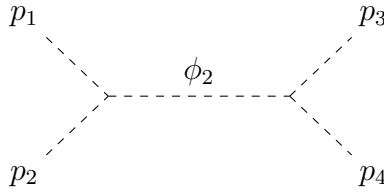
s -channel: $p_1 + p_2$ flows internally



Internal momentum: $q_s = p_1 + p_2$. Contribution:

$$i\mathcal{A}_s = \frac{-\lambda^2}{(p_1 + p_2)^2 - m_2^2}. \quad (106)$$

t -channel: $p_1 - p_3$ flows internally



In this pairing, p_1 and p_3 scatter off each other through the internal line, so $q_t = p_1 - p_3$. Contribution:

$$i\mathcal{A}_t = \frac{-\lambda^2}{(p_1 - p_3)^2 - m_2^2}. \quad (107)$$

u -channel: $p_1 - p_4$ flows internally

With p_1 and p_4 pairing: $q_u = p_1 - p_4$. Contribution:

$$i\mathcal{A}_u = \frac{-\lambda^2}{(p_1 - p_4)^2 - m_2^2}. \quad (108)$$

The full $\mathcal{O}(\lambda^2)$ amplitude is the sum $i\mathcal{A} = i\mathcal{A}_s + i\mathcal{A}_t + i\mathcal{A}_u$.

Mandelstam Variables

Physical Concept 5.4: Mandelstam variables

For a $2 \rightarrow 2$ process with incoming momenta p_1, p_2 and outgoing momenta p_3, p_4 , the three **Mandelstam variables** are:

$$s \equiv (p_1 + p_2)^2 = (p_3 + p_4)^2, \quad (109)$$

$$t \equiv (p_1 - p_3)^2 = (p_2 - p_4)^2, \quad (110)$$

$$u \equiv (p_1 - p_4)^2 = (p_2 - p_3)^2. \quad (111)$$

Remark 21 (Physical meaning). s is the square of the total centre-of-mass energy; $\sqrt{s}$ is the invariant mass of the two incoming particles together. t and u are related to the momentum transfer in the two scattering channels. Because s, t, u are Lorentz scalars, they are the natural kinematic variables for writing amplitudes in a manifestly frame-independent way. In particular, the three diagrams above contribute amplitudes $\propto 1/(s-m^2), 1/(t-m^2), 1/(u-m^2)$ respectively, immediately visible from the Mandelstam variables.

The Mandelstam sum rule

Physical Law 5.1 Mandelstam sum rule

For a $2 \rightarrow 2$ process involving particles with masses m_1, m_2, m_3, m_4 ,

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2. \quad (112)$$

Proof: [Proof (equal-mass case, $m_i = m$ for all i)] Expand each variable using $p_i^2 = m^2$ (all external particles are on-shell):

$$s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = 2m^2 + 2p_1 \cdot p_2, \quad (113)$$

$$t = p_1^2 + p_3^2 - 2p_1 \cdot p_3 = 2m^2 - 2p_1 \cdot p_3, \quad (114)$$

$$u = p_1^2 + p_4^2 - 2p_1 \cdot p_4 = 2m^2 - 2p_1 \cdot p_4. \quad (115)$$

Summing:

$$s + t + u = 6m^2 + 2p_1 \cdot (p_2 - p_3 - p_4). \quad (116)$$

Four-momentum conservation gives $p_1 + p_2 = p_3 + p_4$, i.e. $p_2 = p_3 + p_4 - p_1$. Substituting:

$$p_1 \cdot p_2 = p_1 \cdot (p_3 + p_4 - p_1) = p_1 \cdot p_3 + p_1 \cdot p_4 - m^2. \quad (117)$$

Therefore:

$$p_2 - p_3 - p_4 = (p_3 + p_4 - p_1) - p_3 - p_4 = -p_1. \quad (118)$$

Substituting back into (116):

$$s + t + u = 6m^2 + 2p_1 \cdot (-p_1) = 6m^2 - 2m^2 = 4m^2. \quad (119)$$

Since all four masses are equal, this matches $\sum_{i=1}^4 m_i^2 = 4m^2$. ■

The general proof follows the same steps with $p_i^2 = m_i^2$ and yields (112). The Mandelstam sum rule is a purely kinematic identity — it holds regardless of the dynamics (coupling constants) of the theory.

To Remember

- **Feynman rules for scalar fields:** vertex $\rightarrow i\lambda$; propagator (internal line with momentum q) $\rightarrow i/(q^2 - m^2)$; external states carry no factor; impose four-momentum conservation at each vertex; integrate over each undetermined loop momentum $\int d^4\ell/(2\pi)^4$.
- **On-shell:** $p^2 = m^2$ — satisfied by all physical (external) particles. **Off-shell:** $q^2 \neq m^2$ — the condition for virtual (internal) particles; makes the propagator non-singular for physical processes.
- An interaction ϕ^n generates an n -leg vertex with factor $i\lambda$. Only operators with $n \geq 3$ fields contribute to transitions.
- **Tree-level diagrams** have no loops; all internal momenta are fixed by conservation laws. **Loop diagrams** require integration over undetermined momenta and arise at higher order in the coupling.
- For $2 \rightarrow 2$ scattering with two vertices, there are exactly three inequivalent channels: s , t , and u , named after the Mandelstam variables $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_1 - p_4)^2$.
- **Mandelstam sum rule:** $s + t + u = \sum_i m_i^2$ (summed over all four external particles). This is a kinematic identity that follows from on-shell conditions and four-momentum conservation.

Further Reading

The lecturer noted that Peskin & Schroeder use an unconventional $(-, +, +, +)$ metric sign convention; all other references below use $(+, -, -, -)$ as in this course.

- **Griffiths, Chapter 6.3–6.4** — Feynman rules for scalar theories, diagrammatic computation of $\phi\phi \rightarrow \phi\phi$ scattering, and the three channels. Provides worked calculations close in style to the lecture examples.
- **Peskin & Schroeder, §4.4–4.7** — Feynman rules derived from the path integral and Wick contractions, cross-section calculations, the optical theorem. The standard graduate-level reference (note their metric convention).
- **Halzen & Martin, Chapter 3** — “Electrodynamics of Spinless Particles”: Feynman rules for scalar QED, Mandelstam variables, and explicit cross-section calculations. Very accessible with good worked examples.
- **Tong, *QFT Notes*, Chapter 3, §3.3–3.5** (damtp.cam.ac.uk/user/tong/qft.html): “Feynman Diagrams” — derivation of the Feynman rules including symmetry factors and combinatorics.
- **PDG Kinematics Review** (pdg.lbl.gov/2023/reviews/rpp2023-rev-kinematics.pdf): the canonical reference for s , t , u and all relativistic kinematic identities in the particle-physics notation used throughout this course.

6 Lecture 06 — Connected Diagrams, Loop Divergences, 1PI, and Decay Rates

Date: 26/04/2026

Overview

This lecture completes the foundations of diagrammatic perturbation theory with four important remarks about Feynman diagrams, then transitions to the kinematic side of the problem — Fermi’s Golden Rule for decay rates.

The four diagrammatic notes are:

1. Only **fully connected diagrams** contribute to physical amplitudes; disconnected pieces are absorbed into the normalization.
2. **Vertex corrections**: the effective coupling between particles receives loop corrections, systematically handled by renormalization.
3. **Loop integrals diverge** in the ultraviolet (large loop momentum); two explicit examples demonstrate quadratic and logarithmic divergences. The prescription for dealing with these infinities is **renormalization**.
4. **One-Particle Irreducible (1PI) diagrams** provide a natural organizational tool: they can be resummed into a geometric series that dresses the bare propagator and generates **mass corrections**.

The second half introduces the **general decay rate formula** — the phase-space integral that, combined with the matrix element $|\mathcal{M}|^2$, gives a physical decay rate Γ .

Note 1 — Only Fully Connected Diagrams Contribute

Physical Concept 6.1: Fully connected diagram

A Feynman diagram is **fully connected** if every external leg and every internal line belongs to a single connected piece. Equivalently, there is a path of lines and vertices connecting any two elements of the diagram.

When drawing diagrams for a process, one might imagine configurations where part of the diagram is completely disconnected from the external states — for instance, a free-floating loop that has no legs connecting to the initial or final particles. Such **disconnected** contributions are *never drawn or computed*.

Remark 22 (Why disconnected diagrams can be ignored). In the full path-integral formulation of QFT, the transition amplitude is computed as a ratio of path integrals. The disconnected “vacuum bubble” pieces factorise and appear identically in numerator and denominator, so they cancel exactly in any physical amplitude. The net result is that we only ever need to sum over *fully connected* diagrams. This is not an approximation; it is an exact statement about how the theory is defined.

Worked Example 6.1 (A disconnected diagram for $\phi\phi \rightarrow \phi\phi$)

Consider $\phi\phi \rightarrow \phi\phi$ in ϕ^4 theory. One could imagine drawing a diagram where the two incoming ϕ lines reconnect to the two outgoing ϕ lines without any interaction, alongside a separate closed loop that floats in isolation. This is *not* a fully connected diagram — the loop piece is disconnected from the external legs. We do not include it. Only diagrams in which all four external legs are part of one connected network are included.

Note 2 — Vertex Corrections

The interaction vertices introduced by $\mathcal{L}_{\text{int}}$ are the *bare* building blocks of perturbation theory. However, once we go beyond leading order, we can draw loop diagrams that effectively insert additional structure into a vertex.

Worked Example 6.2 (Effective ϕ^4 vertex at order λ^2)

In ϕ^4 theory, the bare vertex connects four legs with a factor $i\lambda$. At order λ^2 , one can draw a diagram connecting two ϕ^4 vertices by two internal lines forming a loop — an “ice-cream cone” or “lollipop” shape — where all four external legs are still attached. This diagram contributes to the same four-point function as the original vertex, effectively *correcting* the coupling strength. In this sense, the physical (observed) coupling differs from the bare parameter λ in the Lagrangian. The systematic framework for handling this is **renormalization**.

This observation generalises: in any quantum field theory, the parameters appearing in $\mathcal{L}$ (masses and coupling constants) receive perturbative corrections from loop diagrams. The renormalization group encodes how these effective parameters run with the energy scale of the process.

Note 3 — Loop Divergences and Renormalization

Why loops diverge

Every undetermined loop momentum in a Feynman diagram must be integrated over all values from 0 to ∞ (or, in Euclidean space, from 0 to some ultraviolet cut-off Λ). Because the integrand falls off as a power of the loop momentum, large- Q behaviour determines whether the integral converges.

A useful power-counting rule: in $d = 4$ spacetime dimensions, a loop integral with P propagators and V vertices contributing L loops behaves schematically as

$$\int \frac{d^4 Q}{(2\pi)^4} \frac{(\text{numerator factors})}{(Q^2)^P}. \quad (120)$$

Converting to spherical coordinates in 4D, $d^4 Q = Q^3 dQ d\Omega_4$, so the radial integrand goes as $Q^3/(Q^2)^P = Q^{3-2P}$. The integral diverges if $3 - 2P \geq -1$, i.e. $P \leq 2$.

Example 1 — The tadpole diagram (ϕ^4 theory, one propagator)

In ϕ^4 theory, a particle propagating from p to p can emit and reabsorb a virtual pair via the four-point vertex, forming a loop with *one* internal propagator:

$$p \text{ ----- } p$$

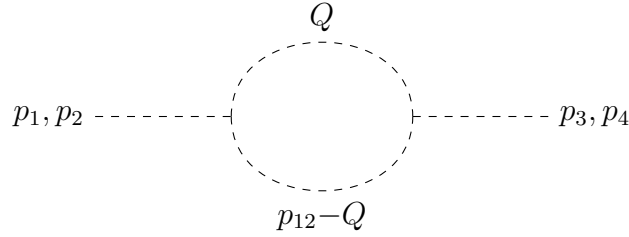
There is one undetermined loop momentum Q , one propagator, and one vertex. Dropping numerical prefactors and focussing on the UV behaviour:

$$\mathcal{A}_{\text{tadpole}} \sim (i\lambda) \int \frac{d^4 Q}{(2\pi)^4} \frac{i}{Q^2 - m^2} \xrightarrow{Q \gg m} \int_0^\Lambda dQ \frac{Q^3}{Q^2} = \int_0^\Lambda dQ Q \sim \Lambda^2. \quad (121)$$

This integral diverges **quadratically** as $\Lambda \rightarrow \infty$.

Example 2 — The “candy” diagram (ϕ^4 theory, two propagators)

A second loop diagram connects two ϕ^4 vertices by *two* internal lines forming a bubble (the professor describes it as shaped like a piece of candy):



Let $p_{12} = p_1 + p_2$. By momentum conservation at v_1 , the two internal momenta are Q and $p_{12} - Q$. There is one undetermined loop momentum Q and two propagators. The UV behaviour:

$$\mathcal{A}_{\text{candy}} \sim (i\lambda)^2 \int \frac{d^4 Q}{(2\pi)^4} \frac{i}{Q^2 - m^2} \frac{i}{(p_{12} - Q)^2 - m^2} \xrightarrow{Q \gg p_{12}, m} \int_0^\Lambda dQ \frac{Q^3}{Q^4} = \int_0^\Lambda \frac{dQ}{Q} \sim \ln \Lambda. \quad (122)$$

This integral diverges **logarithmically**.

Remark 23 (Summary of UV divergence types). In four spacetime dimensions, the two generic types of ultraviolet divergence encountered in loop integrals are:

- **Quadratic divergence:** $\int dQ Q \sim \Lambda^2$ — one propagator loop.
- **Logarithmic divergence:** $\int dQ/Q \sim \ln \Lambda$ — two-propagator loop.

Higher numbers of propagators make the integral more convergent; with $P \geq 3$ the integral is UV-finite in 4D.

Renormalization (brief)

The infinities in (121) and (122) are not a sign that the theory is wrong — they are an artefact of treating the bare parameters of $\mathcal{L}$ as the physical ones. The **renormalization** procedure (or **renormalization group evolution, RGE**) provides a self-consistent prescription for absorbing these divergences into redefined physical parameters. The result is that all measurable quantities are finite. A full treatment is beyond this course, but the key message is: *loops are not ignorable, but their infinities are controllable*.

Note 4 — One-Particle Irreducible Diagrams

Definition and notation

Physical Concept 6.2: One-Particle Irreducible (1PI) diagram

A connected Feynman diagram is **one-particle irreducible (1PI)** if it *cannot* be split into two disconnected pieces by cutting any single internal line. Such diagrams are also called **proper** or **irreducible**. If a diagram *can* be split by cutting one line, it is **one-particle reducible**.

We denote the sum of all 1PI contributions to the two-point function (one particle in, same particle out) by the blob

$$\underbrace{\bigcirc}_{-i\Sigma}, \quad (123)$$

where $-i\Sigma$ is called the **self-energy**. (The factor of $-i$ is a conventional choice that makes the final dressed propagator particularly clean.)

Examples

Consider a single scalar field. The simplest two-point diagrams are:

1. **The tadpole** (drawn in Example 1 above): one vertex with one loop. This *is* 1PI — there is no single internal line whose removal disconnects the diagram into two separate two-point pieces. It contributes to Σ .
2. **Two tadpoles in series**: take two tadpole diagrams and connect them by a single internal propagator. This diagram *can* be split by cutting the internal propagator between them, yielding two individual tadpoles. Hence it is **not** 1PI — it is one-particle reducible. It should *not* be included independently in Σ ; rather, it is automatically generated when we sum 1PI blobs into a series (see below).

The key organisational principle: **we only keep track of 1PI contributions to Σ** . All reducible contributions are reconstructed by the resummation below.

Dressing the Propagator — The Geometric Series

Setting up the sum

Consider the exact two-point function: a particle entering with momentum q and emerging with the same momentum q . In perturbation theory, between the external states we can insert:

- Nothing: just the bare propagator $P \equiv \frac{i}{q^2 - m^2}$.
- One 1PI blob ($-i\Sigma$): a factor $P \cdot (-i\Sigma) \cdot P$.
- Two 1PI blobs: $P \cdot (-i\Sigma) \cdot P \cdot (-i\Sigma) \cdot P$.
- ... and so on to all orders.

Summing all these contributions:

$$\begin{aligned} \Delta(q^2) &= P + P(-i\Sigma)P + P(-i\Sigma)P(-i\Sigma)P + \cdots \\ &= P \sum_{n=0}^{\infty} [(-i\Sigma)P]^n. \end{aligned} \quad (124)$$

Evaluating the geometric series

The sum $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ (valid for $|x| < 1$, analytically continued) gives

$$\Delta(q^2) = \frac{P}{1 - (-i\Sigma)P} = \frac{P}{1 + i\Sigma P}. \quad (125)$$

Now substitute the explicit propagator $P = i/(q^2 - m^2)$:

$$\begin{aligned} \Delta(q^2) &= \frac{\frac{i}{q^2 - m^2}}{1 + i\Sigma \cdot \frac{i}{q^2 - m^2}} = \frac{\frac{i}{q^2 - m^2}}{1 - \frac{\Sigma}{q^2 - m^2}} = \frac{\frac{i}{q^2 - m^2}}{\frac{q^2 - m^2 - \Sigma}{q^2 - m^2}} \\ &= \boxed{\frac{i}{q^2 - m^2 - \Sigma}}. \end{aligned} \quad (126)$$

Physical Concept 6.3: Dressed (full) propagator

The **dressed** or **full propagator**, obtained by resumming all 1PI insertions to all orders, is

$$\Delta(q^2) = \frac{i}{q^2 - m^2 - \Sigma(q^2)}, \quad (127)$$

where $\Sigma(q^2)$ is the (momentum-dependent) self-energy — the sum of all 1PI two-point diagrams.

Physical interpretation: mass corrections

Comparing (127) with the bare propagator $i/(q^2 - m^2)$, we see that the effect of all 1PI insertions is to *shift the pole* of the propagator:

$$m_{\text{bare}}^2 \longrightarrow m_{\text{phys}}^2 = m_{\text{bare}}^2 + \Sigma. \quad (128)$$

The self-energy Σ is a **mass correction** generated by loop diagrams. The physical particle mass is not m from $\mathcal{L}$, but m shifted by quantum corrections. This is another facet of renormalization.

Remark 24 (Why 1PI is the right building block). If we tried to use reducible diagrams as our building blocks, we would be double-counting: a two-tadpole diagram, for example, is already included in the series (124) as the $n = 2$ term, built from two 1PI blobs. The 1PI decomposition avoids any double-counting and is the unique, minimal set of diagrams needed to reconstruct the full two-point function.

Symmetry Protection of Mass Corrections

The 1PI framework has a profound consequence when *symmetries* are present.

Physical Law 6.1 Symmetry protection

Suppose a theory possesses a symmetry that **forbids** a mass term for a given field. Then the self-energy $\Sigma = 0$ to all orders in perturbation theory.

Worked Example 6.3 (Shift symmetry)

Consider a theory invariant under $\phi \rightarrow \phi + a$ (a constant shift). A mass term $m^2\phi^2/2$ is *not* invariant under this shift ($\phi^2 \rightarrow \phi^2 + 2a\phi + a^2 \neq \phi^2$), so symmetry forbids it in $\mathcal{L}$. If loop corrections generated a nonzero $\Sigma \neq 0$, the dressed propagator (127) would exhibit a pole at $q^2 = \Sigma \neq 0$, which is equivalent to the particle acquiring a mass — thereby breaking the shift symmetry. Since the symmetry is exact, this is impossible: every diagram contributing to Σ must be individually included in a sum that vanishes:

$$\Sigma = 0 \quad \text{to all orders,} \quad (129)$$

even without explicit computation of each diagram.

Remark 25 (Power of symmetry). This is one of the deepest uses of symmetry in particle physics. Instead of computing each loop diagram (which may be extremely involved), knowing that a symmetry forbids the mass immediately tells you that the entire class of 1PI self-energy diagrams sums to zero. Conversely, if you *do* compute them and obtain a vanishing sum, it is a strong hint that there is an underlying symmetry at work — even one you have not yet identified.

Fermi's Golden Rule — The General Decay Rate Formula

We now turn from the computation of amplitudes to the connection between amplitudes and *observable rates*. The goal is to derive Γ , the decay rate of a particle at rest.

Setup

We consider a particle of mass m_1 at rest decaying into $n - 1$ final-state particles:

$$1 \rightarrow 2 + 3 + \cdots + n, \quad (130)$$

where particle i has four-momentum p_i and mass m_i . The particle is at rest, so $p_1 = (m_1, \mathbf{0})$.

The general decay rate formula — pedagogical form

The decay rate in the rest frame of particle 1 is:

$$\Gamma = \frac{1}{2m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}\left(p_1 - \sum_{i=2}^n p_i\right) \prod_{j=2}^n \left[2\pi \delta(p_j^2 - m_j^2) \theta(p_j^0) \frac{d^4 p_j}{(2\pi)^4} \right]. \quad (131)$$

Let us identify every factor:

- $|\mathcal{M}|^2$: the squared matrix element for the process $1 \rightarrow 2 \cdots n$, computed using Feynman rules. This is where all the dynamics lives.
- $(2\pi)^4 \delta^{(4)}(p_1 - \sum_{i=2}^n p_i)$: enforces **four-momentum conservation** — the total outgoing four-momentum must equal the initial four-momentum.
- $\delta(p_j^2 - m_j^2)$: enforces that each final-state particle j is **on its mass shell**, $p_j^2 = m_j^2$. These are physical particles, not virtual ones.
- $\theta(p_j^0)$: enforces **positive energy** for each outgoing particle, $E_j = p_j^0 > 0$.
- $\int d^4 p_j / (2\pi)^4$: integrates over all four-momentum components of each outgoing particle.
- $1/(2m_1)$: normalization from the initial state; see below.

Three kinematic constraints

1. **On-shell condition:** $\delta(p_j^2 - m_j^2)$ — final-state particles are real, with $p_j^2 = m_j^2$.
2. **Positive energy:** $\theta(p_j^0)$ — energy is positive; there are no negative-energy outgoing particles.
3. **Four-momentum conservation:** $(2\pi)^4 \delta^{(4)}(p_1 - \sum p_i)$ — total four-momentum is conserved.

The integral in (131) is over *all* kinematically allowed final-state configurations, weighted by the dynamics $|\mathcal{M}|^2$.

Normalization factor $1/(2m_1)$

Recall from the discussion of external states that a one-particle state with momentum k is defined as

$$|k\rangle = \sqrt{2\omega_k} a_{\mathbf{k}}^\dagger |0\rangle, \quad (132)$$

with the Lorentz-invariant normalisation $\langle k'|k\rangle = 2\omega_k (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}')$. For the initial particle at rest, $\omega_{k=0} = m_1$, so the initial-state normalisation contributes a factor $1/(2m_1)$.

Derivation of the standard form

We now perform the integrals over the zeroth (energy) component p_j^0 of each final-state particle, using the on-shell delta functions.

Key identity. For any momentum p with mass m :

$$\delta(p^2 - m^2) = \delta((p^0)^2 - |\mathbf{p}|^2 - m^2) = \delta((p^0)^2 - E_p^2), \quad E_p \equiv \sqrt{|\mathbf{p}|^2 + m^2}. \quad (133)$$

Using the identity $\delta(x^2 - a^2) = \frac{1}{2|a|} [\delta(x - a) + \delta(x + a)]$ with $a = E_p > 0$:

$$\delta(p^2 - m^2) = \frac{1}{2E_p} [\delta(p^0 - E_p) + \delta(p^0 + E_p)]. \quad (134)$$

The $\theta(p^0)$ factor kills the second spike (since $-E_p < 0$), leaving

$$\delta(p^2 - m^2) \theta(p^0) = \frac{1}{2E_p} \delta(p^0 - E_p). \quad (135)$$

Substituting into (131) and integrating dp_j^0 using the delta function sets $p_j^0 = E_j \equiv \sqrt{|\mathbf{p}_j|^2 + m_j^2}$ everywhere it appears:

$$\int \frac{dp_j^0}{2\pi} 2\pi \delta(p_j^2 - m_j^2) \theta(p_j^0) = \int \frac{dp_j^0}{2\pi} \frac{2\pi}{2E_j} \delta(p_j^0 - E_j) = \frac{1}{2E_j}. \quad (136)$$

Standard form of the decay rate

After performing all dp_j^0 integrations, the general decay rate takes the **standard form**:

$$\Gamma = \frac{1}{2m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}\left(p_1 - \sum_{i=2}^n p_i\right) \prod_{j=2}^n \frac{d^3 p_j}{(2\pi)^3 2E_j}. \quad (137)$$

Here p_j^0 has been set equal to $E_j = \sqrt{|\mathbf{p}_j|^2 + m_j^2}$ in all remaining factors (including $|\mathcal{M}|^2$, which generally depends on the four-momenta). The integration is now only over the three-momenta $\mathbf{p}_j$ of the final-state particles.

Remark 26 (Lorentz-invariant phase space). The combination $d^3p_j/(2E_j)$ is the **Lorentz-invariant phase-space measure** for a single on-shell particle with mass m_j . It appears universally in all decay rate and cross-section formulas. Its Lorentz invariance follows directly from (135): $\delta(p^2 - m^2)\theta(p^0) d^4p$ is manifestly Lorentz invariant (the δ and θ are scalars and d^4p is invariant), and integrating over p^0 produces $d^3p/(2E_p)$.

Remark 27 (Structure of the formula). Formula (137) factorises cleanly into two pieces:

- **Dynamics:** $|\mathcal{M}|^2$ — computed from Feynman diagrams using the rules developed in Lecture 05.
- **Kinematics:** the phase-space integral $\int (2\pi)^4 \delta^{(4)}(\dots) \prod_j d^3p_j / [(2\pi)^3 2E_j]$ — a purely kinematic measure over all allowed final states, independent of the theory's coupling constants.

This factorisation is the practical power of the formula: one can evaluate the kinematic phase space independently of the dynamics, and vice versa.

Preview: two-body decays

The simplest and most common case is $n = 2$: particle 1 decays to two particles, $1 \rightarrow 2 + 3$. In this case, four-momentum conservation (in 3D) leaves only the direction of the decay as a free variable — the magnitudes $|\mathbf{p}_2|$ and $|\mathbf{p}_3|$ are fully fixed by kinematics. The phase-space integral reduces to a single angular integral, and the decay rate becomes particularly tractable. This will be worked out explicitly in Lecture 07.

To Remember

- **Fully connected diagrams only:** disconnected pieces cancel in the normalization and never contribute to physical amplitudes.
- **Vertex corrections:** bare couplings receive loop corrections; this is handled by renormalization. Physical couplings differ from bare ones.
- **UV divergences:** loop integrals in $d = 4$ can diverge. Two types: quadratic ($\sim \Lambda^2$, from one-propagator loops) and logarithmic ($\sim \ln \Lambda$, from two-propagator loops). Renormalization provides a finite, self-consistent prescription.
- **1PI diagrams:** a diagram is 1PI if no single internal line cut disconnects it. All 1PI two-point contributions are collected into the self-energy $-i\Sigma$.
- **Dressed propagator:** resumming all 1PI insertions as a geometric series gives

$$\Delta(q^2) = \frac{i}{q^2 - m^2 - \Sigma},$$

showing that Σ shifts the physical mass: $m_{\text{phys}}^2 = m_{\text{bare}}^2 + \Sigma$.

- **Symmetry protection:** if a symmetry forbids a mass term, then $\Sigma = 0$ to all orders. This is a powerful tool: entire classes of loop diagrams must vanish without explicit computation.
- **General decay rate** (standard form):

$$\Gamma = \frac{1}{2m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}\left(p_1 - \sum p_i\right) \prod_{j=2}^n \frac{d^3p_j}{(2\pi)^3 2E_j},$$

with $E_j = \sqrt{|\mathbf{p}_j|^2 + m_j^2}$. The three kinematic constraints (on-shell, positive energy, four-momentum conservation) are built into the phase-space measure.

- The normalization $1/(2m_1)$ comes from the Lorentz-invariant initial-state normalization $\sqrt{2\omega_k}$ evaluated at rest, $\omega = m_1$.
- **Next:** two-body decay $1 \rightarrow 2 + 3$ as an explicit worked example; then scattering cross-sections; then fermions.

Further Reading

The lecturer explicitly recommended the following:

- **Griffiths, Chapter 6.2** — “The Feynman Rules for a Toy Theory”: the decay rate formula and Fermi’s Golden Rule in the simplest scalar setting. Cited directly for the phase-space discussion in the second half of this lecture.
- **Peskin & Schroeder, Chapter 4** — Full derivation of the S -matrix, LSZ reduction formula, decay rates and cross-sections, the role of disconnected/vacuum-bubble diagrams, and 1PI structures. Recommended by the lecturer for the entire content of this lecture.
- **PDG Review of Kinematics** (pdg.lbl.gov/2023/reviews/rpp2023-rev-kinematics.pdf): concise review of relativistic kinematics, phase space, decay rates, and cross-section formulas. The lecturer described this as “very good for bottom lines”.

Additional recommended reading:

- **Tong, *QFT Notes*, Chapter 3, §3.5–3.6** (damtp.cam.ac.uk/user/tong/qft.html): “Decay Rates and Cross Sections” and “Connected Diagrams and the Vacuum Bubble” — covers both main topics of this lecture with full derivations.
- **Peskin & Schroeder, §7.1** — The full two-point function and 1PI self-energy (Källén–Lehmann representation). More advanced, but directly extends the 1PI resummation derived in this lecture to the full non-perturbative propagator.
- **Coleman, *Aspects of Symmetry*, Chapter 3** ([Cambridge UP](https://www.cambridge.org/9780521876223)): an elegant treatment of how symmetries constrain loop corrections and protect masses, directly relevant to the symmetry-protection argument in §6.
- **Halzen & Martin, §4.1** — General decay rate and two-body phase space in the scalar theory, before fermions are introduced.

7 Lecture 07 — Two-Body Decays, Cross-Sections, Symmetry Factors, and Fermions

Date: 30/04/2026

Overview

This lecture completes the treatment of phase space and observables, then opens the chapter on fermions.

1. **Two-body decay kinematics:** explicit formulae for the energies and three-momenta of the daughter particles in the rest frame; integration over the solid angle; the compact two-body decay rate formula.
2. **General scattering cross-section:** the phase-space formula for $1 + 2 \rightarrow 3 + \dots + n$; the Lorentz-invariant prefactor; the standard form of $d\sigma$.
3. **$2 \rightarrow 2$ scattering in the center-of-mass frame:** working out s, t, u for ϕ^3 theory; the equal-mass simplification; the full differential cross-section.
4. **Symmetry factors via contraction counting:** why the three channels in $\phi\phi \rightarrow \phi\phi$ enter with equal weight; the concept of contractions.
5. **Dimensional analysis of matrix elements:** $[\mathcal{M}] = \text{mass}^{4-n}$ for an n -particle process; sanity checks on Γ and σ .
6. **Summary of perturbative QFT:** free vs. interacting fields; role of Feynman diagrams.
7. **Introduction to fermions:** the Dirac equation; gamma matrices and Clifford algebra; γ^5 ; Dirac conjugate $\bar{\psi}$; the Dirac Lagrangian; Weyl spinors.

Two-Body Decay Kinematics

Physical intuition: When a particle at rest decays to exactly two daughters, four-momentum conservation in 3D leaves no freedom in the magnitude of the momenta — the daughters must fly back-to-back with equal and opposite three-momenta, completely determined by the three masses. The only remaining freedom is the direction of the decay axis, which is integrated over to obtain the total rate.

Setup

Consider particle 1 of mass M_1 at rest decaying to particles 2 and 3:

$$1 \xrightarrow{\text{rest}} 2 + 3.$$

The four-momentum of particle 1 is $p_1 = (M_1, \mathbf{0})$, since it is at rest.

Energies and momenta of the daughters

Conservation of four-momentum, $p_1 = p_2 + p_3$, together with the on-shell conditions $p_i^2 = m_i^2$ uniquely determines:

$$E_2 = \frac{M_1^2 + m_2^2 - m_3^2}{2M_1}, \quad (138)$$

and (by the same calculation with $2 \leftrightarrow 3$)

$$E_3 = \frac{M_1^2 + m_3^2 - m_2^2}{2M_1}. \quad (139)$$

Because three-momentum is conserved and the parent is at rest, $\mathbf{p}_2 + \mathbf{p}_3 = \mathbf{0}$, so $|\mathbf{p}_2| = |\mathbf{p}_3| \equiv p_*$. The magnitude is:

$$p_* = \frac{\sqrt{[M_1^2 - (m_2 + m_3)^2][M_1^2 - (m_2 - m_3)^2]}}{2M_1}. \quad (140)$$

This can be written more compactly using the triangle (Källén) function $\lambda(a, b, c) = (a - b - c)^2 - 4bc$, but the form above makes the physics transparent: $p_* > 0$ requires $M_1 > m_2 + m_3$ (the decay threshold).

Integration over the solid angle

From Lecture 06, the standard decay rate formula (with the p_j^0 integrals already done) gives the differential decay rate:

$$d\Gamma = \frac{1}{32\pi^2} |\mathcal{M}|^2 \frac{p_*}{M_1^2} d\Omega_2, \quad (141)$$

where $d\Omega_2 = \sin\theta_2 d\theta_2 d\phi_2$ is the solid angle of particle 2. If $|\mathcal{M}|^2$ is independent of the decay direction (true for scalar particles), we integrate:

$$\int d\Omega = \int_{-1}^1 d(\cos\theta) \int_0^{2\pi} d\phi = 4\pi.$$

The total two-body decay rate is then:

$$\Gamma_{1 \rightarrow 23} = \frac{p_*}{8\pi M_1^2} |\mathcal{M}|^2. \quad (142)$$

Remark 28 (Kinematics fully determined). For a two-body decay, all momenta are uniquely determined by the masses up to the overall direction. In contrast, for three-body and higher decays there is a continuous distribution of final-state configurations — the Dalitz plot. The two-body case is therefore both the simplest and the most predictive.

General Scattering Cross-Section

Physical intuition: The cross-section is the effective area that a target particle presents to an incoming beam. It is defined so that the rate equals the flux times the cross-section. The factor $1/(2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2|)$ in the formula is exactly this flux factor, and the rest is the same phase-space integral as in the decay rate.

The general formula

For the $2 \rightarrow n$ scattering process $1 + 2 \rightarrow 3 + \dots + n$, the total cross-section is:

$$\sigma = \frac{1}{2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2|} \int |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}\left(p_1 + p_2 - \sum_{j=3}^n p_j\right) \prod_{j=3}^n \frac{d^3 p_j}{(2\pi)^3 2E_j}, \quad (143)$$

where $E_j = \sqrt{|\mathbf{p}_j|^2 + m_j^2}$ throughout. The three kinematic constraints — on-shell, positive energy, and four-momentum conservation — are built into the measure, exactly as in the decay rate.

Origin of the prefactor

The prefactor $1/(2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2|)$ has two origins:

- $1/(2E_1) \cdot 1/(2E_2)$: normalization of the two initial-state wave functions (same $\sqrt{2\omega_k}$ normalization used for the decay rate).
- $1/|\mathbf{v}_1 - \mathbf{v}_2|$: the cross-section is obtained by dividing the transition rate by the flux of incoming particles; the flux is proportional to the relative velocity $|\mathbf{v}_1 - \mathbf{v}_2|$.

Lorentz-invariant form of the prefactor

While individual E_i and $|\mathbf{v}_i|$ are frame-dependent, the combination

$$2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2| = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} \quad (144)$$

is Lorentz invariant (it is a Lorentz scalar). This makes the total cross-section Lorentz invariant, as it must be: the total number of scattering events is a physical observable, independent of frame.

$2 \rightarrow 2$ Scattering: Differential Cross-Section

General result

For $1 + 2 \rightarrow 3 + 4$, the differential cross-section with respect to the solid angle in the center-of-mass (CM) frame is:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{CM}} = \frac{1}{2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2|} \cdot \frac{|\mathbf{p}_3|}{(2\pi)^2 4\sqrt{s}} |\mathcal{M}(1 + 2 \rightarrow 3 + 4)|^2, \quad (145)$$

where $\sqrt{s}$ is the CM energy.

Equal-mass simplification

When all four particles have the same mass m (including the massless limit $m \rightarrow 0$), the CM momenta satisfy $|\mathbf{p}_3| = |\mathbf{p}_1| \equiv p_{\text{CM}}$, and (145) simplifies to:

$$\boxed{\left. \frac{d\sigma}{d\Omega} \right|_{\text{CM}} = \frac{|\mathcal{M}|^2}{64\pi^2 s}} \quad (146)$$

This is the form most frequently used in practice.

Worked Example: $\phi\phi \rightarrow \phi\phi$ in ϕ^3 Theory

Physical intuition: In ϕ^3 theory, two scalar particles can scatter by exchanging a virtual ϕ , and the intermediate virtual particle can carry different four-momenta depending on which pair of external legs it connects. The three distinct topologies (s , t , u channels) correspond to these three choices, and in the CM frame they differ by how the scattering angle θ enters.

Review of the three diagrams

The interaction Lagrangian is $\mathcal{L}_{\text{int}} = \lambda\phi^3$. For $\phi\phi \rightarrow \phi\phi$ there are three Feynman diagrams at order λ^2 , labeled by the Mandelstam variable of the propagator:

$$\mathcal{M}_s \sim \frac{\lambda^2}{s - m^2}, \quad s = (p_1 + p_2)^2, \quad (147)$$

$$\mathcal{M}_t \sim \frac{\lambda^2}{t - m^2}, \quad t = (p_1 - p_3)^2, \quad (148)$$

$$\mathcal{M}_u \sim \frac{\lambda^2}{u - m^2}, \quad u = (p_1 - p_4)^2. \quad (149)$$

The total amplitude is $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$.

Kinematics in the center-of-mass frame

Choose the z -axis along the beam. The four four-momenta are:

$$p_1 = (E, 0, 0, +p_{\text{CM}}), \quad (150)$$

$$p_2 = (E, 0, 0, -p_{\text{CM}}), \quad (151)$$

$$p_3 = (E, p_{\text{CM}} \sin \theta, 0, +p_{\text{CM}} \cos \theta), \quad (152)$$

$$p_4 = (E, -p_{\text{CM}} \sin \theta, 0, -p_{\text{CM}} \cos \theta), \quad (153)$$

where all particles have equal mass m , so $E^2 = p_{\text{CM}}^2 + m^2$.

Computing the Mandelstam variables

Computing s . We have $p_1 + p_2 = (2E, \mathbf{0})$, so:

$$s = (p_1 + p_2)^2 = 4E^2. \quad (154)$$

Computing t . We compute $p_1 - p_3$ component by component:

$$p_1 - p_3 = (0, -p_{\text{CM}} \sin \theta, 0, p_{\text{CM}}(1 - \cos \theta)).$$

Then:

$$\begin{aligned} t &= (p_1 - p_3)^2 = 0^2 - p_{\text{CM}}^2 \sin^2 \theta - p_{\text{CM}}^2 (1 - \cos \theta)^2 \\ &= -p_{\text{CM}}^2 [\sin^2 \theta + 1 - 2 \cos \theta + \cos^2 \theta] \\ &= -p_{\text{CM}}^2 [2 - 2 \cos \theta] \\ &= \boxed{-2p_{\text{CM}}^2 (1 - \cos \theta)}. \end{aligned} \quad (155)$$

(Note: the Minkowski metric introduces an overall minus sign on the spatial part; the zero component is zero here since all masses are equal.)

Computing u . Similarly, $p_1 - p_4 = (0, p_{\text{CM}} \sin \theta, 0, p_{\text{CM}}(1 + \cos \theta))$, giving:

$$u = -p_{\text{CM}}^2 (1 + \cos \theta)^2 - p_{\text{CM}}^2 \sin^2 \theta = \boxed{-2p_{\text{CM}}^2 (1 + \cos \theta)}. \quad (156)$$

Remark 29 (Sanity check: $s + t + u = 4m^2$). One can verify $s + t + u = 4E^2 - 2p_{\text{CM}}^2 (1 - \cos \theta) - 2p_{\text{CM}}^2 (1 + \cos \theta) = 4E^2 - 4p_{\text{CM}}^2 = 4m^2$, as required by the general identity $s + t + u = \sum_i m_i^2$.

The full differential cross-section

Substituting into $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$ and then into (146):

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \left| \frac{\lambda^2}{s - m^2} + \frac{\lambda^2}{t - m^2} + \frac{\lambda^2}{u - m^2} \right|^2. \quad (157)$$

Explicitly, using $s = 4E^2$, $t = -2p_{\text{CM}}^2(1 - c)$, $u = -2p_{\text{CM}}^2(1 + c)$ where $c \equiv \cos \theta$:

$$|\mathcal{M}|^2 = \lambda^4 \left(\frac{1}{4E^2 - m^2} + \frac{1}{-2p_{\text{CM}}^2(1 - c) - m^2} + \frac{1}{-2p_{\text{CM}}^2(1 + c) - m^2} \right)^2. \quad (158)$$

Each channel depends differently on $\cos \theta$: the s -channel is flat in angle, while the t - and u -channels peak in the forward and backward directions respectively.

Symmetry Factors via Contraction Counting

Physical intuition: When writing down $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$, one must ensure that the three diagrams enter with the correct relative numerical coefficients. These come from counting the number of distinct ways to “wire up” the external legs to the interaction vertices, using a procedure called *Wick contractions*.

The method

Each ϕ^3 vertex contributes $\lambda\phi\phi\phi$ — three distinguishable leg “slots”. For $\phi\phi \rightarrow \phi\phi$ at order λ^2 there are two vertices, each with three legs, giving $3 + 3 = 6$ slots in total. The four external states (p_1, p_2, p_3, p_4) and the one internal propagator must each connect a specific pair of slots.

To count the symmetry factor for a given diagram, proceed as follows:

1. For each external leg, count the number of available vertex slots it can connect to (given the topology of the diagram), and write down that number.
2. For the internal propagator, count the number of ways the two remaining slots can connect to each other (usually 1, since they are forced by topology once the external legs are placed).
3. Multiply all the numbers together.

s -channel: diagram A

External legs p_1 and p_2 both connect to vertex 1; p_3 and p_4 both connect to vertex 2; the internal line connects the two vertices.

Leg	Vertex	Available slots
p_1	vertex 1	3
p_2	vertex 1	2 (one slot taken)
p_3	vertex 2	3
p_4	vertex 2	2 (one slot taken)
internal	v1–v2	1 (fully determined)

Total: $3 \times 2 \times 3 \times 2 \times 1 = 36$.

***t*-channel: diagram B**

External legs p_1 and p_3 connect to vertex 1; p_2 and p_4 connect to vertex 2. The same counting gives:

$$3 \times 2 \times 3 \times 2 \times 1 = 36.$$

***u*-channel: diagram C**

External legs p_1 and p_4 connect to vertex 1; p_2 and p_3 connect to vertex 2. Again:

$$3 \times 2 \times 3 \times 2 \times 1 = 36.$$

Conclusion

All three diagrams have the same symmetry factor (36), so they enter $\mathcal{M}$ with equal coefficients — no extra numerical prefactors between channels. This confirms that the sum $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$ with unit relative weights is correct.

Remark 30 (Interchange of identical vertices). One might worry about the factor arising from swapping the two vertices (i.e., relabeling vertex 1 as vertex 2). In full perturbation theory, every n -th order term in the expansion of $e^{iS_{\text{int}}}$ comes with a factor $1/n!$ from the Taylor expansion. For two vertices, this $1/2!$ exactly cancels the 2 ways of labeling the vertices. Therefore, **one never needs to include an extra factor for interchanging identical vertices**; it is automatically cancelled by the perturbative expansion.

Physical Concept 7.1: Wick contraction

The procedure of assigning external states to vertex legs and connecting remaining legs via internal propagators is called a **Wick contraction**. Each distinct complete contraction corresponds to one Feynman diagram, and the number of ways to perform a given contraction gives the numerical (symmetry) factor for that diagram.

Dimensional Analysis of Matrix Elements

Physical intuition: Natural units relate mass, energy, time, and length. A quick dimensional check — before evaluating a single loop integral or performing any computation — can immediately reveal algebra errors.

Units of observables

In natural units ($\hbar = c = 1$):

- **Decay rate** Γ : has units of $[\text{time}]^{-1} = [\text{mass}]^1$.
- **Cross-section** σ : has units of area $= [\text{length}]^2 = [\text{mass}]^{-2}$.

Remark 31 (The barn). Cross-sections are sometimes quoted in **barns**: $1 \text{ barn} = 10^{-24} \text{ cm}^2$. This is a convenient unit for nuclear and particle physics cross-sections.

Mass dimension of the matrix element

Physical Law 7.1 Dimension of $\mathcal{M}$

For a process involving n external particles (initial plus final), regardless of how they are distributed between initial and final states, the matrix element has mass dimension:

$$[\mathcal{M}] = \text{mass}^{4-n}. \quad (159)$$

Examples:

- $1 \rightarrow 2 + 3$ (decay, $n = 3$): $[\mathcal{M}] = \text{mass}^1$.
- $1 + 2 \rightarrow 3 + 4$ (scattering, $n = 4$): $[\mathcal{M}] = \text{mass}^0$ (dimensionless).
- $1 \rightarrow 2 + 3 + 4$ (three-body decay, $n = 4$): $[\mathcal{M}] = \text{mass}^0$ (dimensionless).

Consistency check

One can verify these dimensions are consistent with the formulas for Γ and σ :

- **Two-body decay** (142): $[\Gamma] = [p_*]/[M_1^2] \cdot [\mathcal{M}]^2 = \text{mass} \cdot \text{mass}^{-2} \cdot \text{mass}^2 = \text{mass}^1$. ✓
- **$2 \rightarrow 2$ cross-section** (146): $[\frac{d\sigma}{d\Omega}] = [\mathcal{M}]^2/[s] = \text{mass}^0/\text{mass}^2 = \text{mass}^{-2}$. ✓

Summary: Structure of Perturbative QFT

The framework we have built can be summarized as follows:

- **Quadratic (free) terms** in $\mathcal{L}$: terms with exactly two fields, $\mathcal{L} \supset \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$. These describe **free fields** — particles that propagate without interacting. Free particles cannot be created or destroyed; there are no nontrivial scattering or decay processes.
- **Higher-order (interaction) terms**: terms with three or more fields, e.g., $\mathcal{L} \supset \lambda\phi^3$ or $\lambda\phi^4$. These are treated as **perturbations** of the free theory, valid when the coupling λ is small. They introduce interactions: particles can now be created, destroyed, scattered, or decay.
- **Feynman diagrams**: a systematic graphical tool for computing the transition amplitude (matrix element $\mathcal{M}$) for any process, order by order in λ . Each diagram corresponds to a term in perturbation theory.
- **From $\mathcal{M}$ to observables**: the matrix element squared $|\mathcal{M}|^2$, combined with the appropriate phase-space integral, gives physical observables — decay rates Γ and cross-sections σ — that can be compared to experiment.
- **Experiment**: once calculations are in hand, one runs experiments to measure Γ or σ and compare with theory to test whether the theory describes nature.

Fields We Care About

Before diving into fermions, it is useful to catalog the types of fields that appear in the Standard Model and in any relativistic QFT:

Spin	Field type	Symbol	Example
0	Scalar	ϕ	Higgs boson H
1/2	Fermion (Dirac spinor)	ψ, ψ_L, ψ_R	Quarks, leptons (electrons, neutrinos, etc.)
1	Vector (gauge boson)	A^μ	Photon γ , $W^\pm$, Z^0 , gluons

Scalars and vectors are in some sense analogous in their treatment; fermions require substantially more formalism. The gauge bosons (spin-1) will be covered when we discuss local (gauge) symmetries later in the course.

Introduction to Fermions

Physical intuition: For bosons (spin-0, spin-1), we started from the Klein–Gordon equation. For fermions (spin- $\frac{1}{2}$), the appropriate starting point is the Dirac equation. Fermions obey anti-commutation relations (rather than commutation relations) for their creation/annihilation operators, and their fields carry spinor indices.

The Dirac equation

The equation of motion for a free spin- $\frac{1}{2}$ field is the **Dirac equation**:

$$\boxed{(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.} \quad (160)$$

The field $\psi(x)$ is called a **Dirac fermion** (or Dirac spinor); it obeys the Dirac equation because of the specific structure of γ^μ .

The Dirac spinor ψ

$\psi(x)$ is a **four-component** field:

$$\psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}. \quad (161)$$

Important: ψ is *not* a four-vector. A four-vector transforms under Lorentz boosts and rotations as $V^\mu \rightarrow \Lambda^\mu_\nu V^\nu$; ψ transforms differently (as a spinor representation of the Lorentz group). The four components of ψ are internal degrees of freedom — two spin states times two particle/antiparticle states. ψ is also called a **Dirac spinor**.

From the quantum-mechanical perspective: fermions are excitations of fields whose mode expansions use *fermionic* harmonic oscillator operators, satisfying anti-commutation relations $\{a, a^\dagger\} = 1$ rather than commutation relations.

Gamma matrices (Dirac matrices)

Physical intuition: In non-relativistic quantum mechanics, spin- $\frac{1}{2}$ particles are described using Pauli matrices σ^i . The Dirac matrices γ^μ are the relativistic generalization: four 4×4 matrices (one for each spacetime direction) that encode both the Pauli matrices and the relativistic kinematics.

The index μ in γ^μ runs from 0 to 3, so γ^μ represents four matrices:

$$\gamma^0, \gamma^1, \gamma^2, \gamma^3.$$

Each γ^μ is a 4×4 matrix. In the **standard (Dirac) basis**, they are written in 2×2 block form:

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad (162)$$

where

$$\sigma^\mu \equiv (\mathbf{1}_{2 \times 2}, \sigma^1, \sigma^2, \sigma^3), \quad (163)$$

$$\bar{\sigma}^\mu \equiv (\mathbf{1}_{2 \times 2}, -\sigma^1, -\sigma^2, -\sigma^3), \quad (164)$$

and σ^i are the three 2×2 Pauli matrices. Explicitly:

$$\gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad i = 1, 2, 3. \quad (165)$$

Remark 32 (Different bases). The representation (162) is sometimes called the **Weyl** or **chiral basis**. Peskin & Schroeder use this basis. Other texts (e.g., Griffiths) use a different basis (the **Dirac basis**) related by a unitary transformation. Physical results are basis-independent.

Clifford algebra

The gamma matrices satisfy the fundamental anti-commutation relation:

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu} \mathbf{1}_{4 \times 4}, \quad (166)$$

where $g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$ is the Minkowski metric. This is the **Clifford algebra**. It is the defining property of the Dirac matrices — any 4×4 matrix representation satisfying (166) gives an equivalent formulation of the Dirac equation.

The γ^5 matrix

An additional important matrix is:

$$\gamma^5 \equiv i \gamma^0 \gamma^1 \gamma^2 \gamma^3. \quad (167)$$

In the standard basis:

$$\gamma^5 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix}. \quad (168)$$

Key properties:

- $(\gamma^5)^2 = \mathbf{1}$ (it squares to the identity).
- $(\gamma^5)^\dagger = \gamma^5$ (it is Hermitian).
- $\{\gamma^5, \gamma^\mu\} = 0$ for $\mu = 0, 1, 2, 3$ (it anti-commutes with all γ^μ).

γ^5 is the chirality operator. Its eigenvalues ± 1 define right-handed (+) and left-handed (−) chirality states (the Weyl spinors, discussed below).

Dirac conjugate $\bar{\psi}$

For building Lorentz-invariant bilinears, one defines the **Dirac conjugate**:

$$\bar{\psi} \equiv \psi^\dagger \gamma^0. \quad (169)$$

One always works with $\bar{\psi}$ rather than $\psi^\dagger$; the factor of γ^0 is needed to ensure Lorentz invariance. For example, $\bar{\psi}\psi$ is a Lorentz scalar, while $\psi^\dagger\psi$ is not.

Dirac Lagrangian

The Lagrangian that yields the Dirac equation as its equation of motion is:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi. \quad (170)$$

Applying the Euler–Lagrange equation with respect to $\bar{\psi}$ (treated as an independent degree of freedom) gives:

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}} = (i\gamma^\mu \partial_\mu - m)\psi = 0,$$

which is precisely the Dirac equation (160). We treat ψ and $\bar{\psi}$ as independent degrees of freedom, exactly as one treats ϕ and $\phi^\dagger$ independently for a complex scalar.

Weyl spinors

The Dirac spinor ψ decomposes into two **two-component Weyl spinors**:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (171)$$

where ψ_L (the upper two components) is called the **left-handed Weyl spinor** and ψ_R (the lower two components) is the **right-handed Weyl spinor**. Each is a two-component object.

Physical Concept 7.2: Weyl spinors

The left- and right-handed Weyl spinors $\psi_{L,R}$ are the eigenstates of γ^5 :

$$\gamma^5 \psi_L = -\psi_L, \quad \gamma^5 \psi_R = +\psi_R. \quad (172)$$

They are the irreducible representations of the Lorentz group for $\text{spin-}\frac{1}{2}$ fields. Under Lorentz transformations, ψ_L and ψ_R transform into themselves (not into each other), making them the basic building blocks of any spinor field.

Remark 33 (Why Weyl spinors are fundamental). Under Lorentz boosts and rotations, ψ_L and ψ_R transform separately (though with related parameters). The Dirac spinor ψ is therefore a reducible representation: it is the direct sum of a left-handed and a right-handed Weyl representation. In the Standard Model, it turns out that the matter fields (quarks and leptons) have different quantum numbers for ψ_L and ψ_R — left and right handedness are treated differently by the weak force — so the Weyl spinors are the physically natural variables.

To Remember

- **Two-body decay rate:** $\Gamma_{1 \rightarrow 23} = \frac{p_*}{8\pi M_1^2} |\mathcal{M}|^2$, where p_* is the CM momentum of either daughter, fixed by the three masses.
- **General cross-section prefactor:** $1/(2E_1 2E_2 |\mathbf{v}_1 - \mathbf{v}_2|)$; its Lorentz-invariant form is $1/(4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2})$.
- **Equal-mass $2 \rightarrow 2$ cross-section:** $d\sigma/d\Omega = |\mathcal{M}|^2/(64\pi^2 s)$.
- **Mandelstam variables in CM frame** (equal mass m): $s = 4E^2$, $t = -2p_{\text{CM}}^2(1 - \cos\theta)$, $u = -2p_{\text{CM}}^2(1 + \cos\theta)$.

- **Symmetry factors via contraction counting:** count the number of ways to wire external legs to vertex slots; equal factors for all channels means equal relative weights.
- **Dimension of $\mathcal{M}$:** $[\mathcal{M}] = \text{mass}^{4-n}$ for an n -particle process.
- **Dirac equation:** $(i\not{\partial} - m)\psi = 0$.
- **ψ is a four-component Dirac spinor**, not a four-vector.
- **Gamma matrices** satisfy the Clifford algebra: $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$.
- $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$: squares to $+1$, anti-commutes with γ^μ , eigenvalues ± 1 define chirality.
- **Dirac conjugate:** $\bar{\psi} = \psi^\dagger \gamma^0$. Always use $\bar{\psi}$, not $\psi^\dagger$.
- **Dirac Lagrangian:** $\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi$.
- **Weyl decomposition:** $\psi = (\psi_L, \psi_R)^T$; ψ_L and ψ_R are the irreducible Lorentz representations, each two-component.

Recommended Reading

- **Griffiths, Chapter 6.3–6.4** — Two-body and three-body decay rates in the toy scalar theory; cross-section formula. The worked kinematics are directly parallel to this lecture.
- **Peskin & Schroeder, Chapter 3** — The Dirac equation, Dirac matrices, Clifford algebra, Weyl spinors, and the Dirac Lagrangian. The lecturer explicitly recommended Chapter 3 for the fermion material in this and subsequent lectures.
- **Halzen & Martin, Chapter 4–5** — Phase-space kinematics, two-body scattering and decay, followed by the introduction of spinors and Dirac fermions. A very accessible treatment.
- **Tong, *Lectures on Particle Physics*, Chapter 3** (damtp.cam.ac.uk/user/tong/pp.html): phase space and cross-sections with the same conventions used in this course; Chapter 4 covers the Dirac equation.
- **PDG Review of Kinematics** (pdg.lbl.gov/2023/reviews/rpp2023-rev-kinematics.pdf): compact reference for two-body decay kinematics, Mandelstam variables, and cross-section formulas.

Part III

Symmetries & the Standard Model

8 Lecture 08 — Fermion Bilinears, Renormalizability, and the Road to the Standard Model

Date: 03/05/2026

Overview

This lecture completes the foundational treatment of Dirac fermions and introduces the guiding principles behind the Standard Model Lagrangian.

1. **Projection operators:** extracting ψ_L and ψ_R from the Dirac spinor via $P_{L,R}$.
2. **Dirac equation in Weyl form:** coupling of chiralities by the mass; massless decoupling into Weyl equations.
3. **Fermion bilinears:** the complete set of Lorentz-covariant objects bilinear in ψ ; scalar, vector, pseudo-vector, pseudo-scalar, and tensor.
4. **Feynman slash notation:** $\not{A} = \gamma^\mu A_\mu$; compact form of the Dirac equation and Lagrangian.
5. **Component notation:** spinor index structure of the Dirac Lagrangian; distinguishing spinor indices from Lorentz indices.
6. **Mass dimensions:** $[\psi] = \frac{3}{2}$, $[\phi] = 1$; renormalizability limits interactions to at most two fermion fields.
7. **Dirac mass term:** structure $m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L)$; Majorana mass briefly mentioned.
8. **Fermion propagator:** $i(\not{p} + m)/(p^2 - m^2)$; comparison with the scalar propagator.
9. **Chirality and parity:** handedness is reversed under parity.
10. **Requirements on the SM Lagrangian:** locality, translational invariance, Poincaré symmetry, analyticity, renormalizability, naturalness, internal symmetries.
11. **The Standard Model:** the statement $SU(3) \times SU(2) \times U(1)$ spontaneously broken gauge symmetry accounts for almost all experimental data.

Projection Operators and Chirality Decomposition

Physical Motivation

The Dirac spinor ψ is a reducible representation of the Lorentz group: it decomposes into a left-handed and a right-handed two-component Weyl spinor, each of which transforms irreducibly. To isolate each piece we need a projector that commutes with Lorentz transformations. The matrix γ^5 (whose eigenvalues ± 1 distinguish the two chiralities) provides exactly such a projector.

Definition

Given the γ^5 matrix (defined as $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, with $(\gamma^5)^2 = \mathbf{1}$), the projection operators onto left- and right-handed Weyl subspaces are:

$$\boxed{P_L \equiv \frac{1 - \gamma^5}{2}, \quad P_R \equiv \frac{1 + \gamma^5}{2}.} \quad (173)$$

In the standard (Weyl/chiral) basis,

$$\gamma^5 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix},$$

so

$$P_L = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & 0 \end{pmatrix}, \quad P_R = \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{1} \end{pmatrix}.$$

Physical significance: P_L and P_R extract, respectively, the top two and bottom two components of ψ — precisely the left-handed and right-handed Weyl spinors ψ_L and ψ_R introduced in Lecture 07.

Properties

The projectors satisfy the standard idempotency and orthogonality relations:

$$P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = P_R P_L = 0, \quad P_L + P_R = \mathbf{1}. \quad (174)$$

Acting on the four-component Dirac spinor:

$$\psi_L \equiv P_L \psi = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}, \quad \psi_R \equiv P_R \psi = \begin{pmatrix} 0 \\ \psi_R \end{pmatrix}. \quad (175)$$

Dirac Equation in Weyl Component Form

Physical Motivation

Writing the Dirac equation in terms of ψ_L and ψ_R reveals how the mass term couples the two chiralities. In the massless limit the equation decouples into two independent equations — the Weyl equations — one for each chirality. This decomposition is physically important because the weak force acts *only on* ψ_L , so the Weyl basis is the natural language for the Standard Model.

Matrix form

In the Weyl basis, writing $\psi = (\psi_L, \psi_R)^T$, the Dirac equation $(i\gamma^\mu \partial_\mu - m)\psi = 0$ becomes:

$$\begin{pmatrix} -m & i\sigma^\mu \partial_\mu \\ i\bar{\sigma}^\mu \partial_\mu & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0, \quad (176)$$

where $\sigma^\mu = (\mathbf{1}, \sigma^i)$ and $\bar{\sigma}^\mu = (\mathbf{1}, -\sigma^i)$ (recall Lecture 07). Explicitly, this gives two coupled equations:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m \psi_R, \quad (177)$$

$$i\sigma^\mu \partial_\mu \psi_R = m \psi_L. \quad (178)$$

The mass m couples ψ_L and ψ_R : the kinetic part drives each chirality forward in spacetime while the mass “flips” the chirality.

Massless limit: Weyl equations

When $m = 0$, equations (177) and (178) decouple completely:

$$\boxed{i\bar{\sigma}^\mu \partial_\mu \psi_L = 0, \quad i\sigma^\mu \partial_\mu \psi_R = 0.} \quad (179)$$

Explicitly

$$\begin{cases} i \left[I \cdot \frac{\partial}{\partial t} + \sum_{i=1}^3 \sigma^i \frac{\partial}{\partial x^i} \right] \psi_R = 0 & \text{(Massless Right-Handed Weyl Equation)} \\ i \left[I \cdot \frac{\partial}{\partial t} - \sum_{i=1}^3 \sigma^i \frac{\partial}{\partial x^i} \right] \psi_L = 0 & \text{(Massless Left-Handed Weyl Equation)} \end{cases} \quad (180)$$

These are the **Weyl equations**. Each is an equation for a single two-component (Weyl) spinor.

Physical significance: Neutrinos in the original Standard Model were modeled as *massless* Weyl fermions; the discovery of neutrino masses is therefore a sign of physics beyond the SM. The Weyl equations are the minimal relativistic wave equations for massless spin- $\frac{1}{2}$ particles.

Fermion Bilinears and Their Lorentz Properties

Physical Motivation

When constructing interaction terms in a Lagrangian, we need Lorentz-invariant combinations of fermion fields. A single ψ transforms non-trivially; only specific bilinears $\bar{\psi} \Gamma \psi$ (where Γ is a combination of γ matrices) have definite Lorentz transformation properties. Cataloguing all such bilinears gives the complete menu of fermion interactions allowed by Lorentz symmetry.

The complete basis

The set $\{\mathbf{1}, \gamma^\mu, \gamma^5, \gamma^\mu \gamma^5, \Sigma^{\mu\nu}\}$ spans the full space of 4×4 matrices (16 independent elements), where the **rank-2 antisymmetric** tensor is:

$$\Sigma^{\mu\nu} \equiv \frac{i}{2} [\gamma^\mu, \gamma^\nu] = \frac{i}{2} (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu). \quad (181)$$

Note $\Sigma^{\mu\nu}$ is antisymmetric: $\Sigma^{\mu\nu} = -\Sigma^{\nu\mu}$, so it has $\binom{4}{2} = 6$ independent components.

The bilinears and their Lorentz transformation properties are summarized in the following table:

Bilinear	Lorentz type	Components	Notes
$\bar{\psi}\psi$	Scalar	1	Invariant under all Lorentz transf.
$\bar{\psi}\gamma^\mu\psi$	Vector	4	Transforms as $V^\mu \rightarrow \Lambda^\mu{}_\nu V^\nu$
$\bar{\psi}\gamma^5\psi$	Pseudo-scalar	1	Extra – sign under parity
$\bar{\psi}\gamma^\mu\gamma^5\psi$	Pseudo-vector (axial)	4	Extra – sign under parity
$\bar{\psi}\Sigma^{\mu\nu}\psi$	Tensor	6	Antisymmetric in $\mu\nu$

Total: $1 + 4 + 1 + 4 + 6 = 16$ independent components, forming a complete basis for 4×4 matrices. ✓

Remark 34 (Pseudo- means extra minus sign under parity). The prefix “pseudo” indicates that the object transforms like its non-pseudo counterpart under proper Lorentz transformations (rotations and boosts), but acquires an *additional minus sign* under the parity transformation $\mathbf{x} \rightarrow -\mathbf{x}$. For example, $\bar{\psi}\gamma^\mu\gamma^5\psi$ is called an **axial vector**: it transforms as a vector under boosts and rotations, but changes sign under parity, unlike the vector $\bar{\psi}\gamma^\mu\psi$.

Physical significance: The vector bilinear $\bar{\psi}\gamma^\mu\psi$ is the electromagnetic current j_{EM}^μ . The axial-vector bilinear $\bar{\psi}\gamma^\mu\gamma^5\psi$ appears in weak interactions; the combination $V - A$ (vector minus axial) is the hallmark of parity violation in the weak force, which is one of the most surprising experimental facts about Nature.

Feynman Slash Notation

A compact notation for contracting a four-vector with the gamma matrices:

$$\not{A} \stackrel{\text{def}}{=} \gamma^0 A_0 + \gamma^1 A_1 + \gamma^2 A_2 + \gamma^3 A_3 \quad (182)$$

This applies to any object with a Lorentz index μ . For example:

- $\not{\partial} = \gamma^\mu \partial_\mu$ (derivative operator).
- $\not{p} = \gamma^\mu p_\mu$ (four-momentum on a fermion line).
- $\not{A} = \gamma^\mu A_\mu$ (gauge boson field A^μ , relevant for QED).

The Dirac equation and the Dirac Lagrangian in slash notation:

$$(i\not{\partial} - m)\psi = 0, \quad (183)$$

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi} (i\not{\partial} - m) \psi. \quad (184)$$

The kinetic term is $\bar{\psi} i\not{\partial} \psi$ and the mass term is $-m\bar{\psi}\psi$. Both are Lorentz invariant: $\bar{\psi}\psi$ is a Lorentz scalar (from the bilinear table above), and $\bar{\psi}\gamma^\mu\partial_\mu\psi$ is the contraction of a Lorentz vector with a four-derivative, which is also a scalar.

Component Notation for the Dirac Lagrangian

Physical Motivation

It is useful to write the Dirac Lagrangian with explicit spinor (component) indices to understand what kind of object γ^μ is and what indices it carries. A common source of confusion is conflating the spinor index of ψ (which runs 1 to 4 over spin and particle/antiparticle degrees of freedom) with the Lorentz index μ (which runs 0 to 3 over spacetime directions).

Explicit index structure

Let $\alpha, \beta = 1, 2, 3, 4$ label spinor components. Then:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}_\alpha \left[i (\gamma^\mu)_{\alpha\beta} \partial_\mu - m \delta_{\alpha\beta} \right] \psi_\beta, \quad (185)$$

where $(\gamma^\mu)_{\alpha\beta}$ is the (α, β) matrix element of the 4×4 matrix γ^μ . Key points:

- ψ_β is a four-component column vector (spinor), with index β running over spinor space.
- $(\gamma^\mu)_{\alpha\beta}$ is a 4×4 matrix acting on spinor space, and μ is its Lorentz index — not a spinor index.
- ∂_μ carries only a Lorentz index; it acts on the spacetime argument of ψ , independent of the spinor index.

- The spinor indices α and β are summed over (Einstein convention on spinor indices), just like the Lorentz index μ is summed.

Remark 35 (Two separate “fours”). The Dirac spinor has four components, but these four have nothing to do with the four spacetime directions. One set of “four” lives in spinor space; the other lives in Minkowski space. Keeping the two distinct prevents errors in Lorentz transformation arguments: a Lorentz boost does not mix the spinor- α index, it mixes the μ index.

Mass Dimensions of Fermionic Fields

Physical Motivation

In natural units the action $S = \int d^4x \mathcal{L}$ must be dimensionless (it enters e^{iS}). Since $[d^4x] = [\text{mass}]^{-4}$, the Lagrangian density must satisfy $[\mathcal{L}] = [\text{mass}]^4$. Reading off the dimension of ψ from the Lagrangian then constrains what interaction terms are allowed: this is the origin of *renormalizability*.

Derivation of $[\psi] = 3/2$

From the kinetic term $\bar{\psi} i \not{\partial} \psi$:

$$[\bar{\psi} \not{\partial} \psi] = [\psi]^2 [\partial] = [\psi]^2 \cdot 1 \stackrel{!}{=} 4,$$

so $[\psi]^2 = 3$ and:

$$\boxed{[\psi] = \frac{3}{2}} \quad (186)$$

Independently, from the mass term $m \bar{\psi} \psi$: $[m][\psi]^2 = 1 \cdot 3 = 4$. ✓

For comparison, the scalar kinetic term $(\partial\phi)^2$ gives $[\phi]^2 \cdot 1^2 = 4$, so:

$$[\phi] = 1. \quad (187)$$

Physical significance: Fermionic fields have a higher mass dimension (3/2) than scalar fields (1). This makes fermion interactions more constrained: each fermion “uses up” more of the dimension budget.

Renormalizability and the Two-Fermion Constraint

Physical Motivation

A fundamental principle is that the full theory of nature should be **renormalizable**: all interaction terms in the Lagrangian should have mass dimension ≤ 4 . (Terms of dimension > 4 come with inverse powers of some large mass scale Λ , signaling that the theory is only an effective description valid at energies $\ll \Lambda$.) With $[\psi] = 3/2$, this severely limits how many fermion fields can appear in one interaction term.

The constraint

A renormalizable Lagrangian contains only terms with mass dimension at most 4:

$$\boxed{\text{Renormalizable: all terms have } [\text{fields} + \text{derivatives}] \leq 4.} \quad (188)$$

For fermions, two fields already contribute $2 \times \frac{3}{2} = 3$ to the dimension. A third fermion would require dimension $3 + \frac{3}{2} = \frac{9}{2} > 4$. Therefore:

$$\boxed{\text{At most two fermion fields per term in a renormalizable Lagrangian.}} \quad (189)$$

Moreover, a single fermion cannot appear alone (it would be odd under Lorentz transformation and not Lorentz invariant), so fermions always enter in pairs: either $\bar{\psi}\psi$ (dim 3) or $\bar{\psi}\psi$ multiplied by a scalar field or a single derivative (up to dim 4). Possible renormalizable terms involving two fermions include:

- $m\bar{\psi}\psi$ — mass term (dim 4 with m of dim 1).
- $y\phi\bar{\psi}\psi$ — Yukawa coupling to scalar ϕ (dim $1 + 3 = 4$, coupling y dimensionless).
- $\bar{\psi}i\not{D}\psi$ — kinetic term (dim $3/2 + 1 + 3/2 = 4$).

Physical significance: This constraint is why the Standard Model Lagrangian has its specific form. Four-fermion interactions (Fermi theory of weak decays) have dim 6, signaling that Fermi theory is an effective theory — valid at low energies but breaking down near the W boson mass, where the full renormalizable SM takes over.

Structure of the Dirac Mass Term

Physical Motivation

We saw from the Weyl-component form of the Dirac equation that the mass term couples ψ_L and ψ_R . Let us verify this explicitly and understand why this is inevitable for a Lorentz-invariant mass term.

Chirality structure of $m\bar{\psi}\psi$

Using $\psi = \psi_L + \psi_R$ (with $\psi_L = P_L\psi$, $\psi_R = P_R\psi$) and $\bar{\psi} = \psi^\dagger\gamma^0$:

$$\begin{aligned} \bar{\psi}\psi &= (\bar{\psi}_L + \bar{\psi}_R)(\psi_L + \psi_R) \\ &= \bar{\psi}_L\psi_L + \bar{\psi}_R\psi_R + \bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L. \end{aligned} \quad (190)$$

One can show (using the γ^5 eigenvalue properties) that $\bar{\psi}_L\psi_L = \bar{\psi}_R\psi_R = 0$, so:

$$\boxed{m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L).} \quad (191)$$

This is the **Dirac mass term**: it necessarily couples left- and right-handed fields.

Physical significance: A Dirac mass term requires both a left-handed and a right-handed component to exist. In the SM, the left-handed quarks and leptons carry different electroweak quantum numbers than their right-handed counterparts. Mass terms that couple L and R therefore break the electroweak symmetry — which is why fermion masses in the SM can only arise after electroweak symmetry breaking (the Higgs mechanism).

Remark 36 (Majorana mass). A **Majorana mass term** involves only one chirality (e.g., only ψ_L or only ψ_R). Such a term is allowed for electrically neutral particles (e.g., neutrinos) because the charge conjugate of ψ_L can serve as the missing chirality. Majorana masses are not covered in detail in this course.

Fermion Propagator

Physical Motivation

In Feynman diagram computations, every internal fermion line contributes a propagator. The fermion propagator is obtained from the free Dirac Lagrangian in the same way as the scalar propagator, but its numerator carries the momentum $\not{p} + m$ because of the first-order nature of the Dirac equation.

Result

The Feynman propagator for a Dirac fermion with momentum p and mass m is:

$$\boxed{\langle \psi \bar{\psi} \rangle_F = \frac{i(\not{p} + m)}{p^2 - m^2}.} \quad (192)$$

In a Wick contraction:

$$\overbrace{\psi \cdots \bar{\psi}} = \frac{i(\not{p} + m)}{p^2 - m^2}.$$

Compare with the scalar propagator:

$$\text{Scalar: } \frac{i}{p^2 - m^2}; \quad \text{Fermion: } \frac{i(\not{p} + m)}{p^2 - m^2}.$$

Physical significance: The extra factor $(\not{p} + m)$ in the numerator means the fermion propagator falls off as $1/p$ for large p , compared to $1/p^2$ for the scalar. This different UV behavior leads to different divergence structures for loop diagrams involving fermions vs. scalars, with important consequences for renormalization.

Chirality and Parity

The **chirality** (or **handedness**) of a Weyl spinor is its γ^5 eigenvalue: $+1$ for ψ_R , -1 for ψ_L .

Under a **parity transformation** ($\mathbf{x} \rightarrow -\mathbf{x}$, $t \rightarrow t$):

$$\boxed{\psi_L \xrightarrow{\text{parity}} \psi_R, \quad \psi_R \xrightarrow{\text{parity}} \psi_L.} \quad (193)$$

Chirality is reversed under parity. This is why a theory that treats ψ_L and ψ_R differently (as the SM does for the weak interaction) *violates parity* — an experimentally confirmed fact (Wu experiment, 1956).

Requirements on the Lagrangian: Toward the Standard Model

Physical Motivation

We are now ready to ask: what is the Lagrangian that describes nature? Rather than guessing, we impose a set of physical requirements that any Lagrangian must satisfy. Each requirement restricts the allowed terms and guides us toward the Standard Model.

The requirements

The Lagrangian $\mathcal{L}[\phi_i, \partial_\mu \phi_i]$ must satisfy:

1. **Locality and analyticity:** $\mathcal{L}$ depends on fields and their derivatives *at a single spacetime point* x^μ . This defines a **local field theory**. Moreover, $\mathcal{L}$ should be analytic (polynomial) in the fields — a necessary condition for perturbation theory to apply.
2. **Translational invariance:** $\mathcal{L}$ does not depend *explicitly* on x^μ (only implicitly through the fields). By Noether's theorem, this guarantees conservation of energy and momentum.

3. **Poincaré invariance:** $\mathcal{L}$ must be invariant under the full **Poincaré group** = (Lorentz transformations) + (spacetime translations). This is the fundamental spacetime symmetry of special relativity. It is Poincaré symmetry — not just Lorentz symmetry — that underlies the concepts of **mass** and **spin**: these quantum numbers label the irreducible representations of the Poincaré group.
4. **Renormalizability:** $\mathcal{L}$ contains only terms with mass dimension ≤ 4 . Higher-dimensional terms signal an effective theory with a UV cutoff Λ ; a truly fundamental theory should be renormalizable. (Non-renormalizable terms are suppressed by powers of $E/\Lambda \ll 1$.)
5. **Naturalness:** Every term in $\mathcal{L}$ that is *not forbidden* by a symmetry *should* appear. Stated differently, if symmetries allow a term, it must be present — one should not arbitrarily set coefficients to zero.
6. **Internal symmetry groups:** Beyond spacetime symmetry, one may require $\mathcal{L}$ to be invariant under global or local internal symmetry groups (e.g., $U(1)$, $SU(2)$, $SU(3)$). By Noether’s theorem, each symmetry implies a conserved current and conserved charge.

Summary of constraints

Requirement	Consequence
Locality	Fields interact at a point; no action at a distance
Translational invariance	Energy-momentum conservation
Lorentz (Poincaré) invariance	Well-defined mass and spin
Renormalizability	$\text{Dim} \leq 4$ terms only; at most 2 fermions per term
Naturalness	No fine-tuned cancellations without a symmetry reason
Internal symmetries	Conserved charges; gauge interactions

The Standard Model

Applying all of the requirements above, one arrives at a remarkably constrained set of possible Lagrangians. The verdict from over 50 years of experiment is:

Statement of the Standard Model:

Almost all experimental data for elementary particles and their interactions can be explained by a **spontaneously broken** $SU(3) \times SU(2) \times U(1)$ **gauge theory**.

Each factor of the gauge group has a physical meaning:

- $SU(3)$ — the strong force (QCD); gluons are the gauge bosons; quarks carry colour charge.
- $SU(2) \times U(1)$ — the electroweak force; the $W^\pm$, Z^0 , and photon are the gauge bosons; spontaneous symmetry breaking by the Higgs field gives them mass.

The word *almost* is important: the Standard Model does not account for dark matter, dark energy, gravity, neutrino masses (in its minimal form), or the matter-antimatter asymmetry. The outstanding goal of modern particle physics is to find experimental deviations from SM predictions and identify the more fundamental theory.

Physical significance: The rest of the course will be devoted to understanding each word in this statement — what “spontaneous symmetry breaking” means, how gauge symmetries generate interactions, and why this particular group $SU(3) \times SU(2) \times U(1)$ describes nature.

To Remember

- **Projection operators:** $P_L = \frac{1}{2}(1 - \gamma^5)$, $P_R = \frac{1}{2}(1 + \gamma^5)$; $\psi_{L,R} = P_{L,R}\psi$.
- **Weyl equations** (massless limit): $i\bar{\sigma}^\mu \partial_\mu \psi_L = 0$, $i\sigma^\mu \partial_\mu \psi_R = 0$.
- **Dirac mass couples chiralities:** $m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L)$.
- **Fermion bilinears:** scalar ($\bar{\psi}\psi$), vector ($\bar{\psi}\gamma^\mu\psi$), pseudo-scalar ($\bar{\psi}\gamma^5\psi$), axial vector ($\bar{\psi}\gamma^\mu\gamma^5\psi$), tensor ($\bar{\psi}\Sigma^{\mu\nu}\psi$ with 6 components).
- **Pseudo** = extra $-$ sign under parity; **axial** = pseudo-vector.
- **Slash notation:** $\not{A} = \gamma^\mu A_\mu$; Dirac eq. $(i\not{\partial} - m)\psi = 0$.
- **Mass dimensions:** $[\psi] = \frac{3}{2}$, $[\phi] = 1$, $[\mathcal{L}] = 4$.
- **Renormalizability** $\Rightarrow$ at most two fermion fields per term.
- **Fermion propagator:** $i(\not{p} + m)/(p^2 - m^2)$; falls as $1/p$, not $1/p^2$.
- **Chirality reverses under parity:** $\psi_L \leftrightarrow \psi_R$.
- **Requirements on $\mathcal{L}$:** locality, translational invariance, Poincaré invariance, analyticity, renormalizability, naturalness, internal symmetries.
- **The Standard Model:** spontaneously broken $SU(3) \times SU(2) \times U(1)$ gauge theory accounts for almost all particle physics data.

Recommended Reading

- **Peskin & Schroeder, Chapter 3** — Projection operators, chirality, Weyl spinors, fermion bilinears, and the Dirac propagator. The reference explicitly used in this course for fermion material.
- **Tong, *Lectures on Particle Physics*, Chapter 1 (draft)** (damtp.cam.ac.uk/user/tong/pp.htm) Chapter 1 covers the requirements on the SM Lagrangian (locality, renormalizability, gauge symmetry) and states the SM gauge group — directly parallel to the second half of this lecture.
- **Halzen & Martin, Chapter 5** — Fermion bilinears, parity, the $V - A$ structure of weak interactions; accessible treatment.
- **Griffiths, Chapter 9** — Introduction to the Standard Model gauge group; complements the lecture’s closing statement.

9 Lecture 09 – Model Building, Global Symmetries, and Charges

Date: 07/05/2026

Overview

This lecture starts the systematic model-building part of the course. The main point is that a Lagrangian is not guessed term by term. Instead, one specifies the degrees of freedom, the symmetries, and how the fields transform under those symmetries, and then writes the most general local, renormalizable Lagrangian consistent with those inputs.

1. **Review of simple renormalizable Lagrangians:** real scalar fields, Dirac fermions, and Yukawa couplings.
2. **Model building:** the Lagrangian is the endpoint of theory construction and the starting point for phenomenology.
3. **Symmetries:** spacetime symmetries versus internal symmetries; the role of invariance of the Lagrangian.
4. **Discrete symmetry example:** a $\mathbb{Z}_2$ symmetry $\phi \rightarrow -\phi$ removes the cubic scalar interaction and gives conservation of ϕ -parity.
5. **Global continuous symmetries:** fields transform in an internal mathematical space; allowed terms are singlets.
6. **Complex scalar and $U(1)$:** phase rotations $\phi \rightarrow e^{i\theta}\phi$; equivalence to $SO(2)$ rotations of the real and imaginary parts.
7. **Charges:** the coefficient q in $\phi \rightarrow e^{iq\theta}\phi$ is the charge; only relative charges matter when several fields transform.
8. **Two complex scalars:** with charges $q_1 = 1$ and $q_2 = 3$, the symmetry fixes which interactions are allowed.
9. **Noether perspective:** symmetry implies conserved current and conserved charge; invariant interaction terms have total charge zero.

Why Lagrangians Are the Language of Model Building

Physical Motivation

In particle physics, a Lagrangian is both an answer and a starting point. It is the answer to the model-building question: “which theory is compatible with the symmetries and field content we want?” Once written, it becomes the starting point for phenomenology: we quantize it, extract propagators and vertices, compute amplitudes, and compare those amplitudes with experiments.

The general model-building strategy is:

symmetries field content transformation rules	$\implies$	most general allowed $\mathcal{L}$	$\implies$	parameters to measure.
---	------------	------------------------------------	------------	------------------------

(194)

Physical significance: The word “general” is crucial. If a term is allowed by all imposed symmetries and by renormalizability, it should be included with an independent coefficient unless there is a further symmetry reason to remove it.

The output Lagrangian contains parameters such as masses and coupling constants. The theory does not usually predict their numerical values by itself; those parameters must be measured. The theory is tested by checking whether one set of measured parameters consistently predicts many other observables.

Renormalizable Building Blocks Already Seen

Physical Motivation

Before imposing new symmetries, we need to know the menu of terms that Lorentz invariance and renormalizability already allow. The lecture reviewed the simplest scalar and fermion terms because these are the raw ingredients from which later Standard Model interactions will be built.

Real scalar field

For a real scalar field ϕ , with mass dimension $[\phi] = 1$, the most general renormalizable self-interaction polynomial is:

$$\mathcal{L}_\phi = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\mu}{2\sqrt{2}} \phi^3 - \frac{\lambda}{4} \phi^4. \quad (195)$$

Here m is the scalar mass parameter, μ has mass dimension 1, and λ is dimensionless. The numerical factors are conventional and chosen to match the reference notes; they do not change the physics.

Physical significance: Renormalizability stops the polynomial at ϕ^4 . A term ϕ^n with $n \geq 5$ has operator dimension $n > 4$, so its coefficient would carry negative mass dimension and the term would be non-renormalizable.

Two possible terms were deliberately not written:

- A **constant term** in $\mathcal{L}$ does not affect the classical equation of motion for ϕ , because differentiating $\mathcal{L}$ with respect to ϕ or $\partial_\mu \phi$ kills it.
- A **linear term** can be removed by a field redefinition. Concretely, one shifts the integration variable, for example $\phi(x) = \phi'(x) + a$, and chooses the constant a so that the term proportional to one power of the new field ϕ' disappears. This is what the lecture meant by “rotating away” the term: it is just a change of variables, not a spacetime rotation.

Dirac fermion

For a Dirac fermion, with $[\psi] = 3/2$, the free Lagrangian is:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi} i \not{\partial} \psi - m \bar{\psi} \psi = \bar{\psi} (i \not{\partial} - m) \psi. \quad (196)$$

The slash notation means $\not{\partial} = \gamma^\mu \partial_\mu$, so the Lorentz index μ is contracted between the gamma matrix and the derivative.

Physical significance: Fermions cannot appear singly in Lorentz-invariant Lagrangian terms. They enter through bilinears such as $\bar{\psi}\psi$ or $\bar{\psi}\gamma^\mu\psi$. Four-fermion terms have dimension 6 and are therefore outside the renormalizable Lagrangian.

Yukawa coupling

If the theory contains both a scalar ϕ and a Dirac fermion ψ , the simplest renormalizable interaction between them is the Yukawa coupling:

$$\boxed{\mathcal{L}_{\text{Yukawa}} = -y \bar{\psi}\psi \phi.} \quad (197)$$

The operator $\bar{\psi}\psi\phi$ has dimension $3 + 1 = 4$, so the Yukawa coupling y is dimensionless.

Physical significance: Yukawa terms are the prototype for how scalars couple to fermions. In the Standard Model, fermion masses arise from Yukawa couplings to the Higgs field after electroweak symmetry breaking.

Combining the pieces gives the general schematic form

$$\boxed{\mathcal{L} = \mathcal{L}_\phi + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{Yukawa}}} \quad (198)$$

before additional internal symmetries are imposed.

Physical significance: This is the first example of the model-building rule: write every term that is allowed by Lorentz invariance, mass dimension, and later by the chosen internal symmetries.

What Is a Symmetry?

Physical Motivation

A symmetry is an invariance of the equations describing a physical system. In Lagrangian language, the practical criterion is that the action, or at least the Lagrangian density up to harmless total derivatives, is unchanged under a specified transformation of the fields. Symmetries are not decorative; they are the rules that decide which terms may appear.

Modern particle physics treats symmetries as input principles. The observed conservation laws, selection rules, and interaction patterns are then consequences of those inputs.

$$\boxed{\text{symmetry of } \mathcal{L} \implies \text{restricted interactions and conserved quantities.}} \quad (199)$$

Physical significance: Translational invariance gives energy-momentum conservation. Internal symmetries similarly give conserved charges, and later local internal symmetries will determine gauge interactions.

Inputs and outputs

To construct a Lagrangian one specifies:

1. the symmetries to impose;
2. the degrees of freedom, or field content;
3. how each field transforms under each symmetry.

Then one writes the most general local, analytic, renormalizable Lagrangian invariant under those symmetries. Usually “renormalizable” means keeping all operators of dimension ≤ 4 :

$$\boxed{\text{renormalizable model building in } 3 + 1 \text{ dimensions: } [\mathcal{O}] \leq 4.} \quad (200)$$

Physical significance: Truncating at dimension four can accidentally create extra symmetries. These are called accidental symmetries: they were not imposed, but they appear because all lower-dimensional symmetry-breaking operators happen to be absent.

Spacetime Symmetries and Internal Symmetries

Physical Motivation

Not all symmetries act in the same space. Lorentz transformations and translations act on spacetime. Internal symmetries act on labels or components of fields at a fixed spacetime point. Separating these ideas prevents a common confusion: an internal “rotation” is usually not a physical rotation of the coordinate axes.

Spacetime symmetries

The symmetries of special relativity form the Poincare group: translations, rotations, and boosts. In this course they are always imposed. The Lagrangian must be built from Lorentz scalars, with all Lorentz indices contracted properly.

There are also important discrete spacetime-related transformations:

- **Parity P :** spatial inversion $\mathbf{x} \rightarrow -\mathbf{x}$, with t unchanged.
- **Time reversal T :** reversal of time direction.
- **Charge conjugation C :** replacement of particles by antiparticles.

The combined CPT transformation is believed to be an exact symmetry of local Lorentz-invariant quantum field theory. However, C , P , and T separately can be violated, and the course will later discuss this mostly through the language of CP violation.

Internal symmetries

Internal symmetries act on the mathematical space generated by the fields, not on spacetime itself. For example, a complex field may be multiplied by a phase without moving the spacetime point x^μ .

Allowed Lagrangian terms must be singlets under the imposed internal symmetry:

$$\boxed{\text{allowed terms in } \mathcal{L} = \text{combinations of fields that are internal-symmetry singlets.}} \quad (201)$$

Physical significance: “Invariant”, “scalar”, and “singlet” are the same idea in this context: the object does not transform under the symmetry. Model building is therefore the art of forming all possible singlets from the available fields.

A First Discrete Example: $\mathbb{Z}_2$

Physical Motivation

The simplest useful internal symmetry is a sign flip of a real scalar field. This example shows how a symmetry removes an interaction term and produces a selection rule for physical processes.

Start from the real-scalar Lagrangian in Eq. (195) and impose:

$$\boxed{\phi \longrightarrow -\phi.} \quad (202)$$

This is a $\mathbb{Z}_2$ symmetry: applying the transformation twice gives back the original field.

Physical significance: The field ϕ is called $\mathbb{Z}_2$ -odd because it changes sign. The Lagrangian must be $\mathbb{Z}_2$ -even because it must transform into itself.

Which terms survive?

Under $\phi \rightarrow -\phi$,

$$\partial_\mu \phi \partial^\mu \phi \rightarrow \partial_\mu \phi \partial^\mu \phi, \quad \phi^2 \rightarrow \phi^2, \quad \phi^3 \rightarrow -\phi^3, \quad \phi^4 \rightarrow \phi^4. \quad (203)$$

Therefore the cubic term is forbidden. Equivalently, invariance requires $\mu = 0$. The resulting Lagrangian is:

$$\boxed{\mathcal{L}_{\phi, \mathbb{Z}_2} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4} \phi^4.} \quad (204)$$

Physical significance: The symmetry removes every term containing an odd number of ϕ fields. Thus interactions can change the number of ϕ particles only by an even number.

ϕ -parity conservation

Define the ϕ -parity of a state by

$$\boxed{P_\phi \equiv (-1)^{N_\phi},} \quad (205)$$

where N_ϕ is the number of external ϕ particles in the state.

The only interaction vertex is now a four- ϕ vertex. Each vertex changes N_ϕ by an even number. Hence

$$\boxed{P_\phi^{\text{initial}} = P_\phi^{\text{final}}.} \quad (206)$$

Physical significance: The $\mathbb{Z}_2$ symmetry gives a selection rule. Processes with an odd change in the number of ϕ particles are forbidden because they would violate ϕ -parity.

In group-theory language the multiplication rule is:

$$\boxed{\text{even} \times \text{even} = \text{even}, \quad \text{odd} \times \text{odd} = \text{even}, \quad \text{even} \times \text{odd} = \text{odd}.} \quad (207)$$

Physical significance: A Lagrangian term must multiply to an even object. This is the prototype of the more general singlet-building rule.

Global Continuous Symmetries

Physical Motivation

The $\mathbb{Z}_2$ example had only two transformations: do nothing or flip a sign. Continuous symmetries have a continuous transformation parameter. The simplest example is a phase rotation of a complex scalar field.

The word **global** means that the transformation parameter is the same at every spacetime point:

$$\boxed{\theta = \text{constant}, \quad \partial_\mu \theta = 0.} \quad (208)$$

Physical significance: Global symmetries lead to conserved charges. Later, when θ becomes spacetime-dependent, the symmetry becomes local and will require gauge fields.

One Complex Scalar and a $U(1)$ Symmetry

Physical Motivation

A complex scalar has two real degrees of freedom. It can be viewed either as ϕ and $\phi^\dagger$, or as real and imaginary components. A phase rotation in the complex plane mixes these two real degrees of freedom while preserving $\phi^\dagger\phi$.

The $U(1)$ transformation is:

$$\boxed{\phi \longrightarrow e^{i\theta} \phi, \quad \phi^\dagger \longrightarrow e^{-i\theta} \phi^\dagger.} \quad (209)$$

The most general renormalizable Lagrangian invariant under this phase rotation is:

$$\boxed{\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2.} \quad (210)$$

Physical significance: The invariant building block is $\phi^\dagger\phi$. Every allowed term contains equal numbers of ϕ and $\phi^\dagger$, so the total phase cancels.

Equivalence to $SO(2)$

Write

$$\boxed{\phi = \frac{1}{\sqrt{2}}(\phi_R + i\phi_I),} \quad (211)$$

where ϕ_R and ϕ_I are real scalar fields. The same transformation can be written as a rotation of the two real components:

$$\boxed{\begin{pmatrix} \phi_R \\ \phi_I \end{pmatrix} \longrightarrow \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \phi_R \\ \phi_I \end{pmatrix}.} \quad (212)$$

Physical significance: $U(1)$ phase rotations of one complex field and $SO(2)$ rotations of two real fields are two equivalent descriptions of the same symmetry.

For this one-field example, writing instead

$$\boxed{\phi \longrightarrow e^{iq\theta} \phi} \quad (213)$$

does not change the allowed Lagrangian, provided $q \neq 0$. One can absorb q into the definition of the parameter θ .

Physical significance: With a single charged field, the absolute normalization of charge is conventional. With multiple fields, only one charge can be normalized this way; relative charges become physical.

Charges and Several Fields

Physical Motivation

When several fields transform under the same $U(1)$, the important data are their relative phase rotations. These relative rotations are called charges. Charges are part of the model-building input: they determine which interactions are allowed and, for local symmetries, how strongly the fields couple to gauge bosons.

For two complex scalars ϕ_1 and ϕ_2 , impose the simultaneous transformations

$$\boxed{\phi_1 \rightarrow e^{iq_1\theta} \phi_1, \quad \phi_2 \rightarrow e^{iq_2\theta} \phi_2,} \quad (214)$$

with conjugates

$$\boxed{\phi_1^\dagger \rightarrow e^{-iq_1\theta} \phi_1^\dagger, \quad \phi_2^\dagger \rightarrow e^{-iq_2\theta} \phi_2^\dagger.} \quad (215)$$

The number q_i is the $U(1)$ charge of ϕ_i . One may set one nonzero charge to 1 by convention, but in general one cannot set all charges to 1 simultaneously. The ratio q_2/q_1 is physical.

$$\boxed{\text{charge of a product} = \text{sum of charges of its factors, with } q(\phi_i^\dagger) = -q(\phi_i).} \quad (216)$$

Physical significance: A term is invariant precisely when its total charge is zero. This turns symmetry checking into simple charge bookkeeping.

Real fields cannot carry this kind of $U(1)$ charge, because multiplying a real field by a nontrivial phase would make it complex:

$$\boxed{\text{continuous phase charges require complex fields.}} \quad (217)$$

Physical significance: Electromagnetic charge is a familiar example of this idea. Charged fields are complex; neutral real fields do not pick up a $U(1)$ phase.

Example: Two Complex Scalars with $q_1 = 1$, $q_2 = 3$

Physical Motivation

This example is the first nontrivial model-building exercise with charges. We choose field content and charges, then write every renormalizable term whose total charge is zero. The result shows how symmetry allows some interactions and forbids others.

The transformations are:

$$\boxed{\phi_1 \rightarrow e^{i\theta} \phi_1, \quad \phi_2 \rightarrow e^{3i\theta} \phi_2.} \quad (218)$$

Allowed kinetic and mass terms

The kinetic and mass terms are:

$$\boxed{\partial_\mu \phi_1^\dagger \partial^\mu \phi_1 + \partial_\mu \phi_2^\dagger \partial^\mu \phi_2 - m_1^2 \phi_1^\dagger \phi_1 - m_2^2 \phi_2^\dagger \phi_2.} \quad (219)$$

The two masses are independent. Nothing in the imposed symmetry requires $m_1 = m_2$.

Physical significance: A symmetry relates parameters only when it relates the corresponding fields. Here ϕ_1 and ϕ_2 have different charges, so their masses are not forced to be equal.

Allowed quartic interactions

The charge-neutral quartic interactions built only from $\phi_i^\dagger \phi_i$ are:

$$\boxed{-\lambda_{11}(\phi_1^\dagger \phi_1)^2 - \lambda_{22}(\phi_2^\dagger \phi_2)^2 - \lambda_{12}(\phi_1^\dagger \phi_1)(\phi_2^\dagger \phi_2).} \quad (220)$$

In addition, the special charge assignment $q_2 = 3q_1$ permits:

$$\boxed{-\eta \phi_1^3 \phi_2^\dagger - \eta^* (\phi_1^\dagger)^3 \phi_2.} \quad (221)$$

Indeed,

$$q(\phi_1^3 \phi_2^\dagger) = 3q_1 - q_2 = 3 - 3 = 0. \quad (222)$$

Physical significance: The interaction $\phi_1^3 \phi_2^\dagger$ converts charge-three worth of ϕ_1 quanta into one ϕ_2 quantum, or the reverse through the Hermitian conjugate. It changes particle numbers, but it conserves total $U(1)$ charge.

Full renormalizable Lagrangian

Putting the allowed terms together gives:

$$\mathcal{L} = \partial_\mu \phi_1^\dagger \partial^\mu \phi_1 + \partial_\mu \phi_2^\dagger \partial^\mu \phi_2 - m_1^2 \phi_1^\dagger \phi_1 - m_2^2 \phi_2^\dagger \phi_2 - \lambda_{11} (\phi_1^\dagger \phi_1)^2 - \lambda_{22} (\phi_2^\dagger \phi_2)^2 - \lambda_{12} (\phi_1^\dagger \phi_1) (\phi_2^\dagger \phi_2) - \eta \phi_1^3 \phi_2^\dagger - \eta^* (\phi_1^\dagger)^3 \phi_2. \quad (223)$$

Physical significance: Every term is both renormalizable and charge-neutral. The Lagrangian is therefore the most general one compatible with the chosen field content and global $U(1)$ charge assignment, up to the conventional normalization of couplings.

Charge Conservation and Noether's Theorem

Physical Motivation

The word charge has two connected meanings. First, it labels how a field transforms under a symmetry. Second, it is the conserved quantity implied by that symmetry. Noether's theorem is the bridge between the two meanings.

For a continuous global symmetry,

$$\text{continuous global symmetry} \implies \partial_\mu j^\mu = 0 \implies Q = \int d^3x j^0 \text{ is conserved.} \quad (224)$$

Physical significance: The Lagrangian-level statement that each term has total charge zero becomes the physical statement that interactions cannot change the total charge between initial and final states.

For the two-scalar example, every allowed interaction satisfies:

$$Q_{\text{total}}^{\text{initial}} = Q_{\text{total}}^{\text{final}}. \quad (225)$$

Physical significance: Particles may be created or annihilated, but only in combinations whose total charge is unchanged. This is the same logic as electric charge conservation, before the gauge-field interpretation is introduced.

The lecture emphasized two aspects of charge:

1. **Global symmetry aspect:** charge is conserved. This follows from Noether's theorem.
2. **Local symmetry aspect:** charge controls the coupling strength to gauge bosons, such as the photon. This will be developed later when θ is allowed to depend on x^μ .

Terminology: Charges, Quantum Numbers, Representations

In later lectures the words **charge**, **quantum number**, and **representation** will often appear close together. They are not always identical, but in model-building language they play the same role: they tell us how a field transforms under a symmetry group.

$$\text{field content} + \text{symmetry representations} \implies \text{allowed singlet operators in } \mathcal{L}. \quad (226)$$

Physical significance: In the Standard Model, particles are distinguished not only by mass and spin, but also by their quantum numbers under $SU(3) \times SU(2) \times U(1)$. Those quantum numbers determine which interactions exist.

To Remember

- A model is specified by symmetries, field content, and transformation rules.
- The Lagrangian is the most general local, renormalizable singlet built from those fields.
- For a real scalar, imposing $\phi \rightarrow -\phi$ forbids odd powers of ϕ and conserves ϕ -parity.
- For a complex scalar, $U(1)$ phase symmetry means $\phi \rightarrow e^{i\theta}\phi$ and forces terms to contain equal total phase.
- “Global” means the parameter θ is constant in spacetime.
- Charges are the coefficients in the phase rotation, $\phi_i \rightarrow e^{iq_i\theta}\phi_i$.
- A product of fields is allowed only if the sum of charges in the product is zero.
- With several charged fields, relative charges are physical; one can normalize only one nonzero charge by convention.
- Noether’s theorem connects continuous symmetries to conserved currents and conserved charges.
- Local symmetries, where $\theta = \theta(x)$, will lead to gauge interactions.

Recommended Reading

- **Tong, *Lectures on Particle Physics*, Chapters 1–2** (damtp.cam.ac.uk/user/tong/pp.html): model-building logic, renormalizable Lagrangians, and the first discussion of internal symmetries.
- **Peskin & Schroeder, Sections 2.2–2.4**: Noether’s theorem, conserved currents, and symmetries in field theory.
- **Schwartz, *Quantum Field Theory and the Standard Model*, Chapters 8 and 25**: internal symmetries, charges, and the transition toward gauge theory.
- **Georgi, *Lie Algebras in Particle Physics*, Chapters 1–2**: group-theory language behind singlets, representations, and charge assignments.

10 Lecture 10 – Product Groups, Accidental Symmetries, and Local Gauge Invariance

Date: 10/05/2026

Overview

This lecture extends the model-building framework in two directions. First, we allow multiple independent $U(1)$ symmetries acting on multiple fields and discover that the renormalizability constraint can accidentally produce *more* symmetry than was imposed. Second, we promote the transformation parameter θ from a constant to a function of spacetime, arriving at the concept of a **local** (gauge) symmetry, the covariant derivative, and the necessity and properties of massless gauge bosons.

1. **Product groups:** multiple independent $U(1)$ symmetries and how to assign charges under each.
2. **Example:** two complex scalars each charged under a different $U(1)$.
3. **Accidental symmetries:** truncating at dimension four can accidentally give a larger symmetry.
4. **Global versus local:** what changes when $\theta \rightarrow \theta(x)$.
5. **Covariant derivative:** how to restore invariance of kinetic terms under local $U(1)$.
6. **Gauge field:** its transformation law and kinetic term.
7. **Masslessness of gauge bosons:** why local gauge invariance forbids a mass term.
8. **Multiple gauge factors:** each independent $U(1)$ factor requires its own gauge field.

Product Groups

Physical Motivation

In Lecture 09, all fields rotated under a single parameter θ . Nature, however, imposes several independent phase rotations: the Standard Model gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$ already contains multiple independent factors. We therefore need the language to handle several commuting $U(1)$ symmetries simultaneously.

A **product group** $U(1)_A \times U(1)_B \times \dots$ is a collection of independent symmetries, each with its own transformation parameter. Every field carries a separate charge under each factor.

Two fields, two independent $U(1)$ s

Take two complex scalar fields ϕ_1 and ϕ_2 , and require invariance under two independent phase rotations parametrised by θ_A and θ_B . The transformations are:

$$\boxed{\phi_i \longrightarrow e^{iQ_{iA}\theta_A + iQ_{iB}\theta_B} \phi_i, \quad i = 1, 2,} \quad (227)$$

where Q_{iA} is the charge of ϕ_i under $U(1)_A$ and Q_{iB} is the charge under $U(1)_B$.

Physical significance: In contrast to a single $U(1)$, here there are two independent angles. One *cannot* in general write $Q_{iA}\theta_A + Q_{iB}\theta_B$ as a single angle times a single charge for both fields simultaneously, unless the charge ratios happen to coincide.

A concrete charge assignment

Choose:

	Q_A	Q_B
ϕ_1	1	0
ϕ_2	0	1

(228)

meaning ϕ_1 rotates only under $U(1)_A$ and ϕ_2 rotates only under $U(1)_B$:

$$\phi_1 \rightarrow e^{i\theta_A} \phi_1, \quad \phi_2 \rightarrow e^{i\theta_B} \phi_2. \quad (229)$$

Most general renormalizable Lagrangian

A term is allowed if and only if its total charge under *each* $U(1)$ factor is zero. With the charge assignment above:

- Under $U(1)_A$: only $\phi_1^\dagger \phi_1$ combinations are invariant; ϕ_2 is neutral.
- Under $U(1)_B$: only $\phi_2^\dagger \phi_2$ combinations are invariant; ϕ_1 is neutral.

The most general renormalizable Lagrangian is:

$$\begin{aligned} \mathcal{L}_{\text{cocoa}} = & \partial_\mu \phi_1^\dagger \partial^\mu \phi_1 + \partial_\mu \phi_2^\dagger \partial^\mu \phi_2 - m_1^2 \phi_1^\dagger \phi_1 - m_2^2 \phi_2^\dagger \phi_2 \\ & - \lambda_{11} (\phi_1^\dagger \phi_1)^2 - \lambda_{22} (\phi_2^\dagger \phi_2)^2 - \lambda_{12} (\phi_1^\dagger \phi_1) (\phi_2^\dagger \phi_2). \end{aligned}$$

(230)

Crucially, a mixed term such as $\phi_1^3 \phi_2^\dagger$ from Lecture 09 is *forbidden* here: its charge under $U(1)_A$ is $3 \cdot 1 - 0 = 3 \neq 0$. The charge assignment dictates which interactions may appear.

Physical significance: The allowed Lagrangian is more restrictive here than in the $q_1 = 1, q_2 = 3$ example of Lecture 09, because the two fields do not talk to each other through the symmetry. Their only coupling is the portal term $\lambda_{12} (\phi_1^\dagger \phi_1) (\phi_2^\dagger \phi_2)$, which is charge-neutral under both $U(1)$ s.

Accidental Symmetries

Physical Motivation

The Lagrangian $\mathcal{L}_{\text{cocoa}}$ was derived by imposing $U(1)_A \times U(1)_B$. A natural question is: does this Lagrangian have more symmetry than we intended? The answer is yes, and understanding why matters for the Standard Model.

Obtaining the same Lagrangian from a single $U(1)$

Consider a single $U(1)$ with charges:

$$\phi_1 \rightarrow e^{i \cdot 1 \cdot \theta} \phi_1, \quad \phi_2 \rightarrow e^{i \cdot 4 \cdot \theta} \phi_2. \quad (231)$$

A potential mixed term would need total charge zero. The only renormalizable candidate of this form would involve four fields; for example $\phi_1^{-4} \phi_2$ has dimension > 4 . Hence, at the renormalizable level, the most general Lagrangian consistent with this single $U(1)$ also contains no mixed ϕ_1 - ϕ_2 interaction terms, and one again obtains $\mathcal{L}_{\text{cocoa}}$.

But $\mathcal{L}_{\text{cocoa}}$ clearly has two independent phase symmetries: rotating ϕ_1 alone or ϕ_2 alone both leave it invariant. So:

$$\text{imposed: } U(1) \xrightarrow{\text{renormalizability}} \mathcal{L}_{\text{cocoa}} \xleftarrow{\text{symmetry of } \mathcal{L}} U(1)_A \times U(1)_B.$$

(232)

Physical Concept 10.1: Accidental Symmetry

An **accidental symmetry** is a symmetry of the renormalizable Lagrangian that was *not* explicitly imposed. It arises because all symmetry-breaking operators consistent with the imposed symmetry happen to have mass dimension > 4 and are therefore absent from the renormalizable theory.

Physical significance: Accidental symmetries are broken by non-renormalizable operators suppressed by a high scale Λ . Famous examples in the Standard Model are **baryon number** B and **lepton number** L : no renormalizable SM operator violates them, so they appear exact at collider energies. However, dimension-six operators (suppressed by $1/\Lambda^2$) do violate B and L , predicting (unobserved so far) proton decay.

Choice of basis for the symmetry generators

The two independent symmetries θ_A and θ_B can equally well be described by any two independent linear combinations. A common choice is:

$$\theta_+ = \theta_A + \theta_B, \quad \theta_- = \theta_A - \theta_B, \quad (233)$$

giving $U(1)_+ \times U(1)_-$. Any two independent linear combinations are physically equivalent.

From Global to Local Symmetry

Physical Motivation

All symmetries discussed so far have been **global**: the parameter θ is the same everywhere in space and time. This is physically strange. Why should a rotation performed “here” be rigidly correlated with one performed at a distant galaxy?

A **local** (or **gauge**) symmetry removes this rigidity by allowing θ to vary independently at each spacetime point. The price is that we must introduce a new dynamical field—the gauge field—to maintain invariance of the kinetic terms. Gauge symmetry is the organizing principle of the Standard Model.

Definition

Physical Concept 10.2: Local $U(1)$ symmetry

A transformation

$$\phi(x) \longrightarrow e^{iq\theta(x)} \phi(x), \quad (234)$$

where $\theta(x) \equiv \theta(x^\mu)$ is an arbitrary smooth function of spacetime, is called a **local** or **gauge** $U(1)$ transformation.

Physical significance: Global means $\partial_\mu \theta = 0$. Local means θ can take any value at any spacetime point; it is not a rigid phase.

Non-derivative terms are automatically gauge invariant

Any term in the Lagrangian that does not involve a spacetime derivative—mass terms, potential terms—transforms as:

$$\phi^\dagger \phi \longrightarrow (e^{-iq\theta(x)} \phi^\dagger)(e^{iq\theta(x)} \phi) = \phi^\dagger \phi. \quad (235)$$

The phases cancel regardless of whether θ depends on x . Therefore:

$$\boxed{\text{if a non-derivative term is globally invariant, it is also locally invariant.}} \quad (236)$$

Physical significance: Only kinetic terms need special treatment when going from global to local symmetry.

The Problem with Kinetic Terms

Physical Motivation

The derivative ∂_μ does not commute with a position-dependent phase. When θ varies in space, differentiating $e^{iq\theta(x)}\phi(x)$ produces an extra term involving $\partial_\mu\theta$. This extra term breaks gauge invariance of the kinetic Lagrangian and must be corrected.

How $\partial_\mu\phi$ transforms

Under the local transformation (234):

$$\begin{aligned} \partial_\mu\phi &\longrightarrow \partial_\mu(e^{iq\theta(x)}\phi(x)) \\ &= e^{iq\theta(x)}\partial_\mu\phi(x) + iq(\partial_\mu\theta)e^{iq\theta(x)}\phi(x). \end{aligned} \quad (237)$$

The first piece is the “good” term (what we would want from a global transformation); the second is an extra piece proportional to $\partial_\mu\theta$. Therefore:

$$\boxed{\partial_\mu\phi \longrightarrow e^{iq\theta(x)}(\partial_\mu\phi + iq(\partial_\mu\theta)\phi).} \quad (238)$$

The kinetic term $(\partial_\mu\phi^\dagger)(\partial^\mu\phi)$ is *not* invariant under the local transformation because of the cross terms involving $\partial_\mu\theta$.

Physical significance: This breakdown of kinetic-term invariance is universal: it happens for scalar and fermion kinetic terms alike. We must repair it.

The Covariant Derivative

Physical Motivation

The fix is to replace the ordinary derivative ∂_μ with a **covariant derivative** D_μ that, when acting on a charged field, transforms the same way under local $U(1)$ as ∂_μ would under a global $U(1)$:

$$\text{want: } D_\mu\phi \longrightarrow e^{iq\theta(x)}D_\mu\phi. \quad (239)$$

If $D_\mu\phi$ transforms this cleanly, then $(D_\mu\phi)^\dagger(D^\mu\phi)$ is gauge invariant by the same argument as $\phi^\dagger\phi$.

Construction of the covariant derivative

Physical Concept 10.3: Covariant derivative

Introduce a new vector field $A_\mu(x)$ and define:

$$\boxed{D_\mu \equiv \partial_\mu + igqA_\mu,} \quad (240)$$

where g is a dimensionless coupling constant (the **gauge coupling**), q is the charge of the field being acted upon, and A_μ is the **gauge field**.

Transformation law of the gauge field

For $D_\mu\phi$ to satisfy requirement (239), the gauge field must transform as:

$$\boxed{A_\mu(x) \longrightarrow A_\mu(x) - \frac{1}{g} \partial_\mu \theta(x).} \quad (241)$$

Verification: Under simultaneous transformations (234) and (241):

$$\begin{aligned} D_\mu\phi &= (\partial_\mu + igqA_\mu)\phi \\ &\longrightarrow \left(\partial_\mu + igq \left(A_\mu - \frac{1}{g} \partial_\mu \theta \right) \right) e^{iq\theta} \phi \\ &= e^{iq\theta} \partial_\mu \phi + iq(\partial_\mu \theta) e^{iq\theta} \phi + igqA_\mu e^{iq\theta} \phi - iq(\partial_\mu \theta) e^{iq\theta} \phi \\ &= e^{iq\theta} (\partial_\mu + igqA_\mu) \phi = e^{iq\theta} D_\mu \phi. \quad \checkmark \end{aligned} \quad (242)$$

Physical significance: The extra term $-\frac{1}{g}\partial_\mu\theta$ in the gauge field transformation precisely cancels the unwanted $\partial_\mu\theta$ term from the derivative. This is the mechanism by which the gauge field “absorbs” the local phase ambiguity.

Physical significance: The transformation law of A_μ is **additive** (a shift), not multiplicative. It resembles a phase shift rather than a rotation. This is a distinctive feature of abelian gauge fields.

Gauge-invariant Lagrangian for a charged scalar

Replacing $\partial_\mu \rightarrow D_\mu$ everywhere in the scalar kinetic term:

$$\boxed{\mathcal{L}_{\text{scalar}} = (D_\mu\phi)^\dagger (D^\mu\phi) - m^2\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2,} \quad (243)$$

which is invariant under the local $U(1)$ by construction. The same replacement works for fermion kinetic terms: $\bar{\psi}i\not{\partial}\psi \rightarrow \bar{\psi}i\not{D}\psi$.

Kinetic Term for the Gauge Field

Physical Motivation

We have introduced a new dynamical field A_μ . Since it is dynamical, it must have a kinetic term in the Lagrangian. This term must be gauge invariant by itself. The ordinary combination $(\partial_\mu A_\nu)(\partial^\mu A^\nu)$ is not gauge invariant because A_μ shifts by a gradient under (241). We need a combination that cancels these gradients.

Field strength tensor

Define:

$$\boxed{F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu.} \quad (244)$$

Under the gauge transformation (241):

$$F_{\mu\nu} \longrightarrow \partial_\mu \left(A_\nu - \frac{1}{g} \partial_\nu \theta \right) - \partial_\nu \left(A_\mu - \frac{1}{g} \partial_\mu \theta \right) = F_{\mu\nu} - \frac{1}{g} (\partial_\mu \partial_\nu \theta - \partial_\nu \partial_\mu \theta) = F_{\mu\nu}, \quad (245)$$

using the commutativity of partial derivatives. Hence $F_{\mu\nu}$ is gauge invariant.

Physical significance: $F_{\mu\nu}$ is the electromagnetic field strength. Its components are the electric and magnetic fields. Gauge invariance ensures that physical observables—which depend on $F_{\mu\nu}$, not on A_μ directly—are unambiguous.

Gauge-invariant kinetic Lagrangian for A_μ

The unique renormalizable (dimension-four) Lorentz scalar built from $F_{\mu\nu}$ is:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (246)$$

The factor $-1/4$ is a conventional normalization (ensures a positive-definite Hamiltonian with canonically normalized fields). The equation of motion for A_μ derived from $\mathcal{L}_{\text{gauge}}$ alone is:

$$\partial^\mu F_{\mu\nu} = 0, \quad (247)$$

which is precisely Maxwell's equations in vacuum.

Physical significance: The entire structure of electromagnetism—Maxwell's equations, the photon, gauge invariance—follows from demanding a local $U(1)$ symmetry. The gauge field A_μ is the photon field.

Masslessness of Gauge Bosons

Physical Motivation

A_μ has a kinetic term, so its quantum excitations are particles. What is their mass? A mass term for a vector field would normally take the form $m^2 A_\mu A^\mu$. But gauge invariance forbids this, as we now show.

The mass term violates gauge invariance

Under $A_\mu \rightarrow A_\mu - \frac{1}{g} \partial_\mu \theta$:

$$m^2 A_\mu A^\mu \longrightarrow m^2 \left(A_\mu - \frac{1}{g} \partial_\mu \theta \right) \left(A^\mu - \frac{1}{g} \partial^\mu \theta \right) = m^2 A_\mu A^\mu - \frac{2m^2}{g} A^\mu \partial_\mu \theta + \frac{m^2}{g^2} (\partial_\mu \theta)^2 \neq m^2 A_\mu A^\mu. \quad (248)$$

Since this is not invariant, the mass term is *not allowed* in the gauge-invariant Lagrangian. Therefore:

$$\text{local gauge invariance} \implies \text{gauge bosons are massless}. \quad (249)$$

Physical significance: The photon's masslessness is not a measured coincidence. It is a logical consequence of requiring local $U(1)$ gauge invariance. The photon, as the $U(1)_{\text{em}}$ gauge boson, must be exactly massless. A nonzero mass would signal the breaking of gauge invariance.

Degrees of freedom

A massive vector boson has three polarization states; a massless vector boson has only two (transverse polarizations). The masslessness of gauge bosons is consistent with the masslessness of the photon, which is observed to have exactly two polarizations.

Full Gauge-Invariant Lagrangian

Combining the kinetic term for A_μ with the gauge-invariant matter Lagrangian gives the complete theory:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + (D_\mu \phi)^\dagger (D^\mu \phi) - m^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2, \quad (250)$$

with $D_\mu = \partial_\mu + igqA_\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Expanding $(D_\mu\phi)^\dagger(D^\mu\phi)$ produces:

$$(D_\mu\phi)^\dagger(D^\mu\phi) = (\partial_\mu\phi^\dagger)(\partial^\mu\phi) + igqA^\mu(\phi^\dagger\partial_\mu\phi - \phi\partial_\mu\phi^\dagger) + g^2q^2A_\mu A^\mu \phi^\dagger\phi, \quad (251)$$

showing interaction vertices with one or two gauge bosons.

Physical significance: The single-photon vertex (proportional to gq) shows that charge determines the coupling strength of the field to the photon. This is the second role of charge: it governs the interaction rate, not just the phase rotation.

Physical significance: The $U(1)$ gauge field does *not* appear in its own kinetic term with a self-coupling: $F_{\mu\nu}F^{\mu\nu}$ contains no A^4 terms. The photon does not couple to itself. This is a special property of abelian gauge groups; non-abelian gauge fields (gluons, W , Z) do self-interact.

Product Gauge Groups

Physical Motivation

The Standard Model gauge group is $SU(3)_c \times SU(2)_L \times U(1)_Y$. The $U(1)$ factor alone demonstrates the logic: if one has a product gauge group $U(1)_A \times U(1)_B \times \dots$, each factor is an independent local symmetry requiring its own treatment.

Each factor requires its own gauge field and coupling

If the local symmetry decomposes into n independent commuting $U(1)$ factors, then for each factor $\alpha = A, B, \dots$:

- There is an independent parameter $\theta_\alpha(x)$.
- There is an independent gauge field A_μ^α with transformation law

$$A_\mu^\alpha \rightarrow A_\mu^\alpha - \frac{1}{g_\alpha} \partial_\mu \theta_\alpha.$$

- There is an independent dimensionless coupling g_α .
- The covariant derivative acting on a field with charges $(Q_A, Q_B, \dots)$ is:

$$D_\mu = \partial_\mu + ig_A Q_A A_\mu^A + ig_B Q_B A_\mu^B + \dots$$

$$\boxed{U(1)_A \times U(1)_B \times \dots \implies n \text{ massless gauge bosons, } n \text{ independent couplings } g_\alpha.} \quad (252)$$

Physical significance: The coupling constants of different gauge factors are independent: gauge symmetry does not predict the ratio g_A/g_B . Each must be measured. This is why the Standard Model has three independent gauge couplings g_3, g_2, g_1 that are not predicted by gauge symmetry alone—they must be measured from experiment.

Summary Table: Global vs. Local $U(1)$

	Global $U(1)$	Local $U(1)$
Parameter	$\theta = \text{const}$	$\theta = \theta(x)$
Non-derivative terms	Invariant	Invariant
Kinetic terms	Invariant	Not invariant (need D_μ)
New field required	No	Yes: A_μ
Gauge boson mass	—	Must be zero
Consequence	Conserved charge	Gauge interaction

To Remember

- A product group $U(1)^n$ means n independent phase rotations; each field carries n charges.
- Renormalizability (dimension ≤ 4) can produce more symmetry than imposed: this is an **accidental symmetry**.
- Accidental symmetries of the SM include baryon number B and lepton number L ; they are broken by dimension-six operators suppressed by a high scale.
- Going from global to local: only derivative terms (kinetic terms) are affected.
- The covariant derivative $D_\mu = \partial_\mu + igqA_\mu$ restores gauge invariance; A_μ must transform as $A_\mu \rightarrow A_\mu - \frac{1}{g}\partial_\mu\theta$.
- The field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is gauge invariant; its equation of motion gives Maxwell's equations.
- The kinetic Lagrangian for the gauge field is $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$.
- A mass term $m^2 A_\mu A^\mu$ is not gauge invariant, so gauge bosons must be **massless**.
- A product gauge group $U(1)^n$ requires n independent gauge fields and n independent couplings.
- The coupling gq in $D_\mu\phi = (\partial_\mu + igqA_\mu)\phi$ determines how strongly the field interacts with the gauge boson.

Recommended Reading

- **Tong, *Lectures on Particle Physics*, Chapter 2** (damtp.cam.ac.uk/user/tong/pp.html): global and local symmetries, covariant derivatives, gauge fields.
- **Peskin & Schroeder, Section 4.1**: gauge invariance, covariant derivative, $F_{\mu\nu}$.
- **Griffiths, *Introduction to Elementary Particles*, Chapter 9**: gauge invariance and the photon as the gauge boson of $U(1)_{\text{em}}$.
- **Halzen & Martin, Chapter 14**: gauge theories, photon coupling, and QED.

11 Lecture 11 – QED: Interactions, Processes, and Two Generations

Date: 14/05/2026

Overview

This lecture completes the study of local $U(1)$ gauge invariance by explicitly expanding the covariant derivative to expose the interaction vertices it generates, making manifest the physical role of **charge as coupling strength**. We then construct the simplest version of Quantum Electrodynamics (QED), enumerate its spectrum, derive the Feynman vertex rule, and survey the key QED processes. Finally, we generalise to two fermion generations, rotate to the mass basis, and discover an accidental lepton-number symmetry.

1. **Charge as coupling strength:** expanding D_μ for fermions and scalars reveals explicit interaction vertices.
2. **Model-building summary:** symmetry + field content + charge assignments $\rightarrow$ most general Lagrangian.
3. **QED with one generation:** definition, Lagrangian, spectrum, vertex rule, fine-structure constant.
4. **QED processes:** Bhabha, Møller, Compton, pair production/annihilation; loop corrections; parameter counting and predictions.
5. **QED with two generations:** mass matrix, diagonalisation, mass eigenstates (electron and muon), accidental $U(1)_e \times U(1)_\mu$.

Charge as the Coupling Strength

Physical Motivation

We saw in Lecture 10 that a *global* $U(1)$ symmetry implies conservation of a Noether charge Q . A *local* (gauge) $U(1)$ symmetry has an additional consequence: the charge Q assigned to a field determines how *strongly* that field interacts with the gauge boson. This is the second face of charge. It becomes explicit when we expand the covariant derivative inside the kinetic terms.

Recap: the covariant derivative

Recall from Lecture 10 that for a field of charge Q under a local $U(1)$ with coupling constant g and gauge field A_μ :

$$\boxed{D_\mu = \partial_\mu + igQA_\mu.} \quad (253)$$

The combination gQ that appears here — not g and Q separately — controls the strength of every interaction.

Physical significance: D_μ is sensitive to the charge of the field it acts on. If $Q = 0$, the gauge-field term vanishes: the field is invisible to the gauge boson. This is why neutral particles do not couple to the photon.

Fermion kinetic term: the three-point vertex

Physical Motivation

The covariant kinetic term for a Dirac fermion ψ of charge Q is $i\bar{\psi}\not{D}\psi$. Expanding D_μ splits this into a pure kinetic piece (the free theory) and an interaction piece that couples ψ to the gauge boson A_μ .

Derivation

$$\begin{aligned} i\bar{\psi}\not{D}\psi &= i\bar{\psi}\gamma^\mu D_\mu\psi = i\bar{\psi}\gamma^\mu(\partial_\mu + igQA_\mu)\psi \\ &= \underbrace{i\bar{\psi}\gamma^\mu\partial_\mu\psi}_{\text{kinetic term}} + \underbrace{i^2 gQ\bar{\psi}\gamma^\mu A_\mu\psi}_{\text{interaction term}}. \end{aligned} \quad (254)$$

Since $i^2 = -1$:

$$\boxed{\mathcal{L}_{\text{int}}^{(f)} = -gQ\bar{\psi}\gamma^\mu\psi A_\mu.} \quad (255)$$

This is a **three-point interaction**: two fermion legs and one gauge-boson leg, meeting at a single vertex. The corresponding Feynman rule for the vertex factor is $-igQ\gamma^\mu$.

Physical significance: The larger $|Q|$, the stronger the coupling. A field with $Q = 0$ has no interaction term — it decouples completely from the gauge boson at tree level.

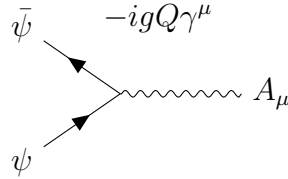


Figure 1: Fermion–gauge-boson vertex arising from the covariant kinetic term. The vertex factor is $-igQ\gamma^\mu$; the coupling scales linearly with the charge Q .

Scalar kinetic term: three-point and four-point vertices

Physical Motivation

For a complex scalar ϕ of charge Q , the covariant kinetic term is $(D_\mu\phi)^\dagger(D^\mu\phi)$. Because A_μ enters D_μ linearly, squaring $(D_\mu\phi)$ produces terms with both one and two gauge-boson fields, yielding *two distinct vertices*.

Derivation

Write $D_\mu\phi = \partial_\mu\phi + igQA_\mu\phi$ and expand:

$$\begin{aligned} (D_\mu\phi)^\dagger(D^\mu\phi) &= (\partial_\mu\phi^\dagger - igQA_\mu\phi^\dagger)(\partial^\mu\phi + igQA^\mu\phi) \\ &= \underbrace{\partial_\mu\phi^\dagger\partial^\mu\phi}_{\text{kinetic}} + \underbrace{igQA^\mu(\phi^\dagger\partial_\mu\phi - \phi\partial_\mu\phi^\dagger)}_{\text{3-pt vertex}} + \underbrace{g^2Q^2 A_\mu A^\mu \phi^\dagger\phi}_{\text{4-pt vertex}}. \end{aligned} \quad (256)$$

$$\boxed{\mathcal{L}_{\text{int},3}^{(s)} = igQA^\mu(\phi^\dagger\partial_\mu\phi - \phi\partial_\mu\phi^\dagger), \quad \mathcal{L}_{\text{int},4}^{(s)} = g^2Q^2 A_\mu A^\mu \phi^\dagger\phi.} \quad (257)$$

The **three-point vertex** (two scalars + one photon) has strength $\propto gQ$ with a momentum factor (from the derivative). The **four-point vertex** (two scalars + two photons, the “seagull” diagram) has strength $\propto (gQ)^2$.

Physical significance: The four-point vertex is unique to scalars and has no fermion analogue. In QED, where the charged matter is the electron (a Dirac fermion), this vertex does not appear. In scalar QED (charged pions, for example), it is essential and restores unitarity in longitudinal-photon scattering.

Model-Building Summary

Physical Motivation

We now have all the ingredients to state the general model-building procedure. This is the algorithm that will produce the Standard Model.

Physical Concept 11.1: Model-building recipe

Given:

1. A **symmetry group** G (e.g. $U(1)$, or $SU(3) \times SU(2) \times U(1)$).
2. A **field content**: a list of all Weyl spinors and complex scalars.
3. **Charge assignments**: how each field transforms under each factor of G .

The output is the most general Lagrangian of mass dimension ≤ 4 invariant under G .

$$\boxed{\text{symmetry} + \text{fields} + \text{charges} \xrightarrow{\text{dim} \leq 4} \mathcal{L}_{\text{most general}}.} \quad (258)$$

If the symmetry is *local* (gauge), one must additionally:

- Add a massless gauge field A_μ^α for each independent $U(1)$ factor α .
- Promote $\partial_\mu \rightarrow D_\mu = \partial_\mu + ig_\alpha Q_\alpha A_\mu^\alpha$ in all kinetic terms.
- Include the kinetic term $-\frac{1}{4}F_{\mu\nu}^\alpha F^{\alpha\mu\nu}$ for each gauge field.

Physical significance: In the Standard Model, *all* imposed symmetries are local gauge symmetries. Global symmetries such as baryon number B and lepton number L are not imposed—they appear accidentally. The reason: gravitational effects at the Planck scale are expected to break any global symmetry, while local symmetries can be exact at all energies.

QED with One Generation

Physical Motivation

We now build the simplest version of Quantum Electrodynamics (QED): a local $U(1)$ theory with a single charged Dirac fermion — the electron.

Definition

Symmetry:	Local $U(1)_{\text{em}}$
Fermions:	e_L (left-handed Weyl, charge $Q = -1$) e_R (right-handed Weyl, charge $Q = -1$)
Gauge field:	A_μ (photon, charge $Q = 0$)
Coupling:	e (electromagnetic coupling constant)

The covariant derivative with coupling constant e and charge Q is:

$$\boxed{D_\mu = \partial_\mu + ie Q A_\mu.} \quad (259)$$

For the electron ($Q = -1$): $D_\mu = \partial_\mu - ie A_\mu$.

Physical significance: The gauge symmetry forces the same coupling constant e to appear in every interaction vertex in QED. We have no freedom to assign different couplings to different vertices — gauge invariance is highly restrictive.

The QED Lagrangian

The most general renormalizable Lagrangian invariant under local $U(1)_{\text{em}}$ for this field content is:

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{e}_L \not{D} e_L + i\bar{e}_R \not{D} e_R - m_e \bar{e}_L e_R - m_e \bar{e}_R e_L, \quad (260)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $\not{D} = \gamma^\mu D_\mu$.

Why is the Dirac mass term allowed? The combination $\bar{e}_L e_R$ has total charge $(-(-1)) + (-1) = 0$ under $U(1)_{\text{em}}$, so it is gauge invariant. Both e_L and e_R have the *same* charge $Q = -1$, which is the necessary condition. Had they carried different charges, the mass term would be forbidden.

Assembling e_L and e_R into a four-component Dirac spinor $e = (e_L, e_R)^T$:

$$\boxed{\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{e} \not{D} e - m_e \bar{e} e, \quad D_\mu = \partial_\mu - ie A_\mu.} \quad (261)$$

Spectrum of QED

Expanding the covariant kinetic term using (259) and $Q = -1$:

$$i\bar{e} \not{D} e = i\bar{e} \gamma^\mu \partial_\mu e + e \bar{e} \gamma^\mu e A_\mu. \quad (262)$$

The first term is the free Dirac kinetic term; the second is the interaction.

$$\boxed{\text{Spectrum: } \underbrace{\text{massive Dirac fermion}}_{\text{electron } e^-, \text{ positron } e^+} (\text{mass } m_e) + \underbrace{\text{massless vector boson}}_{\text{photon } \gamma}.} \quad (263)$$

On the massless photon: The absence of a photon mass is *not* a free choice — it is a prediction. A mass term $m^2 A_\mu A^\mu$ is forbidden by local $U(1)_{\text{em}}$ gauge invariance (as shown in Lecture 10). This prediction is independent of the values of m_e and e .

On the electron/positron: The electron (e^- , charge -1) and positron (e^+ , charge $+1$) are the particle and antiparticle excitations of the same Dirac field e . They have equal mass m_e .

No accidental symmetry: The mass term $m_e \bar{e} e$ mixes e_L and e_R , so there is no independent symmetry rotating them separately. QED with one generation has no accidental symmetry beyond the imposed $U(1)_{\text{em}}$.

The QED vertex rule and the fine-structure constant

From (262), the interaction Lagrangian is $+e \bar{e} \gamma^\mu e A_\mu$. Applying the general rule (255) with $g \rightarrow e$ and $Q = -1$:

$$\boxed{\text{QED vertex rule: } -ie Q \gamma^\mu = +ie \gamma^\mu \quad (Q = -1 \text{ for electron}).} \quad (264)$$

The **fine-structure constant** is:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137}. \quad (265)$$

Physical significance: $\alpha \approx 1/137 \ll 1$ is the small parameter that makes the perturbative expansion in Feynman diagrams reliable. Each additional vertex contributes a factor of $e^2 \sim 4\pi/137 \approx 0.09$, so corrections are genuinely small. This is in contrast to the strong interaction at low energies, where the analogous $\alpha_s \sim 1$ and perturbation theory breaks down.

QED Processes

Physical Motivation

From the single vertex $\bar{e}e\gamma$, we can construct an infinite variety of processes by connecting vertices with propagators. Every amplitude must conserve charge at each vertex: the sum of incoming charges equals the sum of outgoing charges. (This is simply the Ward identity: $\sum Q = 0$ at every vertex.) We survey the most important tree-level and selected loop processes.

Tree-level processes (two vertices, order α^2 in cross-section)

All tree-level QED processes at lowest order involve two vertices; each vertex contributes one power of e , so the amplitude scales as e^2 and the cross-section as α^2 .

Bhabha scattering: $e^+e^- \rightarrow e^+e^-$. Two independent Feynman diagrams contribute:

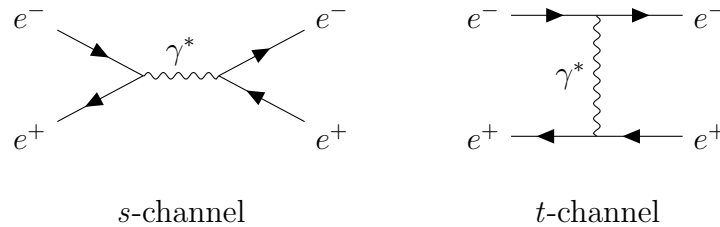


Figure 2: Bhabha scattering $e^+e^- \rightarrow e^+e^-$. The s -channel (left) corresponds to annihilation into a virtual photon followed by pair creation; the t -channel (right) corresponds to photon exchange between the electron and positron. Both contribute at order α^2 .

Charge check (s -channel): At the left vertex, $(-1) + (+1) = 0$; the virtual photon is neutral. At the right vertex, $0 \rightarrow (-1) + (+1) = 0$. Charge conserved at both vertices. ✓

Møller scattering: $e^-e^- \rightarrow e^-e^-$. Two electrons exchange a virtual photon. The s -channel is forbidden: two incoming electrons would need to annihilate into an object of charge -2 , but the photon is neutral — charge conservation fails. Only the t -channel and u -channel (crossed t -channel) contribute.

Compton scattering: $e^-\gamma \rightarrow e^-\gamma$. An electron absorbs a photon and emits another. Two diagrams: in the first, the photon is absorbed before being emitted (direct, with the intermediate virtual electron propagator in the s -channel); in the second, the photon is emitted first (with the virtual electron in the u -channel).

Pair annihilation: $e^+e^- \rightarrow \gamma\gamma$. Electron and positron annihilate into two photons. A single final-state photon is forbidden by energy-momentum conservation. Two diagrams contribute (the electron propagator is in either the t - or u -channel).

Pair production: $\gamma\gamma \rightarrow e^+e^-$. The time-reversed process of pair annihilation. Two real photons collide and materialise as an electron-positron pair.

All tree-level cross-sections scale as:

$$\sigma_{\text{tree}} \propto |\mathcal{M}|^2 \propto e^4 \propto \alpha^2. \quad (266)$$

Higher-order (loop) diagrams

Physical Motivation

Beyond tree level, we add loops: each additional closed loop introduces at least two more vertices, contributing $e^2 \propto \alpha$ to the amplitude squared. An order- α^4 contribution to the Bhabha cross-section (order α^2 in $\mathcal{M}$) requires four vertices total. Representative diagrams include:

- **Vertex correction:** an extra virtual photon emitted and reabsorbed on one of the external electron legs.
- **Photon self-energy** (vacuum polarisation): a fermion loop is inserted inside the virtual photon propagator. This diagram corrects the kinetic term of the photon — equivalently, it rescales the effective gauge coupling, giving rise to the **running of α** .
- **Box diagram:** the electron and positron exchange two virtual photons simultaneously, forming a four-vertex box.
- **Crossed box:** as above, with the photon lines exchanged.

Anomalous magnetic moment. A one-loop correction to the electron–photon vertex contains a virtual photon connecting two points on the electron line. This diagram gives the leading quantum correction to the electron’s magnetic moment:

$$g_e = 2 \left(1 + \frac{\alpha}{2\pi} + O(\alpha^2) + \dots \right). \quad (267)$$

This is the **anomalous magnetic moment**. QED predicts g_e to twelve significant figures, in spectacular agreement with experiment — the most precisely tested prediction of any physical theory.

Physical significance: Loops are not corrections we can choose to ignore. They are required by quantum mechanics (the path integral sums over all field configurations). Their calculation demands renormalisation of e and m_e , after which all predictions are finite and parameter-free.

Parameter counting and experimental tests

Physical Motivation

A theory with n free parameters needs n measurements to fix them; every subsequent measurement is a genuine prediction. How many free parameters does QED have?

The Lagrangian (261) depends on exactly **two parameters**:

$$\text{QED parameters: } m_e \text{ (electron mass), } e \text{ (or equivalently } \alpha = e^2/4\pi). \quad (268)$$

What is not a free parameter?

- The electron charge $Q = -1$ is part of the theory's *definition* (the input charge assignment), not a measurement.
- The photon mass is zero by gauge invariance: $m_\gamma = 0$ is a prediction.

Both parameters have been measured with extraordinary accuracy:

$$m_e = 0.51099895000 \pm 0.00000000015 \text{ MeV}, \quad (269)$$

$$\alpha^{-1} = 137.035999084 \pm 0.000000021. \quad (270)$$

Two measurements fix m_e and $e \implies$ every other QED observable is a parameter-free prediction.

(271)

Physical significance: This is what it means to *test* a theory. After determining m_e and α from, say, measuring the electron rest energy and the Bhabha cross-section, we can then independently predict the Lamb shift, Compton cross-section, anomalous magnetic moment, etc. The agreement of these predictions with experiment to 10+ significant figures is the gold standard of theoretical physics.

QED with Two Fermion Generations

Physical Motivation

Nature provides three generations of charged leptons: electron, muon, and tau. We now generalise QED to two generations (electron + muon) to understand how the particle masses arise from the most general Lagrangian and to identify any emergent symmetry.

Definition

Symmetry:	Local $U(1)_{\text{em}}$
Fermions:	$\ell_{L,i}$ (left-handed, $i = 1, 2$, charge $Q = -1$) $\ell_{R,i}$ (right-handed, $i = 1, 2$, charge $Q = -1$)
Gauge field:	A_μ (photon, $Q = 0$)

All four Weyl spinors carry the *same* charge $Q = -1$, so all can form gauge-invariant mass terms with one another.

The most general mass matrix

Physical Motivation

A Dirac mass term pairs one left-handed and one right-handed field; the product $\overline{\ell_{L,i}} \ell_{R,j}$ is gauge invariant because the charges cancel: $(-Q) + Q = 0$. With two generations, there are $2 \times 2 = 4$ such pairings.

The most general mass Lagrangian is:

$$\mathcal{L}_{\text{mass}} = -\overline{\ell_L}^T M \ell_R + \text{h.c.} = -(\overline{\ell_{L,1}} \quad \overline{\ell_{L,2}}) \underbrace{\begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}}_M \begin{pmatrix} \ell_{R,1} \\ \ell_{R,2} \end{pmatrix} + \text{h.c.}, \quad (272)$$

where M is a general complex 2×2 matrix.

Physical significance: The off-diagonal entries M_{12} and M_{21} mix the two flavours. In the basis (ℓ_1, ℓ_2) , the physical particles do not have definite mass; we must rotate to a better basis.

Rotation to the mass basis

Physical Motivation

Any complex matrix M can be diagonalised by a **bi-unitary transformation**: $M = V_L^\dagger M_{\text{diag}} V_R$. This amounts to independent unitary rotations of the left-handed and right-handed fields. After the rotation, we obtain fields with definite mass — the **mass eigenstates**.

Derivation

Choose unitary matrices V_L and V_R such that:

$$V_L M V_R^\dagger = \begin{pmatrix} m_e & 0 \\ 0 & m_\mu \end{pmatrix}, \quad m_e \leq m_\mu. \quad (273)$$

Define mass-eigenstate fields:

$$\begin{pmatrix} \ell_{L,1} \\ \ell_{L,2} \end{pmatrix} = V_L^\dagger \begin{pmatrix} e_L \\ \mu_L \end{pmatrix}, \quad \begin{pmatrix} \ell_{R,1} \\ \ell_{R,2} \end{pmatrix} = V_R^\dagger \begin{pmatrix} e_R \\ \mu_R \end{pmatrix}. \quad (274)$$

Key point — kinetic terms are invariant: The fermion kinetic terms are sums of diagonal terms:

$$\sum_i i \bar{\ell}_{L,i} \not{D} \ell_{L,i}.$$

A unitary rotation V_L acting on the flavour index i is like a change of orthonormal basis: $\sum_i |\ell_{L,i}\rangle \langle \ell_{L,i}|$ is invariant under V_L . (More concretely: $V_L^\dagger V_L = \mathbf{1}$, so the sum is unchanged.) The same holds for the covariant derivative D_μ since all fields carry the same charge $Q = -1$, so D_μ commutes with the flavour rotation.

$$\boxed{\text{The unitary rotations } V_L, V_R \text{ leave all kinetic terms invariant.}} \quad (275)$$

Lagrangian in the mass basis

In the new basis (e, μ) where the mass matrix is diagonal:

$$\mathcal{L}_{\text{QED},2} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i \bar{e} \not{D} e + i \bar{\mu} \not{D} \mu - m_e \bar{e} e - m_\mu \bar{\mu} \mu, \quad D_\mu = \partial_\mu - ie A_\mu. \quad (276)$$

The two mass eigenstates are:

- The **electron** e : lighter, $m_e \approx 0.511$ MeV, charge -1 .
- The **muon** μ : heavier, $m_\mu \approx 105.7$ MeV, charge -1 .

Both couple to the photon with the same strength e , because both carry $Q = -1$. This is **gauge universality**: the electromagnetic interaction is flavour-blind.

Accidental symmetry: $U(1)_e \times U(1)_\mu$

Physical Motivation

Inspecting $\mathcal{L}_{\text{QED},2}$: there is no term mixing e and μ . Each flavour appears only in its own kinetic term and mass term; the photon couples to each independently. This means we can phase-rotate e and μ with *independent* constant parameters and leave the Lagrangian invariant.

Derivation

Under independent global phase rotations:

$$e \rightarrow e^{i\beta_e} e, \quad \mu \rightarrow e^{i\beta_\mu} \mu, \quad (277)$$

every term in $\mathcal{L}_{\text{QED},2}$ is invariant (the phases cancel in $\bar{e}e$, $\bar{\mu}\mu$, and the kinetic terms). This is an accidental $U(1)_e \times U(1)_\mu$ symmetry:

$$\boxed{G_{\text{accidental}} = U(1)_e \times U(1)_\mu}, \quad (278)$$

even though only a single $U(1)_{\text{em}}$ was imposed.

Why is it accidental? The only renormalizable gauge-invariant operator that could mix e and μ would be a cross-mass term $\bar{e}_L \mu_R$. But we are already in the mass basis, where we have specifically rotated to eliminate such terms. Any cross-mass term that were to appear would require $m_e \neq m_\mu$, and in fact in the mass basis, the mass matrix is diagonal — so no mixing term survives.

Consequences. By Noether's theorem, $U(1)_e$ implies conservation of **electron number** L_e and $U(1)_\mu$ implies conservation of **muon number** L_μ . This forbids processes such as $\mu^- \rightarrow e^- e^+ e^-$ (electron-positron pair in the final state) from happening in pure QED.

Relation to $U(1)_{\text{em}}$. The special case $\beta_e = \beta_\mu \equiv \beta$ is a single common phase rotation — the global limit of the gauge $U(1)_{\text{em}}$. The diagonal subgroup of $U(1)_e \times U(1)_\mu$ is the global electromagnetic symmetry:

$$\boxed{U(1)_{\text{em, global}} \subset U(1)_e \times U(1)_\mu}. \quad (279)$$

Physical significance: Accidental symmetries are always global, never gauged. They are in general broken by higher-dimensional operators (suppressed by a high scale Λ). In nature, L_e and L_μ are broken by neutrino Yukawa couplings in the full Standard Model, allowing processes like $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ (muon decay). However, the total lepton number $L = L_e + L_\mu + L_\tau$ remains approximately conserved at accessible energies.

To Remember

- The covariant derivative $D_\mu = \partial_\mu + igQA_\mu$ generates interaction vertices when expanded: a three-point vertex ($\bar{\psi}\gamma^\mu\psi A_\mu$, strength gQ) for fermions; three-point (strength gQ with momentum) and four-point (seagull, strength $(gQ)^2$) vertices for complex scalars.
- D_μ **knows the charge of the field it acts on.** A $Q = 0$ field has no coupling to the gauge boson.
- Model building: symmetry + fields + charges $\rightarrow$ most general $\dim \leq 4$ Lagrangian; if local, add gauge fields, promote $\partial_\mu \rightarrow D_\mu$.
- QED (one generation) has two parameters: m_e and e . The photon mass zero is a prediction of gauge invariance, not a parameter.
- The QED vertex rule is $+ie\gamma^\mu$ for the electron ($Q = -1$). Fine-structure constant: $\alpha = e^2/4\pi \approx 1/137$.
- Tree-level QED processes (Bhabha, Møller, Compton, pair production/annihilation) each have two vertices; $\sigma \propto \alpha^2$. Once m_e and e are fixed, all others are predictions.

- The massless photon is a prediction; detecting a nonzero m_γ would falsify QED.
- QED with two generations: the 2×2 complex mass matrix is diagonalised by bi-unitary transformations; kinetic terms are unchanged; mass eigenstates are the electron and muon.
- The absence of e – μ mixing in the mass-basis Lagrangian produces an accidental $U(1)_e \times U(1)_\mu$, implying separate conservation of L_e and L_μ in pure QED.
- Accidental symmetries are always global; the global part of $U(1)_{\text{em}}$ is the diagonal of $U(1)_e \times U(1)_\mu$.

Recommended Reading

- **Tong, *Lectures on Particle Physics*, Chapter 3:** QED, covariant derivative, vertex rules, and one-loop corrections.
- **Griffiths, *Introduction to Elementary Particles*, Chapter 9:** QED Lagrangian, the electromagnetic vertex, and scattering processes.
- **Halzen & Martin, Chapter 14:** QED interactions, Bhabha and Møller scattering, parameter counting.
- **Peskin & Schroeder, Sections 5.1–5.5:** Tree-level QED processes, spin sums, and leading radiative corrections.
- **Grossman & Rakshit (course notes), Chapter 3 and Chapter 5:** Charge assignments, local symmetries, and QED spectrum.

Part IV

Non-Abelian Gauge Theory & QCD

12 Lecture 12 – SM Charged Fermions, Accidental Symmetries, and Introduction to Non-Abelian Symmetries

Date: 17/05/2026

Overview

This lecture completes the analysis of accidental symmetries in QED and then uses that analysis as a launching pad to describe the full charged-fermion spectrum of the Standard Model. We then briefly discuss how particle exchange generates potentials in the non-relativistic limit, and close by introducing non-Abelian symmetries — the structure that underlies $SU(3)_c$ and $SU(2)_L$ in the SM.

1. **Accidental symmetries in two-generation QED:** the full $U(1)_e \times U(1)_\mu$ structure, forbidden processes, and how non-renormalizable operators would break these symmetries.
2. **Generalisation to N families:** the mass-basis argument for any number of same-charge Dirac fermions.
3. **SM charged fermion spectrum:** three charge families (charged leptons, up quarks, down quarks), three generations each, the resulting accidental $U(1)^9$.
4. **Potentials from particle exchange:** Coulomb and Yukawa potentials as the non-relativistic limit of field-theoretic amplitudes.
5. **Introduction to non-Abelian symmetries:** what “non-Abelian” means, and why $SU(3)$ and $SU(2)$ appear in the SM.

Accidental Symmetries in Two-Generation QED

Conceptual Background

In Lecture 11 we constructed the two-generation QED Lagrangian in the mass basis and found an accidental $U(1)_e \times U(1)_\mu$ symmetry. Here we make the structure fully explicit, choose the most useful basis of generators, and draw the physical consequences.

The Two Independent Rotations

Recall the mass-basis Lagrangian (Lec. 11, Eq. (276)):

$$\mathcal{L}_{\text{QED},2} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{e}\not{D}e + i\bar{\mu}\not{D}\mu - m_e\bar{e}e - m_\mu\bar{\mu}\mu. \quad (280)$$

Because e and μ appear only in separate kinetic and mass terms, the Lagrangian is invariant under *independent* phase rotations:

$$e \rightarrow e^{i\alpha_e} e, \quad \mu \rightarrow e^{i\alpha_\mu} \mu, \quad (281)$$

with $\alpha_e, \alpha_\mu \in \mathbb{R}$ completely independent. This is the accidental $U(1)_e \times U(1)_\mu$ symmetry.

Any pair of independent linear combinations of these two generators is equally valid. Two particularly useful choices are:

Symmetry	Charge of e	Charge of μ
$U(1)_e$	+1	0
$U(1)_\mu$	0	+1
$U(1)_{e+\mu}$ (diagonal, $= U(1)_{\text{em}}^{\text{global}}$)	+1	+1
$U(1)_{e-\mu}$ (off-diagonal)	+1	-1

The first pair $(U(1)_e, U(1)_\mu)$ is natural in the mass basis. The second pair emphasises that the *diagonal* combination is just the global part of the imposed $U(1)_{\text{em}}$, while the *difference* is a genuinely accidental, “extra” symmetry.

$$\boxed{U(1)_{\text{em}}^{\text{global}} = \text{diagonal of } U(1)_e \times U(1)_\mu.} \quad (282)$$

Significance: The imposed local $U(1)_{\text{em}}$ symmetry already guarantees charge conservation; the accidental $U(1)_e$ and $U(1)_\mu$ provide *additional*, separately conserved quantum numbers — electron number and muon number.

Conservation Laws

By Noether’s theorem, $U(1)_e$ and $U(1)_\mu$ respectively imply:

$$\boxed{L_e \equiv (\#e^-) - (\#e^+) = \text{const}, \quad L_\mu \equiv (\#\mu^-) - (\#\mu^+) = \text{const}.} \quad (283)$$

The imposed $U(1)_{\text{em}}$ gives the *sum* $Q \propto L_e + L_\mu$ (for unit charges); the accidental symmetry additionally conserves each *separately*.

Forbidden Process: $\mu^- \rightarrow e^+ e^- e^-$

Physical Motivation

A concrete payoff of identifying accidental symmetries is that one can immediately rule out processes without drawing a single Feynman diagram.

Analysis

Consider the decay $\mu^- \rightarrow e^+ e^- e^-$. First check the imposed symmetry:

$$\begin{aligned} Q_{\text{initial}} &= -1, \\ Q_{\text{final}} &= (+1) + (-1) + (-1) = -1. \quad \checkmark \end{aligned}$$

Charge is conserved, so this process is not forbidden by $U(1)_{\text{em}}$. Now check the accidental symmetries:

$$\begin{aligned} L_e^{\text{initial}} &= 0, & L_e^{\text{final}} &= (-1) + 1 + 1 = +1. \quad \text{VIOLATED.} \\ L_\mu^{\text{initial}} &= 1, & L_\mu^{\text{final}} &= 0. \quad \text{VIOLATED.} \end{aligned}$$

$$\boxed{\mu^- \rightarrow e^+ e^- e^- \text{ is forbidden in two-generation QED by the accidental } U(1)_e \times U(1)_\mu.} \quad (284)$$

Significance: Without this argument, a student might attempt to draw the diagram — only to find no gauge-invariant topology exists. The accidental symmetry gives the same answer instantly. If we ever observe such a decay, it signals physics beyond this theory.

Non-Renormalizable Operators That Break the Accidental Symmetry

The accidental symmetry is a feature of the *renormalizable* Lagrangian. It can be broken by higher-dimensional operators suppressed by a high scale Λ . The simplest such operator is

$$\mathcal{O}_{\text{LFV}} = \frac{c}{\Lambda^2} \overline{\ell_{L,e}} e_R \bar{\mu}_R \ell_{L,\mu} + \text{h.c.}, \quad (285)$$

where $\ell_{L,e}$ and $\ell_{L,\mu}$ are the left-handed lepton doublets of the electron and muon families, and e_R, μ_R are their right-handed counterparts.

- Lorentz structure: $(\bar{f}_L f_R)(\bar{f}'_R f'_L)$ — two independent Lorentz scalars contracted together. Valid.
- Electromagnetic charge: each bilinear has net charge zero. Valid.
- Mass dimension: $[\psi] = 3/2$, so $[\mathcal{O}_{\text{LFV}}] = 4 \times \frac{3}{2} = 6$. Requires $1/\Lambda^2$ suppression.

Since $[\mathcal{O}] > 4$, this operator is **not included** in the renormalizable Lagrangian. Including it would break $U(1)_e \times U(1)_\mu$ and allow $\mu^- \rightarrow e^+ e^- e^-$ at rate $\sim c^2/\Lambda^4$. Experimental limits on this process constrain $\Lambda/\sqrt{c}$.

$\mu^- \rightarrow e^+ e^- e^- \text{ can only proceed via non-renormalizable (dim-6) operators } \sim \frac{1}{\Lambda^2}; \text{ rate } \propto \frac{1}{\Lambda^4}.$

(286)

Allowed Processes at Tree Level

The two interaction vertices in $\mathcal{L}_{\text{QED},2}$ are $\bar{e} e \gamma$ and $\bar{\mu} \mu \gamma$. Crucially, there is *no* flavour-changing vertex $\bar{e} \mu \gamma$ — this is directly linked to the accidental symmetry.

$e^+ e^- \rightarrow \mu^+ \mu^-$ (s-channel):

$$Q_{\text{initial}} = 0, \quad Q_{\text{final}} = 0; \quad L_e^{\text{initial}} = 0, \quad L_e^{\text{final}} = 0; \quad L_\mu^{\text{initial}} = 0, \quad L_\mu^{\text{final}} = 0. \quad \checkmark$$

The electron-positron pair annihilates into a virtual photon, which then produces a muon-antimuon pair. The t-channel and u-channel are *absent* because they would require a vertex $\bar{e} \mu \gamma$ (coupling an electron to a muon via a photon), which does not exist.

$e^- \mu^- \rightarrow e^- \mu^-$ (t-channel, “Rutherford scattering of muons”):

$$Q_{\text{initial}} = -2, \quad Q_{\text{final}} = -2; \quad L_e, L_\mu \text{ both conserved.} \quad \checkmark$$

The electron and muon exchange a virtual photon (t-channel). The s-channel is absent (would need a neutral $e^- \mu^-$ state, but those carry different lepton numbers and no neutral s-channel intermediate state is available in two-generation QED).

Generalisation to N Same-Charge Fermion Families

Physical Motivation

The two-generation argument was not special to two generations. As long as we add pairs of left-handed and right-handed Weyl spinors all carrying the same $U(1)_{\text{em}}$ charge, the structure is identical.

The General Statement

Suppose we have N pairs $(\ell_{L,j}, \ell_{R,j})$, $j = 1, \dots, N$, all with the same charge Q . The most general mass Lagrangian contains an $N \times N$ complex mass matrix M_{ij} . Since all fields share the same charge, the kinetic terms are invariant under arbitrary $U(N)_L \times U(N)_R$ rotations of the flavour index. We use these rotations to bring M to diagonal form (bi-unitary / singular-value decomposition):

$$\boxed{V_L M V_R^\dagger = \text{diag}(m_1, m_2, \dots, m_N), \quad m_j \geq 0.} \quad (287)$$

In the mass basis the Lagrangian is:

$$\mathcal{L}_{\text{QED},N} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \sum_{j=1}^N \left(i \bar{\ell}_j \not{D} \ell_j - m_j \bar{\ell}_j \ell_j \right), \quad (288)$$

where $D_\mu = \partial_\mu + ieQA_\mu$ is the *same* covariant derivative for all ℓ_j (same charge Q).

Accidental Symmetry

Because every ℓ_j appears only in its own kinetic and mass term, each can be phase-rotated independently:

$$\boxed{\mathcal{L}_{\text{QED},N} \text{ has accidental } U(1)^N \text{ symmetry, conserving } L_j \equiv (\#\ell_j) - (\#\bar{\ell}_j) \text{ for each } j.} \quad (289)$$

The diagonal $U(1)_{\text{em}}^{\text{global}} \subset U(1)^N$ is the global part of the imposed gauge symmetry.

Charged Fermions in the Standard Model

Physical Motivation

In nature, the charged elementary fermions do not all carry the same electromagnetic charge. There are three distinct charge sectors. Within each sector, the general N -family argument applies; between sectors, the rotations are independent.

The Three Charge Sectors

Family	Particles	Q_{em}	Generations
Charged leptons	e, μ, τ	-1	3
Up-type quarks	u, c, t	$+\frac{2}{3}$	3
Down-type quarks	d, s, b	$-\frac{1}{3}$	3

Within each row, the three particles are called **flavours** or **generations**. Heavier generations are conventionally labelled 2nd and 3rd:

1st generation (lightest): e, u, d ;

2nd generation: μ, c, s ;

3rd generation (heaviest): τ, t, b .

The three sectors rotate *independently* to their mass bases, since cross-sector mass terms would mix fields of different charge and are forbidden by $U(1)_{\text{em}}$ invariance.

QED Lagrangian for Nine Fields

Extending (288) to all three sectors:

$$\mathcal{L}_{\text{QED},9} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \sum_f (i\bar{f}\not{D}_f f - m_f\bar{f}f), \quad (290)$$

where f runs over all nine mass eigenstates and $D_\mu^f = \partial_\mu + ieQ_f A_\mu$ carries the appropriate charge $Q_f \in \{-1, +\frac{2}{3}, -\frac{1}{3}\}$.

Accidental Symmetry

$\mathcal{L}_{\text{QED},9}$ has accidental $U(1)^9$ symmetry : one $U(1)_f$ per flavour, conserving L_f .

(291)

More explicitly: since the three charge sectors cannot mix (different Q), the accidental symmetry decomposes as

$$U(1)^9 = \underbrace{U(1)_e \times U(1)_\mu \times U(1)_\tau}_{\text{charged leptons}} \times \underbrace{U(1)_u \times U(1)_c \times U(1)_t}_{\text{up quarks}} \times \underbrace{U(1)_d \times U(1)_s \times U(1)_b}_{\text{down quarks}}. \quad (292)$$

Significance: This is a characteristic of pure QED. In the full Standard Model, weak interactions ($SU(2)_L$ gauge bosons) and the Higgs mechanism generate *flavour-changing* interactions that reduce this accidental symmetry, leaving only three accidental $U(1)$'s: baryon number B and two linear combinations of lepton number.

Potentials from Particle Exchange

Physical Motivation

QFT predicts forces: in the non-relativistic limit where particle production and annihilation can be neglected, the amplitude for two-particle scattering reduces to a potential $V(r)$. The form of V is determined by which particle is exchanged.

Setup

In the limit where particle production and annihilation are negligible, the scattering amplitude defines a

(293)

The connection is:

$$\tilde{V}(\mathbf{q}) = -\frac{1}{(2m_1)(2m_2)} i\mathcal{M}(\mathbf{q}), \quad (294)$$

where $\mathbf{q}$ is the 3-momentum transfer, and $V(\mathbf{r})$ is the Fourier transform of $\tilde{V}(\mathbf{q})$.

Photon exchange — the Coulomb potential. The t-channel amplitude for exchanging a massless photon goes as

$$i\mathcal{M} \propto \frac{1}{q^2} = \frac{1}{|\mathbf{q}|^2 - \omega^2} \xrightarrow{\text{NR}, \omega \rightarrow 0} \frac{1}{|\mathbf{q}|^2}. \quad (295)$$

Fourier-transforming $1/|\mathbf{q}|^2$ back to position space:

$V_{\text{Coulomb}}(r) \propto \frac{1}{r}.$

(296)

Massive scalar exchange — the Yukawa potential. If instead we exchange a scalar of mass m_ϕ with Yukawa coupling y :

$$i\mathcal{M} \propto \frac{y^2}{|\mathbf{q}|^2 + m_\phi^2}. \quad (297)$$

Fourier-transforming:

$$V_{\text{Yukawa}}(r) \propto \frac{e^{-m_\phi r}}{r}. \quad (298)$$

Significance: The range of the force is $\sim 1/m_\phi$. For a massless photon ($m_\phi = 0$), the Yukawa potential reduces to the Coulomb potential — infinite-range. A massive mediator (e.g. the $W^\pm$ or Z^0 boson with $m \sim 80\text{--}90\text{ GeV}$) produces a short-range force, explaining why the weak interaction is confined to sub-nuclear scales.

Introduction to Non-Abelian Symmetries

Physical Motivation

So far all symmetries discussed have been $U(1)$ (Abelian) symmetries. The Standard Model requires two more gauge symmetries, $SU(3)_c$ and $SU(2)_L$, which are **non-Abelian**. Understanding the qualitative difference between Abelian and non-Abelian symmetries is the gateway to the rest of the course.

Abelian vs. Non-Abelian: the Key Distinction

Recall $U(1)$ transformations: a field transforms as $\phi \rightarrow e^{i\theta}\phi$. Two successive rotations by angles θ_1 and θ_2 commute:

$$e^{i\theta_2}(e^{i\theta_1}\phi) = e^{i(\theta_1+\theta_2)}\phi = e^{i\theta_1}(e^{i\theta_2}\phi). \quad (299)$$

The result is independent of the order. This is the **Abelian** (commutative) property.

For a **non-Abelian** symmetry, the transformations are represented by matrices acting on a multi-component field, and in general:

$$U(\theta_1)U(\theta_2)\phi \neq U(\theta_2)U(\theta_1)\phi. \quad (300)$$

The result *depends on the order* of the transformations.

Remark 37. Rotation in 3D space is a familiar example of a non-Abelian operation: rotating first about $\hat{x}$ then about $\hat{z}$ gives a different final orientation than rotating first about $\hat{z}$ then about $\hat{x}$.

Non-Abelian Symmetries in the Standard Model

The two non-Abelian gauge symmetries in the SM are:

$$SU(3)_c : \quad \text{strong (colour) force; 8 gauge bosons (gluons)}. \quad (301)$$

$$SU(2)_L : \quad \text{weak isospin force; 3 gauge bosons } (W^+, W^-, W^0). \quad (302)$$

The full SM gauge group is:

$$G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y. \quad (303)$$

Significance: The subscript L in $SU(2)_L$ means the symmetry acts only on *left-handed* fields — an explicit violation of parity. This is why the SM is chiral and why left- and right-handed fields can (and do) carry different charges.

The language of non-Abelian symmetries requires the formalism of **Lie groups** and **Lie algebras** (generators, structure constants, representations). This will be developed in Lecture 13 onward.

To Remember

- The accidental $U(1)_e \times U(1)_\mu$ in two-generation QED conserves L_e and L_μ separately. The diagonal subgroup is the global $U(1)_{\text{em}}$.
- $\mu^- \rightarrow e^+ e^- e^-$ conserves charge but violates L_e and L_μ — forbidden in renormalizable QED; can only occur via dim-6 operators $\sim 1/\Lambda^2$.
- For N Dirac fermions of the same charge, the mass-basis Lagrangian has accidental $U(1)^N$. Particles of different charge cannot mix (forbidden by gauge invariance), so each charge sector is diagonalised independently.
- SM charged fermions: 9 Dirac fields in three charge sectors ($Q = -1, +\frac{2}{3}, -\frac{1}{3}$) with 3 generations each. QED with these 9 fields has accidental $U(1)^9$.
- Coulomb potential ($\sim 1/r$) arises from massless photon exchange; Yukawa potential ($\sim e^{-mr}/r$) arises from massive scalar exchange. Range of force $\sim 1/m_{\text{mediator}}$.
- Non-Abelian means order of group operations matters. $SU(3)_c \times SU(2)_L$ are the two non-Abelian factors of the SM gauge group $SU(3) \times SU(2) \times U(1)$.

Recommended Reading

- **Tong, *Lectures on Particle Physics*, Chapter 4:** Non-Abelian gauge theories; the language of Lie groups and algebras.
- **Grossman & Rakshit (course notes), Appendix A:** Group theory review — generators, representations, $SU(N)$ — essential for Lecture 13.
- **Griffiths, *Introduction to Elementary Particles*, Chapter 1 and Appendix C:** The particle zoo; the SM field content and charge assignments.
- **Halzen & Martin, Chapter 2:** The Yukawa potential and the range of forces; photon exchange and the Coulomb potential.

13 Lecture 13 – Non-Abelian Symmetries: Representations, Invariants, and Gauge Theory

Date: 28/05/2026

Overview

Lecture 12 introduced the *idea* of a non-Abelian symmetry through the single statement “the order of operations matters.” This lecture makes that idea operational. We build the full machinery a physicist actually uses: **multiplets**, **generators**, **structure constants**, and the two most important **representations** (fundamental and adjoint). We then learn the central practical skill of the course — deciding which field combinations are *allowed* in a Lagrangian by asking whether they contain a **singlet**. Finally we gauge the symmetry: promoting $SU(N)$ to a *local* symmetry forces a gauge field into the adjoint representation, and — unlike $U(1)$ — this gauge field *talks to itself*. This self-interaction is the seed of the entire theory of the strong force.

1. **Multiplets and generators**: how a field transforms under a non-Abelian group; the Lie algebra and structure constants.
2. **Fundamental and adjoint representations**; the $SU(2)$ doublet and $SU(3)$ triplet; the Gell-Mann matrices.
3. **Product groups**: assigning one index per simple factor.
4. **Building invariants**: contracting representations into singlets; the “make an invariant” game; QCD bound states (mesons, baryons, glueballs).
5. **Local non-Abelian invariance**: covariant derivative, gauge field in the adjoint, non-Abelian field strength, gauge self-interactions, and masslessness.

This lecture follows **Chapter 4** of the course notes and the much-loved **Appendix A** (group theory).

13.1 From $U(1)$ to $SU(N)$: Multiplets and Generators

Conceptual Background

For an Abelian $U(1)$ symmetry (Lec. 9–11) the story was *per field*: every field ϕ carried a charge q and transformed by an independent phase,

$$\phi \rightarrow e^{iq\theta} \phi. \quad (304)$$

Each field rotated separately and the order of two rotations never mattered. For a non-Abelian group the basic object is no longer a single charged field but a *collection* of fields that the symmetry rotates *into one another*.

Physical Concept 13.1: Representation and Multiplet

A field ϕ **sits in a representation** R of a symmetry group G , of **dimension** n , if under the symmetry ϕ is an n -component vector

$$\phi = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_n \end{pmatrix}, \quad i = 1, \dots, n,$$

and the symmetry acts by an $n \times n$ **unitary** matrix U that mixes the components:

$$\phi_i \rightarrow U_{ij} \phi_j.$$

This n -component object is called a **multiplet**. The Abelian case is the special case $n = 1$ (a “singlet” carrying only a phase).

The unitary matrix is written as an exponential of Hermitian **generators** T^a :

$$\phi_i \rightarrow (e^{iT^a\theta^a})_{ij} \phi_j \quad a = 1, \dots, N^2 - 1 \text{ (for } SU(N)). \quad (305)$$

Significance: The transformation law is now defined *for the whole multiplet at once*, not field-by-field. The single number q of the Abelian case is replaced by a *set of matrices* T^a .

Note:

Notation for Eq. (305): the index $i, j = 1, \dots, n$ label *components of the multiplet* (which generator-matrix row/column we are in); the index $a = 1, \dots, N^2 - 1$ labels *which generator* and is summed over (repeated-index convention). The θ^a are real continuous parameters of the transformation.

Mathematical Result 13.1 The Lie Algebra and Structure Constants

The generators of $SU(N)$ are Hermitian matrices obeying the closed commutation relations

$$[T^a, T^b] = if^{abc} T^c.$$

This is the **algebra** of the group. The real numbers f^{abc} are the **structure constants**; they are totally antisymmetric and encode the non-commutativity. For an Abelian group all $f^{abc} = 0$.

In a representation R of dimension n , the generators T^a are concrete $n \times n$ matrices. The *same* abstract algebra $[T^a, T^b] = if^{abc} T^c$ holds in every representation — only the size of the matrices changes.

13.2 The Two Representations You Must Know**Mathematical Setup**

Out of the infinitely many representations of $SU(N)$, two appear everywhere in physics.

Physical Concept 13.2: Fundamental Representation

The **fundamental** representation is the smallest non-trivial irreducible representation. For $SU(N)$ it has dimension N . For $SU(2)$ this is the **doublet** (the familiar spin- $\frac{1}{2}$); for $SU(3)$ it is the **triplet**.

Physical Concept 13.3: Adjoint Representation

The **adjoint** representation is built directly out of the structure constants f^{abc} themselves. Its dimension equals the number of generators, i.e. the dimension of the group:

$$\dim(\text{adjoint}) = N^2 - 1.$$

Significance: The adjoint will turn out to be the home of the *gauge bosons*. The fact that gluons live in the 8 of $SU(3)$ is exactly the statement “gauge fields sit in the adjoint.”

Example 1 — the $SU(2)$ doublet. A field ϕ that is a doublet of $SU(2)$ sits in the **2** (the fundamental). It has two components and transforms as

$$\phi \rightarrow \exp\left(i \frac{\tau^a}{2} \theta^a\right) \phi, \quad a = 1, 2, 3, \quad (306)$$

where τ^a are the Pauli matrices and the generators in the doublet are $T^a = \tau^a/2$. There are $2^2 - 1 = 3$ of them.

Example 2 — the $SU(3)$ triplet. A field ϕ that is a triplet of $SU(3)$ sits in the **3**. It has three components and transforms as

$$\phi \rightarrow \exp\left(i \frac{\lambda^a}{2} \theta^a\right) \phi, \quad a = 1, \dots, 8, \quad (307)$$

where the generators are $T^a = \lambda^a/2$ and the λ^a are the eight **Gell-Mann matrices** ($3^2 - 1 = 8$), the $SU(3)$ analogue of the Pauli matrices. They are 3×3 traceless Hermitian matrices:

$$\begin{aligned} \lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda^5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda^7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda^8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \end{aligned} \quad (308)$$

Claim 13.1 Embedded $SU(2)$ subalgebras

Notice $(\lambda^1, \lambda^2, \lambda^3)$ act only on the first two components — they are exactly the Pauli matrices padded with a zero row/column. They generate an $SU(2)$ *subalgebra* sitting inside $SU(3)$. In fact $SU(3)$ contains **three** such $SU(2)$ subalgebras (acting on the $(1, 2)$, $(1, 3)$, and $(2, 3)$ pairs of components). Verifying this is a homework exercise.

13.3 Product Groups: One Index per Factor

Motivation

Realistic symmetry groups are products of *simple* factors, e.g. $G = SU(3) \times SU(2) \times U(1)$. The rule is mechanical: we must state how a field transforms under *each* factor separately, because a transformation in one factor does not touch the others.

Worked Example 13.1 (A field in $SU(3) \times SU(2)$)

Let ϕ be a **triplet** of $SU(3)$ and a **doublet** of $SU(2)$. Then it carries *two* indices, one per factor:

$$\phi_{\alpha i}, \quad \alpha = 1, 2, 3 \text{ (} SU(3) \text{ triplet index),} \quad i = 1, 2 \text{ (} SU(2) \text{ doublet index).}$$

Under an $SU(3)$ rotation only the α index moves:

$$\phi_{\alpha i} \rightarrow (e^{i\lambda^a \theta^a / 2})_{\alpha\beta} \phi_{\beta i} \quad (\text{the } i \text{ index is untouched}).$$

Under an $SU(2)$ rotation only the i index moves:

$$\phi_{\alpha i} \rightarrow (e^{i\tau^a \theta^a / 2})_{ij} \phi_{\alpha j} \quad (\text{the } \alpha \text{ index is untouched}).$$

Key insight: “one index per factor.” Conceptually simple, but keeping the bookkeeping straight is what saves you from confusion later.

13.4 Building Invariants: the Singlet Rule

Conceptual Background

How do we know which terms are allowed in a Lagrangian? For a global $U(1)$ (Lec. 9) the rule was: each term must have total charge zero, so the phases cancel and the term returns to itself. The non-Abelian generalisation replaces “charge sums to zero” with “the product of representations contains a **singlet**.”

Claim 13.2 The Singlet Rule

A Lagrangian invariant under a (global) non-Abelian symmetry consists of terms that are products of representations contracted into a **singlet** — an object that does not transform. A singlet is the non-Abelian way of saying “invariant.” A term is allowed if and only if its field content *can* be contracted into a singlet, under *every* imposed symmetry factor simultaneously.

Note:

In this course we care only about *which combinations can give a singlet*, not about writing the explicit index contractions. (Writing the contractions explicitly is a job for a dedicated group-theory or QFT course.) Whenever we write down a term, it is *implied* that we take the singlet contraction. If no singlet contraction exists, the term is simply not allowed.

The decomposition rules we will be handed

To play the game we only need the “multiplication table” of representations, which will always be provided. For $SU(2)$ (real/pseudoreal representations):

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}, \quad \mathbf{3} \otimes \mathbf{3} \supset \mathbf{1}. \quad (309)$$

The first is nothing but the familiar “ $\text{spin-}\frac{1}{2} \otimes \text{spin-}\frac{1}{2} = \text{triplet} \oplus \text{singlet}$.”

For $SU(3)$ (which has *complex* representations, so $\mathbf{3} \neq \bar{\mathbf{3}}$):

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}, \quad (310)$$

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}, \quad (311)$$

$$\mathbf{3} \otimes \mathbf{8} = \mathbf{15} \oplus \bar{\mathbf{6}} \oplus \mathbf{3}, \quad (312)$$

$$\mathbf{8} \otimes \mathbf{8} \supset \mathbf{1}. \quad (313)$$

Here $\mathbf{8}$ is the adjoint (the “octet”). Note that a singlet appears in $\mathbf{3} \otimes \bar{\mathbf{3}}$ but *not* in $\mathbf{3} \otimes \mathbf{3}$.

Worked Example 13.2 (Two $SU(2)$ doublets, and adding a triplet)

Take two $SU(2)$ doublets ρ, σ (both in the $\mathbf{2}$).

- **Quadratic term $\rho\sigma$:** since $\mathbf{2} \otimes \mathbf{2} = \mathbf{3} \oplus \mathbf{1}$, a singlet exists. The explicit contraction (which we usually do *not* write) is $\varepsilon_{ij}\rho^i\sigma^j$. *Allowed*.
- **Cubic term $\Delta\rho\sigma$ with a triplet Δ (the $\mathbf{3}$):** contract $\rho \otimes \sigma$ into their *triplet* piece; then $\mathbf{3} \otimes \mathbf{3} \supset \mathbf{1}$, so a singlet exists. *Allowed*.

Key insight: you are free to route the contraction through whichever intermediate representation makes a singlet possible. The term is allowed as long as *some* route works.

The “make an invariant” game

Impose $SU(3) \times SU(2) \times U(1)$ and assign each field its transformation properties, written as $(R_{SU(3)}, R_{SU(2)})_{q_{U(1)}}$:

$$Q = (\mathbf{3}, \mathbf{2})_{+1}, \quad U = (\mathbf{3}, \mathbf{1})_{+4}, \quad D = (\mathbf{3}, \mathbf{1})_{-2}, \quad H = (\mathbf{1}, \mathbf{2})_{+3}. \quad (314)$$

For each candidate term we check all three factors. The $U(1)$ check is the easiest (charges must sum to zero); the non-Abelian checks ask whether a singlet exists.

Term	$SU(3)$	$SU(2)$	$U(1)$	Invariant?
$H^\dagger H$	$\mathbf{1}$ trivially	$\mathbf{2} \otimes \bar{\mathbf{2}} \supset \mathbf{1}$	$3 - 3 = 0$	yes
H^3	—	—	$3 + 3 + 3 = 9 \neq 0$	no
$HQU^\dagger$	$\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$	$\mathbf{2} \otimes \mathbf{2} \supset \mathbf{1}$	$3 + 1 - 4 = 0$	yes

Note:

For $SU(2)$ the anti-fundamental is equivalent to the fundamental, $\bar{\mathbf{2}} \simeq \mathbf{2}$ (the doublet is pseudoreal). That is why $H^\dagger H$ already gives a singlet, and why we may freely contract $H \otimes Q$ as $\mathbf{2} \otimes \mathbf{2}$. The bar on $U^\dagger$ merely denotes the complex conjugate field, not a different fermion.

The third row, $HQU^\dagger$, is a genuine invariant under all three factors — it is precisely a **Yukawa interaction** of the type that gives quarks their mass in the Standard Model. Your homework will have you run through ~ 1000 more such combinations.

13.5 $SU(3)$ Colour and the Bound States of QCD

Physical Motivation

$SU(3)$ is not just a mathematical example: a *gauged* version of $SU(3)$ — called $SU(3)_c$, “colour” — is the symmetry of **Quantum Chromodynamics** (QCD), the theory of the strong force. The fields are:

- **quarks** q : triplets, the $\mathbf{3}$ of colour;
- **antiquarks** $\bar{q}$: the $\bar{\mathbf{3}}$;
- **gluons** g : the adjoint, the $\mathbf{8}$ (these are the gauge bosons — next section).

QCD at low energy is **confined**: free quarks are never seen, only colour-singlet bound states. So the question “which combinations can be observed as bound states in nature?” is *the same question* as “which combinations are colour singlets?”

Worked Example 13.3 (Which colour combinations are observable?)

Using Eqs. (310)–(313):

Combination	Decomposition	Physical state
$q\bar{q}$	$\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$	mesons (pions, kaons) — observed
qq	$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$, no $\mathbf{1}$	not allowed
qg	$\mathbf{3} \otimes \mathbf{8}$, no $\mathbf{1}$	not allowed
gg	$\mathbf{8} \otimes \mathbf{8} \supset \mathbf{1}$	glueball
$q\bar{q}g$	contains a $\mathbf{1}$	hybrid state
qqq	$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \supset \mathbf{1}$	baryons (proton, neutron) — observed

For qqq : combine two quarks via $\mathbf{3} \otimes \mathbf{3} \supset \bar{\mathbf{3}}$, then $\bar{\mathbf{3}} \otimes \mathbf{3} \supset \mathbf{1}$.

Significance: The very existence of mesons ($q\bar{q}$) and baryons (qqq) as the building blocks of matter is a direct consequence of which colour products contain a singlet. “Why three quarks in a proton?” — because that is what makes a colour singlet.

Open Problem 13.1: Glueballs and hybrids

Pure-gluon singlets (gg , “glueballs”) and hybrid states ($q\bar{q}g$) are *allowed* by colour symmetry and are expected on theoretical grounds, but have **not yet been unambiguously observed**. Because QCD at low energy is strongly coupled, perturbation theory fails and predictions for these states are hard to make precise; they are typically hunted experimentally by scanning for resonances and identifying their quantum numbers. Confirming glueballs remains an open experimental question.

13.6 Local Non-Abelian Invariance: Gauging $SU(N)$

Conceptual Background

So far the parameters θ^a were constants (a *global* symmetry). We now make them spacetime-dependent, $\theta^a \rightarrow \theta^a(x)$ — a *local* (gauge) symmetry. As in the Abelian case (Lec. 10), most terms survive untouched, but kinetic terms do not.

- Terms built from scalar/fermion fields *without derivatives*: if they were invariant under the global symmetry, they remain invariant under the local one.
- **Kinetic terms** (which contain ∂_μ) are *not* invariant, because ∂_μ hits the spacetime-dependent $\theta^a(x)$ and produces an extra term.

The cure, exactly as for $U(1)$, is to replace the ordinary derivative by a **covariant derivative** containing a new **gauge field**.

The covariant derivative

We demand a derivative D_μ such that $D_\mu\phi$ transforms the *same way* as ϕ itself, $D_\mu\phi \rightarrow U D_\mu\phi$ with $U = e^{iT^a\theta^a(x)}$. This is achieved by

$$D_\mu = \partial_\mu + i g T^a A_\mu^a, \quad a = 1, \dots, N^2 - 1 \quad (315)$$

where g is the **gauge coupling**, T^a are the generators in the representation of the field ϕ being acted on, and A_μ^a is the gauge field. The index a runs over the generators — so *the gauge field carries an adjoint index*, i.e. it sits in the adjoint representation.

Note:

The generators T^a in D_μ take whatever form is appropriate to the representation of the field they act on: $\tau^a/2$ when acting on an $SU(2)$ doublet, $\lambda^a/2$ on an $SU(3)$ triplet, and so on. The gauge field A_μ^a itself, however, always lives in the adjoint.

Transformation law of the gauge field

For $D_\mu\phi$ to transform correctly, the gauge field must transform as

$$A_\mu^a \rightarrow A_\mu^a - f^{abc} \theta^b A_\mu^c - \frac{1}{g} \partial_\mu \theta^a \quad (316)$$

Comparing with the Abelian photon ($A_\mu \rightarrow A_\mu - \frac{1}{g} \partial_\mu \theta$): the first and last terms are the familiar $U(1)$ -like pieces. The *middle* term $-f^{abc} \theta^b A_\mu^c$ is genuinely new — it is present only because the algebra is non-trivial ($f^{abc} \neq 0$). It is the structure constants reappearing, and it is responsible for everything that distinguishes a non-Abelian gauge theory from electromagnetism.

Significance: The non-Abelian gauge field transforms *in the adjoint representation* of the group (the structure-constant piece is exactly an adjoint rotation). For $U(1)$ the “adjoint” is trivial (one uncharged field, the photon); here it is an $(N^2 - 1)$ -component multiplet that transforms non-trivially.

The non-Abelian field strength

The gauge field is dynamical, so it needs its own kinetic term, built from a field-strength tensor. The Abelian $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ generalises to

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - g f^{abc} A_\mu^b A_\nu^c \quad (317)$$

The first two terms are the photon-like part (now carrying an adjoint index a). The last term $-g f^{abc} A_\mu^b A_\nu^c$ is *new* compared to the Abelian case — it is quadratic in the gauge field. The gauge-invariant kinetic Lagrangian is

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} \quad (318)$$

($F_{\mu\nu}^a$ carries three group indices in the structure constant but is a Lorentz scalar after contraction.)

Significance: Because F^a contains a piece $\sim AA$, squaring it in $\mathcal{L}_{\text{gauge}}$ produces terms with *three and four gauge fields* — the gauge bosons interact *with each other*. This never happens for the photon.

Consequence 13.2 Gauge self-interactions

A non-Abelian gauge field couples to itself. Expanding $-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}$ gives, beyond the free part, a **three-gauge-boson** vertex and a **four-gauge-boson** vertex:

$$\mathcal{L}_3 \sim -g f^{abc} (\partial_\mu A_\nu^a) A^{b\mu} A^{c\nu}, \quad (319)$$

$$\mathcal{L}_4 \sim -\frac{1}{4} g^2 f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (320)$$

These are absent in the Abelian (photon) case. For $SU(3)_c$ they are the gluon self-interactions that make the strong force “strong.”



Figure 3: The self-interactions of a non-Abelian gauge field. *Left*: the three-gauge-boson vertex ($\propto g f^{abc}$, Eq. (319)). *Right*: the four-gauge-boson vertex ($\propto g^2 f^{abc} f^{ade}$, Eq. (320)). Neither exists in $U(1)$ (the photon does not couple to itself).

Gauge bosons are massless**Physical Law 13.3** Local non-Abelian invariance $\Rightarrow$ massless gauge bosons

A naive mass term $m^2 A_\mu^a A^{a\mu}$ is *not* invariant under the gauge transformation (316). Therefore local non-Abelian gauge invariance **forbids** a gauge-boson mass:

$$m_A = 0.$$

The gauge bosons of an unbroken non-Abelian symmetry are massless.

Proof: The field A_μ^a is dynamical, so its excitations are physical particles. To give them mass we would add $\frac{1}{2}m^2 A_\mu^a A^{a\mu}$. Under (316) the shift $-\frac{1}{g}\partial_\mu\theta^a$ (and the structure-constant rotation) change this term, so it is not invariant. Hence it cannot appear in a gauge-invariant Lagrangian, and $m_A = 0$. (Exactly as for the photon in the Abelian case — but here it applies to an entire adjoint multiplet.) ■

Significance: The masslessness of the gluon is *not* a choice — it is forced by gauge invariance, just as $m_\gamma = 0$ was. If $SU(3)_c$ is a good (unbroken) symmetry of nature, the gluon *must* be massless.

Several gauge factors**Consequence 13.4** One gauge field and coupling per simple factor

If the gauge group has several commuting factors, each factor gets its *own* gauge field in *its own* adjoint representation, and its *own* independent coupling constant.

Worked Example 13.4 (Gauging $SU(3) \times SU(2) \times U(1)$)

This is the Standard-Model gauge group. We must introduce three gauge fields:

Gauge field	$SU(3)$	$SU(2)$	$U(1)$	Coupling	SM identity
G_μ^a (gluon)	8	1	0	g_3	$SU(3)_c$ gluons
W_μ^a	1	3	0	g_2	$W^\pm, W^0$
B_μ	1	1	0	g_1	hypercharge $U(1)_Y$

Each gauge field is the adjoint of its own factor, a singlet (uncharged) under the others. The $SU(3)$ gluon is the **8** of colour; the $SU(2)$ gauge field is its **3** (the W bosons we will meet in about a week); the $U(1)$ gauge field is an uncharged singlet, exactly like a photon.

To Remember

- A non-Abelian field is a **multiplet** ϕ_i transforming as $\phi \rightarrow e^{iT^a\theta^a}\phi$; the Hermitian generators obey $[T^a, T^b] = if^{abc}T^c$ with structure constants f^{abc} .
- **Fundamental** rep = smallest non-trivial irrep ($\dim N$): $SU(2)$ doublet ($T^a = \tau^a/2$), $SU(3)$ triplet ($T^a = \lambda^a/2$). **Adjoint** rep is built from f^{abc} , $\dim N^2 - 1$ — home of the gauge bosons.
- **Product groups**: one index per simple factor; a transformation in one factor leaves the others' indices alone.
- **Singlet rule**: a term is allowed iff its representations can be contracted into a singlet under every factor (the non-Abelian version of “charges sum to zero”). We track *which* combinations contain a singlet, not the explicit contractions.
- $SU(3)_c$ colour: $q\bar{q} \supset \mathbf{1}$ (mesons), $qqq \supset \mathbf{1}$ (baryons); qq and $q\bar{q}$ contain no singlet. Glueballs (gg) and hybrids ($q\bar{q}g$) are allowed but not yet observed.
- **Gauging**: $\partial_\mu \rightarrow D_\mu = \partial_\mu + igT^a A_\mu^a$; gauge field A_μ^a sits in the adjoint; field strength $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf^{abc}A_\mu^b A_\nu^c$.
- The extra f^{abc} term gives **gauge self-interactions** (3- and 4-boson vertices) — absent for the photon. Gauge invariance forbids a mass term: non-Abelian gauge bosons are **massless**. Each commuting factor has its own adjoint gauge field and independent coupling.

Recommended Reading

- **Course notes, Chapter 4 & Appendix A**: non-Abelian groups, generators, representations, structure constants.
- **Tong, *Lectures on Particle Physics*, Ch. 4**: non-Abelian gauge theory, covariant derivative, field strength, self-interactions.
- **Peskin & Schroeder, Ch. 15**: non-Abelian gauge theory in full detail (generators, $F_{\mu\nu}^a$, the gauge Lagrangian).

- **Griffiths, *Introduction to Elementary Particles*, Ch. 8–9:** colour, QCD, the quark model, mesons and baryons as colour singlets.

14 Lecture 14 – QCD Interactions, Running Couplings, Confinement, and Accidental Symmetries

Date: 31/05/2026

Overview

Lecture 13 built the machinery of non-Abelian gauge theory and used $SU(3)_c$ as the central example. This lecture applies that machinery to Quantum Chromodynamics (QCD), the theory of strong interactions. We start from the QCD Lagrangian, expand it to identify the quark-gluon vertex and the gluon self-interactions, count the free parameters of the theory, and then connect the sign of the QCD beta function to confinement. The final part of the lecture asks a standard model-building question: after imposing the gauge symmetry, what extra global symmetries accidentally appear in the Lagrangian, and what do they imply?

1. **Running couplings:** why a gauge coupling is a scale-dependent parameter.
2. **QCD interactions:** quark-gluon interaction, three-gluon vertex, and four-gluon vertex, all controlled by the same g_s .
3. **Seven QCD parameters:** six quark masses and the strong coupling $\alpha_s = g_s^2/(4\pi)$.
4. **Confinement:** color triplets are not observed as free asymptotic states; observed states are color singlets.
5. **Accidental symmetries:** $U(6)_L \times U(6)_R$ for the kinetic terms, broken by quark masses to $U(1)^6$, and then by weak interactions to baryon number.

This lecture follows **Chapter 5** of the Grossman–Nir–Buchalla notes.

14.1 Running Coupling Constants

Conceptual Background

In a classical Lagrangian the coupling constant looks like a fixed number. In quantum field theory this number is measured at a scale: the loop corrections depend on the energy or momentum scale of the process, and the coupling one extracts from an experiment therefore depends on that scale. The renormalization group packages this dependence into a beta function.

Physical Concept 14.1: Beta Function

For a dimensionless coupling $g(\mu)$ measured at renormalization scale μ , the beta function is

$$\beta(g) \equiv \frac{\partial g}{\partial \ln \mu} \quad (321)$$

where $\ln \mu$ is the natural logarithm of the energy scale.

Significance: The sign of $\beta(g)$ tells us whether the interaction becomes stronger or weaker as we move to higher energies.

The one-loop beta function only depends on fields that are charged under the gauge symmetry. A charged field can appear in loops and therefore screen or anti-screen the gauge interaction.

Abelian comparison. For a gauged $U(1)$ theory with n_f Dirac fermions of charge magnitude $|q| = 1$, the leading beta function is

$$\beta(g_1) = \frac{n_f g_1^3}{24\pi^2} \quad (322)$$

which is positive.

Significance: In a simple Abelian theory the coupling grows toward higher energies and becomes weaker in the infrared.

Non-Abelian comparison. For an $SU(N)$ gauge theory with n_f fermion fields, the leading beta function quoted in the lecture is

$$\beta(g_N) = \left(\frac{2n_f - 11N}{3} \right) \frac{g_N^3}{16\pi^2} \quad (323)$$

where the $-11N$ term comes from the non-Abelian gauge-boson self-interactions.

Significance: For sufficiently few fermions, the non-Abelian coupling becomes weaker at high energy and stronger at low energy; this is the seed of asymptotic freedom and confinement.

Note:

Perturbation theory is an expansion in powers of the relevant coupling. When the coupling is small, a few diagrams give accurate predictions. When the coupling becomes large, the expansion loses predictive power even though the underlying theory can still be perfectly well defined.

14.2 Defining QCD

Physical Motivation

QCD is the local $SU(3)_c$ gauge theory of the strong force. The subscript c stands for color. The quarks are triplets of $SU(3)_c$, and the gluons are the gauge bosons required by local $SU(3)_c$ invariance.

Physical Concept 14.2: QCD Field Content

The defining ingredients of QCD are

$$SU(3)_c \text{ local gauge symmetry, } Q_{Li}(\mathbf{3}), Q_{Ri}(\mathbf{3}), \quad i = 1, \dots, 6, \quad \text{no scalars.} \quad (324)$$

The six flavors are later called u, d, s, c, b, t after diagonalizing the mass matrix.

Significance: QCD is vector-like because both left-handed and right-handed quarks transform in the same color representation.

Since the symmetry is local, we must introduce a gauge field in the adjoint representation of $SU(3)_c$:

$$G_\mu^a, \quad a = 1, \dots, 8. \quad (325)$$

The eight fields G_μ^a are the gluons. The number eight is not arbitrary: it is the dimension of the adjoint representation of $SU(3)$, namely $3^2 - 1 = 8$.

14.3 The QCD Lagrangian

Mathematical Setup

The non-Abelian field strength for the gluon is the $SU(3)$ specialization of the field strength derived in Lecture 13:

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a - g_s f^{abc} G_\mu^b G_\nu^c \quad (326)$$

where g_s is the strong coupling constant and f^{abc} are the $SU(3)$ structure constants, defined by

$$[L^a, L^b] = i f^{abc} L^c. \quad (327)$$

In the triplet representation,

$$L^a = \frac{\lambda^a}{2}, \quad (328)$$

where the λ^a are the eight Gell-Mann matrices.

The covariant derivative acting on a color-triplet quark is

$$D_\mu = \partial_\mu + i g_s G_\mu^a L^a = \partial_\mu + \frac{i}{2} g_s G_\mu^a \lambda^a \quad (329)$$

with the generator matrix acting on the color index of the quark.

Significance: The same g_s appears in both the quark covariant derivative and the gluon field strength; gauge invariance fixes the relative strength of all strong interactions.

Before choosing the mass basis, the kinetic and mass terms are

$$\mathcal{L}_{\text{kin}} = i \bar{Q}_{Li} \not{D} Q_{Li} + i \bar{Q}_{Ri} \not{D} Q_{Ri} - \frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu}, \quad (330)$$

$$-\mathcal{L}_\psi = m_{ij}^Q \bar{Q}_{Li} Q_{Rj} + \text{h.c.} \quad (331)$$

where $i, j = 1, \dots, 6$ label flavor.

The quark mass matrix can always be diagonalized by a bi-unitary transformation,

$$Q_{Li} \rightarrow (V_L)_{ij} Q_{Lj}, \quad Q_{Ri} \rightarrow (V_R)_{ij} Q_{Rj}, \quad (332)$$

such that

$$V_L m^Q V_R^\dagger = \text{diag}(m_u, m_d, m_s, m_c, m_b, m_t) \quad (333)$$

with the convention $m_u < m_d < m_s < m_c < m_b < m_t$.

Significance: Because the strong interaction is flavor-universal, this rotation changes the mass terms but does not introduce flavor-changing gluon couplings.

In the mass basis, we combine Q_{Li} and Q_{Ri} into Dirac quarks $q_i = (Q_{Li}, Q_{Ri})^T$, and the full QCD Lagrangian is

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} + \sum_{q_i=u,d,s,c,b,t} \bar{q}_i (i \not{D} - m_{q_i}) q_i \quad (334)$$

where $\not{D} = \gamma^\mu D_\mu$.

Significance: Once written in the mass basis, QCD is a theory of six massive Dirac quarks interacting with eight massless gluons.

14.4 Interactions in QCD

Mathematical Setup

The Lagrangian in Eq. (334) looks compact, but it contains several interaction vertices. They come from two places: the quark covariant derivative and the non-Abelian part of the gluon field strength.

Quark-gluon interaction. Expanding the quark kinetic term,

$$\begin{aligned}\bar{q}_i i\gamma^\mu D_\mu q_i &= \bar{q}_i i\gamma^\mu \left(\partial_\mu + \frac{i}{2} g_s G_\mu^a \lambda^a \right) q_i \\ &= i\bar{q}_i \not{\partial} q_i - \frac{g_s}{2} \bar{q}_i \gamma^\mu \lambda^a G_\mu^a q_i.\end{aligned}\quad (335)$$

Therefore the interaction term is

$$\mathcal{L}_{q\bar{q}g} = -\frac{g_s}{2} \sum_{i=u,\dots,t} \bar{q}_i \lambda^a \gamma^\mu G_\mu^a q_i \quad (336)$$

Significance: This is the strong-interaction analogue of the QED vertex, but the photon charge Q is replaced by the color generator $\lambda^a/2$.

Claim 14.1 Quark-Gluon Coupling Structure

The quark-gluon interaction in Eq. (336) is

- **diagonal in flavor:** the gluon couples $\bar{q}_i q_i$, not $\bar{q}_i q_j$ for $i \neq j$;
- **universal in flavor:** the same g_s multiplies every quark flavor;
- **vector-like:** q_L and q_R couple with the same strength because both are color triplets.

Gluon self-interactions. The gluon kinetic term contains the field strength of Eq. (326). To see the interaction structure, split

$$G_{\mu\nu}^a = (\partial_\mu G_\nu^a - \partial_\nu G_\mu^a) - g_s f^{abc} G_\mu^b G_\nu^c. \quad (337)$$

Squaring this object gives a free kinetic piece, a cross term with three gluon fields, and a term with four gluon fields:

$$\mathcal{L}_{ggg} = g_s f^{abc} (\partial^\mu G_\mu^a) G_{b\mu} G_{c\mu} \quad (338)$$

$$\mathcal{L}_{gggg} = -g_s^2 (f^{abc} G_b^\mu G_c^\nu) (f^{ade} G_{d\mu} G_{e\nu}) \quad (339)$$

up to the standard convention-dependent arrangement of identical-field symmetry factors.

Significance: Non-Abelian gauge bosons interact with themselves; this has no analogue in ordinary QED and is central to why QCD behaves so differently from electromagnetism.

Note:

Gauge invariance does more than allow these vertices: it fixes them together. There is no independent “three-gluon coupling” or “quark-gluon coupling” to tune separately. They are all consequences of the single local $SU(3)_c$ symmetry.

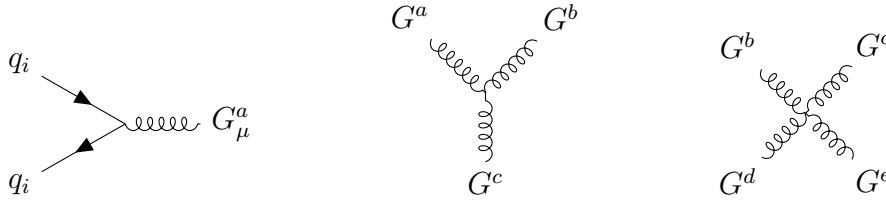


Figure 4: The three basic strong-interaction vertices in QCD: quark-gluon, three-gluon, and four-gluon. Their strengths are controlled by the same coupling g_s ; the four-gluon vertex is proportional to g_s^2 .

14.5 The Parameters of QCD

Physical Motivation

After writing the most general renormalizable Lagrangian allowed by the imposed symmetry, the remaining numbers are parameters that must be measured. For QCD, the answer is unusually economical.

Physical Law 14.1 Seven Parameters of QCD

The QCD Lagrangian has seven free parameters:

$$m_u, m_d, m_s, m_c, m_b, m_t, \quad g_s \text{ or equivalently } \alpha_s \equiv \frac{g_s^2}{4\pi} \quad (340)$$

Significance: Once these seven quantities are measured at a chosen scale and in a chosen scheme, all high-energy QCD predictions are fixed.

Representative quark masses quoted in the notes are

$$\begin{aligned} m_u &= 2.2_{-0.3}^{+0.5} \text{ MeV}, & m_d &= 4.7_{-0.2}^{+0.5} \text{ MeV}, & m_s &= 93_{-5}^{+11} \text{ MeV}, \\ m_c &= 1.27 \pm 0.02 \text{ GeV}, & m_b &= 4.18_{-0.02}^{+0.03} \text{ GeV}, & m_t &= 172.9 \pm 0.4 \text{ GeV}. \end{aligned} \quad (341)$$

They span roughly five orders of magnitude.

Open Problem 14.1: The Flavor Hierarchy Problem

Why are the quark masses so hierarchical? The QCD Lagrangian accepts these masses as input parameters; it does not explain why m_t is enormous compared with m_u . Understanding such patterns belongs to flavor physics.

The strong coupling is commonly quoted as

$$\alpha_s(m_Z^2) = 0.1181 \pm 0.0011, \quad m_Z \simeq 91 \text{ GeV} \quad (342)$$

where m_Z is the Z -boson mass scale.

Significance: Quoting α_s without a scale is incomplete: the measured value depends on the energy scale at which the coupling is defined.

14.6 Confinement and the Running of g_s

Physical Motivation

Experimentally, we do not observe free quarks. We observe hadrons: color-singlet bound states such as mesons and baryons. The confinement hypothesis is the statement that this is not an accident of detector technology but a property of the theory.

Physical Concept 14.3: Confinement Hypothesis

All observable asymptotic states in QCD are singlets of $SU(3)_c$. Equivalently,

$$\begin{array}{l} \text{free color triplets are not asymptotic states;} \\ \text{quarks are confined inside color-singlet bound states.} \end{array} \quad (343)$$

Significance: The physical particle spectrum at low energy is not the same as the Lagrangian field content: quark and gluon fields are useful UV degrees of freedom, while hadrons are the IR degrees of freedom.

The confinement picture is consistent with the sign of the QCD beta function. Above the top mass threshold, all six quarks are relevant and Eq. (323) with $N = 3$ and $n_f = 6$ gives

$$\beta(g_s)(\mu > m_t) = -\frac{7g_s^3}{16\pi^2} < 0 \quad (344)$$

Significance: A negative beta function means that g_s decreases toward high energies and increases toward low energies.

At lower scales, not all quark flavors are active. Let n_q^{light} be the number of quarks with mass below the scale μ . Then

$$\beta(g_s) = -\frac{(33 - 2n_q^{\text{light}})g_s^3}{48\pi^2} \quad (345)$$

which becomes even more negative as heavy quarks drop out.

From Negative Beta Function to Confinement

The beta function is

$$\frac{\partial g_s}{\partial \ln \mu} < 0. \quad (346)$$

Now lower the scale, so $\ln \mu$ decreases. Since the derivative is negative, the change in g_s is positive:

$$\Delta g_s \simeq \frac{\partial g_s}{\partial \ln \mu} \Delta \ln \mu > 0 \quad (\Delta \ln \mu < 0). \quad (347)$$

Therefore g_s grows in the infrared. At long distances, pulling a quark and antiquark apart does not lead to two isolated quarks. Instead, once enough energy is stored in the color field, it is energetically favorable to create a new $q\bar{q}$ pair from the vacuum, producing color-singlet hadrons.

Physical Concept 14.4: Λ_{QCD}

The scale where QCD crosses from the perturbative UV regime into the strongly coupled IR regime is called Λ_{QCD} :

$$\Lambda_{\text{QCD}} \sim 300 \text{ MeV}, \quad \text{roughly where } g_s \text{ becomes strong.} \quad (348)$$

Perturbative QCD is reliable for processes with

$$q^2 \gg \Lambda_{\text{QCD}}^2 \quad (349)$$

where q^2 is the characteristic momentum-transfer scale of the process.

Significance: Λ_{QCD} is not an extra fundamental parameter; it is an emergent scale determined by the running of the coupling and the quark thresholds.

Worked Example 14.1 (Color-Singlet Hadrons)

The confinement hypothesis says that observed hadrons must be color singlets. The most familiar possibilities are

- **mesons:** $q\bar{q}$, since $\mathbf{3} \otimes \bar{\mathbf{3}} \supset \mathbf{1}$;
- **baryons:** qqq , since $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \supset \mathbf{1}$.

Examples include pions, kaons, the proton, and the neutron. These are not fundamental fields in Eq. (334); they are bound states produced by strong dynamics.

14.7 Accidental Symmetries of QCD

Conceptual Background

Once we impose a gauge symmetry and write the most general renormalizable Lagrangian, the Lagrangian may accidentally have additional global symmetries that we did not impose. These accidental symmetries are powerful because they imply selection rules and approximate conservation laws.

The kinetic-term symmetry. If we temporarily ignore the quark mass terms, the six left-handed quarks can be rotated among themselves by a unitary 6×6 matrix, and the six right-handed quarks can be rotated by an independent unitary matrix:

$$\boxed{\mathcal{L}_{\text{kin}} \text{ has accidental global symmetry } U(6)_L \times U(6)_R} \quad (350)$$

because the kinetic term is flavor-universal and does not distinguish u, d, s, c, b, t .

Significance: Large accidental symmetries often appear before mass terms are included; mass terms usually break them.

Including the mass terms. In the mass basis, the quark mass terms are

$$-\mathcal{L}_\psi = m_u \bar{u}u + m_d \bar{d}d + m_s \bar{s}s + m_c \bar{c}c + m_b \bar{b}b + m_t \bar{t}t. \quad (351)$$

Because the six masses are different, arbitrary rotations among the six flavors no longer preserve the Lagrangian. What remains is one independent phase rotation for each Dirac quark:

$$\boxed{U(6)_L \times U(6)_R \xrightarrow{\text{quark masses}} U(1)_u \times U(1)_d \times U(1)_s \times U(1)_c \times U(1)_b \times U(1)_t} \quad (352)$$

or simply $U(1)^6$.

Significance: Pure QCD separately conserves each quark flavor number.

Claim 14.2 Flavor Conservation in Pure QCD

The $U(1)^6$ accidental symmetry implies that each quark flavor number is conserved by pure QCD interactions. For example,

$$u\bar{u} \rightarrow c\bar{c} \quad (353)$$

is allowed because both sides carry zero net $U(1)_u$ and zero net $U(1)_c$ charge, while

$$u\bar{c} \rightarrow c\bar{u} \quad (354)$$

is forbidden in pure QCD because it changes the individual $U(1)_u$ and $U(1)_c$ charges.

Significance: The symmetry does not say quark-antiquark pair production is impossible; it says the net flavor charges must be conserved.

Why the pure-QCD prediction is not exact in nature. In nature, QCD is not the full theory. The weak interaction breaks the $U(1)^6$ symmetry down to a single common quark-number symmetry: baryon number.

Physical Concept 14.5: Baryon Number

Baryon number is the diagonal quark phase symmetry

$$q \rightarrow e^{i\theta} q \quad \text{for every quark flavor } q = u, d, s, c, b, t \quad (355)$$

where all quarks rotate by the same phase. Equivalently, the weak interaction breaks

$$U(1)^6 \longrightarrow U(1)_B \quad (356)$$

Significance: Flavor-changing processes forbidden by pure QCD can occur through weak interactions, but baryon number remains the relevant common conserved quark number in the renormalizable Standard Model.

Note:

At the energy scales of present experiments, weak interactions are much weaker than strong interactions. Therefore the accidental symmetries of the pure QCD Lagrangian are often still useful approximate guides, even though they are not exact symmetries of the full Standard Model.

Even lower-energy approximate symmetries. In the confined regime, the useful degrees of freedom are hadrons rather than free quarks and gluons. There, additional approximate symmetries become useful:

- **isospin** for the nearly degenerate u and d sector;
- $SU(3)$ **flavor** when treating u, d, s as approximately light;
- **heavy-quark symmetry** for sufficiently heavy quarks.

These are not needed for the UV definition of QCD, but they become central tools for low-energy hadron physics.

To Remember

- Couplings run: $\beta(g) = \partial g / \partial \ln \mu$. For $U(1)$ with charged fermions the one-loop beta function is positive; for $SU(N)$ with sufficiently few fermions it is negative.
- QCD is local $SU(3)_c$ with six left-handed and six right-handed color-triplet quarks and no scalars.

- The gluon field G_μ^a lives in the adjoint of $SU(3)_c$, so $a = 1, \dots, 8$.
- The full mass-basis Lagrangian is $\mathcal{L}_{\text{QCD}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \sum_q \bar{q}(i\not{D} - m_q)q$.
- Expanding D_μ gives the quark-gluon vertex; expanding $G_{\mu\nu}^a G^{a\mu\nu}$ gives three-gluon and four-gluon self-interactions.
- All strong-interaction vertices are controlled by the same coupling g_s ; the quark-gluon coupling is diagonal, universal, and vector-like.
- QCD has seven free parameters: six quark masses and g_s (or α_s).
- The negative QCD beta function makes g_s grow in the infrared, consistent with confinement. Perturbative QCD works when $q^2 \gg \Lambda_{\text{QCD}}^2$, with $\Lambda_{\text{QCD}} \sim 300$ MeV.
- The kinetic terms have accidental $U(6)_L \times U(6)_R$; the mass terms reduce this to $U(1)^6$; weak interactions reduce it further to baryon number $U(1)_B$.

15 Lecture 15 – QCD + QED and Spontaneous Symmetry Breaking

Date: 04/06/2026

15.1 Overview

Lecture 14 completed the construction of pure QCD: local $SU(3)_c$, six Dirac quarks in the fundamental representation, eight gluons in the adjoint representation, and the characteristic non-Abelian gluon self-interactions. This lecture first adds electromagnetism and studies the theory

$$SU(3)_c \times U(1)_{\text{EM}},$$

which is a useful intermediate step toward the full Standard Model when weak interactions are neglected. The second and longer part of the lecture starts the next conceptual ingredient: spontaneous symmetry breaking. We begin with a global discrete $\mathbb{Z}_2$ symmetry, then move to a global continuous $U(1)$ symmetry, where the appearance of a massless Goldstone boson becomes visible.

1. **QCD+QED:** field content, charges, covariant derivatives, kinetic terms, interactions, and accidental global symmetries.
2. **Spontaneous symmetry breaking:** the Lagrangian is symmetric, but a chosen vacuum need not be.
3. **Discrete example:** a real scalar with $\mathbb{Z}_2 : \phi \rightarrow -\phi$ and two degenerate vacua.
4. **Continuous example:** a complex scalar with global $U(1)$ symmetry, a Mexican-hat potential, one massive radial mode, and one massless angular mode.
5. **Goldstone theorem:** every broken continuous global generator produces a massless boson.

15.2 Combining QCD with Electromagnetism

Physical Motivation

Pure QCD knows about color but not electric charge. Pure QED knows about electric charge but not color. Nature contains quarks carrying both quantum numbers: an up quark is a color triplet and has charge $+2/3$, while a down quark is a color triplet and has charge $-1/3$. Neglecting the weak interactions, the natural gauge group is therefore the direct product

$$G_{\text{QCD+QED}} = SU(3)_c \times U(1)_{\text{EM}} \quad (357)$$

Significance: This product theory is the part of the Standard Model that already describes colored quarks, gluons, photons, and charged leptons before electroweak interactions are introduced.

Physical Concept 15.1: Quark Field Content under $SU(3)_c \times U(1)_{\text{EM}}$

For generation index $i = 1, 2, 3$, the chiral quark fields transform as

$$u_{Li} \sim (\mathbf{3}, +\frac{2}{3}), \quad u_{Ri} \sim (\mathbf{3}, +\frac{2}{3}), \quad d_{Li} \sim (\mathbf{3}, -\frac{1}{3}), \quad d_{Ri} \sim (\mathbf{3}, -\frac{1}{3}). \quad (358)$$

The first entry is the $SU(3)_c$ representation and the second entry is the electromagnetic charge Q in units of the proton charge $e > 0$.

Significance: Electromagnetism splits the six quarks into two charge sectors, so up-type and down-type quarks can no longer be freely mixed by a single $U(6)$ flavor rotation.

Local gauge invariance forces us to include one gauge field for each generator of the gauge group. The $SU(3)_c$ factor has $3^2 - 1 = 8$ generators, so it gives eight gluons G_μ^a , $a = 1, \dots, 8$. The Abelian $U(1)_{\text{EM}}$ factor has one generator, so it gives one photon A_μ :

$$G_\mu^a \sim (\mathbf{8}, 0), \quad A_\mu \sim (\mathbf{1}, 0). \quad (359)$$

The photon is neutral under electromagnetism because an Abelian gauge boson carries no charge under its own group; the gluons are neutral under $U(1)_{\text{EM}}$ because color and electric charge are independent gauge factors.

Mathematical Setup

Let λ^a be the Gell-Mann matrices, so the fundamental $SU(3)$ generators are $T^a = \lambda^a/2$. Let g_s denote the strong coupling, e the electromagnetic coupling, and Q_f the electromagnetic charge of a field f . A color-triplet quark sees both gauge factors, so its covariant derivative is

$$D_\mu f = \left(\partial_\mu + \frac{i}{2} g_s G_\mu^a \lambda^a + i e Q_f A_\mu \right) f. \quad (360)$$

Significance: In a product gauge theory the covariant derivative is the sum of the gauge connections for each factor, with each term acting only through the representation and charge carried by the field.

For the four quark multiplets in Eq. (358), the charge is

$$Q_{u_L} = Q_{u_R} = +\frac{2}{3}, \quad Q_{d_L} = Q_{d_R} = -\frac{1}{3}. \quad (361)$$

If a field were in a different $SU(3)$ representation, the matrices $\lambda^a/2$ in Eq. (360) would be replaced by the generators of that representation.

The gauge-field strengths are

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a - g_s f^{abc} G_\mu^b G_\nu^c, \quad (362)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (363)$$

where f^{abc} are the $SU(3)$ structure constants defined by $[T^a, T^b] = i f^{abc} T^c$.

The kinetic Lagrangian is

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \sum_{i=1}^3 \left[i \bar{u}_{Li} \not{D} u_{Li} + i \bar{u}_{Ri} \not{D} u_{Ri} + i \bar{d}_{Li} \not{D} d_{Li} + i \bar{d}_{Ri} \not{D} d_{Ri} \right]. \quad (364)$$

Significance: All quark-photon and quark-gluon interactions are already hidden inside this compact kinetic term.

Interaction Vertices

Expanding the fermion term for a quark field f gives

$$\begin{aligned} i\bar{f}\gamma^\mu D_\mu f &= i\bar{f}\gamma^\mu \partial_\mu f + i\bar{f}\gamma^\mu \left(\frac{i}{2}g_s G_\mu^a \lambda^a + ieQ_f A_\mu \right) f \\ &= i\bar{f}\not{\partial}f - \frac{g_s}{2}\bar{f}\gamma^\mu \lambda^a G_\mu^a f - eQ_f \bar{f}\gamma^\mu A_\mu f. \end{aligned} \quad (365)$$

Thus the interaction terms are

$$\mathcal{L}_{q\bar{q}g} = -\frac{g_s}{2} \sum_f \bar{f} \lambda^a \gamma^\mu G_\mu^a f, \quad (366)$$

$$\mathcal{L}_{q\bar{q}\gamma} = -e \sum_f Q_f \bar{f} \gamma^\mu A_\mu f, \quad (367)$$

where f runs over $u_{Li}, u_{Ri}, d_{Li}, d_{Ri}$, or equivalently over the Dirac quarks after forming the mass eigenstates.

Significance: The gluon interaction is universal in flavor but matrix-valued in color, while the photon interaction is color-blind but weighted by the electric charge.

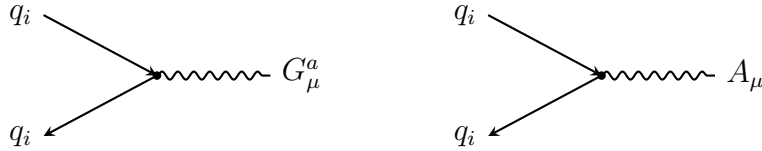


Figure 5: The two gauge interactions of a quark in $SU(3)_c \times U(1)_{\text{EM}}$. The gluon vertex carries the color generator $\lambda^a/2$, while the photon vertex carries the electric charge Q_f .

In the mass basis the quark mass terms are

$$\mathcal{L}_{\text{mass}}^q = -m_u \bar{u}u - m_c \bar{c}c - m_t \bar{t}t - m_d \bar{d}d - m_s \bar{s}s - m_b \bar{b}b. \quad (368)$$

Significance: The mass eigenstates preserve electric charge: u, c, t have $Q = +2/3$, while d, s, b have $Q = -1/3$.

One may also add the three charged leptons $\ell_i = e, \mu, \tau$, which transform as

$$\ell_{Li} \sim (\mathbf{1}, -1), \quad \ell_{Ri} \sim (\mathbf{1}, -1). \quad (369)$$

Since they are $SU(3)_c$ singlets, their covariant derivative contains only the electromagnetic term:

$$D_\mu \ell_i = (\partial_\mu - ieA_\mu) \ell_i. \quad (370)$$

Accidental Global Symmetries

The gauge symmetry is imposed. The global symmetry is what remains accidentally after the allowed kinetic and mass terms are written. In pure QCD the kinetic terms treat all six quark flavors alike, giving $U(6)_L \times U(6)_R$. Once electromagnetism is included, the up-type and down-type sectors have different $U(1)_{\text{EM}}$ charges, so they cannot be rotated into one another without changing the photon coupling. The kinetic terms therefore have

$$G_{\text{global,kin}} = U(3)_{u_L} \times U(3)_{u_R} \times U(3)_{d_L} \times U(3)_{d_R}. \quad (371)$$

Significance: Electromagnetism reduces the accidental flavor symmetry because electric charge distinguishes the two quark sectors.

After diagonal mass terms are included, the independent non-Abelian rotations are broken. If all six masses are nondegenerate, the surviving quark flavor symmetries are the six separate phase rotations

$$G_{\text{global,full}} = U(1)_u \times U(1)_c \times U(1)_t \times U(1)_d \times U(1)_s \times U(1)_b \simeq U(1)^6. \quad (372)$$

Significance: In the absence of weak interactions, each quark flavor number is separately conserved.

15.3 What Spontaneous Symmetry Breaking Means

Conceptual Background

A symmetry of the Lagrangian is a symmetry of the equations of motion and of the full theory. A vacuum is a particular lowest-energy configuration around which perturbation theory is performed. Spontaneous symmetry breaking occurs when the Lagrangian is invariant under a symmetry, but the chosen vacuum is not invariant under that symmetry.

Physical Concept 15.2: Spontaneous Symmetry Breaking

A symmetry transformation g is spontaneously broken if

$$\mathcal{L}[g\phi] = \mathcal{L}[\phi], \quad g\langle\phi\rangle \neq \langle\phi\rangle. \quad (373)$$

Here $\langle\phi\rangle$ denotes the vacuum expectation value, namely the field value in the chosen vacuum.

The reason this matters phenomenologically is that perturbative particles are excitations around the vacuum. If the vacuum is not symmetric, the Lagrangian written in terms of the excitation fields need not display the original symmetry term by term, even though the underlying theory still knows about it through relations among couplings.

Note:

Spontaneous breaking is not the same as explicitly removing the symmetry. If the symmetry is explicit, the most general Lagrangian has more independent parameters. If the symmetry is spontaneous, the broken-phase Lagrangian can contain terms that look asymmetric, but their coefficients remain related by the symmetric parent theory.

15.4 Discrete Example: A Real Scalar with $\mathbb{Z}_2$

Physical Motivation

The simplest example uses one real scalar field $\phi(x)$ and the discrete symmetry

$$\mathbb{Z}_2 : \quad \phi \rightarrow -\phi. \quad (374)$$

Because odd powers change sign, the symmetry forbids terms such as ϕ and ϕ^3 .

Mathematical Setup

The most general renormalizable Lagrangian with this symmetry is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} \mu^2 \phi^2 - \frac{\lambda}{4} \phi^4, \quad \lambda > 0. \quad (375)$$

Significance: The sign of μ^2 decides whether the symmetry is visible in the vacuum or hidden by the vacuum choice.

The potential energy is

$$V(\phi) = \frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}\phi^4. \quad (376)$$

The extrema solve

$$0 = \frac{dV}{d\phi} = \mu^2\phi + \lambda\phi^3 = \phi(\mu^2 + \lambda\phi^2). \quad (377)$$

There are two cases.

Unbroken case: $\mu^2 > 0$. The only real stationary point is $\phi = 0$. The second derivative is

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi=0} = \mu^2 > 0, \quad (378)$$

so $\phi = 0$ is the vacuum. Since $\phi = 0$ is unchanged by $\phi \rightarrow -\phi$, the vacuum respects $\mathbb{Z}_2$.

Broken case: $\mu^2 < 0$. The nonzero stationary points satisfy

$$\phi^2 = -\frac{\mu^2}{\lambda} \equiv v^2, \quad v = \sqrt{-\frac{\mu^2}{\lambda}}. \quad (379)$$

At these points

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi=\pm v} = \mu^2 + 3\lambda v^2 = \mu^2 + 3\lambda \left(-\frac{\mu^2}{\lambda} \right) = -2\mu^2 = 2\lambda v^2 > 0, \quad (380)$$

so the two classical minima are

$$\boxed{\langle \phi \rangle = +v \quad \text{or} \quad \langle \phi \rangle = -v.} \quad (381)$$

Significance: The two vacua are physically equivalent, but either single choice is not invariant under $\phi \rightarrow -\phi$.

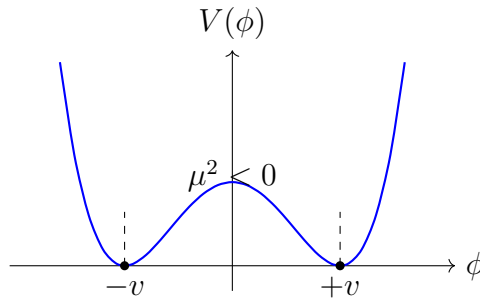


Figure 6: The $\mathbb{Z}_2$ -symmetric potential for $\mu^2 < 0$. The Lagrangian is symmetric under reflection $\phi \rightarrow -\phi$, but choosing either minimum selects one side.

Expansion around the Vacuum

Choose, without loss of generality, the vacuum $+v$. Define the excitation field

$$\phi(x) = v + \phi'(x), \quad \langle \phi' \rangle = 0. \quad (382)$$

The kinetic term is unchanged because v is constant:

$$\partial_\mu \phi = \partial_\mu \phi'. \quad (383)$$

The potential becomes

$$\begin{aligned} V(v + \phi') &= \frac{1}{2}\mu^2(v + \phi')^2 + \frac{\lambda}{4}(v + \phi')^4 \\ &= \frac{1}{2}\mu^2(v^2 + 2v\phi' + \phi'^2) + \frac{\lambda}{4}(v^4 + 4v^3\phi' + 6v^2\phi'^2 + 4v\phi'^3 + \phi'^4) \\ &= \left(\frac{1}{2}\mu^2v^2 + \frac{\lambda}{4}v^4\right) + (\mu^2v + \lambda v^3)\phi' + \left(\frac{1}{2}\mu^2 + \frac{3}{2}\lambda v^2\right)\phi'^2 + \lambda v\phi'^3 + \frac{\lambda}{4}\phi'^4. \end{aligned} \quad (384)$$

Using the minimum condition $\mu^2 + \lambda v^2 = 0$, the linear term vanishes and

$$\frac{1}{2}\mu^2 + \frac{3}{2}\lambda v^2 = -\frac{1}{2}\lambda v^2 + \frac{3}{2}\lambda v^2 = \lambda v^2. \quad (385)$$

Dropping the irrelevant constant vacuum energy, the Lagrangian in terms of the physical excitation is

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi' \partial^\mu \phi' - \frac{1}{2}(2\lambda v^2)\phi'^2 - \lambda v\phi'^3 - \frac{\lambda}{4}\phi'^4. \quad (386)$$

Significance: The broken-phase Lagrangian contains a cubic interaction, but its coefficient is fixed by the same two parameters that controlled the original symmetric theory.

Comparing the quadratic term to $-\frac{1}{2}m_{\phi'}^2\phi'^2$ gives

$$m_{\phi'}^2 = 2\lambda v^2 = -2\mu^2 > 0. \quad (387)$$

Significance: A negative μ^2 in the symmetric-field coordinates is not a physical negative mass squared after expanding around the true vacuum.

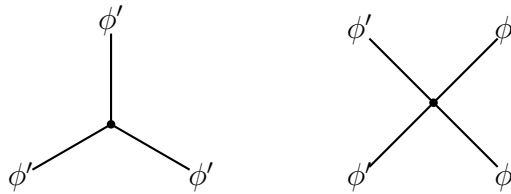


Figure 7: Scalar self-interactions after $\mathbb{Z}_2$ spontaneous symmetry breaking. The cubic vertex appears only after shifting to the vacuum, and its coefficient is related to the quartic coupling and v .

Claim 15.1 Parameter Counting after Spontaneous Breaking

Spontaneous symmetry breaking changes the variables used to describe the theory, but it does not introduce new independent parameters. In the $\mathbb{Z}_2$ example, the broken-phase coefficients $m_{\phi'}^2$, λv , and λ are all functions of the original two quantities μ^2 and λ .

For renormalizable scalar terms in four spacetime dimensions, the coefficients with positive mass dimension can change after shifting by the vacuum, while the dimensionless quartic coupling remains the same. The lecture emphasized this because relations among broken-phase parameters are what make spontaneously broken theories predictive.

15.5 Continuous Example: Global $U(1)$

Physical Motivation

A continuous symmetry has a qualitatively new feature: if the vacuum manifold is continuous, one can move along it without changing the potential energy. That flat direction becomes a massless particle in the quantum theory.

Take one complex scalar field ϕ with global $U(1)$ charge $+1$:

$$U(1) : \quad \phi \rightarrow e^{i\theta} \phi, \quad (388)$$

where θ is a constant real angle.

Mathematical Setup

The renormalizable $U(1)$ -invariant Lagrangian is

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - \mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2, \quad \lambda > 0. \quad (389)$$

Significance: The potential depends only on the magnitude of ϕ , so a nonzero vacuum will have a continuous degeneracy of phases.

It is often useful to write the complex field in terms of two real fields h and χ :

$$\phi = \frac{1}{\sqrt{2}}(h + i\chi), \quad \phi^\dagger \phi = \frac{1}{2}(h^2 + \chi^2). \quad (390)$$

In this language the same symmetry is an $SO(2)$ rotation:

$$\begin{pmatrix} h \\ \chi \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} h \\ \chi \end{pmatrix}. \quad (391)$$

Substituting Eq. (390) into Eq. (389) gives

$$\mathcal{L} = \frac{1}{2} \partial_\mu h \partial^\mu h + \frac{1}{2} \partial_\mu \chi \partial^\mu \chi - \frac{\mu^2}{2} (h^2 + \chi^2) - \frac{\lambda}{4} (h^2 + \chi^2)^2. \quad (392)$$

Significance: Before choosing a vacuum, the two real fields are indistinguishable directions in the same symmetric plane.

For $\mu^2 < 0$, define

$$v^2 \equiv -\frac{\mu^2}{\lambda} > 0. \quad (393)$$

The potential can be written, up to an additive constant, as

$$V(\phi) = \lambda \left(\phi^\dagger \phi - \frac{v^2}{2} \right)^2. \quad (394)$$

Significance: The vacuum condition fixes only the radius in field space, not the angular direction.

The minimum condition is

$$\phi^\dagger \phi = \frac{v^2}{2} \quad \Longleftrightarrow \quad h^2 + \chi^2 = v^2. \quad (395)$$

Thus the set of vacua is a circle of radius v in the (h, χ) plane.

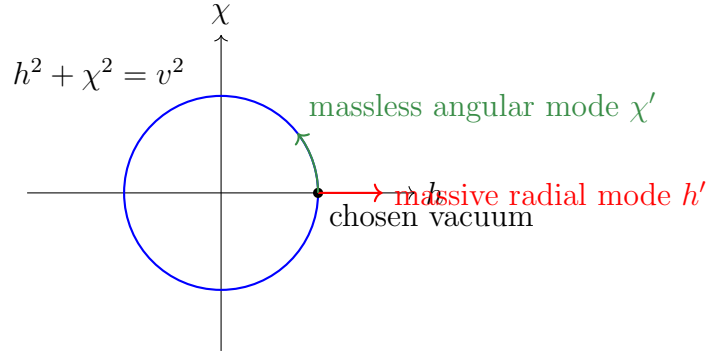


Figure 8: The vacuum manifold of the broken global $U(1)$ theory. Motion radially changes the potential and gives a massive excitation; motion tangentially stays on the degenerate circle and gives a Goldstone boson.

Expansion around a Chosen Vacuum

Choose the vacuum in the positive real direction:

$$\langle h \rangle = v, \quad \langle \chi \rangle = 0. \quad (396)$$

Define fluctuation fields with zero vacuum expectation values:

$$h = v + h', \quad \chi = \chi', \quad \langle h' \rangle = \langle \chi' \rangle = 0. \quad (397)$$

Now

$$h^2 + \chi^2 = (v + h')^2 + \chi'^2 = v^2 + 2vh' + h'^2 + \chi'^2, \quad (398)$$

$$h^2 + \chi^2 - v^2 = 2vh' + h'^2 + \chi'^2. \quad (399)$$

Using

$$V = \frac{\lambda}{4}(h^2 + \chi^2 - v^2)^2,$$

we get

$$\begin{aligned} V &= \frac{\lambda}{4} (2vh' + h'^2 + \chi'^2)^2 \\ &= \frac{\lambda}{4} [4v^2h'^2 + 4vh'(h'^2 + \chi'^2) + (h'^2 + \chi'^2)^2] \\ &= \lambda v^2 h'^2 + \lambda v h' (h'^2 + \chi'^2) + \frac{\lambda}{4} (h'^2 + \chi'^2)^2. \end{aligned} \quad (400)$$

The broken-phase Lagrangian is therefore

$$\mathcal{L} = \frac{1}{2} \partial_\mu h' \partial^\mu h' + \frac{1}{2} \partial_\mu \chi' \partial^\mu \chi' - \lambda v^2 h'^2 - \lambda v h' (h'^2 + \chi'^2) - \frac{\lambda}{4} (h'^2 + \chi'^2)^2. \quad (401)$$

Significance: The original $SO(2)$ rotation symmetry is hidden in the shifted variables, but it still enforces the precise pattern of masses and interactions.

Reading off the quadratic terms gives

$$m_{h'}^2 = 2\lambda v^2 = -2\mu^2, \quad m_{\chi'}^2 = 0. \quad (402)$$

Significance: The radial excitation is massive, while the angular excitation is massless because changing the phase moves along a degenerate set of vacua.

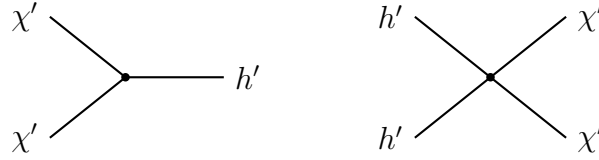


Figure 9: Representative broken-phase scalar interactions in the global $U(1)$ model. The $h'\chi'\chi'$ and $h'h'\chi'\chi'$ interactions are not arbitrary; they are fixed by the same λ and v that determine the radial mass.

Claim 15.2 Choice of Vacuum Does Not Change the Physics

Choosing a different point on the circle $h^2 + \chi^2 = v^2$ merely rotates which linear combination of the original fields is called the radial mode and which is called the angular mode. The spectrum still contains one scalar of mass squared $2\lambda v^2$ and one massless scalar.

15.6 Tachyons and the Meaning of Negative Mass Squared

Conceptual Background

In the original symmetric coordinates, the parameter choice $\mu^2 < 0$ makes the quadratic term look like a negative mass squared around $\phi = 0$. A particle with negative mass squared is often called a tachyon. In this context, however, the tachyon is not a physical asymptotic particle; it is a warning that $\phi = 0$ is not the correct vacuum.

Physical Concept 15.3: Tachyon in a Scalar Potential

A tachyonic quadratic term means that the curvature of the potential at the expansion point is negative:

$$m_{\text{local}}^2 = \left. \frac{d^2V}{d\phi^2} \right|_{\text{expansion point}} < 0. \quad (403)$$

Significance: A tachyonic mass parameter signals an unstable expansion point, not a stable faster-than-light particle in the physical spectrum.

At the true minimum in the $\mathbb{Z}_2$ example, the curvature is instead

$$\left. \frac{d^2V}{d\phi^2} \right|_{\phi=\pm v} = 2\lambda v^2 > 0, \quad (404)$$

so the excitation field ϕ' has an ordinary positive mass squared. This is why perturbative QFT is built around minima of the potential: physical particles are small oscillations around a stable ground state.

15.7 Goldstone Bosons

Physical Motivation

In the global $U(1)$ example, the vacuum manifold is a circle. Once a point on the circle is chosen, there are two qualitatively different infinitesimal motions:

- moving radially changes $h^2 + \chi^2$ and therefore changes the potential energy;
- moving tangentially along the circle keeps $h^2 + \chi^2 = v^2$ and therefore costs no potential energy.

The second motion is a flat direction, and a flat direction has zero restoring force. Since mass squared is the curvature of the potential, zero curvature means a massless scalar.

Physical Law 15.1 Goldstone Theorem

For every spontaneously broken continuous global symmetry generator, the theory contains a massless boson, called a Goldstone boson.

$$N_{\text{Goldstone}} = N_{\text{broken continuous global generators}} \cdot \quad (405)$$

Significance: Massless scalars are not accidental details of the example; they are forced by the geometry of a continuously degenerate vacuum.

For the global $U(1)$ model, there is one generator and it is broken by choosing $\langle\phi\rangle = v/\sqrt{2}$. Therefore Eq. (405) predicts one Goldstone boson, precisely the field χ' found in Eq. (402). In later lectures, gauging the symmetry will modify this conclusion and lead to the Higgs mechanism.

16 Lecture 16 – Non-Abelian SSB, Fermion Masses, and the Higgs Mechanism

Date: 07/06/2026

16.1 Overview

Lecture 15 treated spontaneous symmetry breaking (SSB) of a global $U(1)$, where the entire symmetry was broken and a single Goldstone boson appeared. Today we extend the idea in three directions of increasing physical weight.

1. **Non-Abelian global SSB:** a scalar triplet of $SO(3)$ acquires a vacuum expectation value (vev) that breaks $SO(3) \rightarrow SO(2)$. Only *part* of the symmetry is lost, and the count of Goldstone bosons is dictated by the number of broken generators.
2. **Fermion masses from SSB:** a chiral fermion forbidden from having a bare mass term can nevertheless become massive through a Yukawa coupling to a scalar that condenses. This is the mechanism by which every charged fermion in the Standard Model gets its mass.
3. **The Higgs mechanism:** when the broken symmetry is *local* (gauged), there is no physical Goldstone boson. Instead, the would-be Goldstone is “eaten” by the gauge field, which becomes massive and acquires a longitudinal polarization.

The unifying theme is a single sentence: *the vacuum, not the Lagrangian, decides which symmetries are visible in the particle spectrum*, and every broken continuous generator leaves a fingerprint — a Goldstone boson if global, a massive gauge boson if local.

16.2 Non-Abelian Global SSB: $SO(3) \rightarrow SO(2)$

Physical Motivation

The $SO(3)$ example is the cleanest non-Abelian generalization of last lecture’s $U(1)$ because $SO(3)$ is literally the group of rotations in three dimensions. A scalar in the triplet representation can be pictured as an ordinary vector $\phi = (\phi_1, \phi_2, \phi_3)$ in an internal three-dimensional space. Giving it a vev is the same as pinning that vector to point in a definite direction. Rotations that move the vector are lost; rotations *about* the vector survive. This geometric picture predicts the outcome before any algebra: a three-parameter symmetry collapses to the one-parameter group of rotations in the orthogonal plane.

Mathematical Setup

Let $\phi_a(x)$, $a = 1, 2, 3$, be a real scalar field in the triplet (adjoint) representation of $SO(3)$. Under an infinitesimal transformation with real parameters β^a ,

$$\phi \rightarrow e^{i\beta^a L^a} \phi, \quad (L^a)_{bc} = -i \epsilon_{abc}, \quad (406)$$

where ϵ_{abc} is the totally antisymmetric Levi-Civita symbol and the L^a are the three $SO(3)$ generators in the adjoint representation. The structure constants of $SO(3)$ are themselves ϵ_{abc} ,

which is why the adjoint generators are built from them. The renormalizable, $SO(3)$ -invariant Lagrangian is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi_a \partial^\mu \phi_a - \frac{1}{2} \mu^2 \phi_a \phi_a - \frac{\lambda}{4} (\phi_a \phi_a)^2, \quad \lambda > 0, \quad (407)$$

with summation over the internal index a understood. Only the rotational invariant $\phi_a \phi_a = |\phi|^2$ enters, so the potential depends solely on the length of the internal vector.

Significance: Because the potential sees only $|\phi|^2$, the vacuum can fix the length of ϕ but never its direction, and that leftover directional freedom is the seed of the Goldstone bosons.

Taking $\mu^2 < 0$ as in the $U(1)$ case, the potential is minimized on a sphere of radius v in internal space:

$$\phi_a \phi_a = v^2, \quad v^2 \equiv -\frac{\mu^2}{\lambda} > 0. \quad (408)$$

The direction is a free choice. We exploit this freedom and align the vev entirely along the third internal axis:

$$\langle \phi_a \rangle = (0, 0, v). \quad (409)$$

Significance: Choosing the vev to point along a single axis is a convenience, not physics — any direction gives an equivalent spectrum, but this choice makes the residual $SO(2)$ manifest.

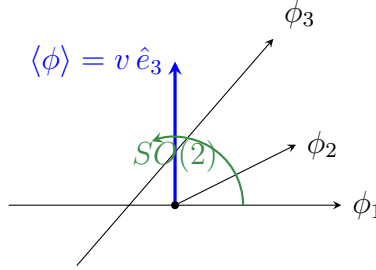


Figure 10: The vev points along $\hat{e}_3$ in internal space. Rotations that tilt the vector (L^1, L^2) are broken; the rotation about the vector (L^3 , in the $\phi_1\phi_2$ plane) leaves it invariant and survives as $SO(2)$.

Expansion around the Vacuum

Shift the field by its vev, introducing fluctuations ϕ'_a with $\langle \phi'_a \rangle = 0$. With the vev along the third axis only the third component is shifted:

$$\phi_1 = \phi'_1, \quad \phi_2 = \phi'_2, \quad \phi_3 = v + \phi'_3. \quad (410)$$

Then

$$\phi_a \phi_a = \phi_1'^2 + \phi_2'^2 + (v + \phi_3')^2 = v^2 + 2v \phi_3' + \phi_1'^2 + \phi_2'^2 + \phi_3'^2. \quad (411)$$

We organize the resulting Lagrangian by the number of fluctuation fields, writing $\mathcal{L} = \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4$ (plus a constant vacuum energy and the kinetic terms, which are unaffected by the constant shift). Substituting Eq. (411) into the potential of Eq. (407) and using $\mu^2 = -\lambda v^2$:

Quadratic terms ($\mathcal{L}_2$). Collect every term with exactly two fluctuation fields. From $\frac{1}{2}\mu^2(\phi_a \phi_a)$ the quadratic part is $\frac{1}{2}\mu^2(\phi_1'^2 + \phi_2'^2 + \phi_3'^2)$. From $\frac{\lambda}{4}(\phi_a \phi_a)^2$, squaring Eq. (411) and keeping quadratic terms gives $\frac{\lambda}{4}[2v^2(\phi_1'^2 + \phi_2'^2 + \phi_3'^2) + 4v^2\phi_3'^2]$. Adding these and using $\mu^2 = -\lambda v^2$, the ϕ_1', ϕ_2' masses cancel and only ϕ_3' survives:

$$V_2 = \left(\frac{1}{2}\mu^2 + \frac{\lambda v^2}{2} \right) (\phi_1'^2 + \phi_2'^2 + \phi_3'^2) + \lambda v^2 \phi_3'^2 = \lambda v^2 \phi_3'^2, \quad (412)$$

so the quadratic Lagrangian and spectrum are:

$$\boxed{\mathcal{L}_2 = -\lambda v^2 \phi_3'^2 \implies m_{\phi_3}^2 = 2\lambda v^2, \quad m_{\phi_1}^2 = m_{\phi_2}^2 = 0.} \quad (413)$$

Significance: The single massive mode is the radial fluctuation along the vev, while the two transverse fluctuations are exactly massless — precisely the two Goldstone bosons demanded by the broken generators.

Cubic terms ($\mathcal{L}_3$).

$$\mathcal{L}_3 = -\lambda v \phi_3' (\phi_1'^2 + \phi_2'^2 + \phi_3'^2) = -\lambda v \phi_3' \phi_a' \phi_a'. \quad (414)$$

A term proportional to ϕ_3' alone (selecting the third internal direction) is manifestly *not* $SO(3)$ invariant: it visibly singles out the broken direction. This is a direct fingerprint of the breaking inside the Lagrangian itself.

Quartic terms ($\mathcal{L}_4$).

$$\mathcal{L}_4 = -\frac{\lambda}{4} (\phi_a' \phi_a')^2, \quad (415)$$

identical in form to the unbroken quartic interaction.

Significance (of $\mathcal{L}_4$): The dimensionless quartic coupling is untouched by SSB — only couplings with positive mass dimension are shifted by the vev, as established last lecture.

Goldstone Counting

This realizes Goldstone's theorem for a non-Abelian group. $SO(3)$ has three generators; the unbroken $SO(2)$ has one (the rotation in the $\phi_1\phi_2$ plane). Hence

$$N_{\text{broken}} = \dim SO(3) - \dim SO(2) = 3 - 1 = 2, \quad (416)$$

matching the two massless scalars ϕ_1', ϕ_2' .

Consequence 16.1 Partial Breaking and Goldstone Count

When a continuous global group G is spontaneously broken to a subgroup H , the number of Goldstone bosons equals the number of broken generators,

$$N_{\text{Goldstone}} = \dim G - \dim H. \quad (417)$$

Significance: The Goldstone count is purely group-theoretic — it depends only on how much symmetry is lost, never on the details of the potential.

The General Criterion for Broken Generators

How do we know *which* generators are broken without expanding the whole Lagrangian? The vacuum supplies the test directly. A generator T^a corresponds to a broken symmetry if the group element it generates fails to leave the vacuum invariant, $e^{i\beta^a T^a} \langle \phi \rangle \neq \langle \phi \rangle$. Infinitesimally this is the statement:

Physical Law 16.2 Broken vs. Unbroken Generator

A generator T^a is *unbroken* if it annihilates the vacuum and *broken* otherwise:

$$\boxed{T^a \langle \phi \rangle = 0 \text{ (unbroken)}, \quad T^a \langle \phi \rangle \neq 0 \text{ (broken)}}. \quad (418)$$

Significance: The unbroken subgroup is exactly the stabilizer of the vacuum, so the entire symmetry-breaking pattern can be read off from a single vector $\langle\phi\rangle$.

For our example, L^3 generates rotations in the $\phi_1\phi_2$ plane, so $(L^3\langle\phi\rangle)_b = -i\epsilon_{3bc}\langle\phi_c\rangle = -i\epsilon_{3b3}v = 0$: the third generator is unbroken. By contrast L^1 and L^2 rotate the third axis into the first and second, giving $L^{1,2}\langle\phi\rangle \neq 0$, so they are broken — consistent with the two Goldstone bosons. The homework explores further breaking patterns with this same criterion.

16.3 Fermion Masses from Spontaneous Symmetry Breaking

Physical Motivation

In the Standard Model the left- and right-handed components of a fermion carry *different* gauge charges — they are chiral. A Dirac mass term $\bar{\psi}_L\psi_R$ couples the two chiralities and therefore generally violates the gauge symmetry, so it is forbidden in the Lagrangian. SSB resolves the tension: if a charged scalar condenses, the very symmetry that forbade the mass term is hidden by the vacuum, and an effective mass appears proportional to the vev. This is how electrons, quarks, and all charged fermions acquire mass.

Mathematical Setup

Take a global $U(1)$ with a left-handed fermion ψ_L , a right-handed fermion ψ_R , and a complex scalar ϕ carrying charges

$$q(\psi_L) = 1, \quad q(\psi_R) = 2, \quad q(\phi) = 1. \quad (419)$$

A bare Dirac mass would be $\bar{\psi}_L\psi_R$, with net charge $-q(\psi_L) + q(\psi_R) = -1 + 2 = +1 \neq 0$. It is not invariant, so

$$\text{no bare mass term: } \bar{\psi}_L\psi_R \text{ is forbidden by } U(1). \quad (420)$$

The fermions are *chiral* under this $U(1)$: left and right transform differently. The only renormalizable, gauge-invariant term linking them uses the scalar — the Yukawa coupling. Lorentz invariance pairs $\bar{\psi}_R\psi_L$, and charge neutrality fixes the scalar that must accompany it:

$$-q(\phi^?) + [-q(\psi_R) + q(\psi_L)] = 0 \Rightarrow \phi\bar{\psi}_R\psi_L \text{ has charge } +1 - 2 + 1 = 0. \quad (421)$$

Hence the allowed interaction is

$$\boxed{\mathcal{L}_{\text{Yuk}} = -y\phi\bar{\psi}_R\psi_L + \text{h.c.} ,} \quad (422)$$

where y is the dimensionless Yukawa coupling and h.c. denotes the Hermitian conjugate $-y^*\phi^\dagger\bar{\psi}_L\psi_R$.

Significance: The Yukawa coupling is the unique renormalizable bridge between the two chiralities, and it is gauge invariant precisely because the scalar carries away the leftover charge that a bare mass term could not.

Generating the Mass

Take $\mu^2 < 0$ so ϕ condenses with $|\langle\phi\rangle| = v/\sqrt{2}$. Choosing the vev along the real direction and expanding,

$$\phi(x) = \frac{1}{\sqrt{2}}(v + h(x) + i\chi(x)), \quad \langle h \rangle = \langle \chi \rangle = 0, \quad (423)$$

with h the radial (massive) fluctuation and χ the Goldstone (massless) fluctuation. Inserting into Eq. (422),

$$\begin{aligned}\mathcal{L}_{\text{Yuk}} &= -\frac{y}{\sqrt{2}}(v + h + i\chi)\bar{\psi}_R\psi_L + \text{h.c.} \\ &= \underbrace{-\frac{yv}{\sqrt{2}}(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R)}_{\text{mass term}} - \underbrace{\frac{y}{\sqrt{2}}(h + i\chi)\bar{\psi}_R\psi_L + \text{h.c.}}_{\text{scalar-fermion interactions}}.\end{aligned}\quad (424)$$

The first piece is exactly a Dirac mass linking the two chiralities into one massive Dirac fermion $\psi = \psi_L + \psi_R$:

$$\boxed{m_\psi = \frac{yv}{\sqrt{2}}}.\quad (425)$$

Significance: A symmetry-forbidden mass becomes allowed once the vacuum breaks the symmetry, and that mass is set by the same vev that breaks it — mass is a measure of how strongly the fermion feels the condensate.

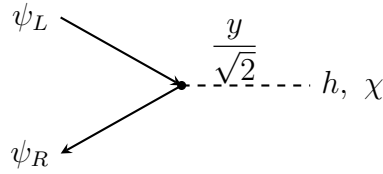


Figure 11: The Yukawa vertex after SSB. The condensate ($\phi \rightarrow v$) supplies the Dirac mass $m_\psi = yv/\sqrt{2}$, while the physical scalars h and χ couple to the fermion with strength $y/\sqrt{2} = m_\psi/v$.

The interaction piece shows that the physical scalars h and χ couple to the fermion with the *same* coupling $y/\sqrt{2}$ that controls the mass. Eliminating y via Eq. (425),

$$\boxed{g_{h\bar{\psi}\psi} = \frac{y}{\sqrt{2}} = \frac{m_\psi}{v}}.\quad (426)$$

Significance: The scalar couples to each fermion in proportion to that fermion’s mass — a defining prediction of the Higgs boson, which interacts most strongly with the heaviest particles.

Claim 16.1 Condition for Mass after Partial Breaking

In the $U(1)$ example the symmetry was broken completely. More generally, when a symmetry is only partially broken, the fermions can acquire a mass from SSB only if their representation under the *unbroken* subgroup is *vector-like* — left and right components transform identically. If they remain chiral under the surviving symmetry, no invariant mass term can form.

Note:

After SSB the surviving Lagrangian is not the same as a theory with no symmetry at all: the broken symmetry still relates couplings (here the mass, the h coupling, and the χ coupling all share the single parameter y). Asking for “the $U(1)$ charge of h or χ ” is meaningless once the vacuum has broken the symmetry — charge is only defined in the symmetric description.

16.4 The Higgs Mechanism: SSB of a Local $U(1)$

Conceptual Background

Everything so far concerned *global* symmetries, whose breaking produces physical massless Goldstone bosons. The decisive new feature arises when the broken symmetry is *local* (gauged). A gauged symmetry has a gauge boson for each generator, and unbroken gauge bosons are necessarily massless (as $U(1)_{\text{EM}}$ forces the photon to be). When such a symmetry is spontaneously broken, the gauge bosons of the broken generators *gain mass*. The bookkeeping that makes this consistent is the Higgs mechanism: the would-be Goldstone boson is absorbed — “eaten” — by the gauge field and reappears as its longitudinal polarization.

Physical Law 16.3 Higgs Mechanism (Statement)

Spontaneous breaking of a *local* symmetry gives mass to the gauge bosons associated with the broken generators. The would-be Goldstone bosons are eaten by those gauge bosons and become their longitudinal components:

$$\boxed{\text{broken local generator} \implies \text{massive gauge boson} + \text{eaten Goldstone} .} \quad (427)$$

Significance: Gauge symmetry forbids an explicit gauge-boson mass term, so the only way a force carrier (like the W and Z) can be massive is for its symmetry to be spontaneously broken — this is why mass and SSB are inseparable in the Standard Model.

Degree-of-Freedom Counting

The mechanism is required by simple counting of physical polarizations:

- A massless gauge boson has 2 transverse polarizations.
- A massive vector has 3 polarizations (two transverse, one longitudinal).
- A real Goldstone scalar has 1 degree of freedom.

The total number of degrees of freedom cannot change; only their organization does:

$$\boxed{\underbrace{2}_{\text{massless gauge}} + \underbrace{1}_{\text{Goldstone}} = \underbrace{3}_{\text{massive gauge}} .} \quad (428)$$

Significance: The “eaten” Goldstone is not lost — it is exactly the longitudinal polarization a massive vector needs, so the mechanism conserves degrees of freedom while turning a massless force carrier into a massive one.

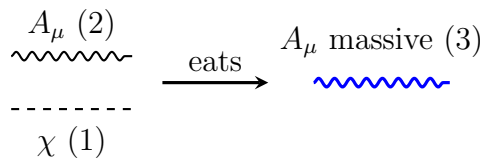


Figure 12: Higgs mechanism as degree-of-freedom rearrangement. The 2 polarizations of the massless gauge boson plus the 1 Goldstone scalar reorganize into the 3 polarizations of a massive vector.

Mathematical Setup

Promote the global $U(1)$ to a local one. A scalar ϕ of charge $+1$ transforms as

$$\phi(x) \rightarrow e^{i\theta(x)}\phi(x), \quad D_\mu\phi = (\partial_\mu + igA_\mu)\phi, \quad (429)$$

with g the gauge coupling and A_μ the gauge field (the analog of the photon). The gauge-invariant Lagrangian is

$$\mathcal{L} = (D_\mu\phi)^\dagger D^\mu\phi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \mu^2\phi^\dagger\phi - \lambda(\phi^\dagger\phi)^2, \quad (430)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the Abelian field strength. This differs from the global theory only by the covariant derivative and the gauge kinetic term.

Significance: The covariant derivative is the entire source of the gauge-boson mass: when ϕ is replaced by its vev inside $(D_\mu\phi)^\dagger D^\mu\phi$, the term $g^2|\langle\phi\rangle|^2 A_\mu A^\mu$ is precisely a photon mass.

For $\mu^2 < 0$ the scalar condenses with $|\langle\phi\rangle| = v/\sqrt{2}$, $v = \sqrt{-\mu^2/\lambda}$, breaking the local $U(1)$. Rather than the linear (real/imaginary) split used earlier, it is now convenient to write the complex scalar in *magnitude-phase* (polar) variables:

$$\phi(x) = e^{i\chi(x)/v} \frac{v + h(x)}{\sqrt{2}}. \quad (431)$$

This is called the *nonlinear realization* of the field: h is the radial (Higgs) excitation and χ is the phase (would-be Goldstone) excitation, both with zero vev. To leading order in the phase, $e^{i\chi/v} \approx 1 + i\chi/v$, so it reduces to the linear realization $\phi \approx \frac{1}{\sqrt{2}}(v + h + i\chi)$ used in Eq. (423).

Significance: In the nonlinear realization the would-be Goldstone χ sits entirely in the *phase*, exactly the part a gauge transformation can rotate — which is what makes “eating” the Goldstone a simple change of gauge.

Outlook: Unitary Gauge

The phase $\chi(x)$ appears only through $e^{i\chi/v}$. Because the symmetry is now *local*, we are free to perform an x -dependent gauge rotation $\theta(x) = -\chi(x)/v$ that removes the phase entirely, leaving

$$\phi(x) \longrightarrow \frac{v + h(x)}{\sqrt{2}} \quad (\text{unitary gauge}). \quad (432)$$

The degree of freedom χ has not vanished; the same gauge transformation shifts A_μ , depositing χ into the gauge field as its longitudinal mode. Carrying this through the covariant derivative produces the explicit gauge-boson mass term — the calculation we complete next lecture (Thursday).

Note:

The contrast with the global case is sharp: for a global symmetry no gauge transformation exists to remove χ , so it survives as a physical massless Goldstone boson. The locality of the symmetry is exactly what converts the Goldstone into a longitudinal polarization.

Part V

Homework Solutions

17 Homework Set 1

17.1 Brief review of relativity

Question 1: Consider two vectors a_μ and b_μ . Write down their dot product that is defined as $a_\mu b^\mu \equiv g^{\mu\nu} a_\mu b_\nu$.

Solution 1. Let a_μ and b_μ be two covariant four-vectors in 4-dimensional spacetime.

- $a_\mu = (a_0, a_1, a_2, a_3)$
- $b_\mu = (b_0, b_1, b_2, b_3)$

$$a_\mu b^\mu = (a_0 \ a_1 \ a_2 \ a_3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

First, we multiply the row vector by the metric matrix (this effectively calculates the contravariant vector $a^\nu = g^{\mu\nu} a_\mu$)

$$a_\mu b^\mu = (a_0 \ -a_1 \ -a_2 \ -a_3) \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Finally, we take the dot product of the resulting row and column matrices to get the invariant scalar:

$$a_\mu b^\mu = a_0 b_0 - a_1 b_1 - a_2 b_2 - a_3 b_3$$

Question 2: Explicitly write down the Euler–Lagrange equation for a classical field theory formulated in 3+1 dimensions.

Solution 2. The E-L equation of classical field theory is given by

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0$$

By using the definition of the operator $\partial_\mu = (\partial_t, \nabla)$

$$\frac{\partial \mathcal{L}}{\partial \phi} - \left[\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial (\partial_t \phi)} \right) + \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}}{\partial (\partial_x \phi)} \right) + \frac{\partial}{\partial y} \left(\frac{\partial \mathcal{L}}{\partial (\partial_y \phi)} \right) + \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}}{\partial (\partial_z \phi)} \right) \right] = 0$$

Expressing the spatial derivatives compactly using the divergence operator and denoting the time derivative as $\dot{\phi} \equiv \partial_t \phi$, the equation takes the form:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial (\nabla \phi)} \right) = 0$$

Question 3: Consider the following quantities: $A_{\mu\nu}$, b_μ and c_μ . Write down all Lorentz scalars you can make out of them that involve at most three quantities. You can assume that A is symmetric.

Solution 3. We are given $A_{\mu\nu}$ (symmetric), and covariant vectors b_μ, c_μ .

First-Order Scalars (1 quantity):

$$g^{\mu\nu} A_{\mu\nu} = A^\mu{}_\mu \equiv \text{Tr}(A)$$

Second-Order Scalars (2 quantities): By contracting vectors:

$$b^\mu b_\mu, \quad c^\mu c_\mu, \quad b^\mu c_\mu$$

By contracting the tensor with itself:

$$A^{\mu\nu} A_{\mu\nu}$$

(Note: we can also form the trivial product $(A^\mu{}_\mu)^2$)

Third-Order Scalars (3 Quantities): If we attempt to combine three vectors (e.g., b, b, c), we have 3 indices. An odd number of indices cannot be fully contracted, so no scalar can be formed solely from the vectors.

Contracting the tensor with the vectors yields:

$$A^{\mu\nu} b_\mu b_\nu, \quad A^{\mu\nu} c_\mu c_\nu, \quad A^{\mu\nu} b_\mu c_\nu$$

(Due to the symmetry of A , $A^{\mu\nu} b_\nu c_\mu = A^{\mu\nu} b_\mu c_\nu$).

Contracting three tensors yields the trace of the cube:

$$A^\mu{}_\nu A^\nu{}_\rho A^\rho{}_\mu$$

Finally, we can construct "disconnected" scalars by multiplying our lower-order scalars. For combinations involving exactly three quantities, we have:

$$A^\alpha{}_\alpha (b^\mu b_\mu), \quad A^\alpha{}_\alpha (c^\mu c_\mu), \quad A^\alpha{}_\alpha (b^\mu c_\mu), \quad A^\alpha{}_\alpha (A^{\mu\nu} A_{\mu\nu}), \quad (A^\alpha{}_\alpha)^3$$

(Note on Levi-Civita: Attempting to use $\epsilon_{\mu\nu\rho\sigma}$ to form a scalar like $\epsilon_{\mu\nu\rho\sigma} A^{\mu\nu} b^\rho c^\sigma$ identically vanishes because the contraction of the symmetric $A^{\mu\nu}$ with the antisymmetric $\epsilon_{\mu\nu\rho\sigma}$ is zero).

17.2 Natural Units

Question 4: The width of a particle is defined as the inverse of its lifetime. The mean lifetime for the B^+ meson is $\tau \approx 1.64 \times 10^{-12}$ sec. What is its width in eV?

Solution 4. The decay width Γ is related to the mean lifetime τ by the uncertainty principle:

$$\Gamma = \frac{\hbar}{\tau}$$

Using the value of the reduced Planck constant in electron-volts, $\hbar \approx 6.582 \times 10^{-16}$ eV · s, and the given lifetime $\tau \approx 1.64 \times 10^{-12}$ s, we find:

$$\Gamma = \frac{6.582 \times 10^{-16} \text{ eV} \cdot \text{s}}{1.64 \times 10^{-12} \text{ s}}$$

$$\Gamma \approx 4.01 \times 10^{-4} \text{ eV}$$

Therefore, the width of the B^+ meson is approximately 0.401 meV.

Question 5: Consider a particle A with a width of $\Gamma = 2.3$ eV. Recall that in the lab frame $t = \gamma\tau$. What is the average distance an A particle travels for $\gamma = 100$? (Note that since γ is large you can use $\beta \approx 1$.) $d = \beta ct = \beta c(\gamma\tau)$

Solution 5. In natural units ($c = 1, \hbar = 1$), the decay width Γ is the inverse of the rest-frame lifetime τ , so $\tau = 1/\Gamma$.

The average distance traveled in the lab frame, assuming $\beta \approx 1$, is:

$$d = \gamma\tau = \frac{\gamma}{\Gamma}$$

Substituting the given values, we find the distance entirely in units of inverse energy:

$$d = \frac{100}{2.3 \text{ eV}} \approx 43.48 \text{ eV}^{-1}$$

To convert this purely kinematic result into a measurable physical length, we multiply by the Energy-Length bridge, $\hbar c \approx 197 \text{ MeV} \cdot \text{fm} = 1.97 \times 10^{-7} \text{ eV} \cdot \text{m}$:

$$d = (43.48 \text{ eV}^{-1}) \times (1.97 \times 10^{-7} \text{ eV} \cdot \text{m})$$

$$d \approx 8.56 \times 10^{-6} \text{ m}$$

The particle travels an average distance of $8.56 \mu\text{m}$.

17.3 Relativistic kinematics: Colliders vs. fixed targets

Question 6: Two proton beams, each with 7 TeV of energy, colliding head-on (this is what the LHC does)

Solution 6. The four-momenta for two particles colliding head-on are:

$$p_1 = (E, \vec{p}), \quad p_2 = (E, -\vec{p}) \quad (433)$$

The total four-momentum of the system is the sum of the individual four-momenta:

$$p_{tot} = p_1 + p_2 = (E + E, \vec{p} - \vec{p}) = (2E, \vec{0})$$

Therefore, we are able to compute the invariant mass squared s as follows:

$$s = p_{tot}^2 = (2E)^2 - |\vec{0}|^2 = 4E^2$$

Taking the square root yields the center-of-mass energy:

$$\sqrt{s} = \sqrt{4E^2} = 2E$$

Given that the energy of each beam is $E = 7$ TeV:

$$\sqrt{s} = 2(7 \text{ TeV}) = 14 \text{ TeV}$$

Question 7: One proton beam with 14 TeV of energy hitting a proton target that is sitting at rest in the lab.

Solution 7. The four-momenta for two colliding particles in a frame where one proton is stationary are given by:

$$p_1 = (E, \vec{p}), \quad p_2 = (m_p, \vec{0}) \quad (434)$$

Thus, the total four-momentum of the system is:

$$p_{tot} = p_1 + p_2 = (E + m_p, \vec{p})$$

The invariant mass squared is given by the square of the total four-momentum:

$$\begin{aligned} s &= p_{tot}^2 = (E + m_p)^2 - |\vec{p}|^2 \\ &= E^2 + 2Em_p + m_p^2 - |\vec{p}|^2 \end{aligned}$$

Using the mass-shell condition $E^2 - |\vec{p}|^2 = m_p^2$, we can substitute to find:

$$\begin{aligned} s &= m_p^2 + 2Em_p + m_p^2 \\ &= 2m_p^2 + 2Em_p = 2m_p(m_p + E) \end{aligned}$$

Taking the square root:

$$\sqrt{s} = \sqrt{2m_p(m_p + E)}$$

Given $E = 14 \text{ TeV} = 14 \times 10^3 \text{ GeV}$ and $m_p \approx 0.938 \text{ GeV}$, the system is highly ultra-relativistic ($E \gg m_p$). The m_p^2 mass term is negligible, allowing the approximation:

$$s \approx 2Em_p$$

$$\sqrt{s} \approx \sqrt{2(14000 \text{ GeV})(0.938 \text{ GeV})} = \sqrt{26264 \text{ GeV}^2} \approx 162.06 \text{ GeV}$$

17.4 Lagrangian formalism of E&M

Question 8: show that the two inhomogeneous Maxwell equations $\nabla \cdot \mathbf{E} = \rho, \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J}$ can be written compactly as $\partial_\alpha F^{\alpha\beta} = J^\beta$.

Solution 8. First, let's review the components:

Four-current:

$$J^\beta = (\rho, \mathbf{J})$$

Field tensor components:

$$F^{0i} = E^i, \quad F^{i0} = -E^i, \quad F^{ij} = -\epsilon^{ijk} B^k$$

Four-derivative:

$$\partial_\alpha = (\partial_t, \nabla)$$

Gauss's law ($\beta = 0$): Compute:

$$\partial_\alpha F^{\alpha 0} = \partial_0 F^{00} + \partial_i F^{i0}$$

Where $F^{00} = 0$ and $F^{i0} = -E^i$ so:

$$\partial_\alpha F^{\alpha 0} = \partial_i(-E^i) = -\nabla \cdot \mathbf{E}$$

Thus:

$$\partial_\alpha F^{\alpha 0} = -\nabla \cdot \mathbf{E}$$

To match Maxwell:

$$\nabla \cdot \mathbf{E} = \rho$$

we identify:

$$J^0 = \rho$$

Ampère–Maxwell law ($\beta = i$): Compute:

$$\partial_\alpha F^{\alpha i} = \partial_0 F^{0i} + \partial_j F^{ji}$$

First term:

$$\partial_0 F^{0i} = \partial_t E^i$$

Second-term:

$$F^{ji} = -\epsilon^{jik} B^k$$

$$\partial_j F^{ji} = -\epsilon^{jik} \partial_j B^k$$

Using the identity: $(\nabla \times \mathbf{B})^i = \epsilon^{ijk} \partial_j B^k$ we get:

$$\partial_j F^{ji} = (\nabla \times \mathbf{B})^i$$

Combine:

$$\partial_\alpha F^{\alpha i} = \partial_t E^i + (\nabla \times \mathbf{B})^i$$

Rewriting:

$$(\nabla \times \mathbf{B})^i - \partial_t E^i = J^i$$

Both components together give:

$$\boxed{\partial_\alpha F^{\alpha\beta} = J^\beta}$$

which is the covariant form of the inhomogeneous Maxwell equations.

Question 9: We assume the following Lagrangian: $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$ and consider each of A_μ to be an independent DOF. Using the E-L equation show that this Lagrangian gives the two inhomogeneous equations.

Solution 9. We start with the Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu,$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

and treat each component A_μ as an independent degree of freedom.

Euler–Lagrange Equations

The Euler–Lagrange equations for fields are

$$\partial_\alpha \left(\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} \right) - \frac{\partial \mathcal{L}}{\partial A_\beta} = 0.$$

Compute $\frac{\partial \mathcal{L}}{\partial A_\beta}$

Only the interaction term depends explicitly on A_β :

$$\frac{\partial \mathcal{L}}{\partial A_\beta} = -J^\beta.$$

Compute $\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)}$

Using

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu},$$

we apply the chain rule:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -\frac{1}{4} \cdot 2F^{\mu\nu} \frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)}.$$

Now,

$$\frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)} = \delta_\mu^\alpha \delta_\nu^\beta - \delta_\nu^\alpha \delta_\mu^\beta.$$

Thus,

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -\frac{1}{2}F^{\mu\nu}(\delta_\mu^\alpha \delta_\nu^\beta - \delta_\nu^\alpha \delta_\mu^\beta).$$

Contracting indices:

$$= -\frac{1}{2}(F^{\alpha\beta} - F^{\beta\alpha}).$$

Using antisymmetry $F^{\beta\alpha} = -F^{\alpha\beta}$, we get:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -F^{\alpha\beta}.$$

Plug into Euler–Lagrange Equation

$$\partial_\alpha(-F^{\alpha\beta}) - (-J^\beta) = 0.$$

Simplify:

$$-\partial_\alpha F^{\alpha\beta} + J^\beta = 0 \Rightarrow \partial_\alpha F^{\alpha\beta} = J^\beta.$$

We obtain

$$\boxed{\partial_\alpha F^{\alpha\beta} = J^\beta},$$

which is the covariant form of the inhomogeneous Maxwell equations:

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mathbf{J}.$$

Question 10: Homogeneous Maxwell's equations

Solution 10. The homogeneous Maxwell equations follow from the definition of the field tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

1. Show that $\nabla \cdot \mathbf{B} = 0$

The magnetic field is defined as

$$B^i = \frac{1}{2}\epsilon^{ijk}F_{jk}.$$

Thus,

$$\nabla \cdot \mathbf{B} = \partial_i B^i = \frac{1}{2}\epsilon^{ijk}\partial_i F_{jk}.$$

Substitute $F_{jk} = \partial_j A_k - \partial_k A_j$:

$$\partial_i F_{jk} = \partial_i \partial_j A_k - \partial_i \partial_k A_j.$$

Hence,

$$\nabla \cdot \mathbf{B} = \frac{1}{2} \epsilon^{ijk} (\partial_i \partial_j A_k - \partial_i \partial_k A_j).$$

Since partial derivatives commute,

$$\partial_i \partial_j A_k = \partial_j \partial_i A_k,$$

the two terms cancel due to antisymmetry of ϵ^{ijk} , giving

$$\nabla \cdot \mathbf{B} = 0.$$

the remaining homogeneous equation:

$$B_i = \frac{1}{2} \epsilon_{ijk} F^{jk}.$$

Then

$$\frac{\partial B_i}{\partial t} = \frac{1}{2} \epsilon_{ijk} \partial_0 F^{jk}.$$

Using $F^{jk} = \partial^j A^k - \partial^k A^j$:

$$\frac{\partial B_i}{\partial t} = \frac{1}{2} \epsilon_{ijk} \partial_0 (\partial^j A^k - \partial^k A^j).$$

Rewriting:

$$\frac{\partial B_i}{\partial t} = \frac{1}{2} \epsilon_{ijk} (\partial^j \partial_0 A^k - \partial^k \partial_0 A^j).$$

Now consider the electric field:

$$E_i = F_{i0} = \partial_i A_0 - \partial_0 A_i.$$

Compute the curl:

$$(\nabla \times \mathbf{E})_i = \epsilon_{ijk} \partial^j E^k = \epsilon_{ijk} \partial^j (\partial^k A^0 - \partial^0 A^k).$$

Expand:

$$(\nabla \times \mathbf{E})_i = \epsilon_{ijk} (\partial^j \partial^k A^0 - \partial^j \partial^0 A^k).$$

The first term vanishes because $\partial^j \partial^k$ is symmetric while ϵ_{ijk} is antisymmetric. Thus:

$$(\nabla \times \mathbf{E})_i = -\epsilon_{ijk} \partial^j \partial^0 A^k.$$

Comparing with the expression for $\partial_t B_i$, we obtain:

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}.$$

Both homogeneous equations can be written compactly as

$$\boxed{\partial_\alpha \tilde{F}^{\alpha\beta} = 0,}$$

where the dual tensor is

$$\tilde{F}^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\mu\nu} F_{\mu\nu}.$$

These equations follow identically from the definition of $F_{\mu\nu}$ in terms of the potential A^μ . They are not dynamical equations but instead express the fact that

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

i.e., the electromagnetic field is derived from a 4-potential.

Thus, the homogeneous Maxwell equations are simply a restatement of this definition.

18 Homework Set 2

Question 11: Fermi's Golden Rule

Read in any book or website about the Fermi golden rule. Very briefly explain when it is applied and what assumptions must be met to use it.

Solution 11. Fermi's Golden Rule states that the transition rate from an initial state $|i\rangle$ to a final state $|f\rangle$ due to a perturbation V is

$$\Gamma_{i \rightarrow f} = 2\pi |\langle f|V|i\rangle|^2 \rho(E_f), \quad (435)$$

where $\rho(E_f)$ is the density of final states at the transition energy. It is derived from time-dependent perturbation theory and applies when (i) the perturbation V is weak (first-order in V is sufficient), and (ii) the observation time t is an *intermediate* timescale: $t \gg \hbar/\Delta E$ (or $t \gg 1/\Delta E$ in natural units) so the sinc profile is sharply peaked and enforces energy conservation, but still short enough that depletion of the initial state is negligible. This keeps first-order perturbation theory self-consistent while yielding a linear-in-time transition probability and a well-defined rate. In particle physics it underlies both decay-rate and cross-section calculations.

Question 12: Harmonic Oscillator Perturbation Theory

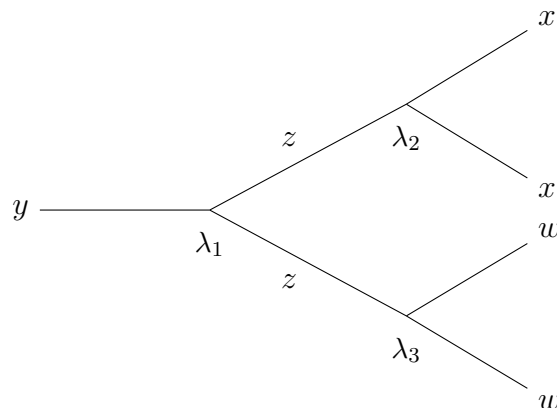
Consider a case with four oscillators, called w, x, y, z and assume that $y \rightarrow 2x2w$ is allowed by energy conservation. We use the following interaction

$$\lambda_1 y z^2 + \lambda_2 z x^2 + \lambda_3 z w^2.$$

1. Draw the diagram and calculate the transition matrix element using the Feynman rules and the correct normalisation.
2. (Optional) Calculate the amplitude using perturbation theory. Find the 6 intermediate states that contribute, calculate the 6 matrix elements and add them up. Verify that the two methods agree.

Solution 12 (Part 1 — Feynman diagram and matrix element). **The diagram**

The process $y \rightarrow 2x + 2w$ requires three vertices at lowest order: vertex λ_1 destroys the incoming y quantum and creates two intermediate z quanta; vertex λ_2 converts one z into two x quanta; and vertex λ_3 converts the remaining z into two w quanta. The tree-level Feynman diagram is



Feynman rules

Each oscillator coordinate is written as

$$\hat{\phi}_i = \frac{a_i + a_i^\dagger}{\sqrt{2\omega_i}}, \quad (436)$$

so that the interaction Hamiltonian is $H' = \lambda_1 \hat{y} \hat{z}^2 + \lambda_2 \hat{z} \hat{x}^2 + \lambda_3 \hat{z} \hat{w}^2$. For the energy-domain Feynman calculation of $\langle 2_x, 2_w | S^{(3)} | 1_y \rangle$, use:

- (i) **External line k** : factor $1/\sqrt{2\omega_k}$.
- (ii) **Vertex j** : factor $-i\lambda_j$.
- (iii) **Internal z propagator** with energy E : $i/(E^2 - \omega_z^2)$.

Combinatorics and normalization

From Wick contractions of $T\{(\lambda_1 y z^2)(\lambda_2 z x^2)(\lambda_3 z w^2)\}$:

- z^2 at the λ_1 vertex gives a factor 2 (two routings of the two internal z lines into the λ_2, λ_3 vertices).
- For identical final quanta, $\langle 2_x | x^2 | 0 \rangle = \sqrt{2}/(2\omega_x)$ and $\langle 2_w | w^2 | 0 \rangle = \sqrt{2}/(2\omega_w)$.

Energy flow

At the two lower vertices, energy conservation fixes internal-line energies:

$$E_a = 2\omega_x, \quad E_b = 2\omega_w, \quad \omega_y = 2\omega_x + 2\omega_w. \quad (437)$$

Amplitude from strict Feynman rules

Collecting external factors, symmetry/combinatorics, couplings, and the two Feynman propagators:

$$\begin{aligned} \mathcal{M} &= \left(\frac{1}{\sqrt{2\omega_y}} \right) \left(\frac{\sqrt{2}}{2\omega_x} \right) \left(\frac{\sqrt{2}}{2\omega_w} \right) \cdot 2 \cdot (\lambda_1 \lambda_2 \lambda_3) \cdot \frac{1}{E_a^2 - \omega_z^2} \cdot \frac{1}{E_b^2 - \omega_z^2} \\ &= \frac{\lambda_1 \lambda_2 \lambda_3}{\omega_x \omega_w \sqrt{2\omega_y} (4\omega_x^2 - \omega_z^2)(4\omega_w^2 - \omega_z^2)}. \end{aligned} \quad (438)$$

Hence

$$\boxed{\mathcal{M} = \frac{\lambda_1 \lambda_2 \lambda_3}{\omega_x \omega_w \sqrt{2\omega_y} (4\omega_x^2 - \omega_z^2)(4\omega_w^2 - \omega_z^2)}}. \quad (439)$$

This is the full Part 1 Feynman-rules result. In TOPT/OFPT language, it corresponds to summing all six time orderings; keeping only forward-time poles would reproduce only a partial-fraction fragment.

Solution 13 (Part 2 (optional) — perturbation-theory check). Use old-fashioned perturbation theory at third order with

$$V_1 = \lambda_1 y z^2, \quad V_2 = \lambda_2 z x^2, \quad V_3 = \lambda_3 z w^2, \quad (440)$$

and

$$\mathcal{A}_{i \rightarrow f}^{(3)} = \sum_{\pi \in S_3} \sum_{n_1, n_2} \frac{\langle f | V_{\pi_3} | n_2 \rangle \langle n_2 | V_{\pi_2} | n_1 \rangle \langle n_1 | V_{\pi_1} | i \rangle}{(E_i - E_{n_1})(E_i - E_{n_2})}. \quad (441)$$

Here $|i\rangle = |1_y, 0_z, 0_x, 0_w\rangle$, $|f\rangle = |0_y, 0_z, 2_x, 2_w\rangle$, and $\omega_y = 2\omega_x + 2\omega_w$.

The six time orderings are

$$(123), (132), (213), (231), (312), (321),$$

where index 1, 2, 3 stands for (V_1, V_2, V_3) respectively. Using the basis ordering

$$|n_y, n_z, n_x, n_w\rangle,$$

the intermediate states and denominators are:

Ordering π	$ n_1^\pi\rangle$	$ n_2^\pi\rangle$	$E_i - E_{n_1^\pi}$	$E_i - E_{n_2^\pi}$
(123)	$ 0, 2, 0, 0\rangle$	$ 0, 1, 2, 0\rangle$	$\omega_y - 2\omega_z$	$\omega_y - \omega_z - 2\omega_x$
(132)	$ 0, 2, 0, 0\rangle$	$ 0, 1, 0, 2\rangle$	$\omega_y - 2\omega_z$	$\omega_y - \omega_z - 2\omega_w$
(213)	$ 1, 1, 2, 0\rangle$	$ 0, 1, 2, 0\rangle$	$-(2\omega_x + \omega_z)$	$\omega_y - \omega_z - 2\omega_x$
(231)	$ 1, 1, 2, 0\rangle$	$ 1, 2, 2, 2\rangle$	$-(2\omega_x + \omega_z)$	$-(\omega_y + 2\omega_z)$
(312)	$ 1, 1, 0, 2\rangle$	$ 0, 1, 0, 2\rangle$	$-(2\omega_w + \omega_z)$	$\omega_y - \omega_z - 2\omega_w$
(321)	$ 1, 1, 0, 2\rangle$	$ 1, 2, 2, 2\rangle$	$-(2\omega_w + \omega_z)$	$-(\omega_y + 2\omega_z)$

Hence

$$\mathcal{A}^{(3)} = \sum_{\pi \in S_3} \frac{\mathcal{N}_\pi}{(E_i - E_{n_1^\pi})(E_i - E_{n_2^\pi})},$$

with $\mathcal{N}_\pi$ the corresponding ladder-operator matrix-element numerators (including the z^2 combinatorics and identical-particle normalization factors). After evaluating these numerators and simplifying with $\omega_y = 2\omega_x + 2\omega_w$, the six terms regroup into

$$\mathcal{A}^{(3)} = \frac{\lambda_1 \lambda_2 \lambda_3}{4 \omega_z^2 \omega_x \omega_w \sqrt{2\omega_y}} \left[\frac{1}{(2\omega_x - \omega_z)(2\omega_w - \omega_z)} - \frac{1}{(2\omega_x + \omega_z)(2\omega_w - \omega_z)} \right. \\ \left. - \frac{1}{(2\omega_x - \omega_z)(2\omega_w + \omega_z)} + \frac{1}{(2\omega_x + \omega_z)(2\omega_w + \omega_z)} \right]. \quad (442)$$

This is precisely the partial-fraction form

$$\frac{1}{E^2 - \omega_z^2} = \frac{1}{2\omega_z} \left(\frac{1}{E - \omega_z} - \frac{1}{E + \omega_z} \right)$$

applied to the two internal z propagators with $E_a = 2\omega_x$ and $E_b = 2\omega_w$. Therefore

$$\mathcal{A}^{(3)} = \frac{\lambda_1 \lambda_2 \lambda_3}{\omega_x \omega_w \sqrt{2\omega_y} (4\omega_x^2 - \omega_z^2)(4\omega_w^2 - \omega_z^2)}, \quad (443)$$

in exact agreement with Part 1.

Question 13: Fermionic Harmonic Oscillator

In this question we study the simple Fermionic harmonic oscillator. The bosonic simple HO is defined with

$$H = \hbar\omega \frac{a^\dagger a + a a^\dagger}{2} = \hbar\omega \left(N + \frac{1}{2} \right), \quad N = a^\dagger a,$$

with $[a, a^\dagger] = 1$, $[a, a] = [a^\dagger, a^\dagger] = 0$.

The Fermionic oscillator replaces commutators by anti-commutators:

$$\{b, b^\dagger\} = 1, \quad \{b, b\} = \{b^\dagger, b^\dagger\} = 0,$$

where $\{f, g\} = fg + gf$. The Hamiltonian is

$$H = \hbar\omega \frac{b^\dagger b - bb^\dagger}{2} = \hbar\omega \left(N - \frac{1}{2}\right), \quad N = b^\dagger b.$$

1. Show that $b^2 = 0$.
2. Assuming the Hilbert space is non-empty, show there exists a state $|0\rangle$ annihilated by b .
3. Define $|1\rangle = b^\dagger|0\rangle$. Show $|1\rangle \neq 0$.
4. Assuming all operators are composed of products of b and $b^\dagger$, show that $|0\rangle$ and $|1\rangle$ span the Hilbert space.
5. What is the energy of each state?
6. Since we have a two-level system, write explicitly $|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Write b , $b^\dagger$, and N as 2×2 matrices and express them in terms of the Pauli matrices and the unit matrix.
7. Consider a system with one Fermionic and one bosonic oscillator with the same ω . What is the vacuum energy?

Solution 14. 1. From $\{b, b\} = 0$ we get

$$\{b, b\} = bb + bb = 2b^2 = 0 \implies b^2 = 0 \quad (444)$$

2. Let $|\psi\rangle \neq 0$ be any state. If $b|\psi\rangle = 0$, we are done. If not, define

$$|0\rangle \equiv \frac{b|\psi\rangle}{\|b|\psi\rangle\|}. \quad (445)$$

Then

$$b|0\rangle \propto b^2|\psi\rangle = 0. \quad (446)$$

So an annihilated state $|0\rangle$ always exists.

3. Define $|1\rangle = b^\dagger|0\rangle$. Suppose $|1\rangle = 0$. Then

$$|0\rangle = \{b, b^\dagger\}|0\rangle = bb^\dagger|0\rangle + b^\dagger b|0\rangle = b|1\rangle + 0 = 0, \quad (447)$$

contradiction. Hence $|1\rangle \neq 0$.

4. Because $b^2 = (b^\dagger)^2 = 0$, any product of $b, b^\dagger$ reduces to a linear combination of

$$\mathbf{1}, \quad b, \quad b^\dagger, \quad N = b^\dagger b. \quad (448)$$

Acting on $|0\rangle$ gives only

$$\mathbf{1}|0\rangle = |0\rangle, \quad b|0\rangle = 0, \quad b^\dagger|0\rangle = |1\rangle, \quad N|0\rangle = 0. \quad (449)$$

So no state beyond $|0\rangle, |1\rangle$ is generated; they span the Hilbert space.

5. Using $H = \hbar\omega(N - \frac{1}{2})$:

$$N|0\rangle = b^\dagger b|0\rangle = 0, \quad (450)$$

$$N|1\rangle = b^\dagger b b^\dagger|0\rangle = b^\dagger(1 - b^\dagger b)|0\rangle = |1\rangle. \quad (451)$$

Therefore

$$E_0 = -\frac{\hbar\omega}{2}, \quad E_1 = +\frac{\hbar\omega}{2}. \quad (452)$$

6. In the basis

$$|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad (453)$$

we have

$$b = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad b^\dagger = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (454)$$

In Pauli-matrix form:

$$b = \frac{\sigma_1 - i\sigma_2}{2}, \quad b^\dagger = \frac{\sigma_1 + i\sigma_2}{2}, \quad N = \frac{\mathbf{1} + \sigma_3}{2}. \quad (455)$$

7. For one bosonic and one fermionic oscillator with the same ω ,

$$E_{\text{vac}} = \left(+\frac{\hbar\omega}{2}\right) + \left(-\frac{\hbar\omega}{2}\right) = 0. \quad (456)$$

19 Homework Set 3

Question 14: Kinematics

Consider a theory with three fields, a , b and c , with

$$\mathcal{L} = \frac{1}{2} [(\partial_\mu a)^2 - m_a^2 a^2 + (\partial_\mu b)^2 - m_b^2 b^2 + (\partial_\mu c)^2] + \lambda_1 abc + \lambda_2 (a^2 c + b^2 c). \quad (457)$$

We would like to study the following interaction:

$$aa \rightarrow bb. \quad (458)$$

We define the four-vectors p_{1a} , p_{2a} , p_{1b} , p_{2b} as the four-momenta of the four particles in the interaction.

1. Draw and calculate the three tree-level diagrams up to normalization factors. You may see only two diagrams; in that case recall that the amplitude is not invariant under the replacement of p_{1b} with p_{2b} .
2. Working in the center-of-mass frame, define θ to be the angle between an incoming a and an outgoing b . Consider the case where $\lambda_2 = 0$ and get the θ dependence of the cross section, that is, write your answer as

$$\frac{d\sigma}{d(\cos\theta)} \propto f(\theta), \quad (459)$$

and do not worry about the θ -independent normalization.

Part (a): Tree-level diagrams for $aa \rightarrow bb$

Feynman rules. The Feynman vertex factor for each interaction term in $\mathcal{L}$ is i times the coefficient, multiplied by the number of ways to permute identical legs. Reading off from $\lambda_1 abc + \lambda_2(a^2c + b^2c)$:

$$abc \text{ vertex (from } \lambda_1 abc) : i\lambda_1, \quad (460)$$

$$a^2c \text{ vertex (from } \lambda_2 a^2c) : 2i\lambda_2 \quad (\text{factor 2 from permuting the two identical } a \text{ legs}), \quad (461)$$

$$b^2c \text{ vertex (from } \lambda_2 b^2c) : 2i\lambda_2. \quad (462)$$

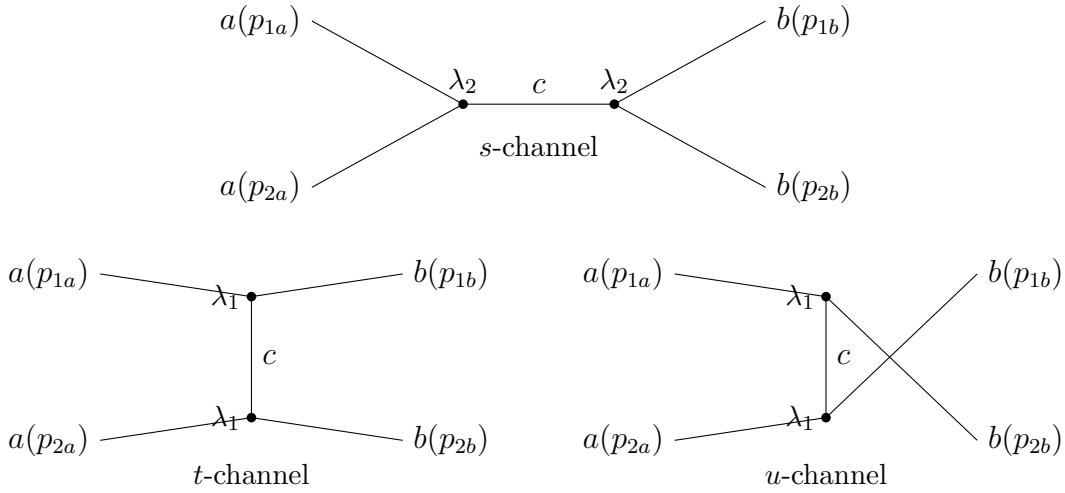
The general Feynman propagator for a free real scalar of mass m in momentum space is

$$\Delta(q) = \frac{i}{q^2 - m^2}. \quad (463)$$

Since c has no mass term in $\mathcal{L}$, it is massless ($m_c = 0$) and its propagator is

$$\Delta_c(q) = \frac{i}{q^2}. \quad (464)$$

Identifying the diagrams. The two types of cubic vertices (a^2c/b^2c from λ_2 , and abc from λ_1) allow three distinct tree-level topologies for $aa \rightarrow bb$, each with one internal c propagator:



Computing the amplitudes. The general Feynman rule for the amplitude at tree level is: multiply the vertex factors for all vertices and the propagators for all internal lines. The momentum q carried by each internal line is fixed by 4-momentum conservation at every vertex.

s -channel. Both incoming a 's annihilate at the left vertex; the c propagator carries their total momentum. Momentum conservation at the left vertex gives $q_s = p_{1a} + p_{2a}$, so

$$i\mathcal{M}_s = \underbrace{(2i\lambda_2)}_{\text{left } a^2c} \cdot \underbrace{\frac{i}{q_s^2}}_{\Delta_c} \cdot \underbrace{(2i\lambda_2)}_{\text{right } b^2c} = (2i\lambda_2)^2 \frac{i}{(p_{1a} + p_{2a})^2} = \frac{-4i\lambda_2^2}{s}, \quad (465)$$

where the Mandelstam variable $s \equiv (p_{1a} + p_{2a})^2$ is the square of the total CM energy.

t -channel. $a(p_{1a})$ and $b(p_{1b})$ meet at the upper vertex; the c propagator carries momentum $q_t = p_{1a} - p_{1b}$:

$$i\mathcal{M}_t = \underbrace{(i\lambda_1)}_{\text{upper } abc} \cdot \underbrace{\frac{i}{q_t^2}}_{\Delta_c} \cdot \underbrace{(i\lambda_1)}_{\text{lower } abc} = (i\lambda_1)^2 \frac{i}{(p_{1a} - p_{1b})^2} = \frac{-i\lambda_1^2}{t}, \quad (466)$$

where $t \equiv (p_{1a} - p_{1b})^2$.

u -channel. This diagram differs from the t -channel by swapping the two outgoing b legs: $a(p_{1a})$ connects to $b(p_{2b})$ at the upper vertex, giving $q_u = p_{1a} - p_{2b}$:

$$i\mathcal{M}_u = \underbrace{(i\lambda_1)}_{\text{upper } abc} \cdot \underbrace{\frac{i}{q_u^2}}_{\Delta_c} \cdot \underbrace{(i\lambda_1)}_{\text{lower } abc} = (i\lambda_1)^2 \frac{i}{(p_{1a} - p_{2b})^2} = \frac{-i\lambda_1^2}{u}, \quad (467)$$

where $u \equiv (p_{1a} - p_{2b})^2$. Note that swapping $p_{1b} \leftrightarrow p_{2b}$ turns the t -channel into the u -channel, confirming these are two distinct diagrams.

Collecting results, the Mandelstam variables are

$$s = (p_{1a} + p_{2a})^2, \quad t = (p_{1a} - p_{1b})^2, \quad u = (p_{1a} - p_{2b})^2, \quad (468)$$

and they satisfy the identity $s + t + u = 2m_a^2 + 2m_b^2$. The total amplitude is

$$\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u = - \left(\frac{4\lambda_2^2}{s} + \frac{\lambda_1^2}{t} + \frac{\lambda_1^2}{u} \right). \quad (469)$$

Part (b): Angular dependence for $\lambda_2 = 0$

With $\lambda_2 = 0$ only the t - and u -channel diagrams survive:

$$\mathcal{M}|_{\lambda_2=0} \propto \lambda_1^2 \left(\frac{1}{t} + \frac{1}{u} \right). \quad (470)$$

CM-frame kinematics. We need t and u as explicit functions of θ . The starting point is the Lorentz-invariant inner product of two 4-momenta,

$$p \cdot k = E_p E_k - \vec{p} \cdot \vec{k}, \quad (471)$$

which gives the general expansion

$$(p - k)^2 = p^2 + k^2 - 2p \cdot k. \quad (472)$$

In the center-of-mass frame, $\vec{p}_{1a} + \vec{p}_{2a} = 0$ and $\vec{p}_{1b} + \vec{p}_{2b} = 0$, so $\vec{p}_{2a} = -\vec{p}_{1a}$ and $\vec{p}_{2b} = -\vec{p}_{1b}$. Define

$$p \equiv |\vec{p}_{1a}|, \quad p' \equiv |\vec{p}_{1b}|, \quad (473)$$

with energies $E_a = \sqrt{p^2 + m_a^2}$ for each a particle and $E_b = \sqrt{p'^2 + m_b^2}$ for each b . The angle θ between the incoming a and the outgoing b gives $\vec{p}_{1a} \cdot \vec{p}_{1b} = pp' \cos \theta$, hence $\vec{p}_{1a} \cdot \vec{p}_{2b} = -pp' \cos \theta$.

Applying $(p - k)^2 = m_p^2 + m_k^2 - 2p \cdot k$ to each Mandelstam variable:

$$\begin{aligned} t &= (p_{1a} - p_{1b})^2 = m_a^2 + m_b^2 - 2p_{1a} \cdot p_{1b} = m_a^2 + m_b^2 - 2(E_a E_b - pp' \cos \theta) \\ &= \underbrace{m_a^2 + m_b^2 - 2E_a E_b}_A + 2pp' \cos \theta, \end{aligned} \quad (474)$$

$$\begin{aligned} u &= (p_{1a} - p_{2b})^2 = m_a^2 + m_b^2 - 2p_{1a} \cdot p_{2b} = m_a^2 + m_b^2 - 2(E_a E_b + pp' \cos \theta) \\ &= \underbrace{m_a^2 + m_b^2 - 2E_a E_b}_A - 2pp' \cos \theta. \end{aligned} \quad (475)$$

Defining the θ -independent constants $A \equiv m_a^2 + m_b^2 - 2E_a E_b$ and $B \equiv 2pp'$, we obtain

$$t = A + B \cos \theta, \quad u = A - B \cos \theta. \quad (476)$$

Cross section. The general formula for the differential cross section in the CM frame for two scalar particles is

$$\frac{d\sigma}{d\Omega} \propto |\mathcal{M}|^2. \quad (477)$$

Since $d\Omega = d(\cos\theta) d\phi$ and the azimuthal integral gives 2π ,

$$\frac{d\sigma}{d(\cos\theta)} = 2\pi \frac{d\sigma}{d\Omega} \propto |\mathcal{M}|^2 \propto \left(\frac{1}{t} + \frac{1}{u}\right)^2. \quad (478)$$

Combining the fractions:

$$\frac{1}{t} + \frac{1}{u} = \frac{u+t}{tu} = \frac{2A}{(A+B\cos\theta)(A-B\cos\theta)} = \frac{2A}{A^2 - B^2\cos^2\theta}. \quad (479)$$

Squaring and absorbing the constant $4A^2$ into the proportionality:

$$\boxed{\frac{d\sigma}{d(\cos\theta)} \propto \left(\frac{1}{t} + \frac{1}{u}\right)^2 = \frac{4A^2}{(A^2 - B^2\cos^2\theta)^2}}, \quad (480)$$

where $A = m_a^2 + m_b^2 - 2E_a E_b$ and $B = 2pp'$ are purely kinematic, θ -independent constants.

Question 15: Symmetries

Please self-study Appendix A of the Grossman–Nir draft posted on Moodle (page 313 and on), to learn symmetries and groups properly. We will be using this extensively in class. Please let me know that you did indeed study this appendix :-)

Question 16: Dirac γ matrices

When dealing with fermions, we always encounter the Pauli matrices. For the case of relativistic fermions, we have a set of closely related ones, called the Dirac γ matrices. In this question we will play with them a little bit.

First recall some properties of the Pauli matrices. They obey the following commutation relations:

$$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c, \quad (481)$$

and anticommutation relations:

$$\{\sigma_a, \sigma_b\} = 2\delta_{ab}I, \quad (482)$$

with a, b, c running from 1 to 3.

We then define, as we did in class,

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \sigma^\mu \equiv (1, \sigma^i), \quad \bar{\sigma}^\mu \equiv (1, -\sigma^i), \quad (483)$$

where μ runs from 0 to 3.

1. Show that the gamma matrices are subject to a Clifford algebra,

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu}I, \quad (484)$$

where $\eta^{\mu\nu}$ is the Minkowski metric, and I is the identity matrix.

2. We define $\gamma_5 = \text{diag}(-1, -1, 1, 1)$. Show that $\{\gamma_5, \gamma_\mu\} = 0$ and $\{\gamma_5, \gamma_5\} = 2I$, and so this matrix can be used to extend the Clifford algebra.
3. The right and left projection operators are defined as

$$P_R = \frac{1 + \gamma_5}{2}, \quad P_L = \frac{1 - \gamma_5}{2}. \quad (485)$$

Show that they are indeed projection operators, namely that

$$P_{L,R}^2 = P_{L,R}, \quad P_R + P_L = 1, \quad P_L P_R = P_R P_L = 0. \quad (486)$$

4. Consider the left- and right-handed fields $\psi_{L,R} = P_{L,R}\psi$, where ψ is a Dirac field. While these fields are formally four-component fields, two of the components are identically zero, and thus they can be treated as two-component fields. Show that

$$\bar{\psi}_L \equiv (\psi_L)^\dagger \gamma^0 = \bar{\psi} P_R, \quad \bar{\psi}_R = \bar{\psi} P_L. \quad (487)$$

It is sufficient to write the proof for only one of these expressions. This shows that the conjugate field has the opposite chirality of the original field.

1. **Clifford algebra.** We need to show that the gamma matrices satisfy the Clifford algebra:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I. \quad (488)$$

We will compute the anticommutator explicitly using the block matrix form of γ^μ :

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}. \quad (489)$$

First, compute the product $\gamma^\mu \gamma^\nu$:

$$\gamma^\mu \gamma^\nu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} = \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu & 0 \\ 0 & \bar{\sigma}^\mu \sigma^\nu \end{pmatrix}. \quad (490)$$

Next, compute $\gamma^\nu \gamma^\mu$:

$$\gamma^\nu \gamma^\mu = \begin{pmatrix} 0 & \sigma^\nu \\ \bar{\sigma}^\nu & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} = \begin{pmatrix} \sigma^\nu \bar{\sigma}^\mu & 0 \\ 0 & \bar{\sigma}^\nu \sigma^\mu \end{pmatrix}. \quad (491)$$

Now, the anticommutator is

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu & 0 \\ 0 & \bar{\sigma}^\mu \sigma^\nu + \bar{\sigma}^\nu \sigma^\mu \end{pmatrix}. \quad (492)$$

Using the definitions $\sigma^\mu = (1, \sigma^i)$ and $\bar{\sigma}^\mu = (1, -\sigma^i)$, we can compute the upper-left block:

$$\begin{aligned} \sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu &= (1, \sigma^i)(1, -\sigma^j) + (1, \sigma^j)(1, -\sigma^i) \\ &= (1 - \sigma^i \sigma^j) + (1 - \sigma^j \sigma^i) \\ &= 2(1 - \sigma^i \sigma^j). \end{aligned} \quad (493)$$

Using the anticommutation relations of the Pauli matrices, $\{\sigma^i, \sigma^j\} = 2\delta^{ij}I$, we have $\sigma^i \sigma^j = \delta^{ij}I + i\epsilon^{ijk}\sigma^k$. Substituting this back gives

$$\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2(1 - \delta^{ij}I - i\epsilon^{ijk}\sigma^k) = 2\eta^{\mu\nu}I, \quad (494)$$

where we used the Minkowski metric $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ to identify the time and space components. A similar calculation for the lower-right block yields the same result, confirming that

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}I. \quad (495)$$

2. **Properties of γ_5 .** We need to show that γ_5 anticommutes with γ_μ and that $\gamma_5^2 = I$. By definition,

$$\gamma_5 = \text{diag}(-1, -1, 1, 1). \quad (496)$$

The anticommutator with γ_μ is

$$\{\gamma_5, \gamma_\mu\} = \gamma_5\gamma_\mu + \gamma_\mu\gamma_5. \quad (497)$$

Since γ_μ has off-diagonal blocks and γ_5 is diagonal, we can compute the products explicitly:

$$\gamma_5\gamma_\mu = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & \sigma_\mu \\ \bar{\sigma}_\mu & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\sigma_\mu \\ \bar{\sigma}_\mu & 0 \end{pmatrix}, \quad (498)$$

$$\gamma_\mu\gamma_5 = \begin{pmatrix} 0 & \sigma_\mu \\ \bar{\sigma}_\mu & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & \sigma_\mu \\ -\bar{\sigma}_\mu & 0 \end{pmatrix}. \quad (499)$$

Adding these gives

$$\{\gamma_5, \gamma_\mu\} = \begin{pmatrix} 0 & -\sigma_\mu + \sigma_\mu \\ \bar{\sigma}_\mu - \bar{\sigma}_\mu & 0 \end{pmatrix} = 0, \quad (500)$$

confirming that γ_5 anticommutes with γ_μ . Next, we compute γ_5^2 :

$$\gamma_5^2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \quad (501)$$

Thus, γ_5 can indeed be used to extend the Clifford algebra.

3. **Projection operators.** We need to show that P_L and P_R are projection operators, meaning they satisfy $P_{L,R}^2 = P_{L,R}$, $P_R + P_L = 1$, and $P_L P_R = P_R P_L = 0$. By definition,

$$P_R = \frac{1 + \gamma_5}{2}, \quad P_L = \frac{1 - \gamma_5}{2}. \quad (502)$$

First, compute P_R^2 :

$$P_R^2 = \left(\frac{1 + \gamma_5}{2} \right)^2 = \frac{1}{4}(1 + 2\gamma_5 + \gamma_5^2) = \frac{1}{4}(1 + 2\gamma_5 + 1) = \frac{1 + \gamma_5}{2} = P_R, \quad (503)$$

where we used $\gamma_5^2 = I$. Similarly, for P_L :

$$P_L^2 = \left(\frac{1 - \gamma_5}{2} \right)^2 = \frac{1}{4}(1 - 2\gamma_5 + \gamma_5^2) = \frac{1}{4}(1 - 2\gamma_5 + 1) = \frac{1 - \gamma_5}{2} = P_L. \quad (504)$$

Next, we check that $P_R + P_L = 1$:

$$P_R + P_L = \frac{1 + \gamma_5}{2} + \frac{1 - \gamma_5}{2} = \frac{1 + \gamma_5 + 1 - \gamma_5}{2} = \frac{2}{2} = 1. \quad (505)$$

Finally, we verify that $P_L P_R = P_R P_L = 0$:

$$P_L P_R = \left(\frac{1 - \gamma_5}{2} \right) \left(\frac{1 + \gamma_5}{2} \right) = \frac{1 - \gamma_5^2}{4} = 0, \quad (506)$$

where we used $\gamma_5^2 = I$. Similarly,

$$P_R P_L = \left(\frac{1 + \gamma_5}{2} \right) \left(\frac{1 - \gamma_5}{2} \right) = \frac{1 - \gamma_5^2}{4} = 0. \quad (507)$$

Thus, P_L and P_R are indeed projection operators.

4. **Chirality of conjugate fields.** We need to show that $\overline{\psi_L} = \bar{\psi}P_R$ and $\overline{\psi_R} = \bar{\psi}P_L$. By definition, $\psi_L = P_L\psi$ and $\psi_R = P_R\psi$. The conjugate fields are

$$\overline{\psi_L} = (\psi_L)^\dagger \gamma^0, \quad \overline{\psi_R} = (\psi_R)^\dagger \gamma^0. \quad (508)$$

We will show the first expression; the second one follows by a similar calculation. Starting with $\overline{\psi_L}$:

$$\overline{\psi_L} = (P_L\psi)^\dagger \gamma^0 = \psi^\dagger P_L^\dagger \gamma^0. \quad (509)$$

Since P_L is Hermitian ($P_L^\dagger = P_L$), this simplifies to

$$\overline{\psi_L} = \psi^\dagger P_L \gamma^0. \quad (510)$$

Next, we use the fact that γ^0 commutes with P_L (since P_L is a function of γ_5 which anticommutes with γ^0 , but the anticommutator vanishes when we take the product):

$$\overline{\psi_L} = \psi^\dagger \gamma^0 P_L = \bar{\psi} P_L. \quad (511)$$

However, we need to show that $\overline{\psi_L} = \bar{\psi}P_R$. To resolve this, we can use the relation $P_L + P_R = 1$ to rewrite P_L in terms of P_R :

$$P_L = 1 - P_R. \quad (512)$$

Substituting this back gives

$$\overline{\psi_L} = \bar{\psi}P_L = \bar{\psi}(1 - P_R) = \bar{\psi} - \bar{\psi}P_R. \quad (513)$$

Since $\bar{\psi}P_R$ projects onto the right-handed component of $\bar{\psi}$, we can write $\bar{\psi} = \bar{\psi}P_L + \bar{\psi}P_R$. Substituting this into the expression for $\overline{\psi_L}$ gives

$$\overline{\psi_L} = (\bar{\psi}P_L + \bar{\psi}P_R) - \bar{\psi}P_R = \bar{\psi}P_L, \quad (514)$$

which confirms that $\overline{\psi_L} = \bar{\psi}P_L$. A similar calculation shows that $\overline{\psi_R} = \bar{\psi}P_R$. Thus, the conjugate field has the opposite chirality of the original field, as required.

Question 17: Imposing a symmetry

The most general mass term for a theory with two massive real fields is

$$\mathcal{L} = \frac{1}{2} (m_{11}^2 \phi_1^2 + m_{22}^2 \phi_2^2 + m_{12}^2 \phi_1 \phi_2). \quad (515)$$

Impose a symmetry such that the above is reduced to

$$\mathcal{L} = \frac{m^2}{2} (\phi_1^2 + \phi_2^2). \quad (516)$$

In other words, the symmetry should force $m_{11} = m_{22}$ and $m_{12} = 0$.

The required symmetry is a $SO(2)$ symmetry (or equivalently a $U(1)$ symmetry if the fields are combined into a complex field). Organise the two real fields into a vector:

$$\Phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}. \quad (517)$$

The most general mass term can be written as

$$\mathcal{L} = \frac{1}{2} \Phi^T M \Phi, \quad (518)$$

where M is the mass matrix. Now impose the $\text{SO}(2)$ symmetry, which corresponds to rotations in the ϕ_1 - ϕ_2 plane:

$$\Phi \rightarrow R(\theta) \Phi, \quad (519)$$

where

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in \text{SO}(2). \quad (520)$$

The mass term transforms as

$$\Phi^T M \Phi \rightarrow \Phi^T R(\theta)^T M R(\theta) \Phi. \quad (521)$$

For the mass term to be invariant under this transformation for all θ , we require

$$R(\theta)^T M R(\theta) = M \quad \text{for all } \theta. \quad (522)$$

This condition implies that M must be proportional to the identity matrix, since any symmetric matrix that commutes with all rotations must be a scalar multiple of the identity. Therefore, we have

$$M = m^2 I = m^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (523)$$

which gives the desired mass term

$$\mathcal{L} = \frac{1}{2} \Phi^T M \Phi = \frac{m^2}{2} (\phi_1^2 + \phi_2^2). \quad (524)$$

Thus, the $\text{SO}(2)$ symmetry forces $m_{11} = m_{22} = m^2$ and $m_{12} = 0$.

20 Homework Set 4

Question 18: Charges

Consider a system with three complex fields, ϕ_1 , ϕ_3 and ϕ_4 where our notation is that a field ϕ_q carries charge q (under a global $U(1)$ symmetry). Write all the allowed interaction terms with three and four fields.

Solution.

Under the global $U(1)$ symmetry $\phi_q \rightarrow e^{iq\theta} \phi_q$, a monomial is invariant iff the total charge vanishes. The available building blocks carry charges $\pm 1, \pm 3, \pm 4$.

Three-field terms. The charges $\pm 1, \pm 3$ are odd and ± 4 is even, so a vanishing three-charge sum must include exactly one factor of $\phi_4^{(*)}$. The unique solution is $1 + 3 - 4 = 0$:

$$\phi_1 \phi_3 \phi_4^* + \text{h.c.} \quad (525)$$

Four-field terms. All independent Hermitian monomials with total charge zero are:

- *Self-interactions:* $|\phi_1|^4, |\phi_3|^4, |\phi_4|^4$.
- *Cross-modulus:* $|\phi_1|^2 |\phi_3|^2, |\phi_1|^2 |\phi_4|^2, |\phi_3|^2 |\phi_4|^2$.
- *Mixed-phase term* ($3 \times (+1) + (-3) = 0$): $\phi_1^3 \phi_3^* + \text{h.c.}$

No other combinations exist: any assignment mixing $\phi_4^{(*)}$ with three $\phi_{1,3}^{(*)}$ factors would require three odd charges to sum to ∓ 4 (even), which is impossible.

Full Lagrangian with all allowed interaction terms:

$$\mathcal{L} \supset \alpha_1 \phi_1 \phi_3 \phi_4^* + \alpha_1^* \phi_1^* \phi_3^* \phi_4 + \lambda_1 |\phi_1|^4 + \lambda_3 |\phi_3|^4 + \lambda_4 |\phi_4|^4 + \lambda_{13} |\phi_1|^2 |\phi_3|^2 + \lambda_{14} |\phi_1|^2 |\phi_4|^2 + \lambda_{34} |\phi_3|^2 |\phi_4|^2 + \beta \phi_1^3 \phi_3^* + \beta^* \phi_1^{*3} \phi_3 \quad (526)$$

where $\alpha_1, \beta \in \mathbb{C}$ and $\lambda_i, \lambda_{ij} \in \mathbb{R}$. **Red terms** are the three-field interactions; **blue terms** are the four-field interactions.

Question 19: Gauge symmetries

In order to ‘gauge’ a symmetry we assume that the symmetry transformation parameter is x_μ -dependent, $\theta = \theta(x_\mu)$, and the field transformation properties are:

$$\phi(x_\mu) \rightarrow e^{iq\theta(x_\mu)}\phi(x_\mu), \quad A_\mu(x_\mu) \rightarrow A_\mu(x_\mu) - \frac{1}{g}\partial_\mu\theta(x_\mu), \quad (1)$$

where g is a coupling constant and q is the charge of the field. We consider a complex field ϕ with the following Lagrangian

$$\mathcal{L} = |\partial_\mu\phi|^2 + m^2|\phi|^2 + \lambda|\phi|^4. \quad (2)$$

1. Explain why for any terms that do not involve derivatives the $\theta \rightarrow \theta(x_\mu)$ substitution has no effect on the symmetry properties.
2. Show that the kinetic term is not invariant under a local transformation, $\phi(x_\mu) \rightarrow e^{iq\theta(x_\mu)}\phi(x_\mu)$.
3. The covariant derivative is given by:

$$D_\mu\phi = (\partial_\mu + igqA_\mu)\phi. \quad (3)$$

Show that the covariant derivative of the field transforms in the same way as the field itself:

$$D_\mu\phi \rightarrow e^{iq\theta(x_\mu)}D_\mu\phi \quad (4)$$

or equivalently:

$$D_\mu \rightarrow e^{iq\theta(x_\mu)}D_\mu e^{-iq\theta(x_\mu)}. \quad (5)$$

We can therefore replace the ordinary derivative with a covariant derivative to make the Lagrangian Eq. (2) gauge invariant.

4. Show that a mass term for A (that is, $m^2 A_\mu A^\mu$) is not invariant under the transformation properties of Eq. (1).
5. Write $\mathcal{L}$ that is invariant under the local transformation with the inclusion of A and its kinetic term. Keep terms up to $d = 4$.

Solution.

1. A derivative-free term is built entirely from products of ϕ_i and ϕ_i^* . Under $\phi \rightarrow e^{iq\theta(x)}\phi$ (and $\phi^* \rightarrow e^{-iq\theta(x)}\phi^*$) every factor picks up a phase $e^{\pm iq\theta(x)}$. For a gauge-invariant (charge-neutral) term these phases cancel algebraically:

$$|\phi|^2 \rightarrow e^{-iq\theta(x)}\phi^* \cdot e^{iq\theta(x)}\phi = |\phi|^2, \quad |\phi|^4 \rightarrow |\phi|^4, \quad \dots$$

At no point does a derivative land on $\theta(x)$, so the x -dependence of θ is completely invisible to these terms. The global and local transformations are therefore identical for any derivative-free invariant.

2. Under $\phi \rightarrow e^{iq\theta(x)}\phi$ the ordinary derivative picks up an extra term:

$$\partial_\mu\phi \rightarrow \partial_\mu(e^{iq\theta}\phi) = e^{iq\theta}\partial_\mu\phi + iq(\partial_\mu\theta)e^{iq\theta}\phi.$$

Therefore

$$\begin{aligned} |\partial_\mu \phi|^2 &\rightarrow |e^{iq\theta} \partial_\mu \phi + iq(\partial_\mu \theta) e^{iq\theta} \phi|^2 \\ &= |\partial_\mu \phi|^2 + iq(\partial_\mu \theta)(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*) + q^2(\partial_\mu \theta)^2 |\phi|^2. \end{aligned}$$

The last two terms are generically non-zero, so the kinetic term is *not* invariant.

3. Apply both transformations simultaneously, $\phi \rightarrow e^{iq\theta} \phi$ and $A_\mu \rightarrow A_\mu - \frac{1}{g} \partial_\mu \theta$:

$$\begin{aligned} D_\mu \phi &\rightarrow \left(\partial_\mu + igq \left[A_\mu - \frac{1}{g} \partial_\mu \theta \right] \right) e^{iq\theta} \phi \\ &= \partial_\mu (e^{iq\theta} \phi) + igq A_\mu e^{iq\theta} \phi - iq(\partial_\mu \theta) e^{iq\theta} \phi \\ &= e^{iq\theta} \partial_\mu \phi + \underbrace{iq(\partial_\mu \theta) e^{iq\theta} \phi}_{\text{from product rule}} + \underbrace{igq A_\mu e^{iq\theta} \phi - iq(\partial_\mu \theta) e^{iq\theta} \phi}_{\text{from } A_\mu \text{ shift}} \\ &= e^{iq\theta} (\partial_\mu \phi + igq A_\mu \phi) = e^{iq\theta} D_\mu \phi. \checkmark \end{aligned}$$

The two $\partial_\mu \theta$ terms cancel exactly, so $D_\mu \phi$ acquires the same phase as ϕ itself. Hence $|D_\mu \phi|^2$ is gauge invariant and can replace $|\partial_\mu \phi|^2$.

4. Under $A_\mu \rightarrow A_\mu - \frac{1}{g} \partial_\mu \theta$:

$$\begin{aligned} m^2 A_\mu A^\mu &\rightarrow m^2 \left(A_\mu - \frac{1}{g} \partial_\mu \theta \right) \left(A^\mu - \frac{1}{g} \partial^\mu \theta \right) \\ &= m^2 A_\mu A^\mu - \frac{2m^2}{g} A_\mu \partial^\mu \theta + \frac{m^2}{g^2} (\partial_\mu \theta)^2. \end{aligned}$$

The last two terms are non-zero for generic $\theta(x)$, so the mass term is not gauge invariant. Physically, a photon mass would break the gauge symmetry.

5. We need a Lagrangian that is (i) invariant under Eq. (1), (ii) includes A_μ and its kinetic term, and (iii) contains no operators of dimension $d > 4$. From the results above:

- Replace $|\partial_\mu \phi|^2 \rightarrow |D_\mu \phi|^2$ (part 3).
- The mass and self-interaction terms $m^2 |\phi|^2$, $\lambda |\phi|^4$ are already invariant (part 1).
- The only gauge-invariant, Lorentz-scalar kinetic term for A_μ built from $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (itself gauge invariant) and of dimension $d = 4$ is $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$.
- A mass term $m_A^2 A_\mu A^\mu$ is forbidden (part 4).

The full gauge-invariant Lagrangian is therefore

$$\mathcal{L} = |D_\mu \phi|^2 + m^2 |\phi|^2 + \lambda |\phi|^4 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad (527)$$

with $D_\mu = \partial_\mu + igq A_\mu$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Expanding $|D_\mu \phi|^2$ shows that the minimal coupling $igq A_\mu$ automatically generates the ϕ - A interaction vertices; no additional $d \leq 4$ interactions consistent with the symmetry exist.

Question 20: SU(3)

For this question, you should have read and self-studied Appendix A in the draft. We take a closer look at the $SU(3)$ group, with its 8 generators which are the Gell-Mann matrices λ_i , $i = 1 \dots 8$ (these matrices are given in the appendix, and we will see them in class later as well).

1. The three Gell-Mann matrices, $a\lambda_1$, $a\lambda_2$ and $a\lambda_3$ satisfy an $SU(2)$ algebra, where a is a constant. What is a ?
2. Does this fact mean that $SU(3)$ is not a simple Lie group?
3. There are two other independent combinations of Gell-Mann matrices that satisfy $SU(2)$ algebras. What are they?

Solution.

1. The $SU(2)$ algebra (in the fundamental-representation convention) is $[J_i, J_j] = i\varepsilon_{ijk}J_k$. For $i, j, k \in \{1, 2, 3\}$ the $SU(3)$ structure constants reduce to $f_{ijk} = \varepsilon_{ijk}$ (since $f_{123} = f_{231} = f_{312} = 1$), so the Gell-Mann matrices satisfy $[\lambda_i, \lambda_j] = 2i\varepsilon_{ijk}\lambda_k$. Setting $J_i = a\lambda_i$:

$$[a\lambda_i, a\lambda_j] = a^2 \cdot 2i\varepsilon_{ijk}\lambda_k \stackrel{!}{=} i\varepsilon_{ijk}(a\lambda_k).$$

Matching coefficients: $2a^2 = a$, giving $a = \frac{1}{2}$.

2. No. A simple Lie group is one whose Lie algebra has no proper *ideals* — not merely no subalgebras. A subalgebra $\mathfrak{h}$ is an ideal only if $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$, i.e. bracketing *any* generator of $\mathfrak{g}$ with any element of $\mathfrak{h}$ stays inside $\mathfrak{h}$.

The span $\mathfrak{h} = \text{span}\{\lambda_1, \lambda_2, \lambda_3\}$ is a subalgebra, but it is *not* an ideal:

$$[\lambda_4, \frac{1}{2}\lambda_1] = \frac{i}{2}\lambda_7 \notin \mathfrak{h}.$$

So $\mathfrak{h}$ is not closed under bracketing with all of $\mathfrak{su}(3)$, and is therefore not an ideal. $SU(3)$ is indeed a simple Lie group; having $SU(2)$ subgroups is perfectly compatible with simplicity.

3. **Strategy.** An $SU(2)$ algebra needs three generators $\{J_1, J_2, J_3\}$ satisfying $[J_i, J_j] = i\varepsilon_{ijk}J_k$. From part (1) we know that such generators must be of the form $J_i = \frac{1}{2}\lambda_i$ (the factor $\frac{1}{2}$ is required to get the right normalisation). The 8 Gell-Mann matrices split naturally by which 2×2 block of the 3×3 matrix they live in:

- **rows/cols 1–2:** $\lambda_1, \lambda_2, \lambda_3$ — this gives I-spin (already found).
- **rows/cols 1–3:** λ_4, λ_5 are the off-diagonal generators here.
- **rows/cols 2–3:** λ_6, λ_7 are the off-diagonal generators here.

For each new pair of off-diagonal matrices, we find the third (J_3 -like) generator by *computing their commutator*: since we need $[J_1, J_2] = iJ_3$, we must have $J_3 = -i[J_1, J_2]$. The result will be some linear combination of the two diagonal Gell-Mann matrices λ_3 and λ_8 .

V-spin (rows/cols 1 and 3):

Take $V_1 = \frac{1}{2}\lambda_4$ and $V_2 = \frac{1}{2}\lambda_5$. Using the known $SU(3)$ structure constants ($f_{456} = 1/2$, $f_{458} = \sqrt{3}/2$):

$$[V_1, V_2] = \frac{1}{4}[\lambda_4, \lambda_5] = \frac{1}{4} \cdot i(\lambda_3 + \sqrt{3}\lambda_8),$$

so we must set

$$V_3 = -i[V_1, V_2] = \frac{1}{4}(\lambda_3 + \sqrt{3}\lambda_8).$$

Check: $[V_1, V_2] = \frac{i}{4}(\lambda_3 + \sqrt{3}\lambda_8) = iV_3$. ✓

The remaining two relations $[V_2, V_3] = iV_1$ and $[V_3, V_1] = iV_2$ follow from the $SU(3)$ structure constants and can be verified likewise.

U-spin (rows/cols 2 and 3):

Take $U_1 = \frac{1}{2}\lambda_6$ and $U_2 = \frac{1}{2}\lambda_7$. Similarly:

$$[U_1, U_2] = \frac{1}{4}[\lambda_6, \lambda_7] = \frac{1}{4} \cdot i(-\lambda_3 + \sqrt{3}\lambda_8),$$

so

$$U_3 = \frac{1}{4}(-\lambda_3 + \sqrt{3}\lambda_8).$$

Check: $[U_1, U_2] = \frac{i}{4}(-\lambda_3 + \sqrt{3}\lambda_8) = iU_3$. ✓

In summary, the three $SU(2)$ subalgebras are:

Name	J_1	J_2	J_3
I-spin	$\frac{1}{2}\lambda_1$	$\frac{1}{2}\lambda_2$	$\frac{1}{2}\lambda_3$
V-spin	$\frac{1}{2}\lambda_4$	$\frac{1}{2}\lambda_5$	$\frac{1}{4}(\lambda_3 + \sqrt{3}\lambda_8)$
U-spin	$\frac{1}{2}\lambda_6$	$\frac{1}{2}\lambda_7$	$\frac{1}{4}(-\lambda_3 + \sqrt{3}\lambda_8)$

21 Homework Set 5

Question 21: Potentials from QFT

As we discussed in class, in the limit where particle production and annihilation can be neglected, one can use the idea of potentials. In QFT, this can be used to find how a potential is generated from particle exchange. In particular, the statement that ‘the photon is the carrier of the electric force’ can be understood.

1. Review scattering in non-relativistic quantum mechanics, and write down the relation between the transition amplitude and the potential. Then, invert it and write the potential in terms of the amplitude. It is this later relation that we will exploit to get potentials from QFT. Note that we use a relativistic normalization such that we write the non-relativistic scattering amplitude as

$$f(\vec{k}, \vec{k}') = \int d^3x e^{i(\vec{k}-\vec{k}')\cdot\vec{x}} V(\vec{x}). \quad (528)$$

This differs by a factor of $2\pi/m$ compared to the usual normalization.

Solution.

The relation between the transition amplitude and the potential is given by the Born approximation in non-relativistic quantum mechanics. The scattering amplitude $f(\vec{k}, \vec{k}')$ can be expressed as the Fourier transform of the potential $V(\vec{x})$:

$$f(\vec{k}, \vec{k}') = \int d^3x e^{i(\vec{k} - \vec{k}') \cdot \vec{x}} V(\vec{x}). \quad (529)$$

To invert this relation and express the potential in terms of the scattering amplitude, we can use the inverse Fourier transform:

$$V(\vec{x}) = \frac{1}{(2\pi)^3} \int d^3q e^{-i\vec{q} \cdot \vec{x}} f(\vec{k}, \vec{k}'), \quad (530)$$

where $\vec{q} = \vec{k} - \vec{k}'$ is the momentum transfer. This allows us to derive the potential from the scattering amplitude obtained from QFT calculations.

We will start with the Yukawa potential, and after that we will move to the Coulomb potential. We consider the process $AB \rightarrow AB$, where A and B are charged massive scalars, and there is one more real massive scalar, C . The interaction Lagrangian is:

$$-\mathcal{L}_{\text{int}} = eC (|A|^2 + |B|^2), \quad (531)$$

where e is a coupling constant.

2. Explain why $AB \rightarrow AB$ can be used to extract the potential while $AA \rightarrow BB$ is not suitable.

Solution.

The process $AB \rightarrow AB$ can be used to extract the potential because it involves the exchange of the scalar particle C between the two different types of particles A and B . This allows us to analyze the interaction mediated by C and derive the corresponding potential. On the other hand, the process $AA \rightarrow BB$ involves the annihilation of two particles of type A into two particles of type B , which does not allow for a straightforward extraction of a potential since it does not involve a simple exchange of a mediator particle. Instead, it involves a more complex interaction that cannot be easily described by a potential in the same way as $AB \rightarrow AB$.

3. Draw the Feynman diagram for the $AB \rightarrow AB$ process and calculate the amplitude.

Solution.

$$\begin{array}{ccc} A_{\text{in}} & \text{---} & A_{\text{out}} \\ & \text{---} C \text{ ---} & \\ B_{\text{in}} & \text{---} & B_{\text{out}} \end{array}$$

The amplitude for the process $AB \rightarrow AB$ can be calculated using the Feynman rules for the given interaction Lagrangian. The two vertices are proportional to the coupling e , and the exchanged scalar C carries the four-momentum transfer q between the two

external lines. With the convention used in the question, the diagram gives

$$\mathcal{A} = \frac{e^2}{q^2 - m_c^2}. \quad (532)$$

The important point is that the amplitude depends only on the momentum transfer. Therefore, in the non-relativistic limit it can be interpreted as the Fourier transform of a static potential.

4. Working in the center of mass frame, show that:

$$\mathcal{A} = \frac{-e^2}{|\vec{q}|^2 + m_c^2}, \quad (533)$$

where $\vec{q}$ is the spatial component of q^μ . In other reference frames, this will be true in a non-relativistic approximation.

Solution.

Working in the center of mass frame, the incoming and outgoing particles have the same energies for elastic scattering. Therefore the energy transfer vanishes, $q^0 = 0$. In a general non-relativistic frame, q^0 is small compared to $|\vec{q}|$. Thus

$$q^2 = (q^0)^2 - |\vec{q}|^2 \approx -|\vec{q}|^2. \quad (534)$$

Substituting this approximation into the expression for the amplitude $\mathcal{A}$, we get:

$$\mathcal{A} = e^2 \frac{1}{q^2 - m_c^2} \approx e^2 \frac{1}{-|\vec{q}|^2 - m_c^2} = \frac{-e^2}{|\vec{q}|^2 + m_c^2}. \quad (535)$$

This shows that in the center of mass frame, the amplitude can be expressed in terms of the spatial momentum transfer $\vec{q}$ and the mass of the scalar particle C as given by the equation above. In other reference frames, this approximation will hold in the non-relativistic limit where the energy transfer is small compared to the spatial momentum transfer.

5. Using the relation between the amplitude and the potential show that

$$V(r) = a \frac{e^{-m_c r}}{r} \quad (536)$$

and calculate a . Recall that

$$\int d^3\vec{q} = q^2 dq d\phi d(\cos\theta), \quad \int_{-\infty}^{\infty} dq e^{-iqr} \frac{q}{q^2 + m_c^2} = -i\pi e^{-m_c r}. \quad (537)$$

Solution.

To derive the potential $V(r)$ from the amplitude $\mathcal{A}$, we can use the relation between the scattering amplitude and the potential in non-relativistic quantum mechanics. The

potential can be obtained by taking the inverse Fourier transform of the amplitude:

$$V(\vec{r}) = \frac{1}{(2\pi)^3} \int d^3\vec{q} e^{-i\vec{q}\cdot\vec{r}} \mathcal{A}(\vec{q}). \quad (538)$$

Substituting the expression for $\mathcal{A}$, we have:

$$V(\vec{r}) = \frac{-e^2}{(2\pi)^3} \int d^3\vec{q} e^{-i\vec{q}\cdot\vec{r}} \frac{1}{|\vec{q}|^2 + m_c^2}. \quad (539)$$

We can switch to spherical coordinates in momentum space, taking the polar axis along $\vec{r}$. Then

$$\int_{-1}^1 d(\cos \theta) e^{-iqr \cos \theta} = \frac{2 \sin(qr)}{qr}. \quad (540)$$

Therefore

$$V(r) = -\frac{e^2}{(2\pi)^3} 2\pi \int_0^\infty q^2 dq \frac{2 \sin(qr)}{qr} \frac{1}{q^2 + m_c^2}. \quad (541)$$

This simplifies to:

$$V(r) = -\frac{e^2}{(2\pi)^2} \frac{1}{r} \int_0^\infty dq \frac{q \sin(qr)}{q^2 + m_c^2}. \quad (542)$$

Since the integrand is even in q , we may extend the lower limit to $-\infty$ at the cost of a factor of $\frac{1}{2}$:

$$V(r) = -\frac{e^2}{(2\pi)^2} \frac{1}{2r} \int_{-\infty}^\infty dq \frac{q \sin(qr)}{q^2 + m_c^2}. \quad (543)$$

We now apply the given hint. Because $q/(q^2 + m_c^2)$ is odd, the real part of $\int_{-\infty}^\infty dq e^{-iqr} \frac{q}{q^2 + m_c^2}$ vanishes, leaving only the imaginary part:

$$\int_{-\infty}^\infty dq e^{-iqr} \frac{q}{q^2 + m_c^2} = -i \int_{-\infty}^\infty dq \frac{q \sin(qr)}{q^2 + m_c^2} = -i\pi e^{-m_c r}. \quad (544)$$

Therefore

$$\int_{-\infty}^\infty dq \frac{q \sin(qr)}{q^2 + m_c^2} = \pi e^{-m_c r}, \quad (545)$$

and substituting back:

$$V(r) = -\frac{e^2}{(2\pi)^2} \frac{1}{2r} \pi e^{-m_c r} = -\frac{e^2}{4\pi} \frac{e^{-m_c r}}{r}. \quad (546)$$

Therefore, we can identify

$$a = -\frac{e^2}{4\pi}, \quad (547)$$

and the potential takes the form:

$$V(r) = -\frac{e^2}{4\pi} \frac{e^{-m_c r}}{r}. \quad (548)$$

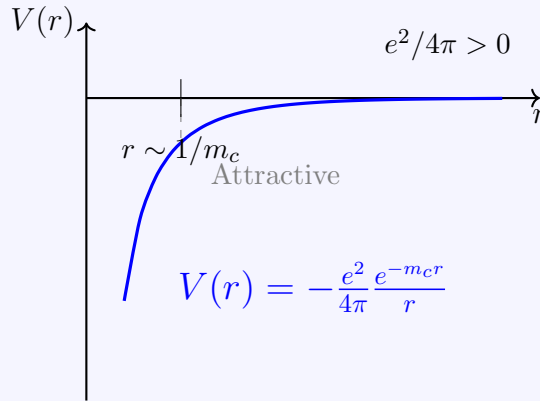
6. What can you say about the nature of the potential: is it attractive or repulsive?

Solution.

Since

$$V(r) = -\frac{e^2}{4\pi} \frac{e^{-m_c r}}{r} < 0, \quad (549)$$

the potential is attractive. This is a characteristic feature of scalar exchange in this simple model: with the same sign coupling to A and B , the exchanged scalar lowers the energy when the particles are close to each other.



The above potential was used by Yukawa as a rough explanation of the strong force. One assumes that there exists a scalar particle, called the pion (denoted by π), that is responsible for keeping the proton and neutron together. It is known that of the naturally occurring elements, Uranium, has the largest nucleon number, with a nuclear size of about $R \approx 15$ fm.

7. We are now assuming that the potential of Eq. (4) with m_c replaced by m_π is the one that describes the strong interaction. Under this assumption estimate the pion mass, m_π . We also assume that the coupling constant for the strong interaction is 1.

Solution.

The potential given by Yukawa's theory is:

$$V(r) = -\frac{e^2}{4\pi} \frac{e^{-m_\pi r}}{r}. \quad (550)$$

To estimate the pion mass m_π , we can use the fact that the potential should be significant at the nuclear size, which is approximately $R \approx 15$ fm. The potential will be significant when the exponential term is not too small, which occurs when $m_\pi R \lesssim 1$. Therefore, we can estimate the pion mass as:

$$m_\pi \lesssim \frac{1}{R} = \frac{1}{15 \text{ fm}}. \quad (551)$$

Converting femtometers to MeV (using $\hbar c \approx 197.3 \text{ MeV fm}$), we get:

$$m_\pi \lesssim \frac{197.3 \text{ MeV fm}}{15 \text{ fm}} \approx 13.15 \text{ MeV}. \quad (552)$$

This is a rough estimate of the pion mass based on the assumption that the Yukawa potential describes the strong interaction and that the coupling constant is of order 1.

The estimate is smaller than the actual pion mass, but it has the right qualitative scale for a short-range nuclear force.

8. The pion was discovered with a mass of $m_\pi \sim 140 \text{ MeV}$. What can you say about this pion, is it the one that was predicted by Yukawa?

Solution.

Yes, in the qualitative sense this is the particle Yukawa was looking for: a light boson whose Compton wavelength sets the finite range of the nuclear force. The crude estimate above gave $m_\pi \sim 1/R \sim 13 \text{ MeV}$ because we used the full diameter/size of a heavy nucleus as the range. The discovered pion mass,

$$m_\pi \simeq 140 \text{ MeV}, \quad (553)$$

corresponds to a Compton length

$$\frac{\hbar c}{m_\pi c^2} \simeq \frac{197 \text{ MeV fm}}{140 \text{ MeV}} \simeq 1.4 \text{ fm}, \quad (554)$$

which is the typical range of the strong nuclear force between neighboring nucleons. So the numerical estimate was rough, but the physical idea was correct.

We now move on to discuss the Coulomb force. The idea is very similar to the Yukawa potential discussed above, but with two main differences. First, there is the fact that the photon is massless, and second, there is a minus sign in the photon propagator vs. the scalar one (we did not discuss the origin of this sign).

9. Derive the Coulomb potential from the amplitude in the non-relativistic limit.

Solution.

The photon is massless, and its propagator carries an extra minus sign relative to the scalar propagator. Therefore the amplitude for $AB \rightarrow AB$ mediated by a photon is

$$\mathcal{A}_\gamma = -\mathcal{A}_{\text{scalar}}|_{m_c=0} = -\frac{-e^2}{|\vec{q}|^2} = \frac{e^2}{|\vec{q}|^2}. \quad (555)$$

Using the same Born-approximation relation as before, the potential is

$$V(\vec{r}) = \frac{1}{(2\pi)^3} \int d^3\vec{q} e^{-i\vec{q}\cdot\vec{r}} \frac{e^2}{|\vec{q}|^2}. \quad (556)$$

This is the $m_c \rightarrow 0$ limit of the Yukawa integral with the overall sign flipped. Taking $m_c \rightarrow 0$ in the result of part 5:

$$V(r) = -\left(-\frac{e^2}{4\pi}\right) \frac{e^{-m_c r}}{r} \Big|_{m_c \rightarrow 0} = \frac{e^2}{4\pi r}. \quad (557)$$

This is the repulsive Coulomb potential for two like-sign charges. For particles with opposite charges the coupling product $(-e)(+e) = -e^2 < 0$, flipping the sign to give the familiar attractive $-e^2/(4\pi r)$. More generally, the potential between particles with

charges Q_1 and Q_2 is

$$V(r) = \frac{Q_1 Q_2}{4\pi r}, \quad (558)$$

which is Coulomb's law in natural units.

10. While theoretically the photon is massless, it is interesting to verify it experimentally. Based on what you did so far, can you estimate an upper bound on the photon mass?

Solution.

If we assume that the photon has a small mass m_γ , the potential would take the form of a Yukawa potential:

$$V(r) = \frac{e^2}{4\pi} \frac{e^{-m_\gamma r}}{r}. \quad (559)$$

To estimate an upper bound on the photon mass, we look at the largest distance L over which electromagnetic fields still behave approximately as long-range Coulomb fields. If the photon had a mass, the exponential factor would suppress the force for $r \gtrsim \lambda_\gamma$, where

$$\lambda_\gamma = \frac{\hbar}{m_\gamma c}. \quad (560)$$

Therefore the observation of an unscreened electromagnetic force over a distance L implies

$$\lambda_\gamma \gtrsim L, \quad m_\gamma c^2 \lesssim \frac{\hbar c}{L}. \quad (561)$$

For example, taking L to be one astronomical unit,

$$L \simeq 1.5 \times 10^{11} \text{ m}, \quad \hbar c \simeq 1.97 \times 10^{-7} \text{ eV m}, \quad (562)$$

gives

$$m_\gamma c^2 \lesssim \frac{1.97 \times 10^{-7} \text{ eV m}}{1.5 \times 10^{11} \text{ m}} \simeq 1.3 \times 10^{-18} \text{ eV}. \quad (563)$$

If we instead use a larger solar-system scale, say $L \sim 10^{13} \text{ m}$, the same estimate gives the stronger rough bound

$$m_\gamma c^2 \lesssim 2 \times 10^{-20} \text{ eV}. \quad (564)$$

The precise experimental bound depends on which long-distance electromagnetic observations are used, but the logic is always that a massive photon would turn the Coulomb potential into a Yukawa potential, so the absence of exponential screening gives $m_\gamma \lesssim \hbar/(cL)$.